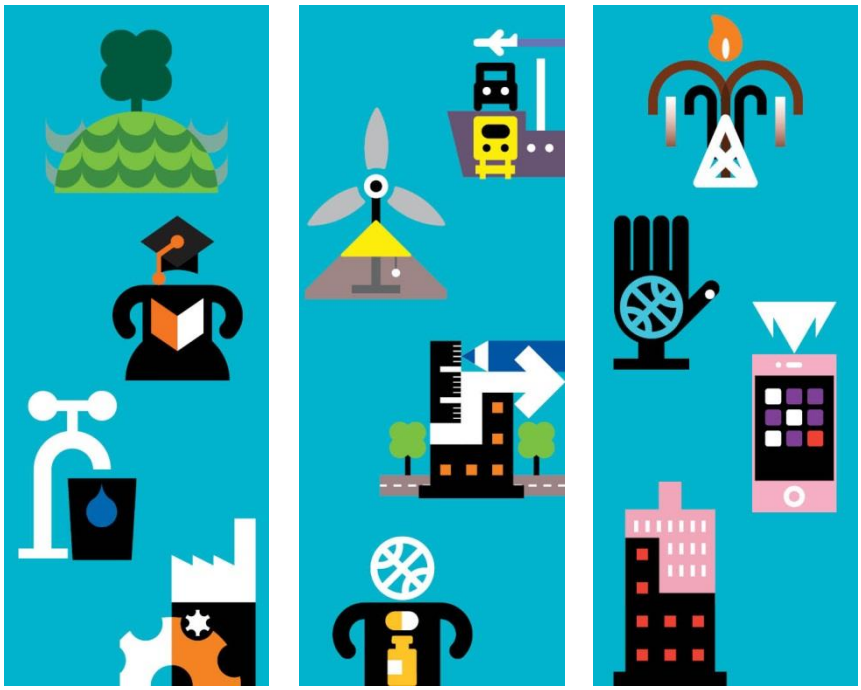


Volume Two: Evaluation of DFID Online Portals and Repositories

Appendices

July 2016

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Volume Two: Evaluation of DFID Online Portals and Repositories

Appendices

July 2016

Mott MacDonald, 10 Fleet Place, London EC4M 7RB, United Kingdom
T +44 (0)20 7651 0300 **F** +44 (0)20 7248 2698 **W** www.mottmac.com

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Appendix A. The Market Research

A.1 Detailed Description of method used

Rationale

The objectives of the market research were:

- To examine how users of research portals and repositories might be segmented;
- To examine the attitudes of Intended Users of research portals and repositories towards using the internet as a channel for finding research evidence;
- To establish the relationship Intended Users of research portals and repositories have with the portals and repositories under investigation and what they value about their services.

In addition to these main objectives, the market research also explores some of the assumptions and hypotheses in the Theory of Change. The questionnaire (see Annex A.2.2) covered the following topic areas: demographics, occupation, ICT context, participants' use of evidence, and specifically how they find, access and use research evidence online.

Developing the Method

The online questionnaire was initially drafted based on reviewing questions used in research on related topics, drawing on the evaluation questions and Ellis framework for structure. It was developed through an extensive process of consultation within the evaluation team (including our Advisory Group and DFID Management teams), with our in-house survey experts and through piloting.

The questionnaire went through several iterations, based on piloting at Inception phase and Stage 2. Feedback from these pilots resulted in improvements being made to the structure of the questionnaire, as well as our definition of evidence and research evidence, whilst still keeping it as short as possible to minimise the risk of respondents dropping out before the end. In addition, early findings from the first country case study, Ghana, prompted the revision of some questions.

The questionnaire was distributed using the industry standard software, Snap Professional[1].

Recruitment and sampling

Our population of interest was development actors worldwide, but with particular emphasis on trying to reach policy-makers in the South, since these groups are less well researched in the literature. Within this broad group we identified[2] the following selection criteria for respondents:

- Has sufficient internet access to enable browsing of websites;
- Is interested in or already seeking research evidence to inform their work;
- Fluent in English;
- Working in one of seven target categories of policy actor[3]. See Table 1 below:

[1] In addition to the creation of professional looking, user-friendly and intuitive questionnaires, the use of our in-house expertise and approved survey software provides a high-level of data security. Online questionnaire hosting via Snap Webhost is ISO 27001 compliant. See http://www.snapsurveys.com/survey-software/security-accessibility-and-professional-outline/.

[2] Based on the original long list provided by DFID in the terms of reference, the literature review, consultation with the Advisory Group and our pilot online questionnaire.

[3] While the seven categories were the focus of our recruitment, the sampling method used resulted in a broader range of

Table 1: Target Intended User groups for the Market Research

South & North	South only
Development worker in civil society (e.g. employed by a national or international non-governmental organisation, or community based organisation)	Elected member of local or national government (e.g. Member of Parliament, member of regional/sub-national legislature or council, Councillor/local council member, government minister)
Academic/Researcher (e.g. researcher or postgraduate student based in a research institute, university, or think tank) with an interest relevant to poverty alleviation or social change in the South	Civil or public servant
Development Consultant	Knowledge broker/intermediary for policy makers[4] (e.g. Parliamentary committee clerk, Parliamentary researcher, Parliamentary librarian, assistant/secretary to Member of Parliament, Government departmental librarian)
	Media professional (e.g. journalist, editor, commentator for online, print or broadcast media)

This table has been revised slightly since it originally featured in our Inception report to avoid any potential overlap between two of our categories. These were originally:

- National Legislature or elected member of local government (e.g. Member of Parliament, member of regional/sub-national legislature or council, Councillor/local council member)
- Executive branch of national or local government (e.g. civil servant, ministerial advisor, local council worker, government minister)

Having concluded during the Inception Phase that random sampling methods would not be practical in this context, we used purposive sampling to recruit respondents within each of the categories of target group. From our investigations in the Inception Phase we assumed an average 10% response rate for this research. Therefore, in order to get the target 50-100 respondents per segment[5] our target sample frame size needed to be 500-1,000 per target group. Our targets aimed to produce a combined sample of up to 700 respondents in the South and 300 in the North. We anticipated this would also allow us to disaggregate by gender and some countries/sub-regions. Section A.2 presents the total number of respondents and how they are distributed across the categories of Intended User, gender, region, etc.

Sample frames

The first sample frame (Wave 1), which had 5,824 contacts, was developed through initially tapping into Mott MacDonald and Open University contact databases e.g. project contacts, consultancy services team database and Open University alumni, and added to through desk-based searching of publicly available email addresses e.g. government directories.

As recommended by DFID, the questionnaire was emailed[6] initially to the first 500 members (batch 1) of the Wave 1 sample frame only (on 3rd November, 2015) to allow a rapid review of responses received to ensure the questionnaire was working as intended. The questionnaire was modified slightly in the light of the responses received and was then sent to the remaining 5,324 email addresses (batch 2) in Wave 1 on 16th November. For this, and each subsequent wave of issuing, the questionnaire

intended and actual users of development research portals having the opportunity to participate in the market research. These responses were included in the final dataset (and even led to an 8th target category being added: Multilateral and donor agency staff based in the South), but those from the target categories were analysed separately.

[4] As distinct from knowledge brokers/intermediaries that are unaffiliated to government.

[5] Based on sub-group size recommendations by the National Audit Office (National Audit Office, Statistical and Technical Team).

[6] Each invitation message contained a unique online questionnaire URL tied to the email address to which it was sent. This means the issuing of reminders was only to those who had not responded. Reminders were scheduled to issue at different times of day, and different days of the week to accommodate the variety in the respondents' working weeks and time zones.

stayed open for at least three weeks and recipients received at least two reminders. Where data were available, the invitation emails were personalised. In cases where open links were used (e.g. when sending to a closed network or set of personal contacts) reminders were not often possible. Section A.2 summarises the mailouts of the questionnaire, the delivery rates and the response rates for each wave.

A.2 Market research metadata

The table below summarises the different waves of respondent recruitment

Table 2: Waves of Market Research respondent recruitment

	Total Sent	Total Received	Total undeliverable	Total completed	Response rate
Wave 1 batch 1[7]	500	425	75	50	12%
Wave 1 batch 2	5,324	4,713	611	659	14%
Wave 2	N/A (6 open links)	Unknown	Unknown	149	Unknown for open links
Wave 3	3,651	2,960	691	79	3%
Wave 4	781+ 3 open links	609	172 (open links unknown)	21+37 from open links	3% for direct mail, unknown for open links
TOTAL	9756 + those sent open links	8282 + those received through open links	1474 (open links unknown)	945	9.2% (unknown for open links)

Wave 1, Batch 2 was built up using OU's alumni database, Mott MacDonald's country offices' contacts and some desk-based research. This sample was a mix of largely academic, government and consultant contacts[8].

Wave 2 was a set of Open Links to closed networks e.g. Evidence Based Policy Development Network.

Wave 3 was a sample included a large proportion of Southern government contacts obtained from Southern government websites.

Wave 4 was a top-up sample of predominately Southern government contacts and knowledge brokers and a set of open links issued to networks of development actors e.g. association of parliamentary librarians.

Almost all questions in the questionnaire were mandatory (with options allowing respondents to effectively skip if preferred e.g. “Don’t know” or “not relevant”). The survey team was only sent data from fully completed and submitted questionnaires and although calculated during pre-testing to take on average 20 minutes to complete, some respondents’ experience was that this could take a lot longer depending on how much information they provided and fast the questionnaire loaded on their browser. These factors will inevitably have affected the response rate.

A questionnaire was marked as undeliverable if the sample member’s email address was no longer working and the response rate is therefore ‘Total completed divided by Total received’. As the sampling method was purposive rather than random, the response rate tells us more about our efficiency in sourcing respondents than the representativeness of the results i.e. we cannot calculate statistical validity of the data regardless of response rate. However, we believe the response rates of individual waves do tell us something else of value:

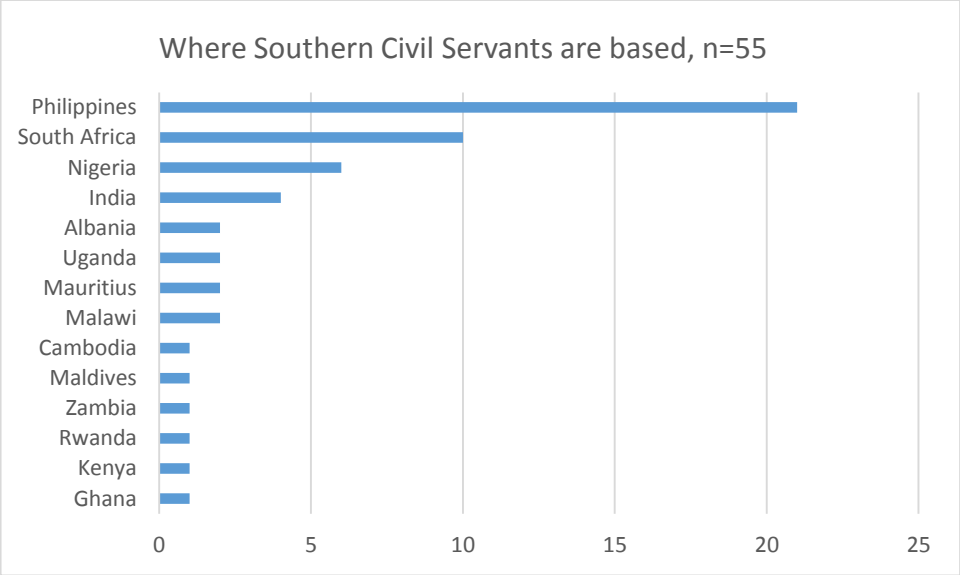
7 This pilot batch was partly reported on in the Stage 2 Interim Report and the full 50 responses were used to inform the development of the coding approach; in fact some early challenges to the original Theory of Change emerged as part of this process where the evaluation team used the ToC for coding and struggled (e.g. replacing Accessible with Discoverable). Their *The findings about Southern Internet* used, albeit not as part of the larger dataset in the main report, but did influence our evaluation approach.

8 Wave 1, Batch 1 and Wave 1, Batch 2 are sub-samples of a single sample (Wave 1); Batch 1 being the 1st 500 members of Wave 1 and used to test the final questionnaire before it was amended and issued to the remaining 5324 members.

a) Southern government contacts are particularly hard to reach through online research. About 1 in 8 of every email sent to Wave 1, Batch 2 resulted in a completed questionnaire being submitted compared to about 1 in 45 of every email sent to Wave 3 (largely sourced via Southern government websites). A contributing factor to this difference was the number of email addresses that were found to be not working among Wave 3 (18.9%).

This is reflected in the distribution of respondents we recruited to the Civil Servant category. The 55 respondents who completed the online questionnaire and met the full sampling criteria for Southern civil servants were from 14 countries; however 10 respondents were from South Africa and 21 from the Philippines - two countries that have made notable advances in e-government.

Figure 1 Southern Civil Servant Respondents by location



b) Online surveys are at risk of being blocked as spam: despite using respected online survey software we know that email providers' spam filters prevented emails from the Survey Team reaching some recipients (where respondents told us they found the email in their Spam folder).

c) A trusted, recognised sender is effective: We have anecdotal evidence that non-recognition of the sender affected response rates for the direct emails from the Survey Team while an open link emailed directly to individual selected members of KM4Dev by a KM4Dev member, with one reminder had a 21% response rate.

How data were analysed

Respondents were first coded into categories of North or South, based on the information provided about their location combined with World Bank data[9]. Using this North/South variable and other classification data they had provided, respondents were then placed into the 10 original categories, plus Southern (Other) and Northern (Other). We identified that the Southern (Other) category had enough respondents within it (63) who could be classified as Multilateral/Donor Agency staff based in the South to create an 11th category of Intended User (see Table 3) capable of being analysed separately.

[9] Countries categorised as High Income, by the World Bank based on its July 2015 categories (World Bank, 2016), were coded as Northern; all others as Southern.

The new North/South and Respondent Category variables and existing classification variables were used for basic frequency and disaggregation reporting for all questions (see Section 5, in main report). However, four groups of Intended User had fewer than 50 respondents thus making them unsuitable for reporting on as separate groups. Their data is included in the wider analysis where appropriate. There remained 652 respondents across the seven larger (Primary) groups of different sizes (from 55 to 163 respondents). This presented a challenge in reporting at a headline level about Intended Users; our early analysis showed some emerging differences in information behaviour between the groups but being of different sizes, there was a risk that a larger group's behaviour could skew our headline findings. Weighting was ruled out because the true sizes of these groups in the wider population are not known; for the same reason, we were not able to weight by gender as information on the gender distribution among Southern academics, for example, is not available.

To resolve this challenge, we present the data in several ways in the next section. For each question we report the overall responses (all who answered the question), but for most questions we also present (and use as headline findings in the main Evaluation Report) the mean average of the percentages across the seven groups that have at least 50 members and refer in these cases to the sample as Primary Intended Users. Thus, when the findings relating to Primary Intended Users are reported, the responses of larger groups (such as Development Consultants in the North and South) do not outweigh those of smaller groups (such as Southern Civil Servants). For comparison purposes we also present the findings from individual groups e.g. civil servants, where appropriate, to illustrate differences between their behaviour.

A.2.1 Market research findings

Introduction to findings

The first set of questions (Q1 to Q14) was asked of all 945 participants of the online market research and tell us the general composition of the sample. Question 14 helped us to identify which of the respondents met the full sampling criteria by asking them if they ever have to find research evidence for work, for themselves or for other people. Those who answered No (95 respondents, 10% of the sample), were taken directly to Question 33 which asks about awareness and use of various portals and repositories likely to be of interest to development actors – skipping questions relating to information behaviour and research evidence.

The analysis of Questions 15 to 22 is based on a sample of 850 respondents (those who replied Yes or Don't Know[10] to Question 14). Of these, 671 fall into the categories of Intended User that we sought to reach through purposive sampling (although not evenly as Table 3 illustrates). Two respondents gave an answer to Question 22 that excluded them from some subsequent questions meaning that Questions 23 to 32 were asked of 848 respondents.

The online market research aimed to obtain responses from 50 to 100 of three categories of development actor in both the North and South, and of four categories in the South only. As the table below shows, and was observed at Interim Reporting stage, some categories proved harder to recruit than others and as was noted in 1, a new category was able to be added.

10 We included the option Don't Know to enable respondents to skip the question if desired or to allow for respondents not understanding the term Research Evidence and the definition provided. However, the answers provided to other questions suggested that those who selected Don't Know did use Research Evidence and tended not to skip other questions.

Table 3: Categories of Development Actor

	Respondents	Who met sampling criteria	% of whom female
Academic/Researcher (North)	66	66	48.5% (32)
Academic/Researcher (South)	81	77	45.5% (35)
Development Consultant (North)	186	163	42.3% (69)
Development Consultant (South)	157	145	33.1% (48)
Development worker in civil society (North)	45	42	57.2% (24)
Development worker in civil society (South)	91	83	31.3% (26)
Elected member of local or national government (South)	15	13	23.1% (3)
Civil servant (South)	65	55	40% (22)
Knowledge broker/intermediary for policy makers (South)	14	14	35.7% (5)
Media (South)	14	13	53.8% (7)
Multilateral/Donor Agency Staff (South)	68	63	41.2% (26)
Other (North)	94	80	55% (44)
Other (South)	49	36	55.6% (20)
TOTAL	945	850	42.5% (361)

The Northern development worker in civil society category fell 5 short of the target, with 45 responses (of which only 40 were asked the wider set of questions about research evidence use) and the following categories only achieved 14 or 15 respondents:

- Elected member of Southern local or national government
- Southern knowledge broker/intermediary for Southern policy makers (e.g. Parliamentary researcher, Government departmental librarian)
- Southern media professional (e.g. journalist or editor of national newspaper)

This makes these four groups unsuitable for reporting on as separate groups in this report, although their data is included in the wider analysis. Their responses have also been treated as further sources of qualitative data about their respective category of Intended User.

In this report, the expression “Intended Users” is used as shorthand for the sub-sample of 734 respondents who meet the full sampling criteria and fall within one of the 11 final groups of Intended User of research portals and repositories (3 categories across North and South – making 6 groups - and 5 unique to the South). The expression “Primary Intended Users” is used to refer to the seven groups for whom we have more than 50 responses and is typically reported as a mean average across their responses to avoid larger groups biasing the results.

Although the sampling method used does not allow for statistical significance to be reported, we have taken as our guide the appropriate margins of error for the sample sizes involved before highlighting differences between groups. We assume there would need to be at least 9 percentage points between the genders, for example, or 21 percentage points between findings from Multilateral/Donor Agency staff in the South, and Southern civil servants in order to point to them as being different.

Summary of Market Research Results by Question

Question 1: Which region are you primarily based in? (Please select one option from the list below) n=945

Figure 2 Respondents by Region

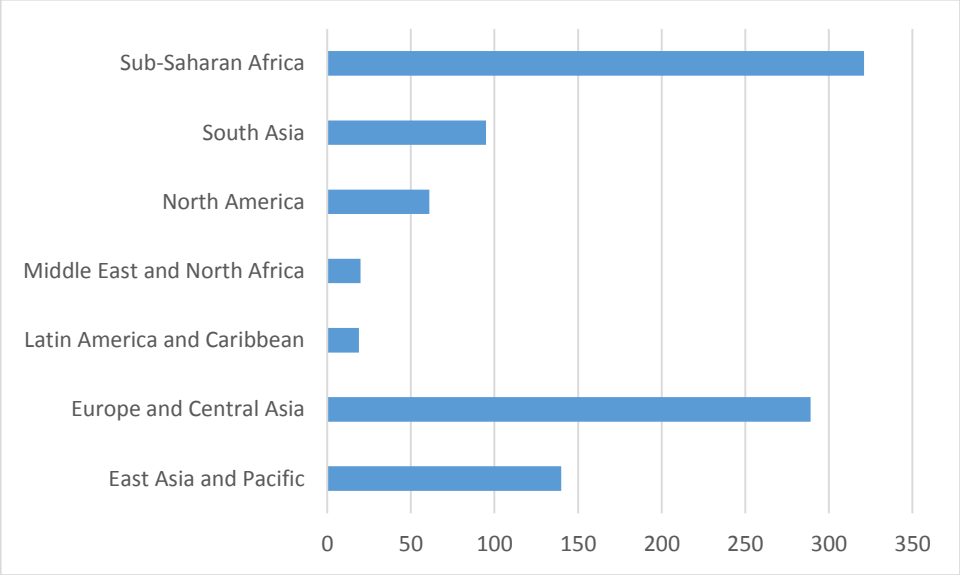


Table 4: Respondents by Region

Region	Count	%
East Asia and Pacific	140	14.8
Europe and Central Asia	289	30.6
Latin America and Caribbean	19	2.0
Middle East and North Africa	20	2.1
North America	61	6.5
South Asia	95	10.1
Sub-Saharan Africa	321	34.0
Total	945	100.0

Which country are you primarily based in?

These responses enable us to classify the respondents as being based in the North or South using the latest World Bank list of High Income countries as the Northern list (World Bank, 2016).
This classification results in the following subsamples:

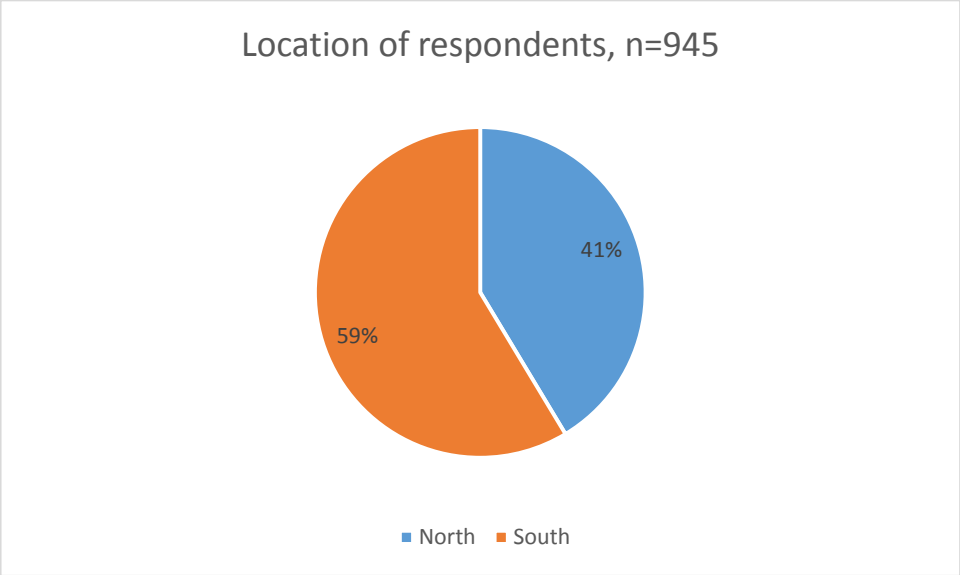
- Respondents based in the North: 391
- Respondents based in the South: 554

The following countries had more than 25 respondents in them:

- UK: 171
- South Africa: 50
- Nigeria: 48
- Kenya: 39
- United States: 40
- Philippines: 38
- Rwanda: 35

- Australia: 34
- India: 29
- Uganda: 28

Figure 3 Location of respondents



Thinking of the last 10 years, for how many of these in total have you been based in [country selected in previous question]? N=945

The data were collected to enable us to check if required against possible bias when looking at the country/region location of respondents.

Figure 4 Respondents' years spent in country based in over the last 10 years

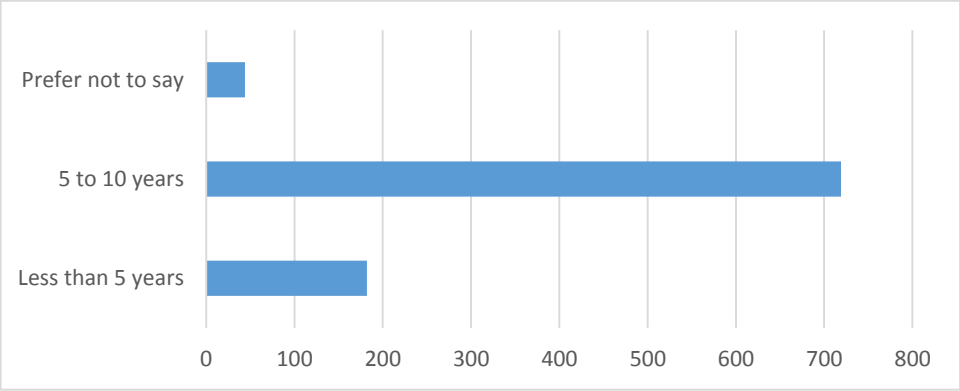


Table 5: Respondents' years spent in country based in over the last 10 years

	Number of respondents	Percentage of total
Less than 5 years	182	19%
5 to 10 years	719	76%
Prefer not to say	44	5%
Total	945	100%

This distribution stays about the same for the different samples, for example, among the 848 respondents who answered questions 23 to 32, 77% Southern respondents had been based in their main country for 5 to 10 of the last 10 years.

Question 2: Which type of organisation do you currently work for (if you are employed by more than one, please select the main one)? N=945

This question, together with the next question on role, was asked to enable us to identify respondents by target group category (through combinations of organisation type, role and country).

Figure 5 Respondents by organisation type

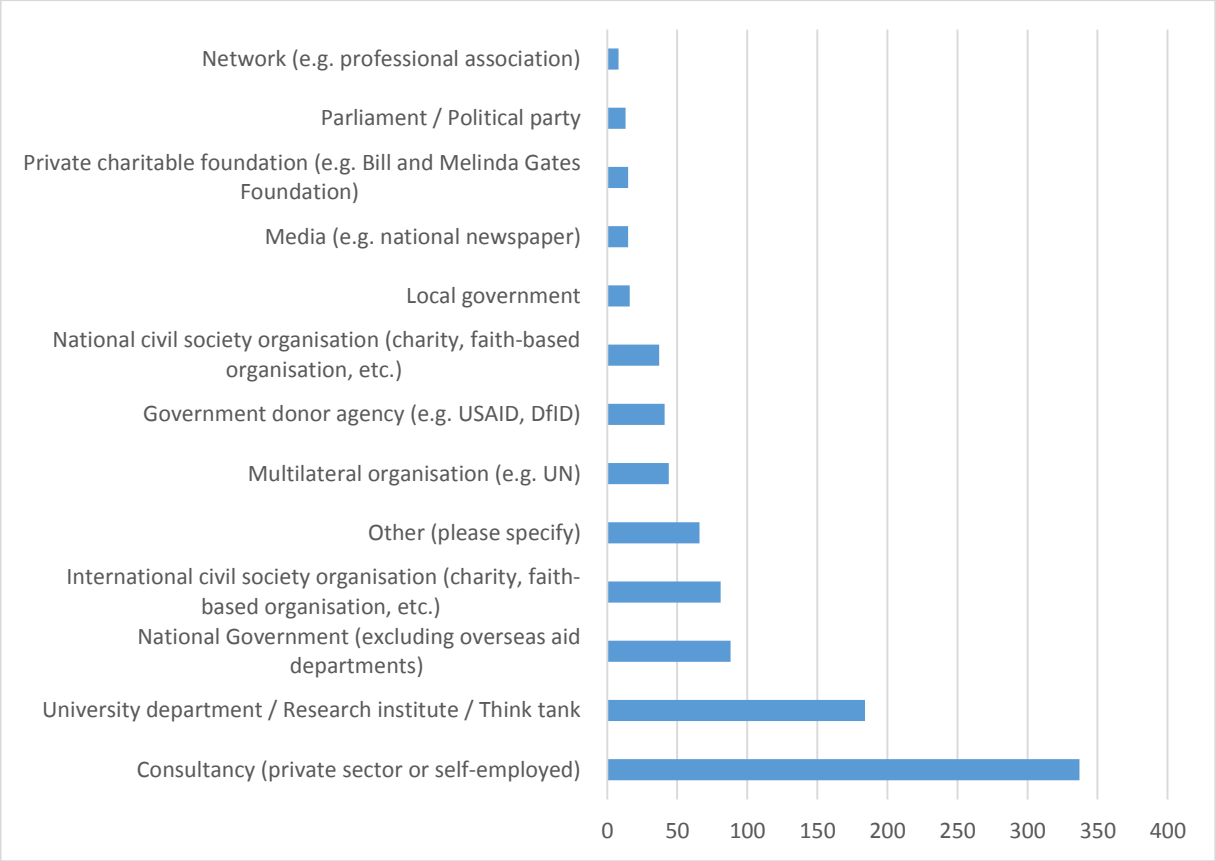


Table 6: Respondents by organisation type

Organisation Type	Number of respondents	Percentage of total
Consultancy (private sector or self-employed)	337	35.7%
University department / Research institute / Think tank	184	19.5%
National Government (excluding overseas aid departments)	88	9.3%
International civil society organisation (charity, faith-based organisation, etc.)	81	8.6%
Other (please specify)	66	7.0%
Multilateral organisation (e.g. UN)	44	4.7%
Government donor agency (e.g. USAID, DFID)	41	4.3%
National civil society organisation (charity, faith-based organisation, etc.)	37	3.9%
Local government	16	1.7%
Media (e.g. national newspaper)	15	1.6%

Organisation Type	Number of respondents	Percentage of total
Private charitable foundation (e.g. Bill and Melinda Gates Foundation)	15	1.6%
Parliament / Political party	13	1.4%
Network (e.g. professional association)	8	.8%
Total	945	100%

Question 3: Which of the following best describes your MAIN role in that organisation? N=945

Figure 6 Respondents by main role

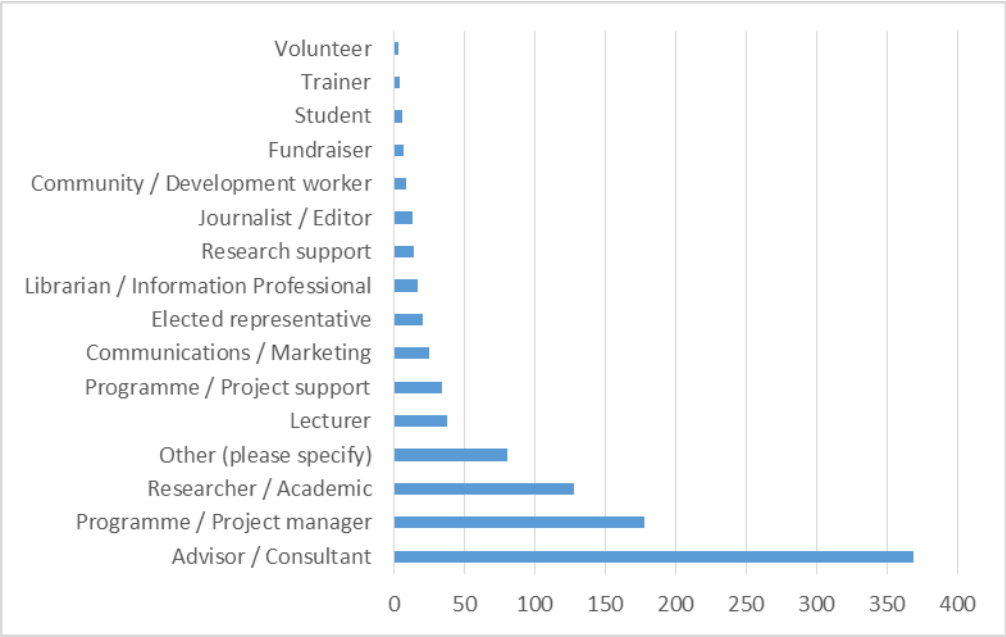


Table 7: Respondents by main role

Main role	Number of respondents	Percentage of total
Advisor / Consultant	369	39%
Programme / Project manager	178	19%
Researcher / Academic	128	14%
Other (please specify)	80	9%
Lecturer	38	4%
Programme / Project support	34	4%
Communications / Marketing	25	3%
Elected representative	20	2%
Librarian / Information Professional	17	2%
Research support	14	2%
Journalist / Editor	13	1%
Community / Development worker	9	1%
Fundraiser	7	<1%
Student	6	<1%
Trainer	4	<1%
Volunteer	3	<1%
Total	945	100%

Question 4: In which sector do you mainly work? n=945

Figure 7 Respondents by sector

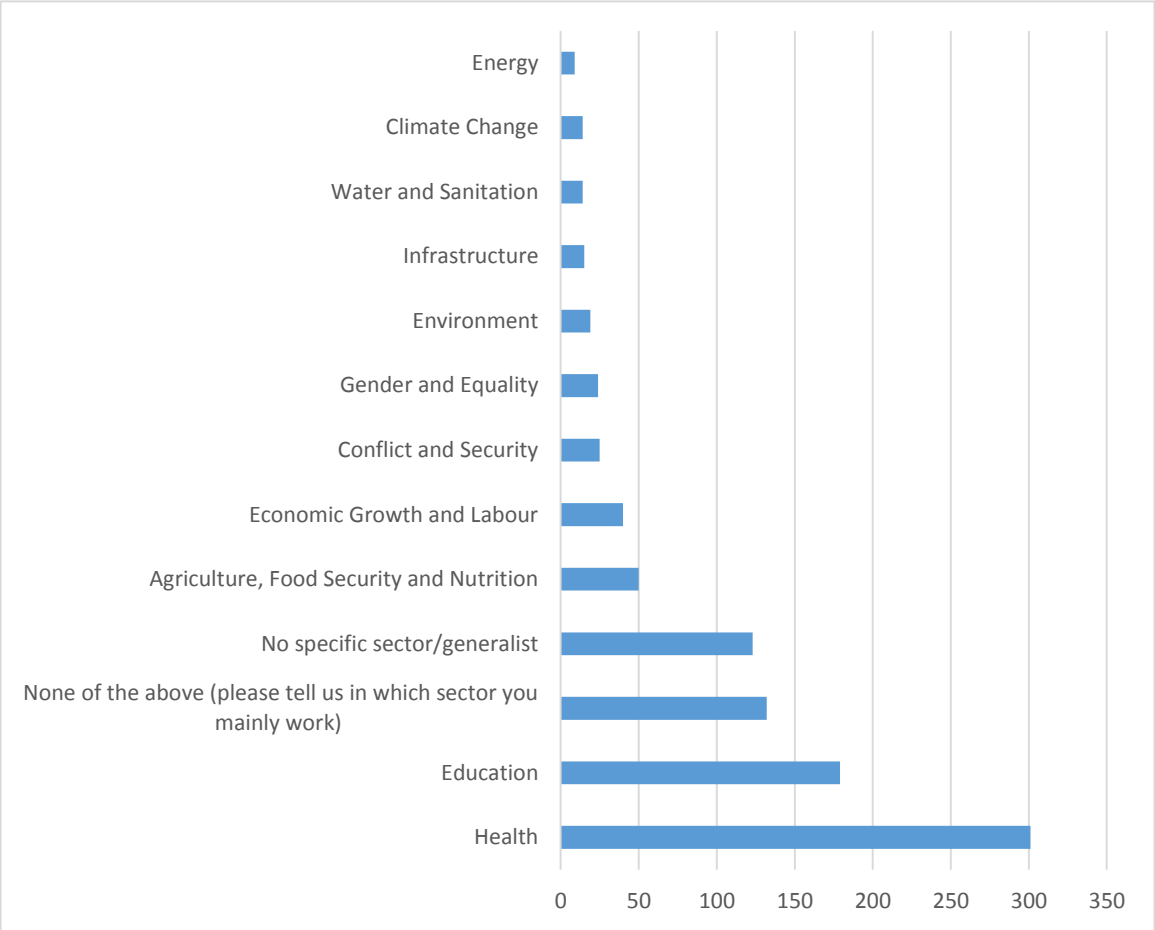


Table 8: Respondents by sector

Main sector	Number of respondents	Percentage of total
Health	301	32%
Education	179	19%
None of the above	132	14%
No specific sector/generalist	123	13%
Agriculture, Food Security and Nutrition	50	5%
Economic Growth and Labour	40	4%
Conflict and Security	25	3%
Gender and Equality	24	3%
Environment	19	2%
Infrastructure	15	2%
Water and Sanitation	14	2%
Climate Change	14	2%
Energy	9	1%
Total	945	100%

Question 5: Which of the following age categories do you fit into? n=945

This question was used to aid analysis of information behaviour (based on an assumption about relationship between age and behaviour). Just over half (55%) of respondents classed themselves as aged 45 to 64 years old.

Figure 8 Respondents by age category

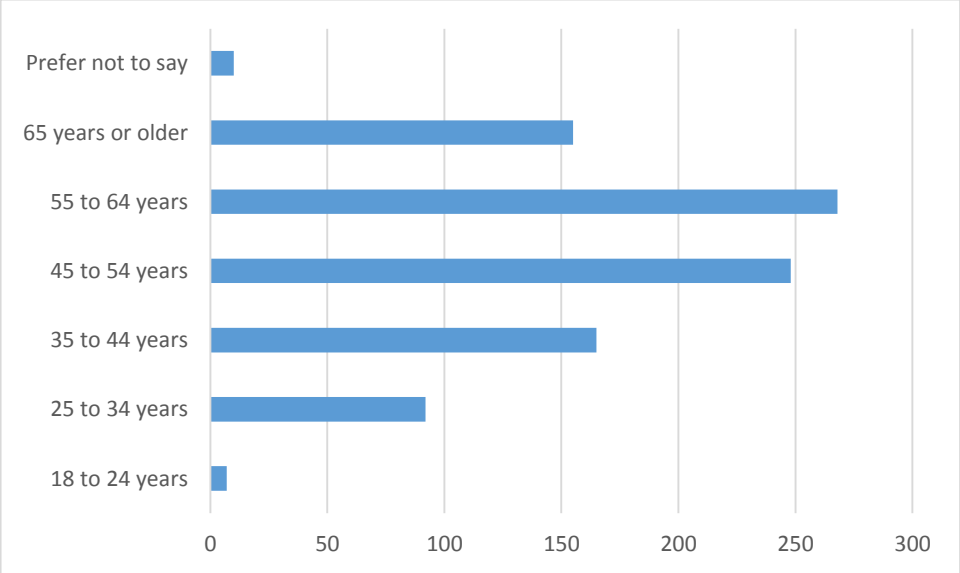


Table 9: Respondents by age category

Age categories	Number of respondents	Percentage of total
18 to 24 years	7	1%
25 to 34 years	92	10%
35 to 44 years	165	18%
45 to 54 years	248	26%
55 to 64 years	268	28%
65 years or older	155	16%
Prefer not to say	10	1%
Total	945	100%

Question 6: What is your highest level of education?

Figure 9 Respondents by highest qualification

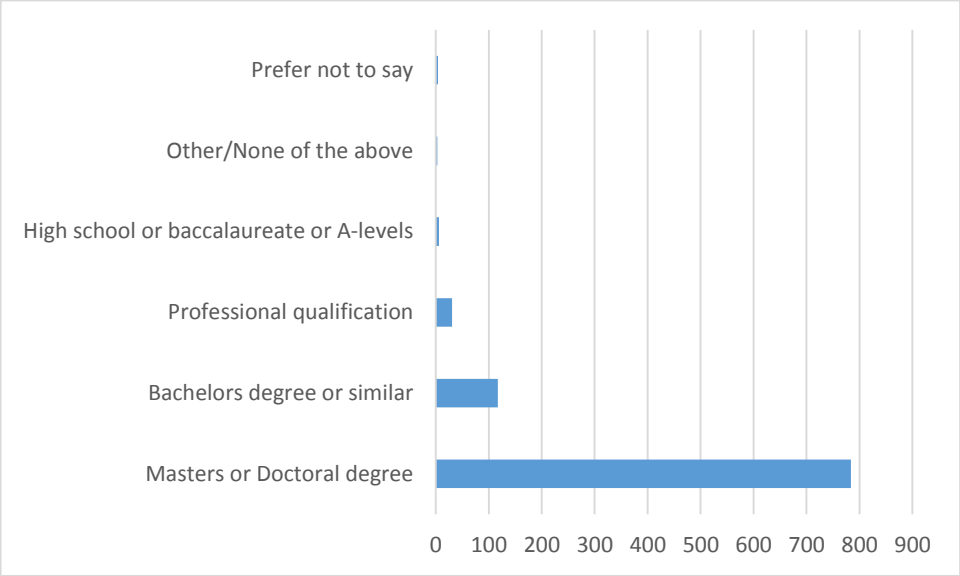


Table 10: Respondents by highest qualification

Highest qualification	Number of respondents	Percentage of total
Masters or Doctoral degree	784	83%
Bachelor's degree or similar	117	12%
Professional qualification	31	3%
High school or baccalaureate or A-levels	6	<1%
Other/None of the above	3	<1%
Prefer not to say	4	<1%
Total	945	100%

Question 7: Are you…? n=945

Female: 399 (42.2%)

Male: 533 (56.4%)

Prefer not to say: 13 (1.4%)

Total: 945

The table overleaf highlights the distribution of genders in each category of Intended User. Women comprised around 40+% of most groups of respondents, but not amongst Southern Development Workers or Southern Development workers[11]. However, it is important to note that because of the sample size, although it looks like there are differences in gender ratios between the Northern Academics and Southern Development Workers in Civil Society they are (probably) not big enough for us to assume that this is the case outside of the people who took part in the Market Research.

[11] Only considering those groups that had a minimum of 50 respondents.

Table 11: Respondents by development actor categorisation

	Female	Male	Prefer not to say	Respondents	% female
Elected member of local or national government (South)	3	12	0	15	20%
Development worker in civil society (South)	28	62	1	91	31%
Development Consultant (South)	52	104	1	157	33%
Knowledge broker/intermediary for policy makers (South)	5	8	1	14	36%
Multilateral/Donor Agency Staff (South)	27	40	1	68	40%
Development Consultant (North)	77	107	2	186	41%
Academic/Researcher (South)	35	45	1	81	43%
Civil servant (South)	28	36	1	65	43%
Academic/Researcher (North)	32	32	2	66	48%
Media (South)	7	6	1	14	50%
Development worker in civil society (North)	27	17	1	45	60%

Question 8: Which of the following best describes how often you access the Internet for any purpose (i.e. for work/leisure etc.)? This should include access from any device (Desktop, Laptop, Tablet or Mobile) and from any location (home, work, internet café or any other location).

(Please tick one box only), n=945

98% of respondents said they access the internet at least a few times a day (Many times or A few times).

Figure 10 Respondents' frequency of accessing the internet

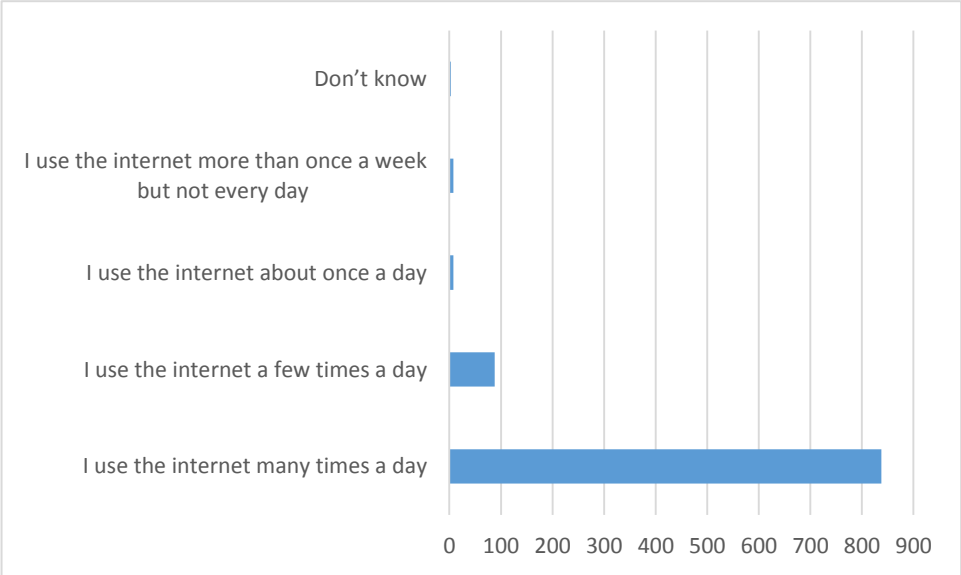


Table 12: Respondents' frequency of accessing the internet

Frequency of accessing the internet	Number of respondents	Percentage of total
I use the internet many times a day	838	88.7%
I use the internet a few times a day	88	9.3%
I use the internet about once a day	8	.8%
I use the internet more than once a week but not every day	8	.8%
Don't know	3	.3%
Total	945	100%

Question 9: If you would like to explain your answer, please use the space below (149 responses)

Some of the answers given indicate that "how often you access the internet" was interpreted by some respondents as meaning having internet connection (the opportunity to access the internet) e.g. *"Always connected to internet during working hours and evenings"*. A selection of answers given by those who selected "Many times a day" illustrates some of the different contexts that may lie behind internet access frequency.

Mainly via laptop but check emails on mobile phone when out of the office. [Development Consultant, Northern]

The cost of the internet connection in PNG [Papua New Guinea] is more than the total salary of most professionals in the country [Academic/Researcher, Southern]

…we have very poor accessibility to academic materials and as such I rely on on-line materials to conduct my work [Academic/Researcher, Southern]

Have my tablet with me all the time so i scroll regularly the [whole] day. I use my laptop and computer when i start working on my projects [and] use internet as [needed]. My cell phone is also always [connected] to my [emails and] with internet connections always [Civil servant, Southern]

I am always on internet through my smart phone that is always on and i use my laptop for at least an hour daily [Civil servant, Southern]

When the network is available, I use the Internet [Elected member of local or national government, Southern]

Question 10: Which of the following devices do you ever use to access the Internet for any purpose (i.e. for work/leisure etc.)?
(Please tick all that apply), n=945

Almost all respondents (97%) said they (ever) use a laptop or desktop computer to access the Internet. 71% of respondents used a laptop or desktop computer, and a mobile; and 40% or respondents used both of these devices plus a computer tablet e.g. iPad. Just 67 respondents (7%) used Smart TV to access the Internet.

Mobile or cell phone use

Focusing just on the 734 respondents in the "Intended Users" sub-sample (i.e. those who meet the sampling criteria with the addition of Multilateral/Donor Agency staff in the South), 74% (542 respondents) said they use a mobile or cell phone to access the Internet.

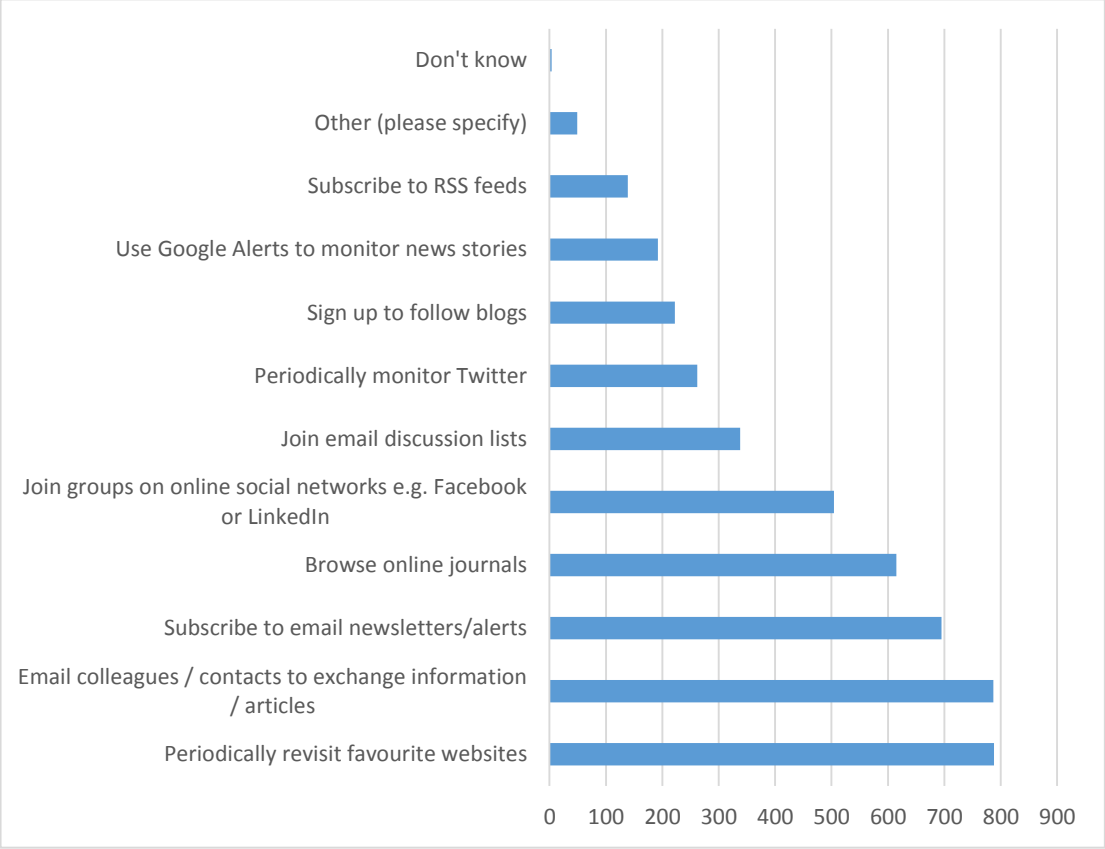
The table below shows there is not a large variation between types of Intended Users[12] with those using a mobile or cell phone to access the Internet ranging between 65% and 79%.

Table 13: % of Target Users that use a mobile phone to access the internet

Target User Group	% that use a mobile phone to access the internet
Academic/Researcher (North)	77%
Academic/Researcher (South)	65%
Development Consultant (North)	76%
Development Consultant (South)	72%
Development worker in civil society (South)	71%
Civil servant (South)	75%
Multilateral/Donor Agency Staff (South)	79%

Question 11: How do you use the internet, if at all, to keep up to date with your sector/profession?
(Please tick all that apply), n=945

Figure 11 Respondents' use of online tools to keep up to date



[12] Focusing just on the categories that have at least 50 respondents

Table 14: Respondents' use of online tools to keep up to date

Online tool used to keep up to date	Number of respondents	Percentage of Total
Periodically revisit favourite websites	788	83%
Email colleagues / contacts to exchange information / articles	787	83%
Subscribe to email newsletters/alerts	695	74%
Browse online journals	615	65%
Join groups on online social networks e.g. Facebook or LinkedIn	504	53%
Join email discussion lists	338	36%
Periodically monitor Twitter	262	28%
Sign up to follow blogs	222	24%
Use Google Alerts to monitor news stories	192	20%
Subscribe to RSS feeds	139	15%
Other (please specify)	49	5%
Don't know	4	.4%

Analysis of the Northern and Southern respondents separately shows there are minor differences between the two groups in the popularity of some of the tools but not large enough to suggest these differences exist in the wider populations from which the samples were drawn.

Looking at gender, there are even fewer differences, with one exception: **80% of the female respondents said they subscribe to email newsletters/alerts to keep up to date with their sector or profession, compared to 69% of male respondents**. The table below illustrates how these differences continue in the North and South.

Table 15: Respondents' subscription to email newsletters/alerts

Gender	North/South	Total in sample	Who subscribe to email newsletters/alerts	Percentage of subsample who subscribe
Female	North	188	159	85%
Female	South	211	161	76%
Male	North	198	144	73%
Male	South	335	223	67%

Question 12: Do you ever have to find EVIDENCE of any kind for yourself or other people as part of your work? (Please tick one box only), n=945

Figure 12 Respondents who ever have to find evidence for work, for themselves or for other people

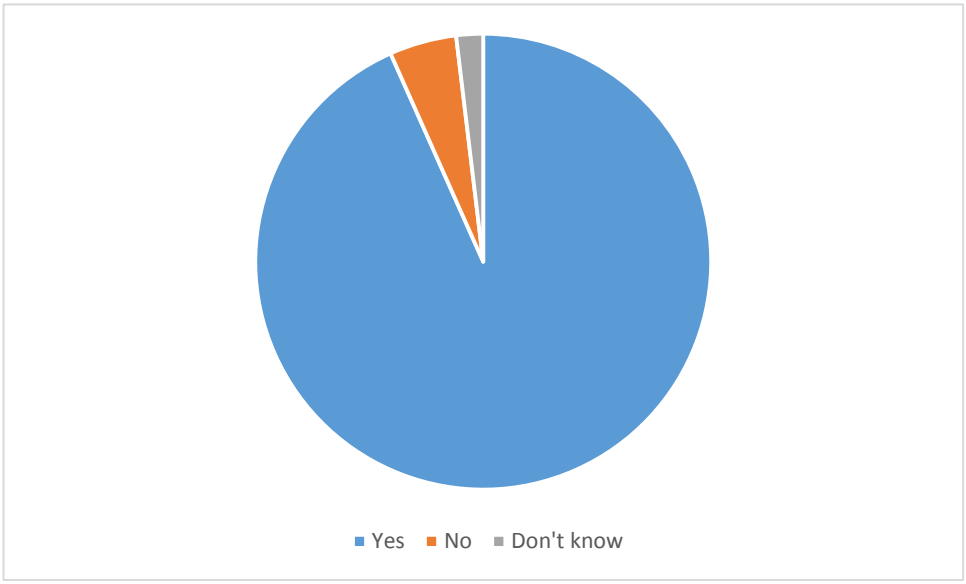


Table 16: Respondents who ever have to find evidence for work, for themselves or for other people

	Number of Respondents	Percentage of Total Sample
Yes	882	93%
No	45	5%
Don't know	18	2%
Total	945	100%

A definition was provided of Evidence and Research Evidence. The percentage of respondents giving Yes as their answer to this question is close to 93% in the gender and North/South sub-samples. 18 said Don't Know (2 North and 16 South).

Question 13: Please think of a time you recently looked for EVIDENCE of any kind for work. What was the subject matter and what type of information or data were you looking for? (Please write in below)

This question was asked to help put analysis of other questions into context (see analysis of Question 14 for example). 830 respondents (out of 882 who said they ever look for evidence) answered this question.

Question 14: Do you ever have to find RESEARCH EVIDENCE for yourself or other people as part of your work?
(Please tick one box only), n=945

This question was included to help identify respondents who matched the full sampling criteria (and could be classed as Intended Users), specifically people who are interested in seeking, or who are already seeking, research evidence to inform their work.

Figure 13 Respondents who ever have to find research evidence for themselves or others

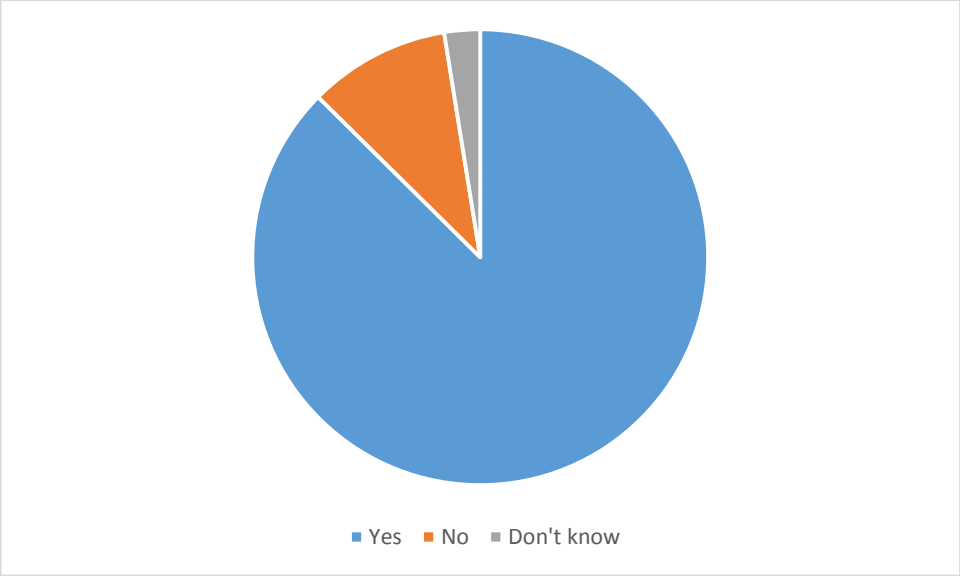


Table 17: Respondents who ever have to find research evidence for themselves or others

	Number of Respondents	Percentage of Total Sample
Yes	826	87%
No	95	10%
Don't know	24	3%
Total	945	100%

Those who answered No (95 respondents, 10% of the overall sample), were not asked questions about information behaviour in relation to obtaining research evidence and were taken directly to Question 33 which asks about awareness and use of various portals and repositories likely to be of interest to development actors. 40 of the 95 were based in the North (10% of the Northern respondents), 55 in the South (10% of the Southern respondents).

Questions 15 to 22 (below) therefore were answered by a sample of 850 respondents (those who replied Yes or Don't Know[13] to Question 14). Of the Don't Knows, 6 were based in the North (1.5% of Northern respondents) and 18 in the South (3.2% of Southern respondents). From this point in the report, we refer to this sample of 850 respondents as "Research Evidence Seekers" (within this sample of 850 we find the subset of "Intended Users" however the 850 includes respondents that also fall into the Other Northern and Other Southern categories).

Table 18: Respondents who answered that they never have to find research evidence for themselves or others

Category of Respondent	Number of responses
Northern development consultant	23
Northern academic/researcher	0
Northern development worker in civil society	3
Southern development consultant	12

13 We included the option Don't Know to enable respondents to skip the question if desired or to allow for respondents not understanding the term Research Evidence and the definition provided. However, the answers provided to other questions suggested that those who selected Don't Know did use Research Evidence and tended not to skip other questions. For example, none of the 24 'Don't Knows" selected the option 'Don't Know' in the next multiple choice question (How often, if ever, they use colleagues as a source to find research evidence).

Category of Respondent	Number of responses
Southern academic/researcher	4
Southern development worker in civil society	8
Southern media professional	1
Southern knowledge broker/intermediary for policy makers	0
Southern national legislature or elected member of local government	2
Southern civil servant	10
Multilateral or donor agency staff based in the South	5
Other (Northern)	14
Other (Southern)	13
Total	**95**

As the table above shows, many of the 95 respondents who said No fell into some of the priority categories of Intended User for research portals and repositories. Furthermore, the 95 who said No included 12 Eldis users, 14 R4D users and 8 SciDev.Net users, so the answers they had provided to Question 13 were examined. One interpretation is that there seems to be a group of people who fit the intended target audience of the portals and repositories but have a need for practical information rather than research evidence, for example respondents were looking for:

What is best practice in medical terms, comparing UK and German practice.

Educational Policy, frameworks, quality assurance guidelines - all related to higher education. Sometimes statistics related to HE institutions or students. Also security briefings for international travel risk assessments

Monitoring influence – methodology

country background information on proposal writing.

The amount of GDP allocated to Health sector in countries across Asia and the pacific

Market-based sanitation Nature and number of sanitation businesses

However, a second interpretation may be that these were 'false negatives' i.e. that these respondents do search for research evidence, but still answered 'no' potentially as the question and/or the definition of research evidence were not written clearly enough.

Question 15: If you would like to explain your answer, please use the space below:

All respondents were given the opportunity to explain the answer they gave to Question 14 (whether Yes, No or Don't know). Of the 95 respondents who said 'No', 10 provided a response as listed below:

Development consultant (North) working in governance and institutional development: *my clients are [rarely] interested in research. also: research takes too [long] to produce findings that are relevant NOW*.

Development Consultant (South), working in Governance: *Research evidence is of value in project design (and proposals) but as this is already determined by the time I am running a project, there is much less need for the level of detail that would [qualify] as "research evidence" in my day-to-day work. I would normally expect HQ support in accessing relevant material for reports, etc., when necessary.*

Civil servant (South), working in Health: *As head of office i usually let my staff do it for me. My office also is not much on research except for [operational] researches.*

Development worker in civil society (South), working in Economic Growth and Labour: *Up to now is not familiar to me.*

Multilateral/Donor agency staff, Generalist, (South): *Not yet. As my work focuses on the bridge between research and policy, where evidence-based policy can mean evidence in the broadest sense, i.e. local wisdom, diagnostic studies, expert opinion and research evidence.*

Other (retired), worked in Health, (North): *research evidence was not part of my role, but would have been performed one layer above, at MOD level.*

Other, working in Education, (North): *Rarely.*

Development consultant, working in Health, (North): *I don't have to, but it might be relevant.*

Other, working in Environment, (South): *I just use internet, read newspapers to collect the information on this subject.*

Development worker in civil society, working in anti-slavery/anti-trafficking, (North): *It isn't necessary though I do use it when relevant material is available; perhaps from reports or studies, I use UN and ILO findings, indices and statistics regularly.*

From reading these comments, and those that the wider group of 95 gave to Question 13 (which asked about a recent search people had made online for evidence), it would seem that some respondents (including two Northern development consultants) said No because they did not understand the term 'Research Evidence', the definition provided, or the distinction between Evidence and Research Evidence. Others gave reasons that suggest they encounter research evidence but do not actively seek it out or that other people supply them with it.

Question 16: How often, if ever, do you use the following sources to find RESEARCH EVIDENCE for yourself or other people as part of your work?
(Please tick one box per row), n=850

Table 19: Frequency of use of sources to find research evidence

Source of research evidence	Frequently	Occasionally	Rarely	Never	Don't know	Total
Ask friends/family to recommend sources	66	224	344	212	4	850
Radio programmes and/or their online sites/apps	68	162	319	295	6	850
Television programmes and/or their online sites/apps	71	176	315	284	4	850
Blogs	75	209	312	239	15	850
Twitter	77	139	191	431	12	850
Help desks / enquiry services (online, telephone or face to face)	96	168	310	270	6	850
Social networking sites e.g. Facebook or LinkedIn	141	234	241	229	5	850
Email discussion lists	181	271	232	160	6	850
National or local newspapers and/or their online sites/apps	195	263	256	135	1	850
Ask colleagues to recommend sources	265	452	124	9	0	850
Email newsletters/alerts	272	321	183	71	3	850
Websites that summarise, profile, link to or report on other people's research, evaluation findings or data	386	324	105	29	6	850
Government websites of the country in which you are primarily based, e.g. national data portals	429	294	98	29	0	850
Specialist journals in print or online	488	254	79	24	5	850
Other organisational websites that make their own research, evaluation findings or data available online e.g. research institutes, bilateral donors, inter-governmental organisations or consultancies	553	242	47	7	1	850
Other (optional question)	37	17	9	107	129	299

From the above table we can see that the category of website that Eldis and SciDev.Net[14] fall into are used frequently by 45% (386/850) of the respondents and the category of website that R4D falls into is used frequently by either 50% (429/850) or 65% (553/850) of respondents depending on whether it is considered a government website or bilateral donor website.

14 We avoided using the term "research portal" because the way it is interpreted is too broad to be useful in this questionnaire. Instead we used the following description "Websites that summarise, profile, link to or report on other people's research, evaluation findings or data".

Figure 14 Frequency of use of sources of research evidence

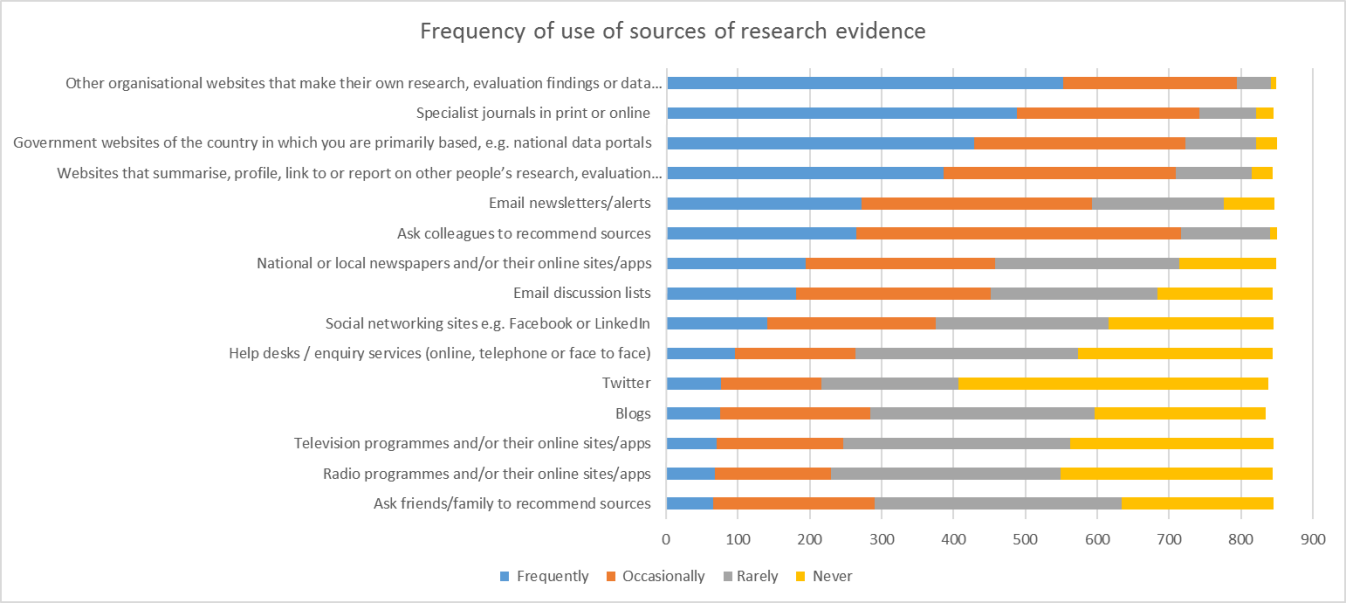
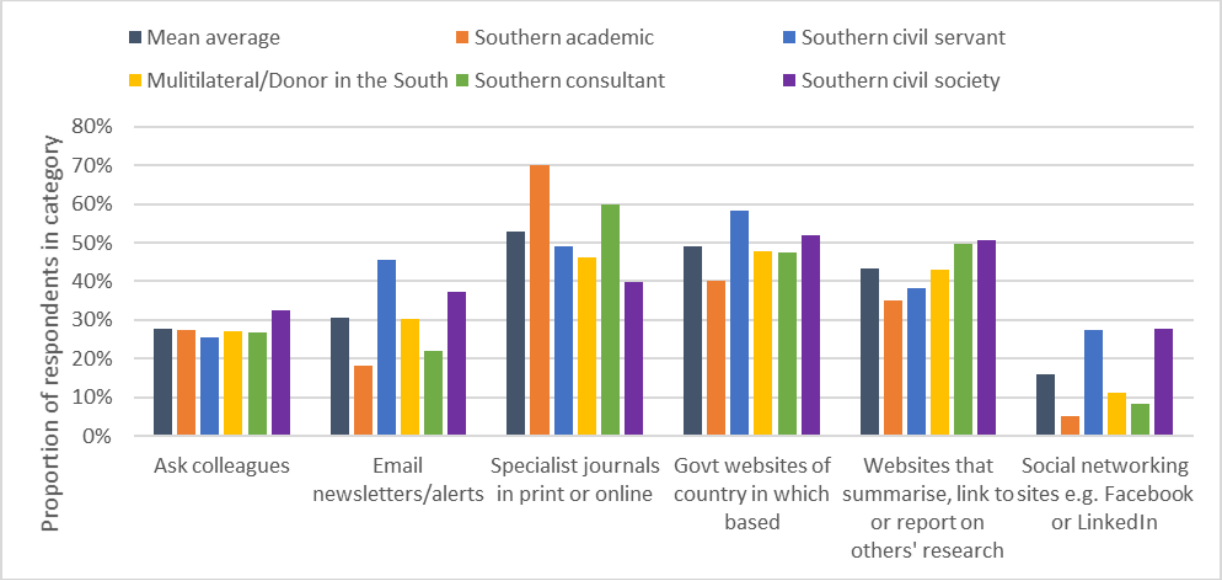


Figure 15 shows the distribution between sources of particular relevance to DFID portals, among Primary Intended Users in the South.

Figure 15 Sources used Frequently by Primary Intended Users to find research evidence (Southern only)



In the analysis of question 11, a difference by gender in use of email newsletters or alerts was observed, with a higher percentage of female respondents making use of them to keep up to date with their sector or profession. This difference is **not noticeable** here i.e. among the "research evidence seekers". The gap is just 5 percentage points: 35% of female respondents use email newsletters/alerts frequently as a source of research evidence compared to 30% of male respondents.

However, there are a couple of differences in use by gender:

- 64% of female respondents said they use specialist journals in print and online Frequently, compared to 52% of male respondents.
- 71% of female respondents said they use Other organisational websites that make their own research, evaluation findings or data available online Frequently, compared to 60% of male respondents.

Question 17: If you would like to explain any of your answers, please use the space below:

85 respondents added a comment to explain the selections they made (of which some were along the lines of "nothing to add").

Question 18: You said you use websites that summarise, profile, link to or report on other people's research, evaluation findings or data [FREQUENTLY / OCCASIONALLY / RARELY]. If these websites were no longer available to you, what difference would that make to your work? (Please write in below)

Only respondents who had indicated in Question 16 that they used "websites that summarise, profile, link to or report on other people's research, evaluation findings or data" at all as sources of research evidence were asked this question and were given the option of selecting "Don't know" rather than writing an answer. Based on reviewing the answers provided, we recommend the question is edited if used in the future, to change the wording to "what difference, if any" and to include a tick box option to say "Little or no difference".

The purpose of Question 18 was to understand the value knowledge brokering websites create for their users and examine the relationship between use and reliance i.e. would people who use these websites miss them if they weren't there, how much and why?

The answers provided by the 709 respondents who did supply one (out of a possible 815) were coded by the extent to which the lack of availability would create a detrimental effect on the respondent:

- **little or no effect** e.g. "Not much. Not my primary source of information".
- **some effect** e.g. "These are so important to me. I need my evidence quickly and easily to drop into the reports I am writing."
- **unclassified effect** - where the meaning of the whole answer is unclear or it is unclear if the consequence would be significant e.g. a respondent who answered "I look for alternate sources" is describing what they would do in response, not how much effort it would cost them.

Table 20: The effects on respondents' work if knowledge brokering websites were no longer available to them

Response coded as:	Frequently	%	Occasionally	%	Rarely	%
Don't know	33	9%	46	14%	27	26%
Little or no effect	22	6%	65	20%	41	39%
Some effect	295	76%	177	55%	31	30%
Unclassified effect	36	9%	36	11%	6	6%
Total	**386**	**100%**	**324**	**100%**	**105**	**100%**

59% of Southern Intended Users (263 respondents) reported there would be some negative effect on their work if websites that summarise, profile, link to or report on other people's research, evaluation findings or data were no longer available to them.

The answers given by these 263 Southern Intended Users were then coded further by the type/s of effect. The coding was developed through a bottom-up approach and with reference to the Theory of

Change. The resulting list (below) can be inverted to identify the benefits that respondents believe that they gain from having these kinds of websites available to them.

Types of effect:

- ***Availability***: the respondent expected that there would be less research evidence available to them.
- ***Longer to find:*** the respondent expected that it would take them longer to find research evidence.
- ***Harder to find:*** the respondent expected that finding research evidence would be harder.
- ***Direct impact on quality of work/decisions:*** the respondent was able to give concrete examples of how not being able to make use of these websites would have negative consequences on the influence of their work, the effectiveness of their decisions, etc.
- ***Lose syntheses/summaries:*** the respondent expected that they would lose access to valued syntheses or summaries of research (the terms are used interchangeably by respondents).
- ***Harder to keep abreast/stay current:*** the respondent expected that their ability to keep up to date with what was new would be affected.
- ***Harder to judge quality:*** the respondent expected that their ability judge the quality of the research evidence they found online would be impaired.
- ***Would be missing sources:*** the respondent was concerned that they would not encounter some sources of research or information as a consequence.
- ***Key starting point***: the respondent would lack one of the main places they use to start their searches for information or research.
- ***Affect ability to connect with others***: the respondent expected that their opportunities to connect with other people such as research authors, or others working in their field, would be reduced.
- ***Risk of duplicating research***: the respondent expected that the risk of them carrying out research already undertaken would be increased.
- ***Less well-informed***: the respondent expected to not know as much about subjects of relevance to them.
- ***Have to turn to physical sources/photocopying:*** the respondent expected having to need to visit libraries and other physical sources more (or even at all) and/or make copies.
- ***Forced to find other sources:*** the respondent anticipated needing to seek out and identify new sources of information/evidence/publications as a result.
- ***Harder to verify/triangulate information/evidence:*** the respondent expected that they would be less able to check information obtained elsewhere.
- ***Less able to support writing with evidence:*** the respondent expected that their reports or other outputs would be more likely to lack evidence as a consequence.

Table 21: Effect on Southern Intended Users' work if knowledge brokering websites were no longer available to them (among 263 who anticipated a negative effect)

Types of negative effect on work	Frequency of mentions within 263 responses	% of respondents citing the effect (of 263 Intended Users based in the South who anticipated one or more negative effects)
Longer to find	73	28%
Availability	57	22%
Uncategorised effect (non-specific)	50	19%
Direct impact on quality of work/decisions	22	8%
Harder to find information/evidence	21	8%
Other (only 1 or 2 respondents per effect)	15	6%
Lose valued syntheses/summaries	13	5%
Harder to keep abreast/stay current	11	4%
Would be missing sources	8	3%
Affect ability to connect with others	7	3%

Types of negative effect on work	Frequency of mentions within 263 responses	% of respondents citing the effect (of 263 Intended Users based in the South who anticipated one or more negative effects)
Less well-informed	6	2%
Have to turn to physical sources/photocopying	6	2%
Lose a key starting point for searches	6	2%
Harder to judge quality of material found online	5	2%
Forced to find other sources	5	2%
Less able to support writing with evidence	3	1%
Harder to verify/triangulate information/evidence	3	1%
Risk of duplicating research	3	1%

Q19: How do you judge whether you can trust RESEARCH EVIDENCE that you find online? (Please write in below), n=945

This question was kept deliberately open rather than supplying a list of options for three main reasons:

- to reduce social desirability bias - respondents putting down criteria they think they "ought" to use
- to allow respondents to use their own language
- to enable unprompted and unexpected criteria to emerge

780 respondents provided an answer to the question (92% of those asked) and 70 selected the option "Don't Know" [referenced as Question 20 in the questionnaire]. These were coded using a bottom up approach guided by consulting existing guidance on judging research evidence and information, including the DFID How To Note on Assessing the Strength of Research Evidence (DFID, 2014) and Lucey's factors of "good information" (Lucey, 2005). After coding the pilot set of responses obtained through Wave 2, it was noted that the methods used by respondents could be broadly categorised by stages of reading material found online:

Do I trust the source? This is a broad category where judgements are made about whether or not the research evidence can be trusted based on provenance, author, publisher, website where it was found, person who made them aware of it, etc. Respondents often used the term "source" or "trusted source". 490 respondents said they use some aspect of the Source to judge whether or not to trust the research evidence they found online. 158 respondents **only** referred to Source as their method for judging trustworthiness.

Does the material appear trustworthy? This category is a set of filters that respondents apply that do not require reading of the main content such as what kind of peer review process has it been through? When was the research undertaken? What are the citation metrics[15]? What references are in the Bibliography? 201 respondents referred to using one or more of these Filtering criteria to judge trustworthiness of research evidence found online and 20 respondents **only** referred to using one or more of them as their method.

Do I trust it having read it? These criteria required the respondent to read the material and included methods such as judging the quality of the research design and analysis, in some cases using a formal checklist to do so, examining how well it was written. 157 respondents mentioned one or more of these criteria in their answer.

15 Respondents were sometimes imprecise about whether the citation metrics referred to the journal it was published in, or the article itself.

167 respondents (21% of those who gave an answer) went beyond this stage and made specific reference to triangulating, cross-checking or comparing the research evidence with what they already knew or with other sources.

The following set of codes emerged:

Table 22: Methods by which respondents judge whether they can trust research evidence found online

Method used by respondent	Number of responses making reference to it	Percentage (of those who answered the question, n=780)
Assessment of source (general)	338	43%
Triangulating/comparing to other sources	167	21%
Author	148	19%
Research quality (methods, analysis, etc.)	147	19%
Peer review	124	16%
Organisation (funder or producer)	109	14%
Journal published in	102	13%
Personal judgement/experience/instinct	82	11%
References made within source/bibliography	51	7%
Citations/impact metrics	50	6%
Consult other people	25	3%
Recommended by others	21	3%
Unclassified response	48	6%
Quality of writing	8	1%
Using external checklist	5	1%
Date research undertaken	5	1%

Primary Intended Users and categories

62% of Primary Intended Users who answered the question said they use the source of research evidence to help judge if they should trust it; 18% of those who answered listed no other criteria. This last finding varied by group being as low as 8% for Northern Academics and as high as 27% for Southern Consultants: "*usually trust the research evidences published in an authentic source. for example; UN, DPs (DFID, USAID, DFAT...), international organisation, national but renowned organisation.. Also the agencies whom the DPs and UN bodies trust"* [Southern Consultant]. 20% of Primary Intended Users (who gave an answer to the question) do not judge research evidence in isolation and made specific reference to triangulating, cross-checking or comparing the research evidence with what they already knew or with other sources.

Question 21: Have you received any formal training in how to use the internet to find RESEARCH EVIDENCE? [Question 20 was the option to select "Don't Know" to Question 19], n=850

Table 23 Research evidence seekers who have been trained in how to use the internet to find research evidence

	Yes	% of Total	No	% of Total	Don't know	% of Total	Total	% of Total
All n=850	**271**	**32%**	**565**	**66%**	**14**	**2%**	**850**	**100%**
Male	131	27%	340	71%	6	1%	477	100%
Female	136	38%	218	60%	7	2%	361	100%
Prefer not to say	4	33%	7	58%	1	8%	12	100%
North	113	32%	234	67%	4	1%	351	100%
South	158	32%	331	66%	10	2%	499	100%

The table above shows that 32% of respondents in this sample ("research evidence seekers") had been trained in how to use the internet to find research evidence. There was no difference when looking at the North and South sub-samples, but a possible difference is noticeable when looking at gender with 38% of women (136 out of 361) saying they had received formal training in how to use the internet to find research evidence, compared to 27% of men (131 out of 477).

Question 22: How often do you search the internet (from any location and using any device) for RESEARCH EVIDENCE for yourself or for other people as part of your work? (Please tick one box only), n=850

Figure 16 Frequency of searching the internet for research evidence

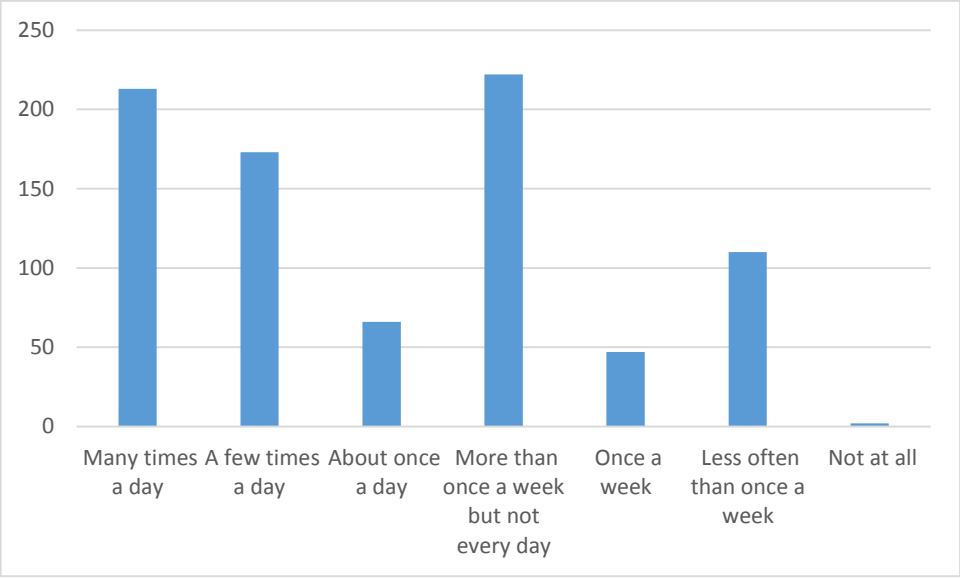


Table 24: Frequency of searching the internet to find research evidence online

Frequency	Number of responses	%
Many times a day	213	26
A few times a day	173	21
About once a day	66	8
More than once a week but not every day	222	27
Once a week	47	6
Less often than once a week	110	13

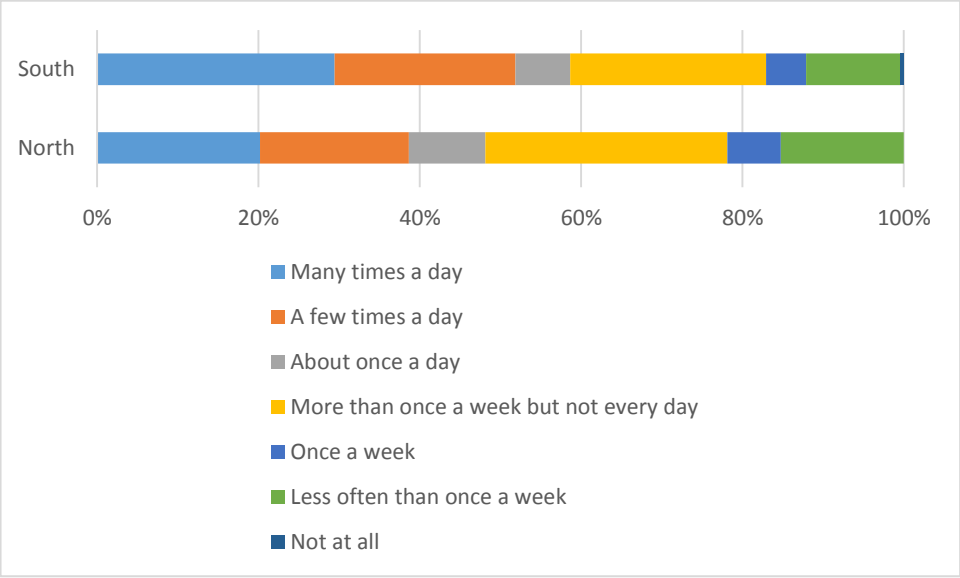
Frequency	Number of responses	%
Not at all	2	0
Total:	833	100
Don't know	17	

The chart above excludes the 17 respondents who selected "Don't know" and shows that about a quarter of this sample search the internet for research evidence for work many times a day and 82% search at least as often as more than once a week. The two respondents who said "Not at all" (i.e. who do not look online for research evidence) were a Southern Researcher/Academic and a Project/Programme Manager for a Government Donor Agency in the South who had explained in earlier questions that they use evidence but other people collect it for them when requested. These two respondents were redirected to Question 33 onwards and the sample for Questions 23 to 32 becomes 848.

46% of respondents who were able to say how often they searched the internet for research evidence, said that they searched a few or many times a day (386 out of 833). There appears to be a difference between North and South for this result:

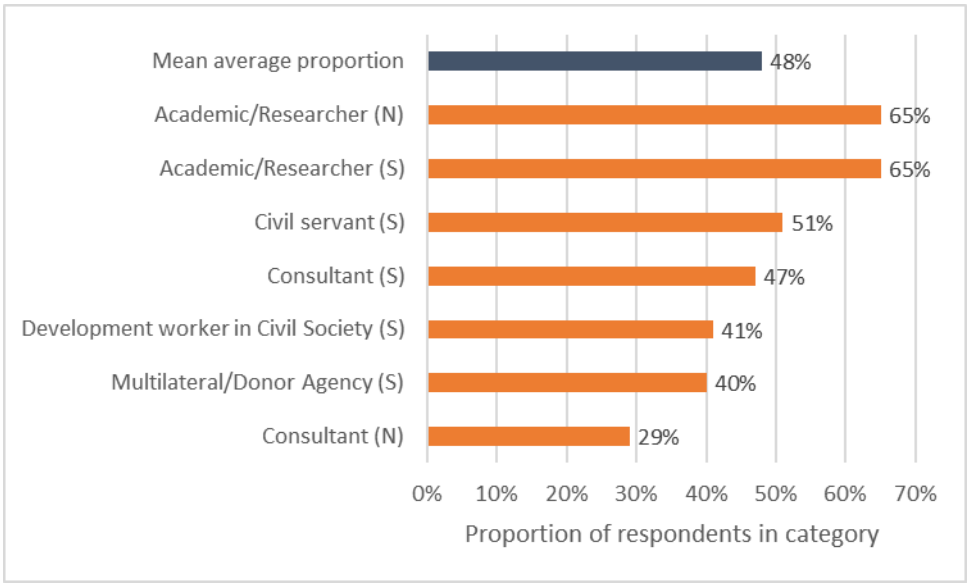
39% of those based in the North answered a few or many times a day compared to 52% of those based in the South, i.e. those in the South tend to search more frequently.

Figure 17 Frequency of searching the internet for research evidence



48% of Primary Intended Users (compared to 46% of all Intended Users) said they searched online for research evidence for work a few or many times a day. The chart below shows how this behaviour varies by category. The numbers of respondents in each Priority Target Group requires large percentage gaps between findings in order for conclusions about differences in behaviour to be inferred but we were able to identify that Academics/Researchers (globally) were more likely to be searching this frequently for research evidence, than any other group.

Figure 18 Primary Intended Users searching online for research evidence a few or many times a day, by category



Question 23: Please think about the last time you searched online for RESEARCH EVIDENCE for yourself or other people for work. Which, if any, of the following did you do? (Please tick all that apply), n=848

Figure 19 Respondents' approach to searching

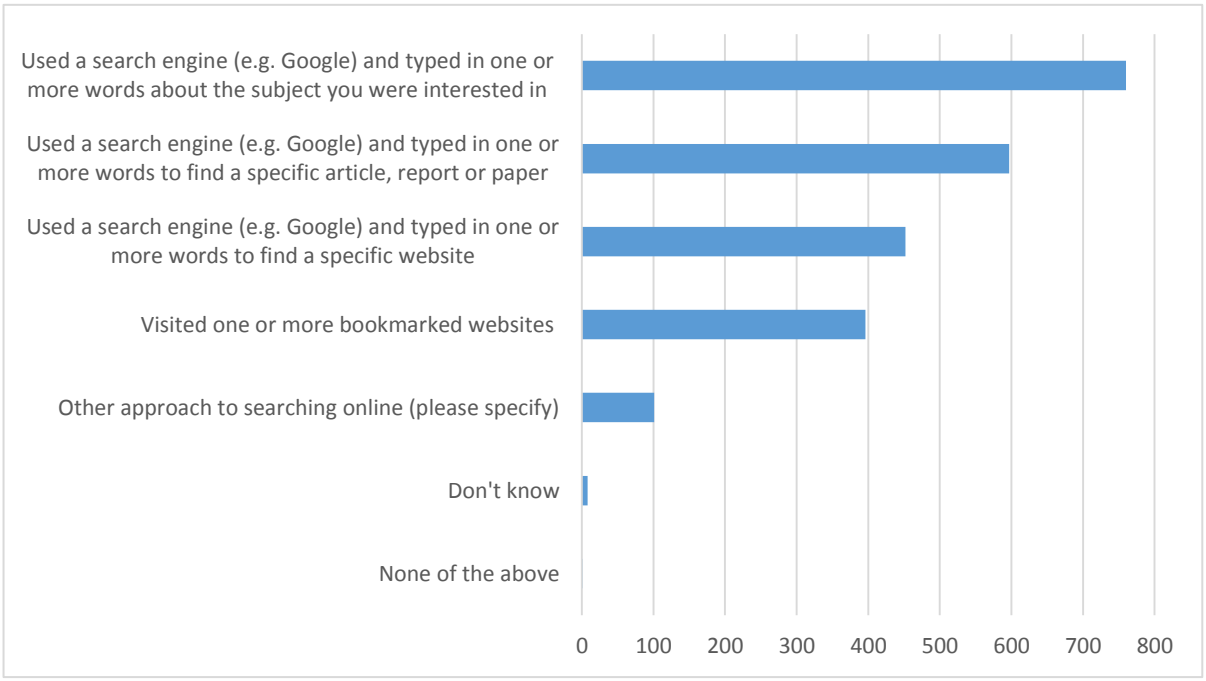


Table 25: Respondents' approach to searching

Approach to searching	Number of respondents	%
Used a search engine (e.g. Google) and typed in one or more words **about the subject** you were interested in	760	90
Used a search engine (e.g. Google) and typed in one or more words **to find a specific article, report or paper**	597	70

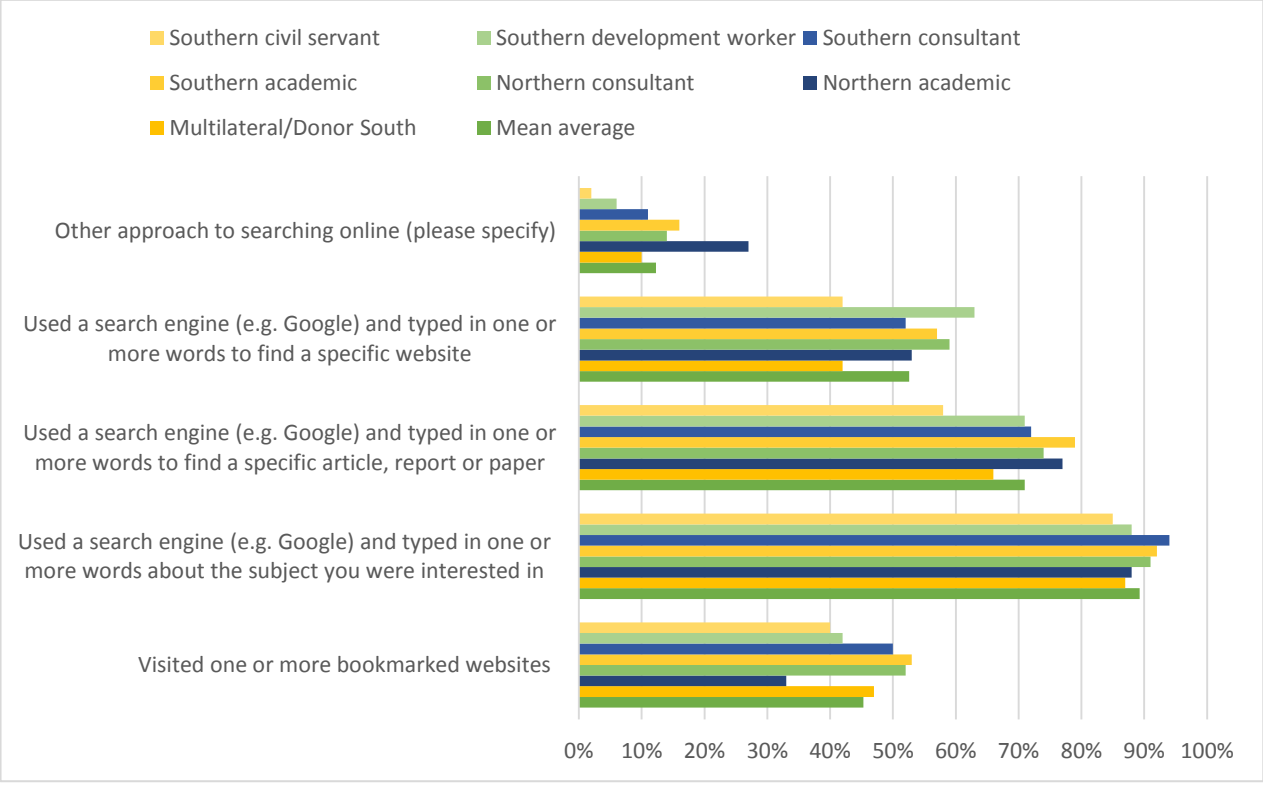
Approach to searching	Number of respondents	%
Used a search engine (e.g. Google) and typed in one or more words **to find a specific website**	452	53
Visited one or more bookmarked websites	396	47
Other approach to searching online (please specify)	101	12
Don't know	8	1
None of the above	1	<1

848 respondents answered the question. 54% used a search engine to navigate their way to a known website which means that a proportion of organic search in website statistics could be accounted for by people who are using a search engine as a substitute for bookmarking or typing a full address into the navigation bar on a browser. If this is the case in the wider population, it implies a need for portals to focus on homepage design and website name promotion.

101 respondents provided one or more other approach to searching. Those occurring most frequently were searching online resources provided by a library, research databases such as ResearchGate, or citation/bibliographic databases (particularly PubMed).

The chart below illustrates the distribution of responses among the Primary Intended Users.

Figure 20 Approaches used by Primary Intended Users, the last time they searched online for research evidence



The table below focuses on responses from Southern Civil Servants who are research evidence seekers, n=55:

Table 26: Southern Civil Servant research evidence seekers' approach to searching

Approach to searching	Number of respondents
Used a search engine (e.g. Google) and typed in one or more words **about the subject** you were	47

Approach to searching	Number of respondents
interested in	
Used a search engine (e.g. Google) and typed in one or more words **to find a specific article, report or paper**	32
Used a search engine (e.g. Google) and typed in one or more words **to find a specific website**	23
Visited one or more bookmarked websites	22
Don't know	3
Other approach to searching online (please specify)	1
None of the above	0

Question 24: On that same occasion, while searching online for RESEARCH EVIDENCE, did you do any of the following?
(Please tick all that apply), n=848

Figure 21 Action taken while searching for research evidence

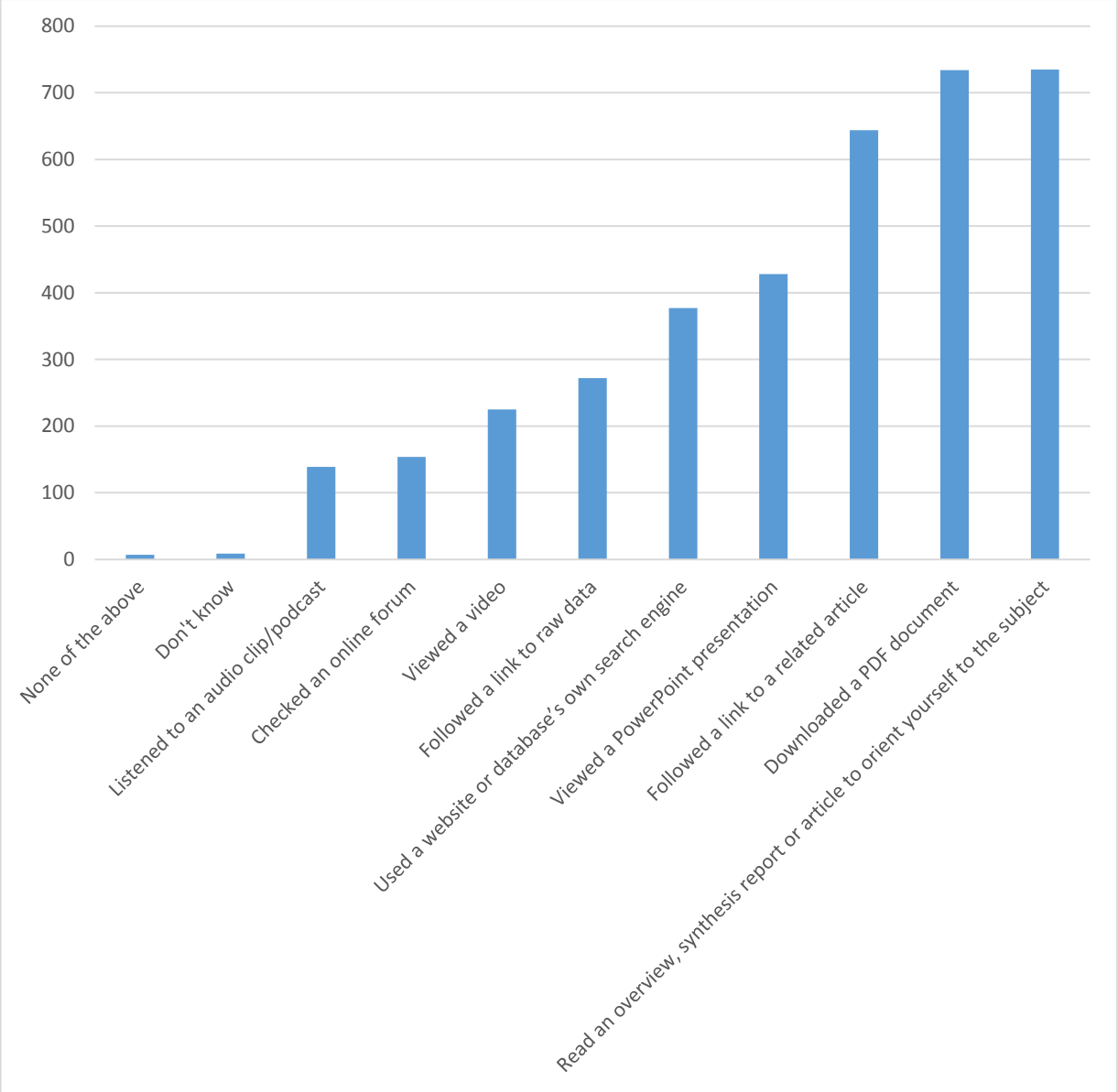


Table 27: Action taken while searching for research evidence[16]

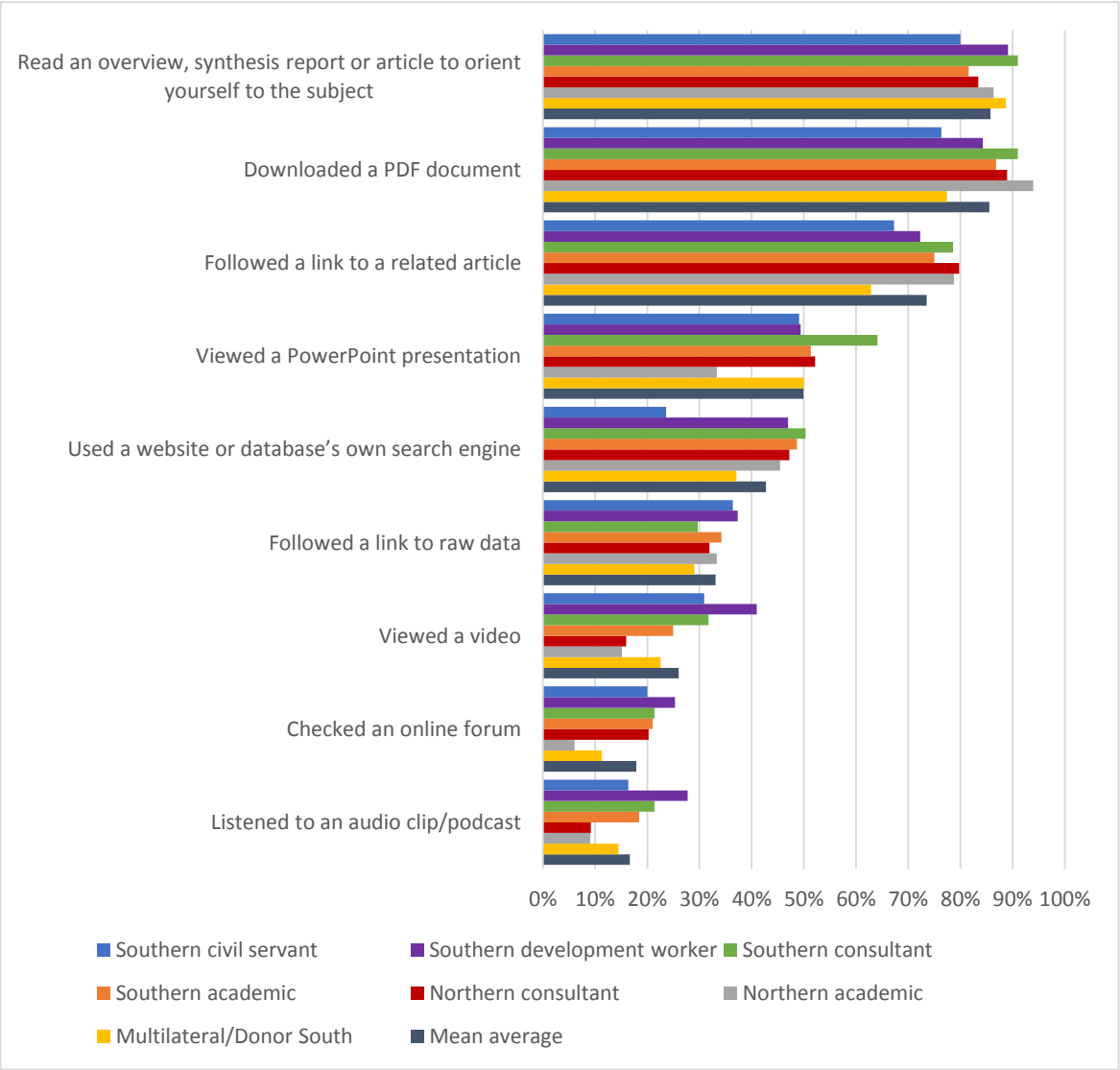
Action taken	Number of respondents	%
Read an overview, synthesis report or article to orient yourself to the subject	735	87%
Viewed a video	225	27%
Viewed a PowerPoint presentation	428	51%
Listened to an audio clip/podcast	139	16%
Followed a link to raw data	272	32%
Followed a link to a related article	644	76%
Used a website or database's own search engine	377	45%
Checked an online forum	154	18%
Downloaded a PDF document	734	87%
None of the above	7	<1%
Don't know	9	1%

The use of internal site search was of particular interest to the Evaluation team so was analysed more closely. There is no obvious gender or North/South difference among those who answered the question but there is a difference based on if respondents said they had received formal training in how to find research evidence online: 52% of people who said they had received formal training used a site search the last time they looked online for research evidence (140/271) compared to 41% of people who said they hadn't had formal training (233/563).

The chart below illustrates the responses provided by Primary Intended Users:

[16] It must be remembered that the Market Research only reports on what people say they remember doing not what they actually did.

Figure 22 Action taken by Primary Intended Users while searching for research evidence



Question 25: On that same occasion what did you do, if anything, with the RESEARCH EVIDENCE you found during your search online?
(Please tick all that apply), n=848

Figure 23 What respondents did with the research evidence they found online

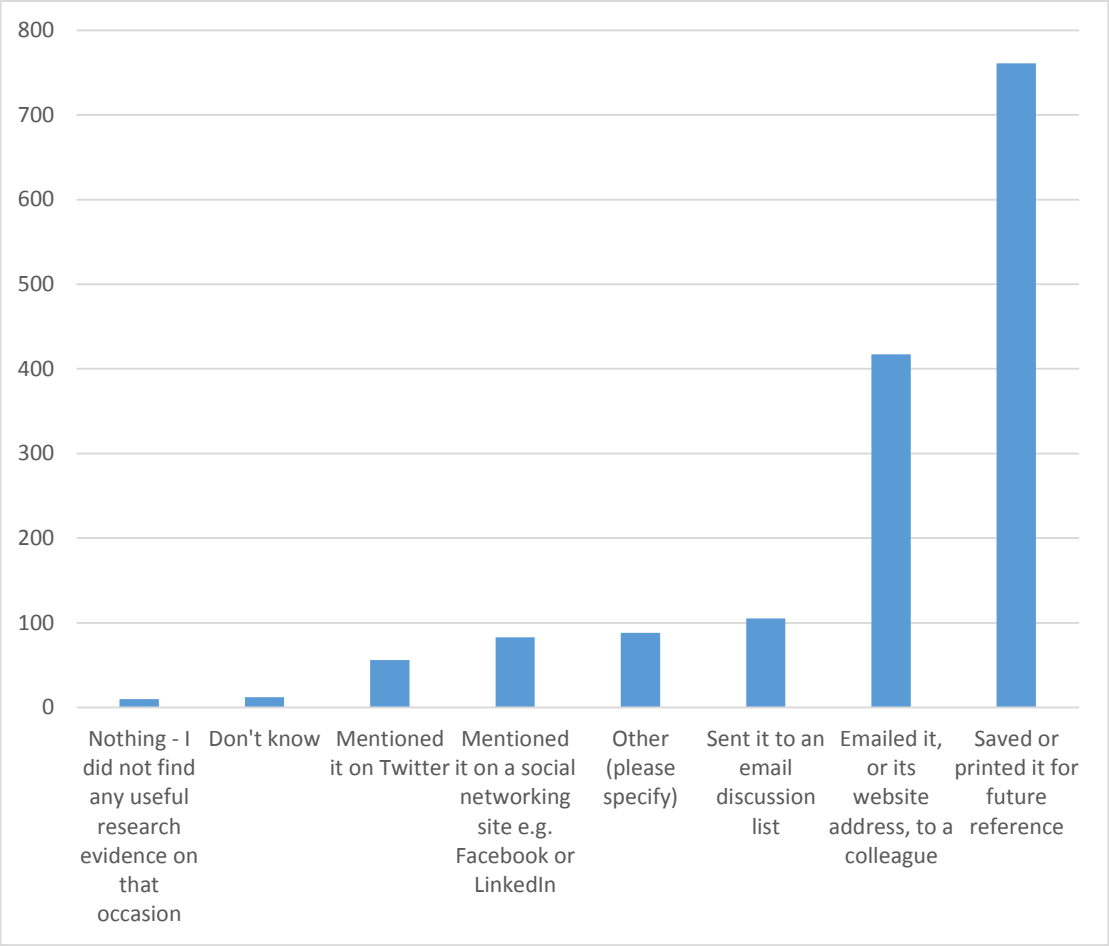


Table 28: What respondents did with the research evidence they found online

Action taken	Number of respondents	%
Emailed it, or its website address, to a colleague	417	49
Saved or printed it for future reference	761	90
Mentioned it on Twitter	56	7
Mentioned it on a social networking site e.g. Facebook or LinkedIn	83	10
Sent it to an email discussion list	105	12
Other (please specify)	88	10
Nothing - I did not find any useful research evidence on that occasion	10	1
Don't know	12	1

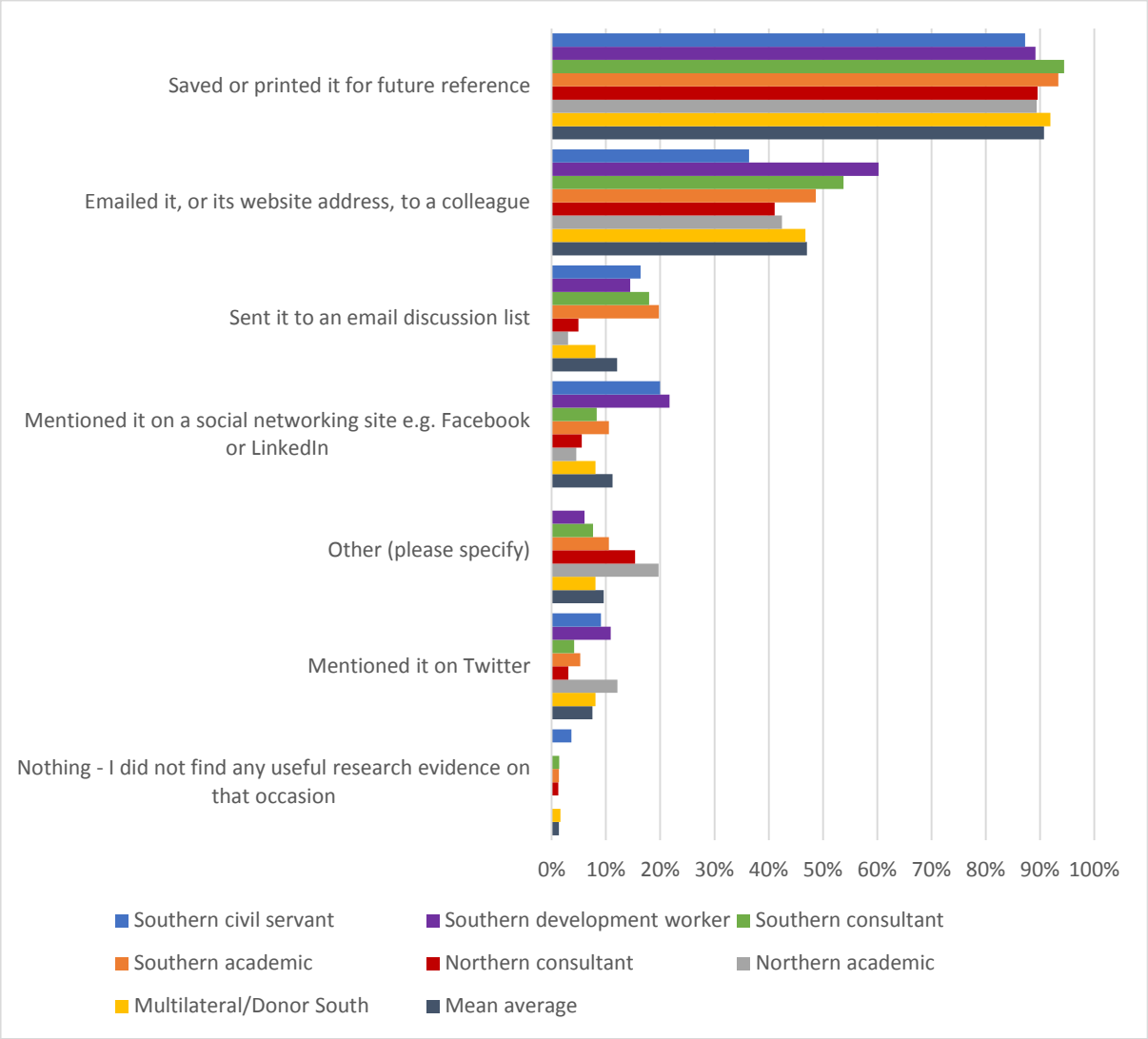
Of the 88 Other responses, the most common next steps taken were:

Used/cited it in a report (31 respondents)

Used/cited it in academic paper/presentation (16 respondents)

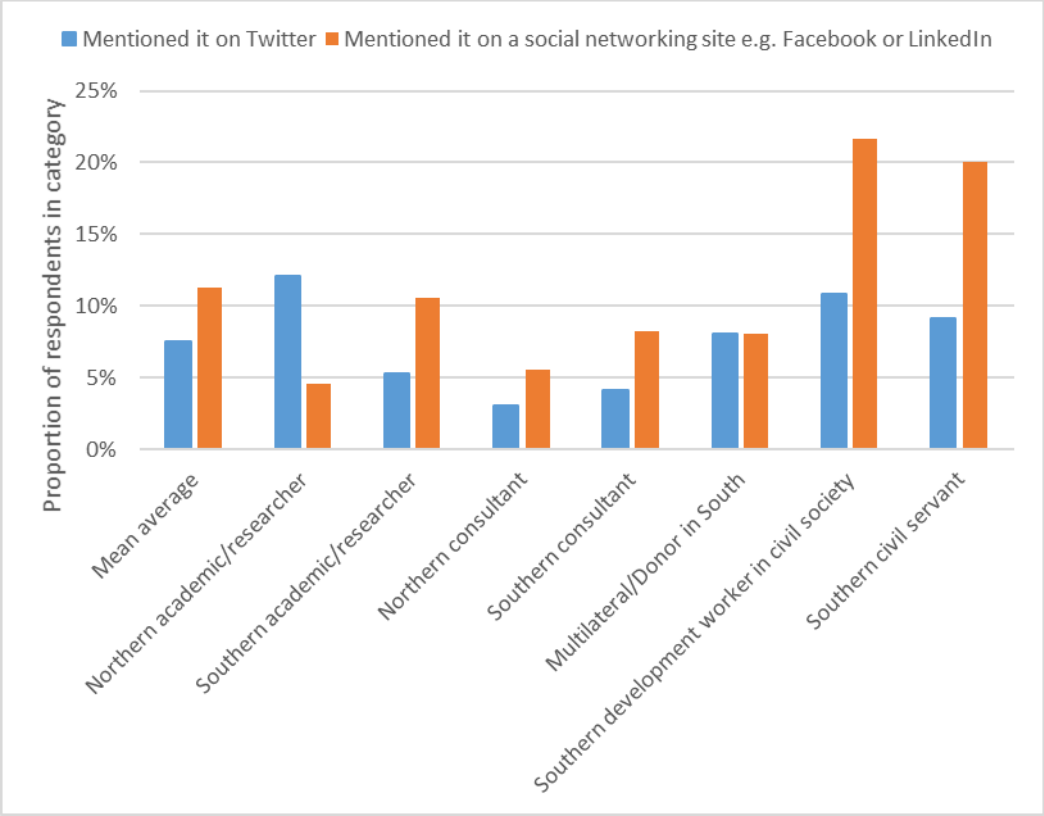
The chart below presents the findings as provided by Primary Intended Users:

Figure 24 What Primary Intended User respondents did with the research evidence they found online



Across the evaluation research methods, there was a focus on investigating the extent to which social media is a means of discovering research evidence. The person making the research evidence available on social media need not be the originator or publisher, as this next chart shows by presenting the proportions of Primary Intended Users who mentioned on either Twitter or a social networking site, the research they found online.

Figure 25 Proportions of Primary Intended Users who mentioned on either Twitter or a social networking site, the research they found online



Q26: If you would like to explain any of your answers, please use the space below:

This question was asked to aid understanding of information behaviour of individual respondents in relation to the preceding three questions. There were 58 responses to this question.

Q27: What, if any, problems do you commonly experience in using the internet to search for, find or access RESEARCH EVIDENCE for yourself or other people, for work? (Please write in below), n=848

653 respondents out of 848 provided an answer; 195 selected the option “Not applicable/don’t know”. 574 of these were Intended Users of the research portals and repositories. 36 of these made comments that confirmed they have few or no problems and 4 comments could not be classified (meaning was too unclear). As a result, **73% of the 732 Intended Users asked this question commonly experience one or more problems in using the internet to search for, find or access research evidence for work, particularly:**

38% (204 of 538 Intended Users reporting a problem) Gated access to webpages or files (requires password, site blocked by organisation or country firewalls, subscription or payment required)

25% (135 of 538 Intended Users reporting a problem) Problems related to searching (don’t know how to construct searches, general search engine problems e.g. Google, too many search results, etc.)

20% (107 of 538 Intended Users reporting a problem) Problems with Internet connection (low bandwidth, power failure, etc.) [14 of these were Northern respondents including those who commented they experienced it when doing work overseas].

3% (14 respondents of 538 Intended Users reporting a problem) Usability of specific websites

Other problems include:

- unable to judge sources for quality
- finding contradicting information within the same search
- being side-tracked (by interesting "stuff") while searching

Out of 497 Southern respondents (not limited to Intended Users), 28% reported problems relating to paywalls and firewalls, 21% reported problems with internet access, and 19% reported problems with searching the internet.

Q28a: Can you give us an example of a website (NOT a search engine) that you frequently use to find or access RESEARCH EVIDENCE for yourself or other people, for work?
(Please write the name of the website below, including the website address if possible), n=848

663 respondents provided one or more examples (out of a possible 848). This question was deliberately positioned before Q33 so that any mentions of Eldis, R4D or SciDev.Net at this point would be unprompted. Some caution should be used in interpreting the findings from this question as some respondents listed more than one website, others named generic types of websites (these were excluded from analysis) and the naming of a website could be linked to how recently it was used. However, the data is useful for investigating what type of websites are frequently used and why (when combined with the data obtained in the next question).

The infographic below shows how frequently different websites were named, unprompted, by Intended Users (the size of the text is relative to the frequency of mentions and limited to the top 50 websites). The most frequently named websites were: WHO[17], and the World Bank, with 42 and 40 mentioned respectively, followed by the PubMed, Lancet and ODI websites (all mentioned at least 20 times). Google, Google Scholar, Cochrane, DFID, ResearchGate and Eldis followed, with at least 10 mentions each. R4D was mentioned by 8 Intended Users and SciDev.Net by 3.

[17] Some respondents named specific sub-websites from these organisations e.g. the World Bank Publications website; these sub-sites were counted separately, and do not appear in the infographic due to the infrequency with which they were mentioned.

Figure 26 Infographic indicates which websites were most commonly named by Intended Users as sources frequently used to find research evidence



Among the responses provided by Southern Civil Servants it was evident that the government websites of the countries in which they are based are key sources.

Q29: Why do you use this website frequently to find or access RESEARCH EVIDENCE for yourself or other people for work?
(Please write in below)

Of the 663 respondents who answered Q28, 472 Intended User respondents gave usable responses to this question. These were analysed to identify any common themes. The most common reasons people gave were:

- Reliable content (trusted/of good quality/reliable) – 175
- Content is relevant – 156
- Comprehensive collection (breadth and volume) – 102
- Website is easy to use or convenient – 58
- Content is up-to-date – 40
- Website provides links to other sources – 25
- Content is well written e.g. good summaries or synthesis – 20
- Full text material is free – 14

If we assume that respondents use “being likely to find relevant content” as a prerequisite for repeated use of a website when looking for research evidence and thus exclude “Content is relevant” from analysis, then the next three most important criteria for respondents are: comprehensive collection, trusted content, easy to use/convenient website. The following websites were all described by one or more respondents as having all three of these criteria:

Maternal Health Task Force (based at Harvard School of Public Health) http://www.mhtf.org/

JSTOR (journal articles accessed via library subscription) http://www.jstor.org/

REACH Resource Centre (for emergency, recovery and development decision-making) http://www.reachresourcecentre.info/

3ie (International Initiative for Impact Evaluation) http://www.3ieimpact.org/

PubMed (biomedical citations database) http://www.ncbi.nlm.nih.gov/pubmed

Institute for Health Metrics and Evaluation (based at University of Washington)http://www.healthdata.org/

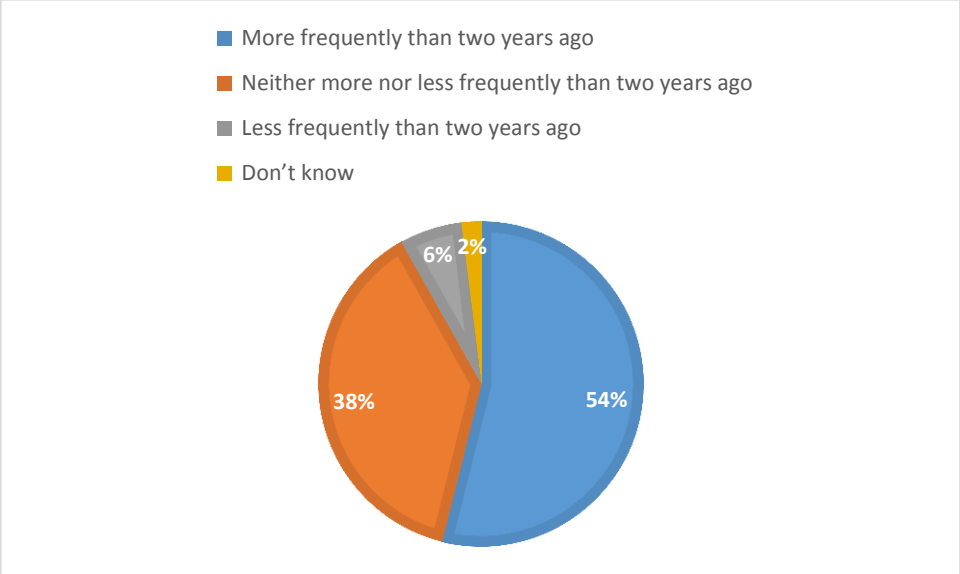
Questions 30 to 32

The free text responses provided for these questions have been analysed and reported on separately in Section 5 of the main Evaluation Report as an input to drawing out plausible pathways between portal use and uptake of evidence in policy and practice.

Q30: Comparing your experience now to two years ago, please select the response which most accurately describes your experience:

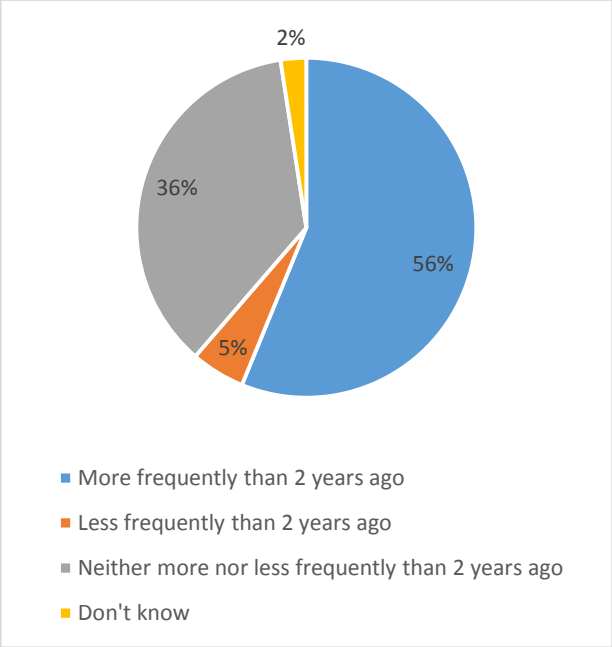
'I look online for RESEARCH EVIDENCE for work....'
(Please tick one box only), n=848

Figure 27 Frequency respondents look online for research evidence in comparison to two years ago



The following chart shows how these proportions change for the Primary Intended Users only.

Figure 28 Frequency Primary Intended User respondents look online for research evidence in comparison to two years ago

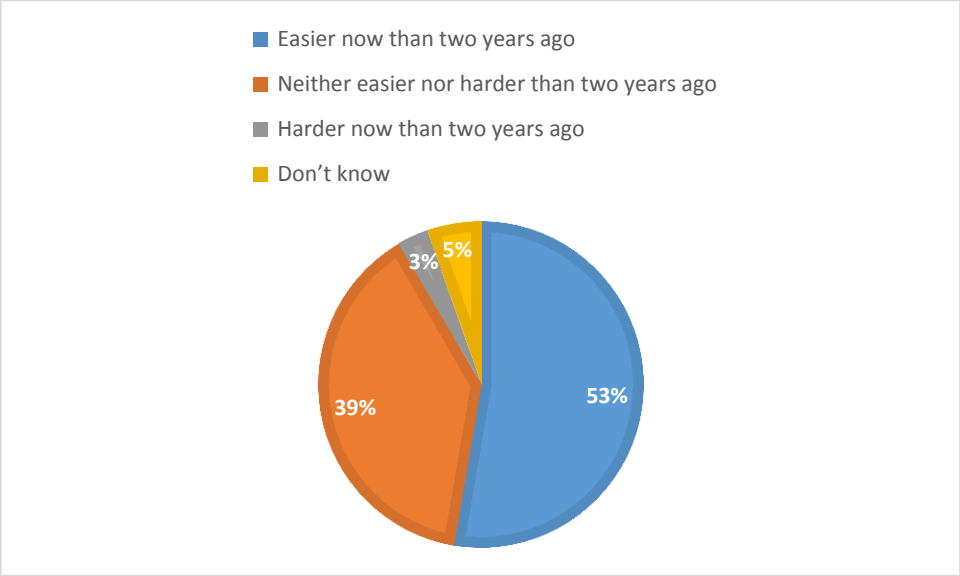


Q30a: You said that you look online for RESEARCH EVIDENCE for work **[RESPONSE to Q30]**. Please use the box below to explain why this is. (Please write in below)

743 people provided additional information.

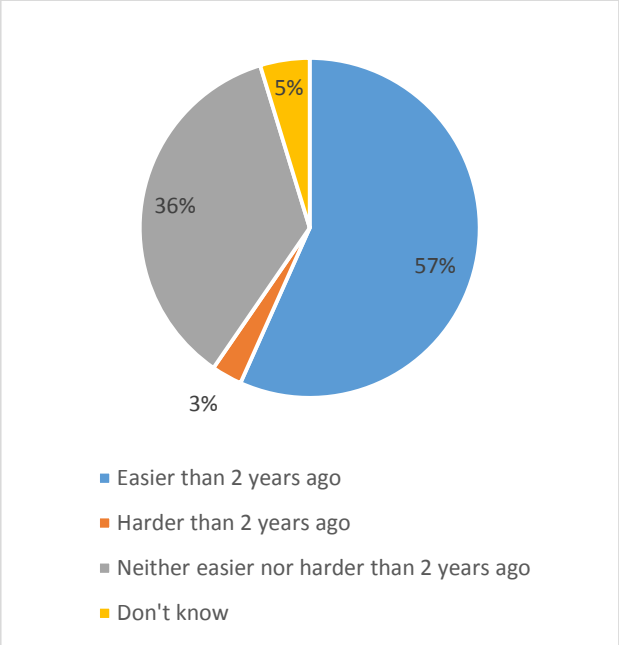
Q31: Comparing your experience now to two years ago, please select the response which most accurately describes your experience: 'Finding RESEARCH EVIDENCE online for work is...' (Please tick one box only), n=848

Figure 29 Ease of finding research evidence online compared to two years ago



The chart below shows how the proportions differ for Primary Intended Users only.

Figure 30 Ease of finding research evidence online compared to two years ago for Primary Intended Users

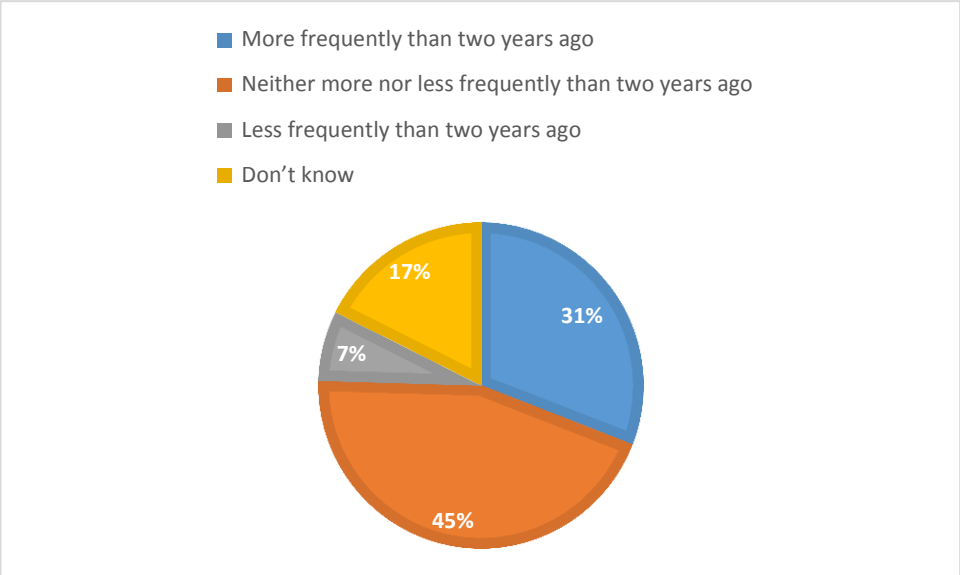


Q31a[18]: You said that finding RESEARCH EVIDENCE online for work is [RESPONSE to Q31]. Please use the box below to explain why this is. (Please write in below)

674 people provided additional information.

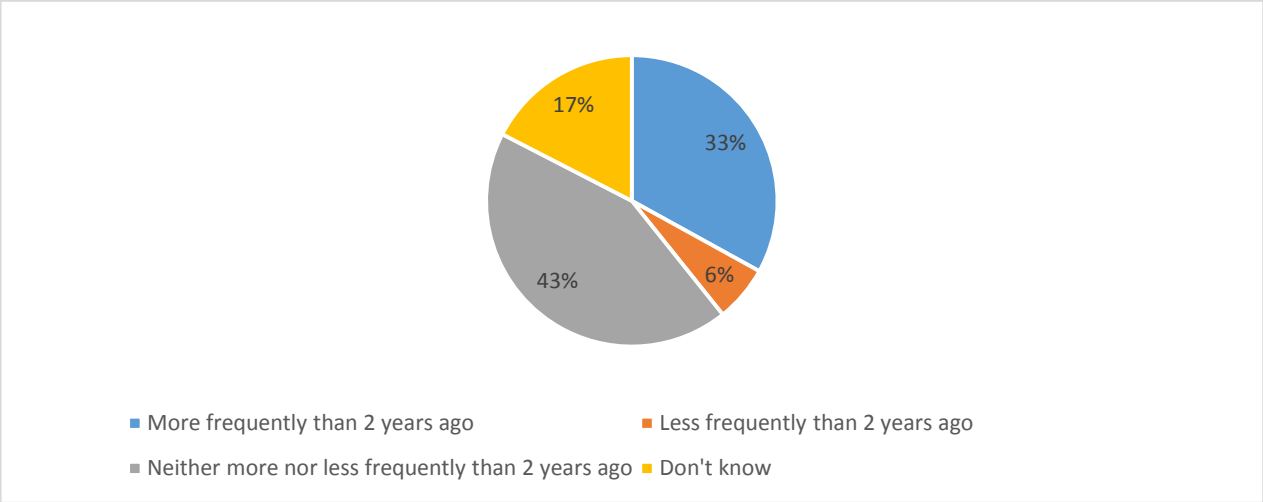
Q32: Comparing your experience now to two years ago, please select the response which most accurately describes your experience: 'My colleagues and I discuss the quality of RESEARCH EVIDENCE obtained online...' (Please tick one box only), n=848

Figure 31 Frequency of discussing quality of research evidence obtained compared to two years ago



Again, the chart below illustrates the proportions when looking at the Primary Intended Users only.

Figure 32 Frequency of discussing quality of research evidence obtained compared to two years ago (Primary Intended Users)



[18] The results of the analysis of responses to Q30a, 31a and 32a can be found in the main report Section 6.

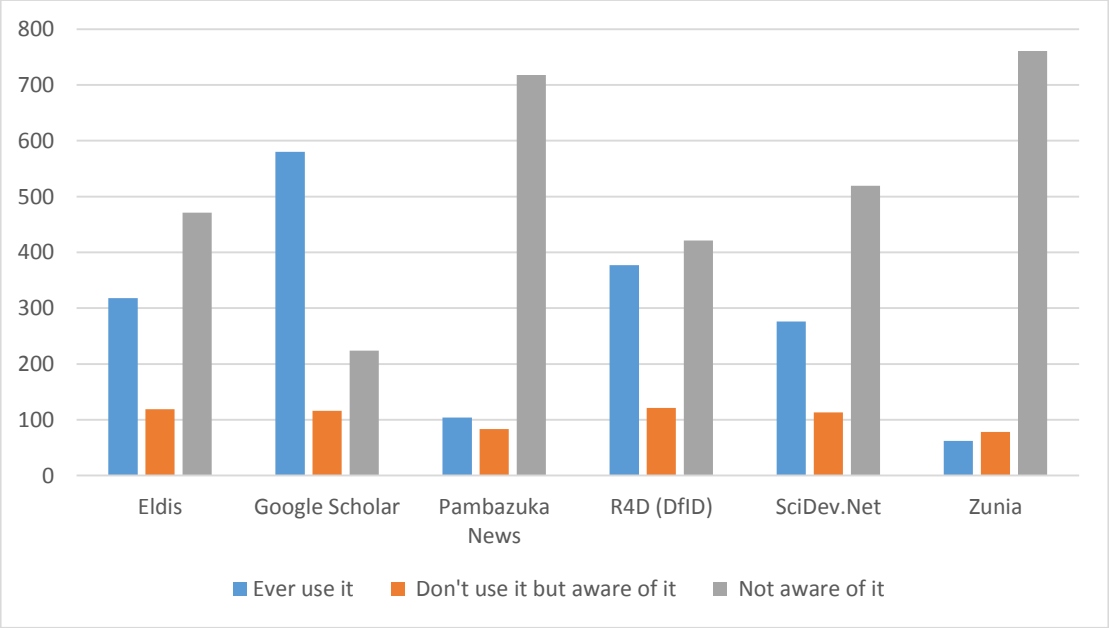
Q32a: You said that you and your colleagues discuss the quality of RESEARCH EVIDENCE obtained online [RESPONSE TO Q32]. Please use the box below to explain why this is. (Please write in below)

517 people provided additional information.

Q33a to Q33n: How often do you use the following websites (for any purpose)? Frequently, Occasionally, Rarely, Don't use it but aware of it, Not aware of it, Not sure (Please tick one box only per row), n=945

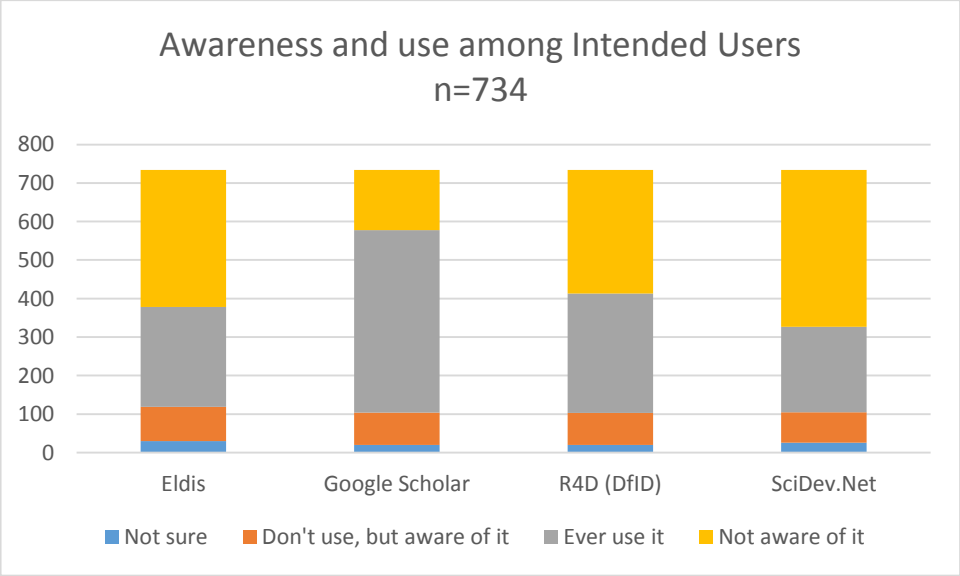
The chart below illustrates the level of use (at all) and awareness among Intended Users, of the three DFID portals, the sites used as comparators in the Value for Money assessment (Section 8 of Main Evaluation Report) and Google Scholar as a benchmark site.

Figure 33 Comparing use and awareness of the DFID funded portals and their comparators



The market research found that among Intended Users, awareness of the DFID portals ranged between 41% (SciDev.Net) to 54% (R4D), with Eldis occupying the middle at 47% as illustrated below. As a benchmark, Google Scholar's awareness among this same group was 76%.

Figure 34 Awareness and use among Intended Users



However, awareness is only part of the story as all the portals had some unmet potential to increase their use with between 21% (R4D) and 26% (SciDev.Net and Eldis) of those who were aware of the portals reported that they did not use them. Among these, just 16 were the same people and cut across several sectors, target group categories and countries. By comparison, Google Scholar's unmet potential for use was 15%. A conclusion one might draw from this analysis is that to attract 100 new users to R4D, for example, you need to make 126 people aware of it. Though, this should be approached with caution as it is unlikely to be as straightforward as this and absolute measures of awareness in themselves are unhelpful (although comparing year on year trends of awareness can be useful if promotional activity is being undertaken). However, it is reasonable to conclude that there is the opportunity to both increase awareness and to increase use by understanding what needs the portals are not meeting for Intended Users and using this understanding to guide decision-making about content, design, promotion, associated services, etc.

The same analysis was carried out on the three sectors that had more than 50 Intended User respondents: Health (considered to be relevant to all three DFID portals), Education and Generalist.

Figure 35 Awareness and use within health sector

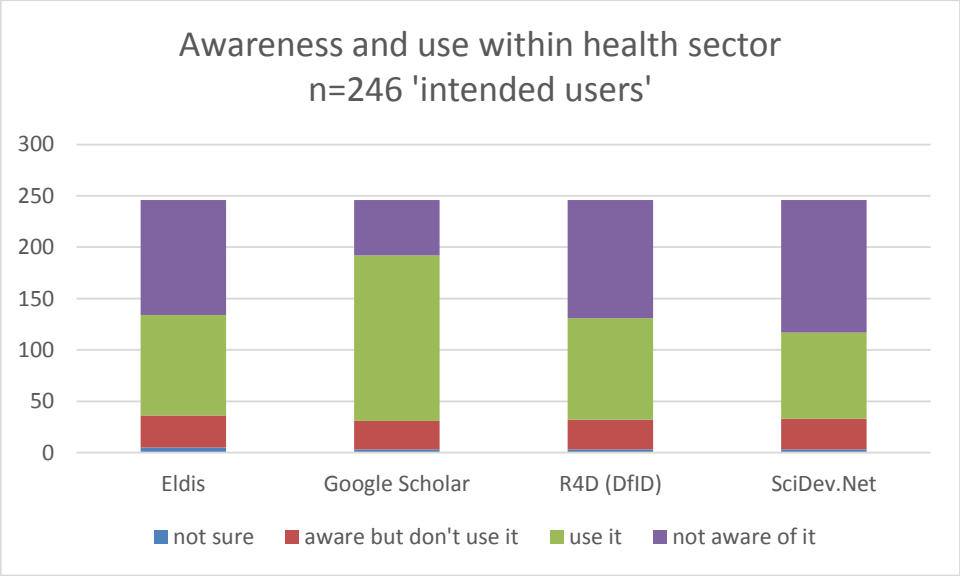


Figure 36 Awareness and use within education sector

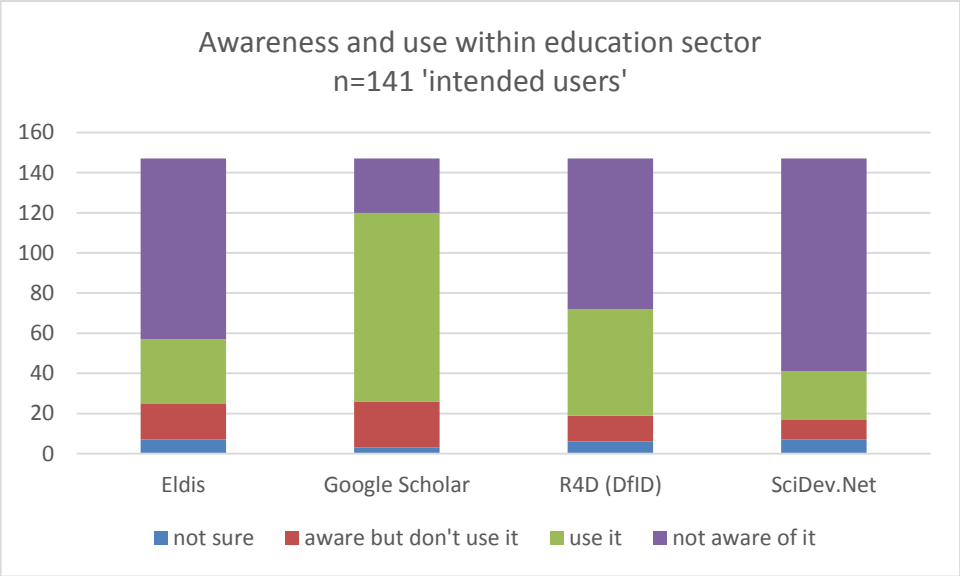
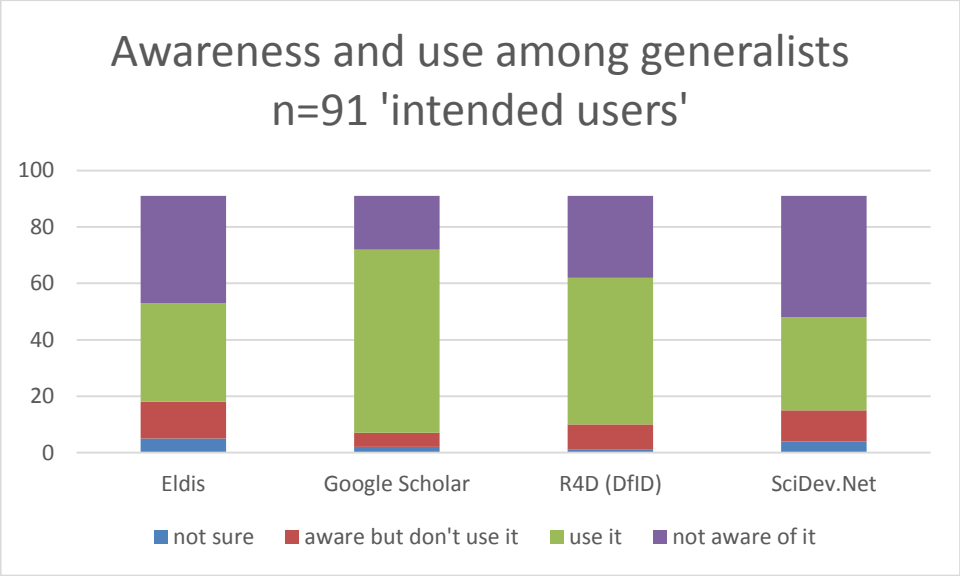
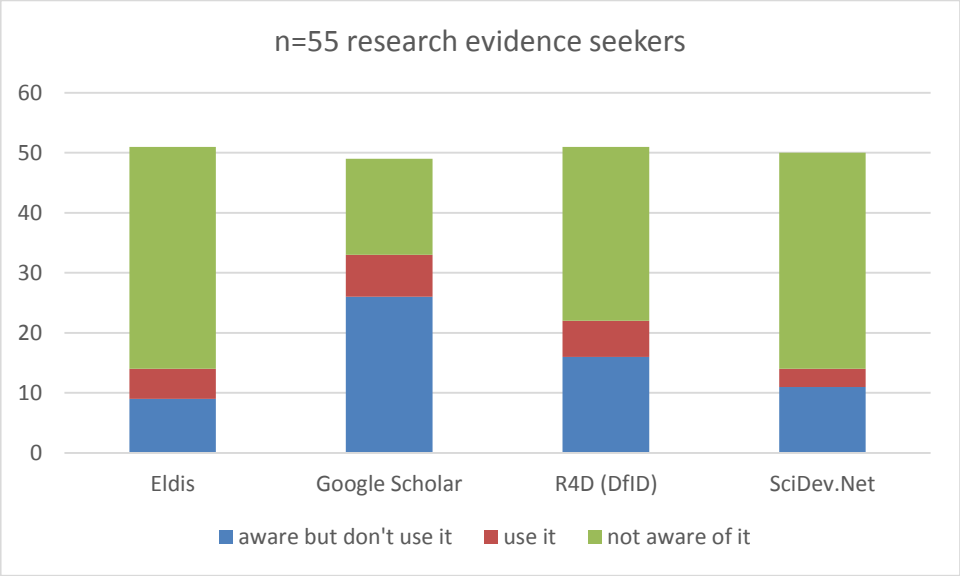


Figure 37 Awareness and use amongst generalists



A similar pattern of awareness and use can be seen among the Southern civil servants (see chart below).

Figure 38 Awareness and use amongst Southern civil servants



Other findings of note from the sample n=734 (Intended Users) include:

- 119 respondents said they use Eldis, R4D and SciDev.Net, including six Southern civil servants (in a mix of sectors and countries).
- 16 respondents said they were aware of but did not use any of Eldis, R4D and SciDev.Net.
- 198 respondents are aware of none of the three websites.

Eldis, R4D and SciDev.Net each had a small proportion of respondents who were aware of their website but did not use it (between 79 and 89 respondents). Among these, just 16 were the same people across several sectors, target group categories and countries.

Questions 34 to 48

Respondents were asked follow-up questions about their perception of the DFID portals depending on the response given in Q33 about their awareness or use of Eldis, R4D or SciDev.Net. Frequent users were asked why they use the website so often, while occasional, rare or non-users (who were aware of the sites) were asked what changes would need to be made, if any, to increase their use. It was not possible to randomise the order of the portals listed in Q33 for technical reasons so there is a slight limitation that those who were subsequently asked about all three portals will have been asked about Eldis first and SciDev.net last, and may have provided less information about the latter portal.

All the comments have been analysed for common themes and these are presented below with comments from Intended Users that illustrate these common themes.

Eldis

Frequent users of Eldis – largely because they appreciate the content (and some comments on ease of use, and from getting the email newsletter.)

It offers a lot of information on development work as well as manuals or handbooks valuable to our policy work. [Southern civil servant]

it really has very current information [Southern development worker in civil society, working in Agriculture, Food Security and Nutrition]

It tends to provide me with the kind of information which I need for my work, and it is more accessible than some of the other sites. [Southern development worker in civil society]

good repository of grey literature as well as published. [Southern academic/researcher]

Because it is really useful, and maybe the most efficient - among the ones i know - in terms of "value for time" and relevance of the findings. [Northern development consultant]

And from people who use it less frequently, or not at all, about what would need to change for them to use Eldis more often or at all (these were mostly about needing more content relevant to the respondent, but then improving the search function and the need to prompt them to use it/being discoverable]

Before I worked on development, and now on health, Eldis don't have much docs on health. [Southern civil servant]

More topics on public financial management and government budgeting [Southern civil servant]

interface. sometimes very hard to get to the information. needs to be indexed better. brings up superfluous records so i would just leave the website because it is just to time consuming to go thru it. [Southern development consultant]

I used to receive regular mailings which would lead me to follow up on interesting articles and then I might browse beyond these. The mailings stopped and I didn't pick up on this to revive them. [Northern development consultant]

I forget about it, ELDIS rarely comes up in M&E discussion groups or in social media [Southern development consultant]

Two respondents observed that the website was down (believed to be on different dates)

Well: I just tried to access it to refresh my memory. The main improvement I could see is making sure it loads at all: I couldn't access it. [Southern development consultant]

Well, at the moment the site appears to be down! My previous impression of Eldis was that it focused on agriculture, and I haven't looked at it recently as that's no longer what I'm working on. However, I'm open to looking again if the site comes back up. [Northern academic]

R4D

Frequent users of R4D - on why they use it so often; again largely because they appreciate the content, but also reputation comes up quite strongly

It offers valuable information for policy support work. [Southern civil servant]

Because it offers an excellent range of information (reports & projectetc.) - it's one of my bookmarked websirtes to refer to [Southern development worker in civil society]

This website provides me with most of the information I require in my day to day work. [Southern development worker in civil society]

Government's perspective; guaranteed quality. [Northern academic/researcher]

And from people who use it less frequently, or not at all, about what would need to change for them to use R4D more often or at all. The largest number of comments are to do with subject coverage not matching the respondents' needs, although given the nature of R4D, this seems to be an explanation for reduced/lack of use, rather than a criticism. A very close second is the search function and other design issues.

Most of the information i require is region specific and DFID is mostly country specific so not sure what change can be made [Southern development worker in civil society]

More structured packaging by region [Southern civil servant]

A better search function. Site is too cluttered and hard to navigate. Site assumes that you know what you're looking for and whether it is project based or not - in reality I want to browse thematically and have better breakdown of publications within each main theme. A summary on the main page would also be useful - it takes time to click through to the website hosting the article and often it is not relevant to what I am looking for. Often the papers are not available anyway. Although the site includes information on whether a paper is peer reviewed or not there is not further information on the credibility of the publisher. I prefer to use sites that I am familiar with and which provide this level of detail. [Northern academic/researcher]

SciDev.Net

Frequent users of SciDev.Net – on why they use it so often; also largely because they appreciate the content, but in this case the next common is being prompted to use it.

Collates information on new researches across a variety of themes and areas with links for further information [Southern civil servant]

Again it is a good way to keep up with research being conducted that relates to African issues [Southern development consultant]

It has good news and timely research [Southern media professional]

And from people who use it less frequently, or not at all, about what would need to change for them to use SciDev.Net more often or at all. No single factor emerges strongest: the three most common are to do with content topic (again, is that fair given it is a specialist site), improving design and reminding people to use it.

scientific information available is sometimes not giving a clear picture of what it is like in countries being looked at [Southern development worker in civil society]

More alerts from them made available including some of the articles in open access Journals [Southern academic/researcher]

Not a bad site but the page designs are too loud and generally distractive with the information in the blocks rolling up and down [Southern civil servant]

Comments also suggest that SciDev.Net was being confused with Science Direct either because of what the respondent says about having to pay for articles or because they admit it, e.g. *Apologies! I read this as science direct. Have never heard of this*. [Southern academic on climate change]

Q49: Any other comments on the questionnaire or topics covered

238 comments were submitted. There was some feedback on the length of the questionnaire but some respondents in every category of Intended User commented on what they had learned through doing the questionnaire, usually that they had learned about useful websites they had not previously known about, but also it had made them think about how they approach their searching.

Q50: We would like to ask a selection of people who have completed this survey a few follow-up questions by email or telephone. Would you like to participate in this further research? (Please tick one box only) Yes, No, n=945

423 respondents agreed to being contacted again and supplied either an email address or telephone number.

Q52: Would you like us to email you the final report of our evaluation? (Please tick one box only) Yes, No

766 respondents requested the final report and several commented that they were particularly interested in seeing the findings.

A.2.2 Market Research Numbered Questionnaire

DFID PORTALS EVALUATION: Online market research

Questionnaire issued to Wave 1, Batch 2 onwards

Numbers have been added to the questions (including sub-questions/answer options where appropriate) based on the numbering used in the dataset. This is the questionnaire that was issued to Wave 1, Batch 2 and subsequent samples, and includes some questions not included in the version received by Wave 1, Batch 1. Some respondents will only have been asked some of the questions, based on the answers they gave earlier in the questionnaire.

Q1: Which region are you primarily based in?

- East Asia and Pacific
- Europe and Central Asia
- Latin America and Caribbean
- Middle East and North Africa
- North America
- South Asia
- Sub-Saharan Africa

Q1a to Q1h: Which country are you primarily based in? [answer choices are a dropdown list dependent on the region selected in Q1]

Q1a_last10years to Q1g_last10years: Thinking of the last 10 years, for how many of these in total have you been based in COUNTRY.?

- Less than 5 years
- 5 to 10 years
- Prefer not to say

Q2: Which type of organisation do you currently work for (if you are employed by more than one, please select the main one)?

- University department / Research institute / Think tank
- Consultancy (private sector or self-employed)
- Multilateral organisation (e.g. UN)
- Parliament / Political party
- National Government (excluding overseas aid departments)
- Government donor agency (e.g. USAID, DFID)
- Local government
- Private charitable foundation (e.g. Bill and Melinda Gates Foundation)
- International civil society organisation (charity, faith-based organisation, etc.)
- National civil society organisation (charity, faith-based organisation, etc.)
- Independent library / information service
- Media (e.g. national newspaper)
- Network (e.g. professional association)
- Other (please specify)

Q2a: please specify

Q3: Which of the following best describes your MAIN role in that organisation?

- Advisor / Consultant
- Communications / Marketing
- Community / Development worker
- Elected representative
- Fundraiser
- Journalist / Editor
- Lecturer
- Librarian / Information Professional
- Programme / Project manager
- Programme / Project support
- Researcher / Academic
- Research support
- Student
- Trainer
- Volunteer
- Other (please specify)

Q3a: Please specify

Q4: In which sector do you mainly work?

- Agriculture, Food Security and Nutrition
- Climate Change
- Conflict and Security
- Economic Growth and Labour
- Education
- Energy
- Environment
- Gender and Equality
- Health
- Infrastructure
- Water and Sanitation
- No specific sector/generalist
- None of the above (please tell us in which sector you mainly work)

Q4a: Please specify

Q5: Which of the following age categories do you fit into?

- Under 18 years
- 18 to 24 years
- 25 to 34 years
- 35 to 44 years
- 45 to 54 years
- 55 to 64 years
- 65 years or older
- Prefer not to say

Q6: What is your highest level of education?

- Masters or Doctoral degree

Bachelors degree or similar
Professional qualification
High school or baccalaureate or A-levels
Other/None of the above
Prefer not to say

Q7: Are you

Female
Male
Prefer not to say

Q8: Which of the following best describes how often you access the Internet for any purpose (i.e. for work/leisure etc.)? This should include access from any device (Desktop, Laptop, Tablet or Mobile) and from any location (home, work, internet café or any other location). (Please tick one box only)

I use the internet many times a day
I use the internet a few times a day
I use the internet about once a day
I use the internet more than once a week but not every day
I use the internet once a week
I use the internet less often than once a week
Don't know

Q9: If you would like to explain your answer, please use the space below:

Q10_1 to Q10_6: Which of the following devices do you ever use to access the Internet for any purpose (i.e. for work/leisure etc.)?
(Please tick all that apply)

Laptop or desktop computer
Computer tablet (e.g. iPad)
Mobile / cell phone
Smart TV
Other device (please specify)
Don't know

Q11_1 to q11_12: How do you use the internet, if at all, to keep up to date with your sector/profession?
(Please tick all that apply)

Periodically revisit favourite websites	Email colleagues / contacts to exchange information / articles
Subscribe to email newsletters/alerts	Join email discussion lists

- Subscribe to RSS feeds
- Sign up to follow blogs
- Join groups on online social networks e.g. Facebook or LinkedIn
- Browse online journals
- Periodically monitor Twitter
- Use Google Alerts to monitor news stories
- Other (please specify)
- Don't know

Q11a – please specify

Q12: Do you ever have to find EVIDENCE of any kind for yourself or other people as part of your work? (Please tick one box only)

- Yes
- No
- Don't know

Q13: Please think of a time you recently looked for EVIDENCE of any kind for work. What was the subject matter and what type of information or data were you looking for?
(Please write in below)

Q14: Do you ever have to find RESEARCH EVIDENCE for yourself or other people as part of your work?
(Please tick one box only)

- Yes
- No
- Don't know

Q15: If you would like to explain your answer, please use the space below:

Q16a to Q16q: How often, if ever, do you use the following sources to find RESEARCH EVIDENCE for yourself or other people as part of your work?
(Please tick one box per row)

Frequently, Occasionally, Rarely, Never, Don't Know

a) Ask colleagues to recommend sources
b) Ask friends/family to recommend sources
c) Help desks / enquiry services (online, telephone or face to face)
d) National or local newspapers and/or their online sites/apps
e) Radio programmes and/or their online sites/apps
f) Television programmes and/or their online sites/apps
g) Email newsletters/alerts
h) Email discussion lists
i) Social networking sites e.g. Facebook or LinkedIn
j) Twitter

k) Specialist journals in print or online

l) Government websites of the country in which you are primarily based, e.g. national data portals

m) Other organisational websites that make their own research, evaluation findings or data available online e.g. research institutes, bilateral donors, inter-governmental organisations or consultancies

n) Websites that summarise, profile, link to or report on other people's research, evaluation findings or data

o) Blogs

p) Other

q) Please specify

Q17: If you would like to explain any of your answers, please use the space below:

Q18: You said you use websites that summarise, profile, link to or report on other people's research, evaluation findings or data FREQUENTLY. If these websites were no longer available to you, what difference would that make to your work?
(Please write in below)

Q18a: Don't know

Q19: How do you judge whether you can trust RESEARCH EVIDENCE that you find online?
(Please write in below)

Q20: Don't know

Q21: Have you received any formal training in how to use the internet to find RESEARCH EVIDENCE?

Yes

No

Don't know

Q22: How often do you search the internet (from any location and using any device) for RESEARCH EVIDENCE for yourself or for other people as part of your work?
(Please tick one box only)

Many times a day

A few times a day

About once a day

More than once a week but not every day

Once a week

Less often than once a week

Not at all

Don't know

Q23_1 to Q23_7: Please think about the last time you searched online for RESEARCH EVIDENCE for yourself or other people for work. Which, if any, of the following did you do? (Please tick all that apply)

- Visited one or more bookmarked websites
- Used a search engine (e.g. Google) and typed in one or more words about the subject you were interested in
- Used a search engine (e.g. Google) and typed in one or more words to find a specific article, report or paper
- Used a search engine (e.g. Google) and typed in one or more words to find a specific website
- Other approach to searching online (please specify)
- None of the above
- Don't know

Q24_1 to Q24_11: On that same occasion, while searching online for RESEARCH EVIDENCE, did you do any of the following?
(Please tick all that apply)

- Read an overview, synthesis report or article to orient yourself to the subject
- Viewed a video
- Viewed a PowerPoint presentation
- Listened to an audio clip/podcast
- Followed a link to raw data
- Followed a link to a related article
- Used a website or database's own search engine
- Checked an online forum
- Downloaded a PDF document
- None of the above
- Don't know

Q25_1 to Q25_8: On that same occasion what did you do, if anything, with the RESEARCH EVIDENCE you found during your search online?
(Please tick all that apply)

- Emailed it, or its website address, to a colleague
- Saved or printed it for future reference
- Mentioned it on Twitter
- Mentioned it on a social networking site e.g. Facebook or LinkedIn
- Sent it to an email discussion list
- Other (please specify)
- Nothing - I did not find any useful research evidence on that occasion
- Don't know

Q25a: Please specify

Q26: If you would like to explain any of your answers, please use the space below:

Q27: What, if any, problems do you commonly experience in using the internet to search for, find or access RESEARCH EVIDENCE for yourself or other people, for work?
(Please write in below)

Q27a Don't know/not applicable

Q28a: Can you give us an example of a website (NOT a search engine) that you frequently use to find or access RESEARCH EVIDENCE for yourself or other people, for work?
(Please write the name of the website below, including the website address if possible)

Q28b: Don't know/not applicable

Q29: Why do you use this website frequently to find or access RESEARCH EVIDENCE for yourself or other people for work?
(Please write in below)

Q29a: Don't know

Q30: Comparing your experience now to two years ago, please select the response which most accurately describes your experience:

'I look online for RESEARCH EVIDENCE for work....'
(Please tick one box only)

- More frequently than two years ago
- Neither more nor less frequently than two years ago
- Less frequently than two years ago
- Don't know

Q30a: You said that you look online for RESEARCH EVIDENCE for work **[RESPONSE to Q30]**. Please use the box below to explain why this is.
(Please write in below)

Q30b Don't know

Q31: Comparing your experience now to two years ago, please select the response which most accurately describes your experience: 'Finding RESEARCH EVIDENCE online for work is...'
(Please tick one box only)

- Easier now than two years ago
- Neither easier nor harder than two years ago
- Harder now than two years ago
- Don't know

Q31a: You said that finding RESEARCH EVIDENCE online for work is **[RESPONSE to Q31].** Please use the box below to explain why this is.
(Please write in below)

Q31b: Don't know

Q32: Comparing your experience now to two years ago, please select the response which most accurately describes your experience: 'My colleagues and I discuss the quality of RESEARCH EVIDENCE obtained online...'
(Please tick one box only)

More frequently than two years ago

Neither more nor less frequently than two years ago

Less frequently than two years ago

Don't know

Q32a: You said that you and your colleagues discuss the quality of RESEARCH EVIDENCE obtained online **[RESPONSE TO Q32].** Please use the box below to explain why this is. (Please write in below)

Q32b: Don't know

Q33a to Q33n: How often do you use the following websites (for any purpose)? (Please tick one box only per row)

Frequently, Occasionally, Rarely, Don't use it but aware of it, Not aware of it, Not sure

a) BLDS Digital Library
b) BRIDGE
c) Communications Initiative
d) DANIDA Research Portal
e) Development Experience Clearinghouse (USAID)
f) Devex
g) Eldis
h) Global Agricultural Research Archive
i) Google Scholar
j) Open Knowledge Repository (World Bank)
k) Pambazuka News
l) R4D (DFID)
m) SciDev.Net
n) Zunia

[Q34-38 are asked depending on response given to Q33g]

Q34: You said that you use Eldis FREQUENTLY. Why do you use this website so often? (Please write in below)

Q34a: Don't know

Q35: You said that you use Eldis [OCCASIONALLY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q35a: Don't know

Q37: You said that you ARE AWARE OF BUT DON'T USE Eldis. What changes, if any, would need to be made to the website for you to use it? (Please write in below)

Q38: Don't know.

Q36: You said that you use Eldis [RARELY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q36a: Don't know

[Q39-44 are asked depending on response given to Q33l]

Q39: You said that you use R4D (DFID) FREQUENTLY. Why do you use this website so often? (Please write in below)

Q39a: Don't know

Q40: You said that you use R4D (DFID) [OCCASIONALLY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q40a: Don't know

Q41: You said that you use R4D (DFID) [RARELY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q41a: Don't know

Q42: You said that you ARE AWARE OF BUT DON'T USE R4D (DFID). What changes, if any, would need to be made to the website for you to use it? (Please write in below)

Q43: Don't know.

[Q44-48 are asked depending on response given to Q33m]

Q44: You said that you use SciDev.Net FREQUENTLY. Why do you use this website so often? (Please write in below)

Q44a: Don't know

Q45: You said that you use SciDev.Net [OCCASIONALLY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q45a: Don't know

Q46: You said that you use SciDev.Net [RARELY]. What changes, if any, would need to be made to the website for you to use it more often? (Please write in below)

Q46a: Don't know

Q47: You said that you ARE AWARE OF BUT DON'T USE SciDev.Net. What changes, if any, would need to be made to the website for you to use it? (Please write in below)

Q48: Don't know.

Q49: Any other comments

Q50: We would like to ask a selection of people who have completed this survey a few follow-up questions by email or telephone. Would you like to participate in this further research? (Please tick one box only) Yes, No

Q51: How would you prefer to be contacted regarding participation in further research? (Please tick one box only) Email, Telephone

Q52: Would you like us to email you the final report of our evaluation?
(Please tick one box only) Yes, No

Appendix B. The Country Case Studies

The original plan was to conduct case studies in Ghana, Tanzania and Bangladesh but security concerns in Dhaka led to a change in the third country which, after consultation with DFID, was agreed to be Nigeria.

B.1 Detailed description of methods used

The purpose of the in-country studies was to gather much more detailed evidence about intended portal users' information-seeking behaviour in their everyday work than could be elicited from the market research alone. The case studies employed a set of qualitative methods to better understand the policy-making context and to evaluate the portals' usefulness in context. The methods used were: semi-structured interview, face-to-face session in-country (contextual inquiry plus portal evaluation), research diaries and finally a re-run of the heuristic evaluation conducted during inception, to take on board what was learned about Intended Users from the in-country studies. This combination of these methods allowed us to capture and analyse behaviour regarding portal interaction, stories of use and uptake that illustrate the broader picture of policymaking within which the portals sit, and an assessment of the usability of the portals from the point of view of the intended audiences (both actual and potential users).

B.1.1 Recruitment and sampling

Participants were chosen using purposive sampling, based on the likelihood that they would provide useful and interesting data for our purposes. These participants came from the agreed Southern user categories but they are not expected to be representative of the wider Intended User population. Our aim was to engage Intended Users who were not current users of the DFID portals. This had several advantages:

- the participants were not over-familiar with the portals under study. Regular users of a portal react differently when using the portal than those who have not used it before;
- we could learn about people who are Intended Users but who have not embraced the portals into their working day;
- we introduced the portals to a new set of people.

Participants were chosen according to the following criteria. Participants are sought who are:

- in one of the seven categories of policy actor listed in Table 3;
- a national of the country under study, or have lived in the country for more than 10 years;
- users who regularly seek and use information to inform policy, programmes and practice;
- users who use the internet regularly for some purpose;
- prepared and able to engage with the project.

Participants fitting the following criteria were excluded from detailed analysis. Potential participants who:

- have lived, studied or worked in the North for more than 5 years within the last 10 years; two of the participants from Ghana have lived outside the country for 6 years – GH7 and GH8. They are used in our discussions when an interesting comparison can be made with other participants;
- seek and use information to inform policy, programmes and practice less than twice a week.

B.1.2 Semi-structured interview

A standard semi-structured interview was conducted from the UK via phone to explore the Intended User's background, role and activities, and their willingness to take part in subsequent aspects of the study. The aim of these interviews was to locate participants who fitted the recruitment criteria above and who could provide recent examples of using research evidence to influence policy, programmes or practice. It was particularly important that they could provide a specific output such as a report or video or presentation that illustrated the use of evidence in this way. The process is summarised below; steps 2, 3 and 4 were piloted three times with volunteers and refined after each pilot.

The interviews provided useful preparation for the participant and for the researchers who were travelling to the country, but in several cases it was not possible to conduct the interview due to time and connectivity constraints. Where this happened, a preliminary phone conversation was held in-country to confirm the participant's availability and establish a suitable example of evidence use.

The semi-structured interview process

- List of suitable people within the country was compiled to find people who use evidence in their everyday work on policy, programmes and practice; we relied on a 'point person' in each country. This person had knowledge of the country, was used to identify likely participants who fit this criteria. Just about all the participants contacted were carrying out relevant work.
- Contact people: An introductory email was sent by the point person to each contact on their list inviting them to take part, attaching a project information sheet, and copying the OU interviewer. The OU interviewer emailed contacts to arrange a time to call and requesting a phone number to call them on. If this didn't get a response, and there was a phone number, the interviewer phoned the contact.
- Confirmation of interview: An email was sent to confirm the interview time, send a consent form to read and instructions for preparing for the interview:

Please prepare for the interview by identifying a couple of recent and typical examples of looking for and using evidence to develop policy, programme or practice. For example "I was drafting a note related to the use of ICT in schools". I would like you to be able to recall the key activities that you carried out during this work such as: what evidence you used; where you found it; why you chose it; and how it informed your thinking. We will discuss these when we talk.

It would be helpful if you can identify a concrete output or outputs such as a report or presentation or sections of a policy document that demonstrates how the research evidence was used and will provide a focus for our discussions.

Interview

A brief introductory discussion checked whether the participant was ready for the interview and asked them if the interview could be audio recorded (most were). Recording was carried out using an Olympus digital voice recorder equipped with telephone pickup microphone placed between the telephone earphone and the interviewer's. Some notes were also written during the session.

The first set of questions determined the participant's suitability. This was followed by a discussion of their example(s) of evidence seeking. The aim here was to ascertain they had a recent, relevant example which they could talk about fluently and that it was not confidential (which couldn't be used in this study). They were asked to email any outputs they mentioned to be used as a focus during the face to face sessions.

At the end of the interview they were asked if they were willing to take part in a face to face session and as many details as possible were established.

Outcomes

The semi-structured interviews lasted between 5 mins (if a participant had a very restricted schedule) to 30 mins. Participants provided 1 to 3 examples, with none being confidential. The times for many face to face sessions had to be arranged when in-country as they were not able to commit to a time 1-3 weeks in advance. A small summary of key points about each participant was written up for use when preparing for and running the face to face sessions. Table 29 shows a summary of contacts made, and how many of the interviewees were seen in-country.

The interviews were designed to locate suitable participants. Detailed findings about their search behaviour were collected during the face to faces sessions. Seven people did not take part in a subsequent face to face session. Their self-reported search behaviour was similar to others in their category. These are combined with the findings from the face to face sessions.

Table 29: Phone Interviewees

Country	Number in list	Number contacted	Number of interviews	Number of these seen in-country	Comments
Ghana	28	17	5	4	Two point people in UK.; process took 3 weeks
Tanzania	60+	11	6	4	Four point people; two in-country and 2 in UK; process took 4 weeks
Nigeria	20	9	9	5	Four point people; all in-country; process took 4 weeks

Challenges

The main challenge was the difficulty faced in arranging the interviews. Reply rates to the emails were low. Mobile phones often didn't connect. This was particularly true in Ghana.

The majority of participants gave mobile phone numbers as the best way to contact them. Some interviews were arranged but due to difficulty getting through on the phone couldn't be carried out in time for country visit. It was often hard to hear the participant clearly – a function of the mobile phones and people's accents. This necessitated asking them to repeat themselves. The phone interview schedule fell across some national holidays and across national elections in Ghana. The elections made it hard to contact the MPs in Ghana. One MP interviewed had a particularly interesting use of information, but was too busy electioneering in his/her constituency to meet for a face to face session.

The process described above wasn't followed when the only way to interview someone was to carry out the interview on the spur of the moment. Some people seemed suitable but were out of the country at the time of the sessions. Some email addresses and phone numbers provided by point people didn't work. A number of people were retired, out of the country or on vacation at the time of the sessions, had changed career, were in a different city (some of these were still interviewed). Two people suffered a bereavement during the process, one person had a baby. A lot of people were happy to be interviewed in evenings or weekends.

The process of recruiting participants worked best when the point person personally knew the person and could encourage them to take part, that it was legitimate etc.

B.1.3 Face-to-face sessions in-country

The face-to-face sessions consisted of two parts. The first part was conducted using Contextual Inquiry and focused on an example where the participant had identified and used evidence recently to influence policy, programmes or practice. The example used was identified either during the semi-structured interview or immediately prior to the face-to-face session being conducted. In a few cases the example was identified during the face-to-face session. Contextual inquiry gathers data to explicate a user's daily tasks and ways of working. It follows an apprenticeship model with the researcher working as an 'apprentice' to the participant in their place of work, using a combination of observation, discussion, and reconstruction of past events (in our case, the recent example of evidence use).

The second part of the session focused on the three live DFID portals (Eldis, R4D and Scidev.net) and asked participants to use the portals to find information relevant to their example topic and specific articles (as set out in evaluation question 2). The specific articles used were chosen randomly and are listed in the user profiles.

Table 30: Specific articles used in the second half of the face-to-face session

Scidev.net	"How to communicate in an interdisciplinary team" by Jessica Thompson, December 2013 "Big data for development: Facts and figures" by Emmanuel Letouzé, April 2014
Eldis	"Strengthening of health systems for equity and development in Africa", Africa Health Strategy 2007-2015 "Three tools to unlock finance for land-use mitigation and adaptation" by Falconer et al, July 2015
R4D	"Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality" 2010-2012 "Using climate information to achieve long-term development objectives for African ports" by Woolhouse and Lumbroso, 2015

The face-to-face session was piloted three times with volunteers and the design evolved during that process. Specifically, the importance of focusing on an output from the example of evidence use was emphasised.

Challenges

Once participants had agreed to take part in the study, there were few challenges. These participants welcomed the researchers in-country and were very willing not only to spend time with us but also to engage with the process.

Outcomes

A summary of the data collected for each participant, and the spread of participants across categories and organisations is provided in the main report (section 5). In some cases, the full face-to-face session was not possible because of internet connectivity or because of limited time on the part of the participant. However we collected the agreed number of full face-to-face sessions (8 per country), and were able to supplement this number with several further sessions and informative interviews with relevant participants. Once in country we followed the purposive sampling approach and took the opportunity to collect data from any participants we could access and who would provide useful input to any of the Evaluation's questions.

The sessions were conducted in the participant's normal place of work, using their normal online environment where possible. A small number of participants came to the researchers' location bringing their own laptops. Sessions lasted between 30 minutes and 4 hours, with the majority lasting between 90 and 120 minutes. All participants agreed to be audio-recorded and all but one agreed to be video-

recorded, but in two of the sessions (GH2 and GH13) internet connectivity failed and so video-recording was not required. The video recordings focused on the participant's computer screen.

Detailed user profiles were developed from the data collected during these face-to-face sessions, and these are included in the sections below.

B.1.4 Research diaries

During the contextual inquiry part of the face-to-face session, participants were asked to re-construct their behaviour in the context of a recent example. This provides an authentic task for us to study, but it is necessarily compressed into a short time. The diary was intended to provide a different perspective on the use of research evidence by our participants, through a snapshot of activity captured over a two-week period. During the diary study, participants were asked to record their activities each day using a short online questionnaire that was mobile-enabled, i.e. it could be completed on a smartphone.

Diaries have several advantages: they do not take up much researcher time to collect data; they do not require special equipment or expertise; and they are suitable for recording data about a user's normal everyday work over a longer time frame than contextual inquiry allows.

All participants who undertook the face-to-face session were asked if they would participate in the diary study, and most agreed to try. However, diary studies rely on participants being reliable and remembering to complete them at the assigned time and the data collected depends entirely on the activity of the participant at the specific time when the diary is run.

The diary process

Initially we expected to use SurveyMonkey to run the diaries, and although SurveyMonkey is a popular platform, the design interface and distribution options are limited compared with Qualtrics, which is a far more powerful platform. The team chose Qualtrics for this project, after taking into account the views of the Open University's Student Survey Office, which distributes online surveys to tens of thousands of participants every year. Other than an unexpected fault in the automated distribution of the journals after a few days of the Diary Study, Qualtrics was a very useful platform. Even then, the daily journals could be distributed manually through the Qualtrics interface.

The exercise was designed to send each participant a journal (short questionnaire) each day to be completed during the course of that day. In general participants completed each journal on the day sent, occasionally on the next morning. Four diary sessions were run, each lasting 10 working days. Diarists were sent instructions and a rehearsal diary beforehand, in order to practice using the survey system. A few people (4) had trouble usually due to not understanding to click on a link that took them to the online journal, but were helped to access it. The study itself consisted of 9 daily journals containing the same questions, and a reflection journal on the 10th day which asked participants to consider their activity over the previous two weeks.

During each 10 day diary session several text reminders were sent, as needed, and a thank you email at the end. Individual emails from participants were replied to promptly so as to not interrupt the diary process.

Although over 30 people agreed to complete a diary when asked during the face to face sessions to taking part, and were sent a journal each day, a few (7) never engaged. This turned out to be due to a number of reasons: cost of internet access, internet not working, travelling for work, life events. The reason for the gap between number of journals opened each day and number completed is that on any particular day a few people would open the journal but make no more progress. This was often due to work commitments, internet access, travel and similar reasons.

The diary was developed iteratively with several pilot runs, and one final pilot that was sent to the rest of the team.

Challenges

The main challenge faced at this stage was to secure sufficient engagement from the participants to complete the 10-day exercise.

Concerns about the quality of internet connectivity in the participant countries were raised during the pilot stage and redesign of the online journals mitigated this to some extent (removal of pop-up windows and survey logic requiring higher bandwidth). An email-only version of the survey was designed at a later stage, as an alternative if internet connectivity was very poor, but in the event, this was not required.

Ideally, a Diary Study includes a final interview with the participant, based upon the diary entries. As this was not possible given the time constraints, a Reflection journal was included on the 10th day. This Reflection journal provoked the most discussion amongst colleagues and was not finalised until some of the questions were relocated to the Market Research. These issues were resolved, although the key meeting was held several days into the Ghana in-country study. Fortunately, there was sufficient time to finalise the journal questions for distribution a few days later.

Outcomes

A summary of the data collected for each participant, and the spread of participants across categories and organisations is provided in the main report (Section 5). Across the three case study countries, 16 'complete' sets of diary entries were collected, where 'complete' means that the participant stayed engaged to the last entry on the 10th day.

Details of the questions asked in the daily and reflection journals, and the diary data are contained in the sections below.

B.1.5 Heuristic Evaluation

This is a well-established expert-based evaluation method. Originally pioneered in the early 1990s it is commonly used in commercial practice and has been adapted for some specific types of online product (although not specifically for portals). Nielsen's set of 10 heuristics form the basis of an expert review of the product that combines an expert view of good interaction design and a clear understanding of the user population (Budui & Nielsen, 2010). In this set of studies we conducted an initial heuristic evaluation of the three DFID portals currently online during the inception stage. After the in-country studies, the exercise was conducted again, this time taking into account what had been learned about Southern users.

The results of this second exercise are contained in the sections below.

B.1.6 Generating the user profiles

A user profile was generated based on the project's evaluation questions and the analysis frameworks being employed. The data from the three methods employed to study the participants' information behaviour and research evidence use was the basis for generating a user profile for each participant who took part in a face-to-face session. Data from the phone interview, the face-to-face session and the diary were combined in this one representation for compactness.

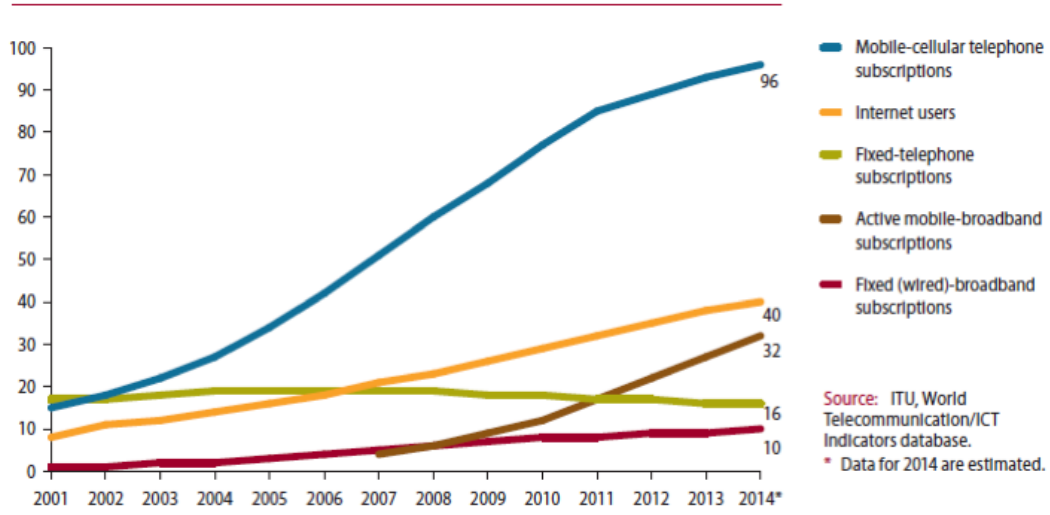
The category numbers used in the user profile are listed below:

1. Elected member of national government, e.g. Member of Parliament
2. Elected member of local or regional government, e.g. member of regional legislature or council, Councillor/local council member
3. Civil servant or public servant; a person in the public sector employed for a government department or agency
4. International aid/development worker in civil society e.g. employed by a national or international non-governmental organisation, or community based organisation
5. Academic/Researcher, e.g. researcher or postgraduate student based in a research institute, university, or think tank
6. Development consultant, e.g. employed by a private sector consultancy company
7. Knowledge broker/intermediary for policy makers, e.g. Parliamentary committee clerk, Parliamentary researcher, Parliamentary librarian, assistant/secretary to Member of Parliament, Government departmental librarian
8. Media professional, e.g. journalist, editor, commentator for online, print or broadcast media

B.2 Country Case Study ICT Context

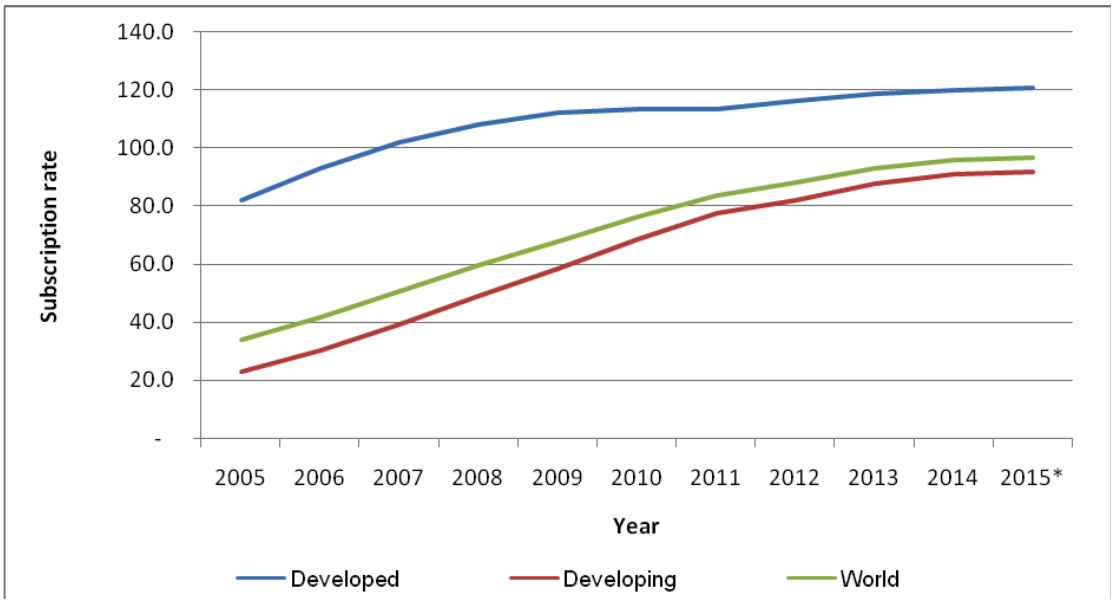
Our case study participants all relied on accessing portals and websites through computers and mobile devices. This section will talk about the current landscape for such access.

Figure 39 Global trends in access to ICT 2001-2014 (penetration rates per 100 inhabitants)



Access to ICTs continues to grow worldwide with mobile-cellular leading the way and while mobile-cellular growth rates are slowing down, they continue to grow most in the developing world (United Nations, 2014). For instance, whereas the subscription rate in developing countries increased from 22.9% in 2005 to 91.8% in 2015 (a 300% rise), that of developed countries only rose from 82.1% to 120.6% within the same period, representing a 40% rise.

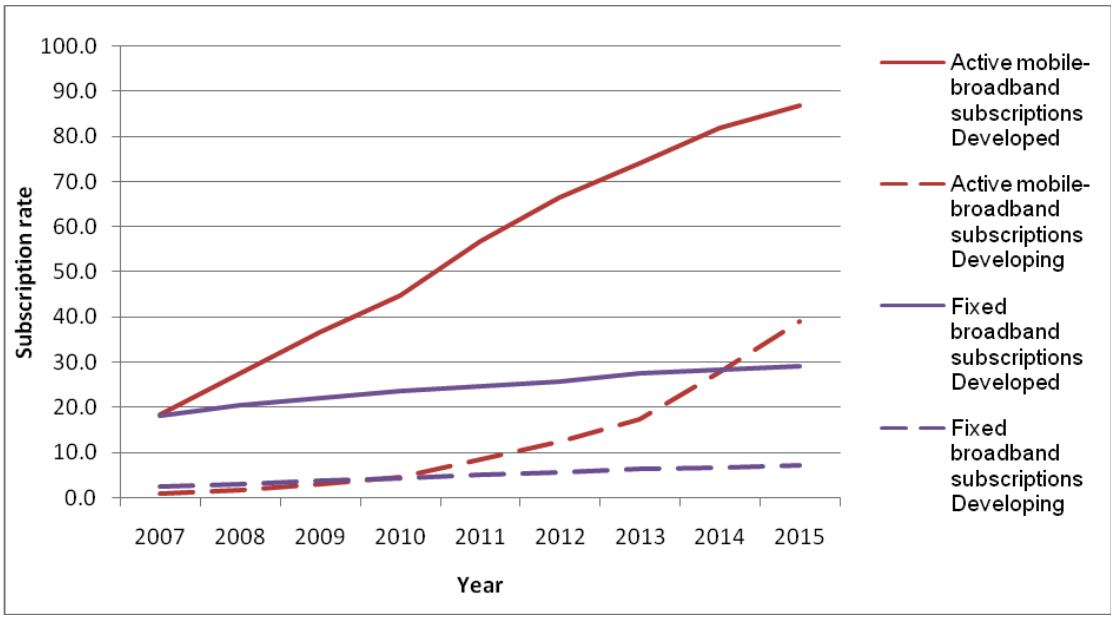
Figure 40 Mobile-Cellular phone subscriptions per 100 inhabitants (ITU, World Telecommunication/ICT indicators database.)



Note: The developed/developing country classifications are based on the United Nations M49
**Data for 2015 are estimated*

Unlike Northern countries where access to the internet for most people was initially through fixed lines (e.g. modem and telephone line) and desktop computers and has more recently been available as mobile broadband, in the developing countries it is the growth of mobile networks that is fuelling Internet usage (United Nations, 2014). The graph in Figure 40 Mobile-Cellular phone subscriptions per 100 inhabitants shows that mobile broadband subscription in developing countries grew from 0.8% in 2007 to almost 40% in 2015.

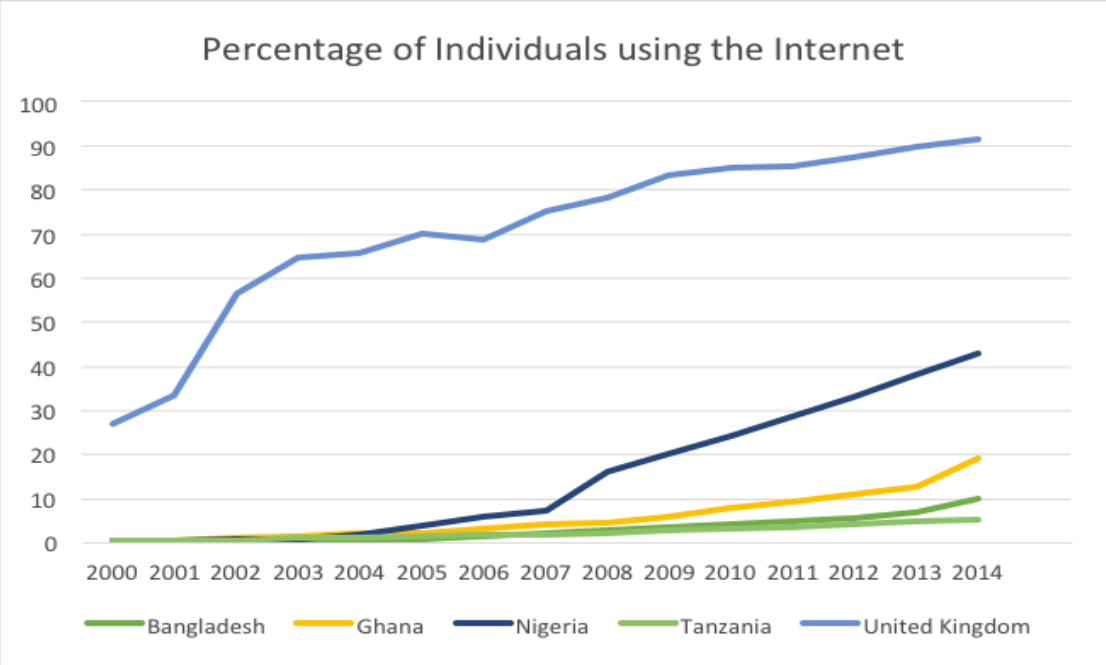
Figure 41 Fixed (wired)-broadband and mobile-broadband subscriptions in developed and developing countries, 2009-2015 (per 100 inhabitants)



Source: ITU, World Telecommunication/ICT indicators database.
Note: The developed/developing country classifications are based on the United Nations M49

There is a similar growth in internet usage in our Case Study Countries with Nigeria leading the way, and as data for each country will show below, this is fuelled by the growth in internet access through mobile broadband. From our own experience with our three case study countries use of internet and contacting participants on mobile phones did seem easier in Nigeria. Individual subscription to broadband requires an access device (computer, smart phone or tablet) and the cost of these can be a barrier. It is for over 80% of people in Nigeria (Nigeria's National Broadband Plan 2013- 2018). In the UK 66% (Q1 2015) of UK adults had a smartphone (Ofcom, 2015).

Figure 42 Percentage of Individuals using the Internet in selected countries[19]



Mobile broadband is provided over 3G and 4G mobile phone signals[20]. It offers high-speed data transmission, enables multimedia communication, improves access to information, and supports high-quality internet connectivity. With mobile prices now very low in many African countries, it is anticipated that mobile operators in these countries will increasingly turn their attention to coverage, network quality and value-added services, as areas where they can differentiate themselves (Policy Paper 4 - Understanding what is happening in ICT in Ghana)[21].

1.1.1.1 Government versus Private Company usage of internet

The World Economic Forum's Network Readiness Index sub-indexes allows comparison of countries on various indicators. Comparing the extent of business internet use against government use shows that business usage is more advanced in Ghana and Nigeria while government usage is more advanced in Tanzania. This will affect the ability of government departments and outside agencies to use research evidence when developing policy.

[19] extracted from http://www.itu.int/en/ITU-D/Statistics/Documents/statistics/2015/Individuals_Internet_2000-2014.xls

[20] 4G is up to five times faster than the UK 3G average (Uswitch)

[21] Research ICT Africa (2012) *Understanding what is happening in ICT in Ghana*

Table 31: World Economic Forum's Network Readiness Index sub-indexes (World Economic Forum, 2016)

	Networked Readiness Index	The political and regulatory environment	Business usage	Government usage
Ghana	101	50	84	92
Nigeria	104	116	79	95
Tanzania	118	83	121	100

Other common characteristics of internet access in the South also hold true for our case study countries:

- It is increasingly widespread but levels vary markedly between countries
- Coverage is better in urban areas
- Access through fixed lines is a smaller percentage and power and equipment problems make it less reliable hence mobile broadband access is increasingly more important
- Cost of access is high relative to salaries
- Despite more mobile-friendly websites being available, access via a mobile device remains more limited for many reasons: small screen size, lack of windows, javascript, speed, latency, cost (Zakas, 2013)

B.2.1 Nigeria ICT Country Context

A regional IT manager for West Africa, working for a large organisation, commented that the use of the Internet has increased dramatically especially in the last 5 years; that people use smartphones to access the Internet and that it "follows you everywhere". Fibre is available in the centre of large cities, and if it doesn't go to an office (it is often too expensive to lay this extra fibre) it is available to the last mile. For a large office, the first choice for Internet access is fibre, although an alternative method is needed as a back-up (satellite access or mobile phones) due to reliability issues. The IT manager expected fibre to become more extensive in cities, but not elsewhere and that here wireless will continue be the method of choice as it is easier to set up the infrastructure (one base station with its own power supply connected to a trunk line can supply an area of up to about 10km radius (Public Telecommunication Networks Unit, 2001)).

From 2012-13 the length of fibre optic lines (land and submarine) increased by 20%, the number of Base Transceiver Stations sites for mobile networks by 22%, and there were more microwave radio links (these provide the link between base stations) (2014 Year End Subscriber/ Network Data Report for Telecommunications Operating Companies in Nigeria). As the Nigeria National Broadband Plan 2013 – 2018 says, "Nigeria's International connectivity landscape has come a long way from a single international submarine cable system with 340 GB total capacity installed in 2001 to a total of four cable systems with international bandwidth capacity of over 9 Tbit/s by 2012 (Nigeria's National Broadband Plan 2013- 2018).

Teledensity (the number of telephone connections for every hundred individuals) has increased from 0.4 (2000) to 99.39 (end 2014) – from one of the lowest in the world to leading in Africa with over 99% of this via mobile phones (2014 Year End Subscriber/ Network Data Report for Telecommunications Operating Companies in Nigeria). Note we found many people with more than one SIM for reliability issues therefore 100% penetration does not mean 100% of the population.

The total Internet subscriptions were 76.5 million as at December 2014, with over 99% via mobile network and only 0.2% by fixed wired/wireless (2014 Year End Subscriber/ Network Data Report for Telecommunications Operating Companies in Nigeria).

Problems with the operating environment reported by mobile companies:

- Serious ecosystem issues: heightened state of insecurity in some parts of the country, the vandalism on telecommunications infrastructure, damage to infrastructure (often by agencies of State, Federal & Local Governments) seeking to exert inordinate taxes
- The high costs of generating power at cell sites (required due to intermittent and unreliable power)
- The high cost of bandwidth subscription to the core/backbone network
- Customer churn (people switching between networks) (2014 Year End Subscriber/ Network Data Report for Telecommunications Operating Companies in Nigeria)

B.2.2 Ghana ICT country context

The comments of the regional IT manager for West Africa working for a large organisation above also apply for Ghana. As with Nigeria, infrastructure is increasing. The capacity of submarine cables doubled from 2012 to 2013 (to 12.3 terabits) (Ministry of Communications, Ghana, 2015). Teledensity is at over 100%, and is carried by the mobile network.

Ghana is among the African countries with the most competitive prices for mobile telephony services and it ranked 4th position in the first quarter of 2013 for prepaid mobile and for ICT services mobile telephony is the dominant platform even in business and residential areas that have fixed lines (Policy Paper 4 - Understanding what is happening in ICT in Ghana).

Table 32: Teledensity in Ghana (Ministry of Communications, Ghana, 2015)

	2010	2011	2012	2013
Teledensity /Penetration Rate:	76.6% (17,714,846)	86.1% (21,450,564)	102% (25,903,408)	108.23% (28,296,904)
Fixed Line	1.2% (277,897	1.1% (284,721)	1.12% (284,981)	1.03% 270,422
Mobile	75.4% (17,436,949)	84.6% (21,165,843)	101.3% (25,618,427)	107.19% (28,026,482)

B.2.3 Tanzania ICT Country Context

In a phone interview, a regional IT manager for East Africa working for a large organisation made very similar comments to the IT manager in West Africa.

Fibre links are available for offices or can be laid for an average cost of $500. Subscription costs for a dedicated link are: 2MB average $500/month, 8MB average $1380/month with a choice of companies to use. It is quite stable, but they do get break downs that can last 2-3 hours per day. To deal with this they switch to alternate connection methods. The causes of interruption of connection are things like power off

at connection point and fibre disconnections both within and outside the country. The coverage for Mobile phones and dongles (MiFi) is good for 3G, 4G is becoming available, and prices are coming down as there is more competition. For example, 1MB was $200/month 4 years ago, but now costs $67/month.

Teledensity in Tanzania is lower than for Nigeria or Ghana at 71% as of 2014, but again the vast majority is carried by the mobile network (The Tanzania Communications Regulatory Authority, 2015). As Research ICT Africa points out, this is a nominal penetration rate as the reality is that an increasing number of mobile users are acquiring second, third and fourth SIMs in order to be able to place on-net calls over multiple networks and take advantage of promotional pricing (Research ICT Africa, 2013).

The Tanzania Communications Regulatory Authority estimates that there are 9 312 272 users of the internet in the country as of 2013, translating into a penetration level of 21% (The Tanzania Communications Regulatory Authority , 2013).

However, RIA point out that “this figure includes estimates of internet café users, organisation/institution use and household and individual use (Research ICT Africa, 2013). The combination of different sources and methodologies makes this estimate very unreliable and, therefore, this high level of penetration is extremely unlikely. The 2012 RIA Tanzania ICT Survey findings put the total number of internet users at 3.5% of the population over the age of 15. This shows how difficult it is to measure any of these figures accurately.

B.2.4 UK ICT Country Context

Similar to Southern countries the rate of growth in mobile data use continues to outstrip that on fixed broadband networks; it grew by a factor of 64% over the past year. However, the volume of data carried over mobile networks is still a small proportion (around 1%) of data carried over all networks (Ofcom, 2015). “Levels of mobile coverage in rural areas continue to be lower than in urban areas. A new study has shown this reflects the higher costs per user of providing coverage in less densely populated areas.”

B.2.5 Internet Coverage by Access method

The following tables are indicative only. Each country measures and reports different figures. For example Ofcom reports premises coverage while Nigeria talks about teledensity (the number of telephone connections for every hundred individuals). In terms of understanding the access premises coverage gives a more accurate picture. Figures for the UK are included for comparison only.

Table 33: Internet Coverage by access method (2G, 3G, 4G)

Three generations of technology used to deliver mobile services. It can be used for voice and/or data	Nigeria (2013) (Nigeria's National Broadband Plan 2013- 2018)	Ghana	Tanzania	UK (2015 data) (Ofcom, 2015)
2G provides voice, text services and capable of transferring data, but only very slowly and unreliably The maximum broadband speed is 473Kb (Marling, 2016)	98 % teledensity	Almost 60% of individuals subscribe to a mobile telephone service but close to one- third (28%) have multiple SIM cards (Policy Paper 4 - Understanding what is happening in ICT in Ghana)	As of 2013, 89% of population is covered to the -95dBm signal level, with 54% of geography covered (Research ICT Africa, 2013)	> 99% premises (outdoors) covered by at least one operator The geographic area cover is 84% No change over last year
3G downloads at over 5Mbps (theoretical limit of 10Mb, but a newer standard called DC-HSDPA allows real-world speeds of 10Mb+ (Marling, 2016)	< 35% and mostly in urban areas. The aim is national 3G wireless coverage to at least 80% of population by 2018.	Exists. Vodafone claims up to 14.4 Mbps on 3G http://www.vodafone.com.gh/personal/internet/mobile-internet	Exists	> 99% premises (outdoors) covered by at least one operator The geographic area cover is 79%
4G Connections speeds of over 10Mbps. Used mostly to provide data access and not voice (But to use 4G requires a relatively new phone or dongle as it is not supported by most older devices (Marling, 2016))		Surfline Communications launched first commercial 4G Frequency Division Duplexing Long Term Evolution (FDD-LTE) network in Accra and Tema and real world speeds are likely to average around 80Mbps (TeleGeography, 2014)	Exists	90% premises (outdoors) covered by at least one operatorGeographic area covered is 48%

Table 34: Internet Coverage by access method (Broadband)

	Nigeria (2013) (Nigeria's National Broadband Plan 2013- 2018)	Ghana	Tanzania	UK (2015 data) (Ofcom, 2015)
Broadband - high-speed internet connection that has surpassed dial-up as the standard way to connect to the internet Provided through: ADSL broadband, cable broadband and 3G and 4G mobile broadband, satellite	Alliance for Affordable Internet shows average broadband speeds of 1Mbps (Alliance for Affordable Internet) Wired: Available 1.5%, Usage 0.5% Wireless: Available 35%, Usage 6% 41.45% of population have access to the internet (Federal Ministry for Communication Technology)	Alliance for Affordable Internet shows average broadband speeds of 4Mbps (Alliance for Affordable Internet) The aim is for 50% Broadband penetration for Ghanaians by 2015 (Alliance for Affordable Internet)	Alliance for Affordable Internet shows average broadband speeds of 0.51 Mbps (Alliance for Affordable Internet)	Coverage of premises, all speeds, is 100% The proportion of adults with broadband in the UK (fixed & mobile) 80% (Q1 2015) (Ofcom). 92% of premises can access more than 10Mps (10Mbit/s appears to be the tipping point beyond which most consumers rate their broadband experience as 'good'.) Those with lower speeds are mostly rural Take-up of any type of broadband is 78% of all premises
Super fast broadband 30Mbps – 300 Mbps	Available through dedicated fibre lines	Vodafone claims to offer ranges from 128k up to 155MB and above. (Vodafone)		Available to 83% of premises with 27% take-up Only available to 37% premises in rural areas Only 68% SME have access
Ultrafast broadband Download of greater than 300 Mbps				2% coverage 0.003% take up

Table 35: Relative cost of 1gb broadband bundle in Ghana, Nigeria and Tanzania

	Average monthly household income (in USD purchasing power parity) (Research ICT Africa, 2013)	Overall NRI Ranking Networked Readiness Index WEF (2011) (Research ICT Africa, 2013) Measures the likelihood of a nation being able to take advantage of ICT to foster economic growth	Government readiness (Research ICT Africa, 2013)	Individuals that own a mobile phone RIA ICT Survey data 2011-12 (Research ICT Africa, 2013)	Cheapest monthly prepaid mobile 1GB broadband bundle per month (in USD) April 2013 (Research ICT Africa, 2013)	Cheapest prepaid product for OECD basket (40 calls per month prepaid (2010)) http://www.researchictafrica.net/prices/Fair_Mobile_PrePaid.php
Ghana	510.66	99 ↑	116	60%	3.72	Q2 2013: 3.84 Q2 2015: 2.39
Nigeria	404.36	104 ↓	123	66%	50.99	Q2 2013: 6.01 Q2 2015: 3.82
Tanzania	245.01	118	117 [22]	36%	10.17	Q2 2013: 5.44 Q2 2015: 4.00

22 this is ranked much higher than that for business or individual because ICTs are recognised as central to the government's vision of the future

B.3 User Profiles and Diary Entries

B.3.1 Ghana User Profiles and Diary Entries

Table 36 summarises our Ghanaian participants and the data collected from them. This section contains the user profiles generated from 10 of those participants a summary of the daily journal entries.

Table 36: Summary of Ghanaian participants and data collected

Participant	Gender	Category	Sector	Data collected (P, CI, DP, D*)
GH1	M	Development Consultant	Health	CI, DP, D8
GH2	M	National legislature or elected member of local government	Politics	f2f interview (internet not available)
GH3	M	Civil/Public Servant	Procurement	CI, DP, D2
GH4	M	Civil/Public Servant	Education	P, CI, DP, D4
GH5	M	Civil/Public Servant	Education	CI, DP, D8
GH6	M	Academic/Researcher	Education	CI, DP, D2
GH7**	M	Academic/Researcher	Education	CI, DP, D2
GH8**	M	Knowledge broker/intermediary for policy makers	Governance	CI, DP, D7
GH9	M	Development worker in civil society	Health	P, CI, DP, D4
GH10	M	Development worker in civil society	Health	P, CI, DP, D9
GH11	F	Development consultant	Gender and Equality	CI, DP, D4
GH12	M	Knowledge broker/intermediary for policy makers	Economic Growth	P, CI, DP, D8
GH13	F	Civil/Public Servant	Legal	f2f interview (internet not available)
GH14	M	National legislature or elected member of local government	Politics	P
14 participants overall	**11 complete f2f sessions**	**7 complete diary sets (i.e. reflection completed)**		

* P = phone interview, CI = contextual inquiry (first half of f2f session), DP = discussion of portals (second half of f2f session), D= Diary, Dx = x number of journal entries returned

** Participant has lived outside Ghana for 6 years out of the last 10, and is not included in main analysis but is useful for comparisons

Knowledge of:

Scidev.net: none

Eldis.org: none

R4D: none

Portal challenges:

- Most internet searches done in Accra because of internet speed
- Uses phone and hotspots for internet connection outside Accra (last year used modem a lot)

Use of information:

Shared online: previous project used a group folder, or email link. In current role, only a two-person team

Shared offline: no

Saved: Yes**,** GH1 has many folders, and saves for two reasons:

1. poor internet access; 2. to avoid repeat search

Repurposed: highlights key sentences in source; extracts to Word; populates outline of target document; uses these to (re)write report

Read: read through report, table of contents, searches for keywords

Participant – GH1 Category 6

GH1 is an Investment/Capacity-Building Specialist. GH1 supervises capacity building activities on a project for USAID. Uses information online a lot but needs to tailor it to clients' needs.

When looking at reports to determine relevance, the first paragraphs and abstract are key to determine if they are relevant. GH1 talks about 'relevance' a lot. Will copy a sentence from a report if they need to re-find it, maybe to check the reference. GH1 always saves download first before reading in case connection is lost.

If GH1 wants specific information and knows where to get it from, they will go to the website where they know it will be, e.g. for statistics they go to the Ghana statistical service website, but these sites will also pop up in the general search. If looking for a specific thing they would use the general search (they use the general search engine incredimail) and type in the title. GH1 is more used to the general search than looking for specific articles.

"most of the work we do uses information online" "downloading takes a bit of time sometimes <laughs>"

Example use of evidence:

Recommendations for UNICEF about using microfinance to provide sanitation. GH1 uses the keywords 'microfinance' and 'WASH' in this task.

Uptake:

Availability of research evidence has improved. Search skills have improved "**It has changed my behaviour and colleagues and how we**

Information behaviour:

- Opens each source in a new tab, so can easily go back and not waste time pressing 'back' or re-searching
- Reads headings then first few sentences
- Uses '+' for searching
- Looks for a page with report
- Might search within website or report

Starting: incredimail search engine, using key words

Chaining: from Google to found sources; from found sources to download reports

Browsing: scans titles in Google, scans reports for keywords, scans website for relevant items

Differentiating: opens websites in different tabs, downloads reports

Monitoring: if comes across information relevant to the sector will save

Extracting: uses both general search and website search

Processing: see repurposed on LHS

Assessing validity/use:

- Relevance – keywords in title/abstract
- Date
- West Africa or Ghana specifically
- Domain knowledge, e.g. a lot of work on latrines is done in Cambodia
- Reports are better because they describe work that has been done rather than just a project website

	interact" Initially you do a lot more on your own, reading what others have done. GH1 believes that now there is a lot more to read about what others have done before starting to do anything. Knowing more and reading more means that you can bully colleagues who are not reading more. Need to read and read, **it has changed attitudes"**. Flow of writing has improved because I have read more.	

Participant – GH1	Category 6 part 2	
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "Big data for development: facts and figures": 1. chooses 'browse type' and reads options 2. chooses 'data visualisations' and scans articles 4. searches on webpage for 'development' 5. finds article at the bottom of the page **Steps to find information:** 1. scrolls down the screen 2. returns to central topic menu and looks at each one, reading the titles 2. goes back to Health, looking for WASH, maybe under Health Systems 3. scans screen and goes to "refine by" and chooses health as well **"if I'm frank with you, I will not continue with this website" "the website is not for what I am looking for"** **Barriers to use:** subject domain ((but GH5's domain was water sanitation)) **Validity/use:** nothing found of interest **Downloading:** nothing downloaded **Interactive functions:** none used	**Steps to find specific article "Africa Health Strategy 2007-2015":** 1. went to Health and Health Systems 2. reads the LH menu items 3. types in search "Health Systems + development + Africa" resulting in 4097 documents 4. scrolls down scanning titles 5. searches again "Africa Health Strategy" and still 4097 documents **"it's more like a headache"** 6. types "Strengthening of health systems" in search. Still 4097 documents 7. gives up on Eldis, goes back to general search in incredimail and types complete title 8. required document is first in the general search **Steps to find information:** 1. straight to search box and enters 'microfinance' 3. reads number of results (494) 4. looks at the first few titles and decides to refine search "microfinance + WASH" 5. this results in 616 documents, and they laugh 6. looks at the list to find microfinance and WASH 7. looks at page 2 of results, **"microfinance and WASH together should come up first, but doesn't"** 8. goes to page 3 of search results, reading titles, then page 4 Finds "microfinance for water supply services" 9. clicks on this link and reads abstract, clicks on	**Steps to find specific article:** 1. chooses 'browse by theme', and scans list (notices that WASH is in the theme list) 2. chooses economic growth 3. enters 'development finance' in the webpage search, but no results (hadn't noticed that the search was on projects as well) 4. goes back to simple search on home page and enters 'development finance institutions and infrastructure' 5. error occurred – technical fault 6. simple search box and checks keywords, 7. same error message 8. goes to advanced search and re-enters long search string 9. it's too much for me – advanced search is too much, maybe have it in two stages because all of it is a lot. 10. scans the search results **"I may be forced to go back to the general search and type in the whole title"** 11. continued looking at the results lists but didn't find the article **Steps to find information:** 1. scans the menu items on LHS 2. moves to Simple Search box and enters 'microfinance + WASH' 3. scans the titles of results and considers whether to look at some of them in more detail

	‘web’ link 10. scans the resultant page. Looking to see if it is key or just interesting 11. on to page 5 of search 12. goes to filter on LHS but no financial services filter, so chooses Health filter. Appears to have lost way – what is this doing? hadn’t noticed that the 57 documents in this search are “within HEALTH”, and hence that it had filtered Goes back to home page – clears the ‘search’ box and presses ‘return’ **Barriers to use:** there is a lot of functionality that people don’t notice, e.g. search has two tabs: documents and organisations. Search is confusing. Use of tabs is confusing – when choose document a new tab opens but GH1 didn’t notice that and was confused **Validity/use:** As for general searches above **Downloading:** downloads as above, using tabs **Interactive functions:** no impact	4. notices ‘sanitation’ but says that they have to go further **“a number of them have microfinance and WASH”** 5. chooses one with ‘sanitation’ – it is a DFID advanced search (111 results) 6. chooses the first three in the list to open in new tab 7. opens one of the tabs to look at it in more detail **“it was a bit misleading to me initially. I didn’t look hard enough because the one I chose was 106 records but others are one-off”** **Barriers to use:** advanced search is confusing **Validity/use:** As for general searches above **Downloading:** downloads as above, using tabs **Interactive functions:** no impact

Knowledge of:

Scidev.net: not tested

Eldis.org: not tested

R4D: not tested

Portal challenges:

None observed or mentioned, except that the internet reception was non-existent in his constituency office

Use of information:

Shared online: none mentioned

Shared offline: including it in debate, and by holding up tablet to show people, but not directly sharing – just the main points

Saved: yes

Repurposed: through debate

Read: yes

Participant – GH2 Category 1

Member of parliament for 3 years. Is on the education committee and so is interested in evidence related to education. GH2 was keen to tell us that MPs don't make policy but they sit between the minister who makes policy and Ghana Education Service (GES) who implement policy.

"I use the tablet so much to get information from the internet to work with"

They draw on primary data collected from schools and research studies, e.g. looking into complementary basic education in order to place children from hard-to-reach areas into mainstream education. The committee GH2 is on compares the budget to what actually happened. GH2 uses data from GES and EMIS (Education management system (http://moe.gov.gh/emis/index.htm). GH2 also accesses information from UNESCO to compare and make a cogent argument. Information comes from individuals and communities who have checked the data on the ground. This is all about checking the data.

"you have to be careful that whatever the content is, it must be a source that is recognised"

GH2 uses Google a lot, even to find the websites they know. The EMIS report tells GH2 what they need to know to debate the situation in parliament, and the core issues in education. Other NGOs produce data and if they are on the net then google will find them. GH2 doesn't visit specific websites, but relies on google to pick out what they need.

"If I find evidence from a source I don't know, because of my credibility I normally ignore it, I would not make it an issue on the floor of parliament but draw the attention of minister on the quiet"

Information behaviour:

Not tested as the connection in the constituency office did not work. Session was restricted to mostly audio recording and no substantial internet use.

Assessing validity/use:

Source, e.g. World Bank, UN etc.

Uptake:

Availability of evidence on the internet is **"Far better, far better than before"**

Having evidence easily available is affecting what GH2 does – an ordinary citizen can challenge what is said now, and in parliament it is happening more. On the floor of parliament I have the tablet. ***"You even see as soon as you quote the source someone puts it in and says no, no."*** You really have to be ok about the evidence itself and others will pick it up, read it and authenticate the issues they are presented with. Fellow MPs might hold their ipad or phone and quote from it.

"It's forcing the credibility issue on us that we must look for the right information if we really want to debate, otherwise

	GH2 is very concerned about their credibility, especially in what they bring to debates in parliament. Information is difficult to get hold of except on the internet. You can't rely on gossip. There is openness of information now but you need to go beyond that because people don't yet have the courage to put the real issues on the web, so you need to ask others.	**you'd better keep your mouth shut"**

Knowledge of:

Scidev.net: none

Eldis.org: none

R4D: none

Portal challenges:

- Some pages returned errors, but GH4 thinks it is a connection issue
- Internet is slow in the afternoon
- It takes a long time to download government data. **"this is taking longer than it should"**

Use of information:

Shared online: Keeps stuff in Dropbox so can access from any physical location. The division has a Dropbox folder to share. But GH4 doesn't have it available to show us

Shared offline: government data also on CD

Saved: usually downloads information, often prints it out – because it's not easy

Participant – GH4 Category 3

GH4 is in the Ministry of Education, working at policy level, both implementing and carrying out monitoring and evaluation.

Goes on line to read all the major headlines from local internet portals (myjoy online, joyfm, peacefm online) to get all that is happening in real time. Also looks internationally e.g. at BBC website – for work and for personal use. Looks for issues that as a ministry they have to respond to. For example GH4's project was being discussed on myjoyonline and people didn't understand so their public relations company put a correction on their website.

The Ministry has its own website (www.moe.gov.gh) and are putting all information about secondary schools on line, as a result of requests for information and to share with their workers. **"This is our website where we have a lot of existing information already"**

Example use of evidence:

As part of the Secondary Education Improvement Project (SEIP) they needed to know issue with supply of places: **"We knew that demand for secondary education was in excess of supply…no documentation…. question arose naturally how do we know the demand for secondary education."** They needed a firm idea of admission patterns over the years to be sure that the additional 200 schools government planned would actually be used. The data came from the ministry's own website. The documents for this project are all very long.

Uptake:

Availability has changed GH4's behaviour **"a lot, a lot"** since starting job in 2005 when connectivity was a problem. In 2009, when studying an online course, used the internet hourly. Online information is helpful but also brought a lot of work, and allowed work to intrude on home. For example the boss can get in touch with you at home and you can be forced to work. Because of this GH4 has a house rule: **"no more browsing in the house during the**

1. Information behaviour:

- Uses government-generated data, e.g. enrolment and admission levels in Ghana (EMIS, myjhsresults.net) to investigate the situation
- When GH4 tried to show us this data through the government website, it had been redesigned and so it could not be found. GH4 then used google to find them **"for all the portals for my own work I go to them through google."**
- Uses non-govt websites depending on the project
- Asks a colleague for a website address

Starting: types URL address directly or uses google to find known website

Chaining: from initial webpage to subpages through links

Browsing: on news websites

Differentiating: some webpages opened in separate tabs

Monitoring: saves and prints out lots of articles and documents

Extracting: none directly

Processing: see repurposed

Assessing validity/use:

- Keywords and relevance

to find things again or the internet doesn't work. **Repurposed:** reads education statistics to find evidence to support impression that there is a gap between applicants and available places. **Read:** if time, the whole document, otherwise the first one to two paragraphs.	**weekend, do all your browsing in the office"** Availability of research evidence on line has improved. **"In the past it was difficult to prove what you were saying, but now in meetings and conference rooms you can access and link to evidence"** It has also affected the way people interact, it keeps them **"at bay if you have enough evidence".**	• Is it current? • Government's own data • If there is a trend running through a number of articles then you begin to appreciate it. • If you are paying for it you get the feeling that it's a well-researched document. • Other people believe information found on the net, even if detrimental, they assume it's correct. **"if something is free, it's a good thing, but sometimes you doubt the information if it's free"**

Participant – GH4	Category 3	part 2

Scidev.net

Steps to find specific article "How to communicate in an interdisciplinary team":

1. typed the full title into the search box
2. reads the number of articles returned (seven)
3. scrolled down them looking for the title
4. clicked on the title (3rd one down)
5. was happy could find it so easily

Steps to find information:

1. scrolled down the home page and back up, trying to appreciate what it is about
2. hovered mouse over 'Browse type' and then over five of the other menu options (not 'Communication'), looking for 'education' **"Things are categorised… It's more obvious to see than R4D"**
3. researcher points out this is a scientific site
4. thought the banner 'Trading in death' looked interesting so clicked through and read first paragraph. Usually reads the whole article if there is time, but if not the first 1 – 2 paragraphs to get an idea of what it is about
5. decides to read it later

Eldis.org

Steps to find specific article "Three tools to unlock finance for land-use mitigation and adaptation":

1. typed full title into search box
2. it was top of the first page
3. chooses the first one to read
4. reads first paragraph **"very nice, you really want to spend more time reading. I like it this way"**

Steps to find information:

1. chooses "Evidence for policy and practice" from LHS menu under Resource Guides.
2. reads first two paragraphs of overview in the middle **"this will set the tone for the whole story"**
3. commented would like to spend a whole day reading from this page **"this is the core of my work… to make sure my work is informed by evidence"**
4. chooses a story titled "Education in Urban contexts"
5. likes the paragraphs describing the document as it saves time to see if it's worth going further, and the bullet points linking research evidence **"it's well organised and I like that it sets to tone right at the beginning… it's a teaser"**
6. downloads PDF, this happened really quickly

R4D

Steps to find specific article "Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality":

1. had trouble finding home page
2. clicked on DFID logo and went to DFID home page
3. clicked on "browse by theme"
4. read sub-categories looking for anything to do with development institutions, finance and infrastructure
5. chose miscellaneous infrastructure category
6. scrolled down page but didn't find anything
7. returned to "browse by theme" page
8. looking for anything to do with global finance, institutions, infrastructure
9. chooses 'Public financial management governance and conflict' as had some of the key words **"so many links"**
10. scrolled down the page, reading headings
11. went to next page, and then next page, and then next page (4th), looking at titles
12. goes to front page
13. chooses advanced search and enters some keywords, scrolls down and says **"I have to fill out these (sighs and**

Barriers to use: none observed **Validity/use:** interest only as GH4 didn't find anything on education **Downloading:** no problems **Interactive functions:** not used	7. saves PDF **Barriers to use:** none observed **Validity/use:** relevance to the work **Downloading:** fast download **Interactive functions:**	**laughs)"** 12. looks down search results **"the numbers are slowing me … there are 3000"** 13. chooses the refine by documents on RHS 14. and tried using that but didn't find the article **Steps to find information:** 1. enters 'www' before R4D URL and researcher points out that it is not needed 2. reads the introduction on the front page 3. reads the options and chooses 'browse by country' to see if there is anything on Ghana 4. chooses 'all projects' 5. looks for projects DFID have implemented in Ghana **"it looks organised if you compare it with other websites, the front page can scare you away"… "I don't see anything on Ghana yet"** 6. Searches again, narrowing to West Africa, but internet connection is **"bad"** **Barriers to use:** commented **"Once you get used to a particular site… you know how things are arranged… it becomes a bit easy. This is my first time here.…. If I'd used it a number of times I would know…"** **Validity/use:** keywords **Downloading:** internet connection is slow **Interactive functions:** not used

Knowledge of:

Scidev.net: none

Eldis.org: none

R4D: none

Portal challenges:

- If it needs to be paid for then needs to go to university to access
- Variable internet connection takes a long time to do anything and portals time out **"if you really need it then you persist"**

Use of information:

Shared online: no

Shared offline: not directly

Saved: saves documents and information so as not to be on internet too long as this costs money. If this can't be saved directly then cut and paste into Word.

Repurposed: read and summarise the content relevant to GH5's purpose, relate the work from other countries

Read: title, abstract or any summaries. Look at any statistics and project outputs. Is influenced by the title

Participant – GH5 Category 3

GH5 works in the non-formal education sector and has been focusing on complementary basic education (CBE).

GH5 demonstrates ERIC (Education Resources Information Center), which was learned about from GH5's lecturer. GH5 uses JSTOR and ERIC because they publish in social sciences and education.

GH5 trusts PDF documents **"If I find something that is not PDF, then someone else might have tampered with it"**

Switches to smartphone hotspot part way through the session as internet connection via the dongle fails. Linking to ERIC is very slow on the laptop and on his phone. ERIC is rendered differently on laptop and on the smartphone. Seems unsure about the navigation of ERIC on the laptop.

GH5 appears to not be confident in using the internet and searching for articles in a portal **"I am not very good with the IT terminologies".**

When accessing through ERIC or JSTOR, GH5 reads the abstract which is usually free, and then goes to the university to get the full article if required. If they can reach the PDF they would download it and save to their machine.

The general search engines make it more difficult to find research evidence. On the portals (such as R4D), material is categorised and more focused which makes it easier.

Example use of evidence:

A report from a project (Complementary basic education) that was to provide literacy to children 8-12 years not in formal education. DFID funded together with the Ghana Education Service. Children may be in

2. Information behaviour:

- GH5 uses the phone to connect to hotspot
- Tries to save information directly from the website, else copies the text and pastes into Word
- GH5 starts by getting an international context and then narrows down to Ghana
- GH5 goes to the University to access journals or portals that have to be paid for

Starting: uses Google or a general search engine to start, will go to portal if looking for journal article, e.g. ERIC or JSTOR

Chaining: on ERIC, browses by category and links through to educational process: social perspectives

Browsing: no evidence of browsing

Differentiating: relevance, source

Monitoring: information is sought for a specific purpose rather than keeping abreast of an area

Extracting: cuts and pastes from website if interesting

Processing: as for repurposed

Assessing validity/use:

- Credibility: published journal articles,

	farming or other occupations that take them away from formal education. **Uptake:** Communication is more prompt. **"I should be able to respond wherever I am because I can download it to my phone"**. Access to research evidence is easier because the portals categorise materials and have pop-ups and menus that direct you to a topic, as opposed to a general search engine.	no news items, JSTOR, source is more important than the form • Author's education and experience (profile) • Relevance: to topic, other parts of Ghana, or other African countries, methods that others have used that Ghana can use

Participant – GH5	Category 3	part 2
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "Big data for development: Facts and figures": **"here the search is hidden"** 1. types 'big data for development : facts and figures' in search box 2. 16 results 3. scrolls down first page of results reading the titles 4. chooses fourth article in list **Steps to find information:** 1. reads sub-menus of each main menu item 2. 'children' under the 'Health' menu may be relevant but moves on to others looking for education 3. sees 'education' under 'Communication' but it's not children's education 4. decides to look at 'education' and see if it relates to children **"education is under 'communication' – why? I feel it is hidden. Maybe I don't understand"** 5. scrolls down articles looking at titles, and observes that this is about higher education, so not relevant 6. goes back to 'children' under 'Health' but decides that it will not talk about education 7. concludes that this website will not offer the kind of information GH5 is looking for **Barriers to use:** not appropriate areas	**Steps to find specific article "Three tools to unlock finance for land-use mitigation and adaptation":** 1. types 'three tools to unlock finance for land-use mitigation and adaptation' into search box on LHS. 2. opens first item in the results list to confirm that it is the right one. **"it's very fast and the portal is friendly because I was able to scan by the guide headings… I could move to a specific topic"** **Steps to find information:** Tries to open on laptop and smartphone to see which is faster – the smartphone loads more quickly but uses the laptop 1. scrolls to bottom of the page 2. reads resource guide list then goes to other topics – chooses education 3. reads the LHS menu under education, and is interested in 'working children and education' 4. chooses this topic as it is relevant to CBE 5. focuses on the titles of articles in 'latest additions' 6. scrolls back to top of page again 7. reads the short abstract that appears in the 'latest additions' list, starting with first article 8. chooses an article on child labour but it won't download the first three attempts 9. reads summary and confirms that this would not be directly relevant because it is not linking to	**Steps to find specific article "Using climate information to achieve long-term development objectives for African ports":** 1. types 'using climate information to achieve long term development objectives for African ports' into simple search box 2. the typed title is in each item on the first page 3. looks for author and date but can't see it in search result information 4. chooses the first item **"comparing the three, Scidev was easier to see in terms of author and when I clicked it was visible. On R4D it was hidden"** **Steps to find information:** 1. reads items in LHS menu 2. types 'complementary basic education' in simple search box 3. points to the first two articles as being relevant 4. chooses the first one as it is about implementation, and second one is about technology (?) 5. reads detailed summary **"it is good to use"** **Barriers to use:** presentation of search results is confusing, especially for the specific article search.. All of the results except the first are advanced searches in themselves which user doesn't realise **Validity/use:** topic relevance

Validity/use: relevance to CBE **Downloading:** does not download **Interactive functions:** not used	education **Barriers to use:** only internet speeds and downloads **Validity/use:** relevance **Downloading:** downloads as above, using tabs **Interactive functions:** not used	**Downloading:** opening items automatically in a separate tab confuses user **Interactive functions:** not used

Knowledge of:

Scidev.net: only newsletter?

Eldis.org: none

R4D: none

Portal challenges:

- High cost of internet

Use of information:

Shared online: Shares websites with students e.g. NCBI database, plant genome database and rice database.

Shared offline: no

Saved: Yes**,** has many folders; saves for two reasons:

1. limited internet access;
2. It enables detailed reading.

Repurposed: GH6 rewrites all the important information they have gathered online and then references them.

Read: Reads through the abstract and findings. Then reads through the entire document if the document is relevant.

Participant – GH6 Category 5

GH6 is a Lecturer at the University of Ghana. GH6 is also the coordinator of research programmes of Francophone West Africa.

GH6 is currently writing a research proposal about Genetic enhancement**.** GH6 uses general search engines [mainly Google] to look for specific articles for the proposal—mainly because it is openly available. GH6 types the full title of the research proposal into a general search engine to look for specific articles and alters the keywords to find the specific article. GH6 also types a question into general search engine when looking for specific article. When selecting a specific article, GH6 places more emphasis on the closely related titles and the source of the article e.g. FAOSTATS and Ministry of Food and Agriculture, Ghana. GH6 reads the abstract and findings of the specific document, and if relevant to their work, downloads it. Gh6 opts for options that allow them to download specific articles through social media e.g. Facebook. GH6 follows the above criteria to download an article titled **"An overview of post-harvest challenges facing tomato production in Africa"** through Facebook.

GH6 then saves the document in a folder. GH6 later on reads the document in detail and then copies the relevant sections to support their work.

Example use of information:

Research proposal on Genetic enhancement.

Output: None

Uptake:

It has fostered growth in knowledge through access to data and tools online. Indicates that **"I am able to access the Genomic database….I was surprised to find that such type**

Information behaviour:

- types full sentence into a general search engine.
- scrolls up and down.
- reads the titles on the first page of the search results.
- opens specific articles in different tabs.
- saves relevant specific article in a folder and reads at a convenient time.

Starting: Enters an important sentence in general search engine.

Chaining: pays attention to how related the title is and then reads abstract and findings. Then copies key references into Google search engine to look for more articles online.

Browsing: GH6 scrolls up and down the titles and sources in general search engine to find the most relevant items.

Differentiating: Downloads relevant reports. GH6 discards those that are irrelevant to their work by closing the portal on which they were found. GH6 considers the credibility of the author, how recent the publication is, and the relevance of the title to the work.

Monitoring: If comes across information relevant to work, will save.

Extracting: downloads quantitative data

	of database exist". Very confident of getting information for any type of job required to be done.	from FAOSTATS and Ministry of Agriculture. **Processing:** see repurposed on LHS. **Assessing validity/use:** • Relevance – Author and source of the article. E.g. FAOSTAT is the most reliable for data on Tomato production in Ghana. • How closely related the title, abstract and findings are to their work.

Participant–GH6 Category 5 part 2

Scidev.net

Steps to find specific article "Big data for development: facts and figures":

1. types the full title of the article topic (as indicated above) into the search box.
2. scrolls down the first page of the search results.
3. finds article in the middle of the first page of the search results and amazed to see an inserted video playing in the article.
4. pauses a while to watch the video on "big data".

Steps to find information:

1. scrolls up and down the home page.
2. clicks on "Global" then "sub-Saharan Africa".
3. Scrolls down to look for topics related to "Postharvest losses in tomatoes", but finds none.
4. Types "Postharvest losses in tomatoes" into search box but gets zero results. Becomes discouraged. Indicates that **"this is not a good website".**
5. reduces the search by typing only "Tomatoes".
6. becomes encouraged because finds information that has been published on the university's website with the title **"New vegetable lab launched to boost food security".**
7. Bookmarks "www.scidev.net".
8. Tries to go back to the homepage but finds it

Eldis.org

Steps to find specific article "Africa Health Strategy 2007-2015":

1. types the whole title into the search box and gets **28853** search results.
2. gets discouraged with the search results—and decides to narrow the search results by clicking a tab called Health.
3. unable to find the article.
4. gives up.

Steps to find information:

1 scrolls up and down the first page of the portal.

2 types keywords "postharvest tomatoes" in the search box and finds 83 search results—is indifferent with the search results **"Its a lot of work going through all them** [83 search results] **but at the same time I am happy that there are a lot of information on tomatoes"**

3 randomly clicks one of the titles on the first page of the search results and downloads the document.

4 looks for the abstract of the document but unable to find it.

5 scans through the document and saves it for future use.

Barriers to use: The several search results makes it difficult to identify the exact information on the

R4D

Steps to find specific article:

1. goes to advanced search and uses 8 minutes to enter the title of the article in the advanced search page.
2. scrolls down and then finds the document at the bottom of the page.
3. downloads the article and saves it.
4. then bookmarks the portal.

Steps to find information:

1. scrolls up and down home page.
2. clicks on "country" but does not find anything relevant.
3. downloads the R4D user guide.
4. reads the R4D user guide cursorily, but finds it time consuming.
5. did not find anything.
6. types **"funding for projects in Ghana"** in the search box and gets **4390** search results. Gives up.
7. clicks on advanced search to find information on tomato.
8. becomes discouraged with the detailed information required on the advance search.
9. results are not relevant to work.
10. finds it difficult to go back to the home page.

Barriers to use:

1. difficulties in identifying where the home tab

confusing. Indicates that **"this site [website] is confusing"**. **Barriers to use:** finds it difficult to navigate to the home page. Asks **"where is the home tab?....I want to go to the home page"** **Validity/use:** finds a document that has initially been published on the university's website. GH6 considers the author. The date of the publication (ideally, it must be a recent publication) and the relevance of the title to the work. **Downloading:** nothing downloaded. **Interactive functions:** none used.	page. **Validity/use:** For general search. Same as that of Scidev.net **Interactive functions:** no impact	for R4D was. 2. gets discouraged to continue with the advanced search page because of the detailed requirements. **Validity/use:** For general search. Same as SciDev.net **Downloading:** downloads specific article. **Interactive functions:** no impact.

Knowledge of: **Scidev.net**: None **Eldis.org**: None **R4D:** None **Portal challenges:** Eldis was very slow. Difficult to navigate to the home page on R4D. **Use of information:** **Shared online:** Shares information with colleagues at work and in the house. **Shared offline:** No **Saved:** Saves articles after reading them. **Repurposed:** Copies online article and rewrites them in own words. **Read:** First reads abstract/summary and if relevant, reads the entire article.	**Participant–GH9** **Category 4** GH9 works at a sexual and reproductive health charity in Ghana. GH9 is currently undertaking a project in collaboration with the national population council. In addition to Google search engine, GH9 uses both local and international portals to search for evidence online. Example of the International Portals includes the World Health Organisation (WHO) and International Planned Parenthood Federation (IPPF). The local portals include Ghana Statistical Service (GSS) and Ghana Health Service (GHS). Uses the GHS and GSS portal to look for Ghana specific and health related online information. For instance, uses both portals for both quantitative and qualitative data. Uses the international portals to search for general international information on health issues, for example GH9 tries to look for average HIV/AIDS prevalence rate on the WHO portal. Opens every online article on a different tab. Prefers to search for publications on these portals. GH9 prefers to copy relevant online information onto Microsoft word and rewrites it in own words. Also uses Google search engine for general search. Does a lot of general search on health related issues. In doing so, GH9 prefers to type in key words into Google search engine. For instance, GH9 prefers to type keywords of a specific article into Google search engine when looking for general information on health. GH9 demonstrates this by typing two key words “adolescent reproductive health”. Gets 7.4 million results. Reads the brief summaries below each title on the first page of the search results to find the most relevant ones. Clicks on the second title. Saves the article. Prefers to read through thoroughly and highlight the salient points in the article. Often uses stickers to identify the most	**Information behaviour:** • Searches for information on various health related portals. This includes WHO, IPPF, and GSS. • Copies relevant articles into word. Prefers to redraft in own words. • Saves relevant article. Seldom prints since it may be expensive. • Becomes uncomfortable when portals are not responding. **Starting:** Starts online search with general search in Google. **Chaining:** No evidence found. **Browsing:** Browses both local and international portals as well as Google search engine. **Differentiating:** Ensures that sections of articles that are read are highlighted and marked for future reference. Opens every online article on a new tab. **Monitoring:** Constantly checks health related portals for new information on health. **Extracting:** Copies new online information identified from recognised health related portals. **Processing:** Rewrites all online evidence that is copied.

	significant parts. Also likes typing names of authors who are authorities in their field. For instance, GH9 demonstrates this by typing the name of a Ghanaian medical professor who has done a lot of work on adolescent health. **Example use of evidence:** Republic of Ghana adolescent reproductive health policy. **Uptake:** Able to get information faster and more easily.	**Assessing validity/use:** • Considers the author. Sometimes types the name of an academic authority in a particular health related field to access valid information. • Also considers the date and source of article.

Participant–GH9 Category 4 part 2		
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. Types the specific article into the search box of the portal. 2. Realises that the portal is not responding as quick as it should. Waits a while. Portal uploads. 3. Types the keywords of the specific article into the search box of the portal. Scrolls down slowly. Finds the title of the specific article on the first page of the portal.	**Steps to find specific article:** 1. Tries to look for specific article. Uses keywords in the title of the specific article as a guide searching. 2. Decides to look for the topic "finance" and "land use mitigation" since they are the keywords in the specific article. 3. Finds **5762** results but the specific article was not found. 4. Then types the keywords of the article into the search box of the article. Finds **10128** search results but still the specific article was not found. 5. GH9 then adds the name of the author and date of publication and then finds the article on the first page of the search result. Clicks on the specific article.	**Steps to find specific article:** 1. Navigates with cursor to the home page of the portal. Clicks on themes in search for specific article. 2. Tries to look for specific article with the keywords in the title. Was not successful. 3. Clicks on advance search and realises it requires a lot of details. Becomes discouraged to continue with the advance search. 4. Types the keywords of title of the specific article into the search box of the portal. 5. GH9 finds the specific article on the first page of the portal. Checks the name of the author and reads the summary below the title.
Steps to find information: 1. Types the URL of the portal into the find bar on the internet. 2. Takes a careful look at the portal. Then clicks on the topic "health" on the menu bar. Further clicks on the sub-topic "HIV/AIDS" on the menu bar. 3. Indicates that the portal is useful but the content of the articles on the portal are not detailed enough. 4. Tries to go back to the home page. Finds it difficult to locate the home tab. Finds it later on. Clicks on it to go back to the home page.	**Steps to find information:** 1. Types the URL of Eldis into the find bar of the portal but the portal fails to upload. Decides to try it for a second time. 2. Eventually gets the portal uploaded. Scrolls up and down. Browses through the resource guide section of the portal. 3. Scrolls further down to "other topics" heading and then clicks on "children and young people".	**Steps to find information:** 1. Types the URL of the portal into the find bar of a laptop. 2. Looks carefully at the portal. Clicks on "Theme". Indicates that the portal is well ordered. 3. GH9 indicates that "**the topics have been put into thematic areas... this makes searching easier**". 4. Was impressed with how the various topics have been broken down into various sub-
Barriers to use: Portal responds very slowly. Also finds it difficult to navigate to the main portal.		
V**alidity/use:** Because portal is a DFID funded portal. **Downloading:** No significant evidence.		

Interactive functions: No significant evidence.	Tries to look for something around family planning and abortion. Clicks on the second title on the first page of the portal. 4. Takes careful look at the abstract, year of publication, name of publisher and author, since that is what attracts GH9 to make further searches. 5. Finds an article related to family planning. Downloads and saves it. **Barriers to use:** Portal did not upload properly. Had to try for the second time. **Validity/use:** Since it is a DFID funded portal. **Downloading:** Downloads a general article. **Interactive functions:** No significant impact.	topics. Indicates that "**this has been broken down into reproductive health, strengthening health systems, HIV/AIDS, ...it is good**" 5. Clicks on the topic "reproductive health". GH9 subsequently clicks on the third title of the first page of the search results. 6. Likes the presentation of the article on the portal. The portal shows the name of the author, date and the fact that DFID is funding the portals makes it credible. 7. Tries to click on the second page to look for more relevant article. Finds no relevant article. 8. Decides to browse the portal by country. 9. Clicks on the heading, "country/region". Finds no relevant information. Was expecting to get health related online articles that are specific to Ghana. 10. Goes back to the home page. **Barriers to use:** No clear barriers realised. **Validity/use:** see SciDev.net on the LHS. **Downloading:** No evidence. **Interactive functions:** no impact.

Knowledge of: **Scidev.net**: Yes, since 2010. Subscribes to their newsletter. Found it through Google. Often uses the portal. **Eldis.org**: None. **R4D:** Familiarity, based on newsletter. **Portal challenges:** Portals are not user friendly. **Use of information:** **Shared online:** Yes **Shared offline:** No **Saved:** Saves only relevant articles. **Repurposed:** uses online data for the preparation of power point presentations. **Read:** Reads full articles online.	**Participant–GH 10** **Category 4** GH 10 is a development civil society worker, working on projects related to sexual experiences of young people, used to educate groups, e.g. the police, develop policy on things like rape. Relies on public information from highly recognised institutions in Ghana. For instance, GH10 uses various local internet portals like Ghana Open data platform and Ghana Statistical Services (GSS) database to look for data on the Ghanaian youth. Uses the Ghana Statistical Service portal a lot because it is easier to get data and policy documents. Finds the GSS portal reliable because it belongs to governments. GH10 types the URL for the Ghana Statistical Service Portal. Scrolls down and then opens two topics on the portal. Opens each of the articles on a different tab. Decides to look for the Ghana Demographic and Health Survey report. Clicks on one of the new tabs. Finds the Ghana Demographic and health survey report. Downloads the report on the portal. Finds the keyword contraceptive by typing it into a search box in PDF and then uses the enter key on the keyboard of the computer to look for the keyword "adolescent" in the entire report. In doing so, GH10 writes down various facts and figures related to the keyword adolescent in the report as personal notes. GH10 downloads data and uses the data for power point preparation and report writing. Shares online information on social network platforms on Facebook called "noyawa". Also shares online information in the form of videos and pictures. GH10 shares URL on the platform as well. This encourages GH10 to look for more online research information online. Example GH10 accesses information on Ghana Demographic and Health survey report from the Ghana Statistical Service website. Also relies on internationally recognised portals like World Health Organisation,	**Information behaviour:** • types the URL for the Ghana Statistical Service Portal. • Opens each of the articles on a different tab. • Finds and save the most relevant article. • Shares articles and URL of relevant article. **Starting:** Starts online search with Google. **Chaining:** No evidence of chaining **Browsing:** Searches for information on GSS portal. Opens relevant articles on a different tab. **Differentiating:** Saves relevant articles and discards unwanted ones. **Monitoring:** Checks the GSS portal regularly to ascertain whether there is any new information related to sexual experience of young people. **Extracting:** Download data from the GSS portal. **Processing:** Prepares power point presentation with new downloaded article. **Assessing validity/use:**

	International Labour Organisation, and Knowledge for health. In determining the validity of information on the portal, GH10 emphasises how current the information is. Also considers how data were organised for the preparation of the document. The source of the online information is also important. **Example use of evidence:** Digital Story telling Project (which includes elements of Sexual Violence) is based on real life stories of some young people in Ghana. https://www.YouTube.com/watch?v=TAbGzMLt_iY **Uptake:** It has increased knowledge acquisition. Information access is now very cheap and easier to access.	• Emphasises on current information. • Considers articles from international sources.

Participant–GH10 Category 4 part 2		
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. Types the name of the author of the specific article in the search box of SciDev.net. 2. Finds the articles on the first page of the portal. 3. Clicks on it and reads the title. **Steps to find information:** 1. Types the phrase **"reproductive health"** into the search box of the www.scidev.net. 2. Finds the search box very easy to use compared to going through the various topics on the portal. 3. Scrolls down and clicks on the fifth title on the first page of the search results. **Barriers to use:** None observed **Validity/use:** DFID funded portals are always reliable because DFID will not compromise its credibility. **Downloading:** Downloads only relevant article. **Interactive functions:** No impact	**Steps to find specific article:** 1. In search for the specific article GH10 types the name of the author into the search box of the portal. Finds no results. Search results are rather related to "**Climate Change"**. 2. Clicks on the topic "**agriculture and food"** to see if the title of the specific article can be found is available. Out of **307** search results, GH10 did not find the specific article. 3. Decides to open the same portal on two different tabs. Types keywords of the specific article in the search box of the Eldis portal on the first tab. **4.** For the portal on the second tab GH10 clicks on **climate change**. This is because the specific article is related to **climate change.** 5. Types the keywords of the article into the search box of the portal. Finds the article on the portal on the second tab. **Steps to find information:** 1. Types the URL of Eldis into the search box of Google search box. 2. Pays attention to the latest documents on the portal. 3. GH10 scrolls down and opens each relevant article on a new tab. Clicks on the last topic on the bottom left of the portal. Finds articles on	**Steps to find specific article:** 1. Uses the advance search topic on the home page of the portal in the first tab to search for specific article. 2. Types keywords title of specific article into advance search but realises that it requires more information. 3. In typing the title of a specific article into the search box, GH10 selects types and puts the full title in quotation. Only two articles were realised on the search results. 4. Realises that the two search results are closely related articles. Identifies the article at the top of the search results as the specific article. 5. Reads the title and the summary. 6. Clicks on the actual document which is attached as PDF. **Steps to find information:** 1. Clicks on the topic country or Region and opens it on a new tab. 2. Does this to keep the home page opened. 3. Selects Ghana and finds the various articles on the search result. 4. Looks for health related topics but does not find an article that is relevant for the current work GH10 is doing. 5. Does not find the portal user friendly—mainly because, of the search results presents a mixture of

	"ICT for development" and **"Mental Health"**. 4. Download the PDF attachment of the document. Scrolls down the downloaded article. Reads the summary under the title "**Education technology topic guide"** on the portal. **Barriers to use:** The portal is slow **Validity/use:** See SciDev.Net on the LHS. **Downloading:** Downloads article in PDF format **Interactive functions:** No clear evidence	"documents" and "projects". 6. Goes to the home page and then clicks on themes. Finds the presentation on the portal orderly but indicates that there are so many topic **Barriers to use:** Not user friendly mainly because of the article **Validity/use:** Did not specify **Downloading:** downloads specific article **Interactive functions:** No impact

Knowledge of:

Scidev.net: none

Eldis.org: none

R4D: none

Portal challenges:

- None observed

Use of information:

Shared online: central repository and email

Shared offline: doesn't print unless has to (e.g. travelling at airport)

Saved: may save document for use later on

Repurposed: reads, takes paper-based notes on key points, maybe copy a sentence electronically to collate in a word document and then re-phrase and re-organise

Read: none mentioned

Participant – GH11 Category 6

GH11 carries out research, monitoring and evaluation with civil society organisations and international development partners, and also designs capacity building programmes, promoting good governance and good advocacy. GH11 has worked on DFID-funded programmes for over 10 years, in different programmes and for different organisations.

GH11's reason for looking for information or research evidence is driven by client's requirements. GH11 goes to specific organisational sites when they know that this organisation has done relevant work. However **"usually I would google for what I want".**

GH11 also uses books and conference materials as sources, and is careful to cite the work properly.

GH11 is surprised that they do not know any of the portals **"Its' interesting, I've worked on DFID's programmes most of my working life".** When going via google GH11 doesn't pay attention to the end source.

GH11 tends to think it's their fault when they can't find things in the portals. GH11 returns to the point many times that they are amazed GH11 has never heard of these DFID-funded portals before nor come across them **"I've never heard of these anywhere. I feel shameful... especially if you are working on a DFID-related project, you want to be efficient"**

Information behaviour:

- GH11 mainly uses Google, even using a general search engine (bing) to find the Google website
- will go to a specific website if GH11 knows they have a lot of good material and they have done relevant work, e.g. ODI, and gets to their website through google
- personal domain knowledge of, e.g. ODI and ITAD, informs GH11's choice

Starting: uses google or a general search engine

Chaining: chooses relevant links from general search or in particular pages

Browsing: no evidence of browsing

Differentiating: content is key, fit to client's requirements

Monitoring: information is sought for a specific purpose rather than just to keep abreast

Extracting: cuts and pastes elements of a website or article if relevant

Processing: as for repurposed

Assessing validity/use:

- Pays more attention to the document and its 'fit' (relevance, content) than to the source (website, author)
- A document that is comparable with what GH11 is trying to develop, e.g. a similar strategy document for a similar organisation
- Fits the requirements of clients to the evidence and other strategies being found
- Country context – choosing a strategy from Europe cannot be imposed on Ghana

	Example use of evidence: A gender strategy for PLAN (international NGO focusing on child rights), to use within their organisation **Uptake:** Agrees that there is a lot of information available but that care is needed to decide where to choose things from. Authenticity is very important, and there are some sites you wouldn't go to. The easy availability of research evidence has also influenced interactions. **"you can refer people to it and it's easy to access … when having a conversation you can easily refer to it."** **In my advocacy training we encourage trainees to look out for data – facts and figures so the arguments are backed up with facts and figures…. you can compare with other countries, e.g. how a particular policy is faring in one country compared to others depending on the context"**	• Titles plus focus/relevance, e.g. legal definitions

Participant – GH11 Category 6 part 2		
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: Not attempted as SciDev.Net focuses on scientific topics	**Steps to find specific article "Strengthening of health systems for equity and development in Africa":** 1. can't find home page 2. enters 'health systems' 3. reads titles on first page 4. goes to search box on results page and adds 'for equity and development' 5. reads the titles on first page of results 6. returns to search box on results page and changes search to 'Africa health strategy 2007-2015' 7. required article is first in the list, GH11 chooses it to check.	**Steps to find specific article "Using climate information to achieve long-term development objectives for African ports":** 1. thinks about what to do for a minute 2. types 'climate information' into simple search box 3. reads first page of results and chooses third in the list 4. reads that it is 'in Rwanda, not African ports' and downloads linked PDF 5. returns to search results page 6. returns to tab with PDF download and notices that it is 'in Rwanda' 7. returns to search results page **"these are not complete sentences so.."** it is hard to see 8. chooses the eighth search result **"some show the author so it's easier"**
Steps to find information:	**Steps to find information:** 1. scrolls half way down front page 2. chooses Evidence for policy and practice on LHS 3. chooses Governance on LHS rapidly. Governance is GH11's key interest 4. reads the first paragraph on Governance 5. scrolls to the bottom of the article list, and back to top again 6. looks at sub-topics on LHS **"they've broken it down and that's interesting"** 7. reads the titles of the article list looking for anything on Ghana 8. notices country profile and chooses More countries **"there's no west Africa**	**Steps to find information:** 1. scrolls down the front page 2. chooses 'Browse by country' 3. enters Ghana in the 'Country' box and Western Africa in the 'Region' box 4. checks the documents and projects options and chooses Browse button 5. scrolls down the first page of results **"It's interesting… I haven't found a need to"** come to this website
Barriers to use: **Validity/use**: **Downloading:** **Interactive functions:**		

	here" 9. chooses Ghana in the Africa south of Sahara category. **"I was expecting to see West Africa"** 10. reads the titles of articles page 1, page 2. **"there's so much information in here"** **Barriers to use:** search is confusing to use **Validity/use:** relevance **Downloading:** no problems **Interactive functions:** not used	6. returns to browse by country options, chooses 'current projects' and 'projects', i.e. not documents 7. reads the titles **"I haven't found the kinds of projects I know"** 8. decides to search for a specific project they know, but can't find how to go back to that search, and can't easily just change the browsing parameters 9. chooses completed projects in Ghana Western Africa 10. when search results are displayed, looks down the refine options on RHS because wants to be able to put in the name of the project 11. goes though the first two pages of the search results but can't find what looking for. Does not want to go through 724 records 12. returns to home page and enters STAR-GHANA in simple search 13. scrolls down results pages, but can't see STAR-Ghana **Barriers to use:** can't find projects they are or have been engaged in **Validity/use:** focuses on projects **Downloading:** not used **Interactive functions:** not used

Knowledge of: **Scidev.net**: None **Eldis.org:** None **R4D:** None **Portal challenges:** • Unreliable internet access. • Difficulty in accessing top quality document on the internet in the office in Parliament. **Use of information:** **Shared online:** Via email to colleagues at work or WhatsApp group. **Shared offline:** No **Saved:** Saves only relevant documents in a folder. **Repurposed:** GH12 makes own paper notes on documents taken from the internet. **Read:** Yes, first reads title and if relevant, reads abstract and conclusion before deciding whether to read the whole document.	**Participant–GH12** **Category 7** GH12 is a research officer at Parliament of Ghana. Works with different parliamentary Select Committees in Ghana and is currently working on an International Trade Bill. In addition to the use of paper books, GH12 also looks for evidence online for the preparation of the bills. GH12 indicates that **"what I do is to get...information on International [Trade] Commission from in other jurisdictions online"**. GH12 uses Google search engine to look for evidence online—mainly because it is easier and accessible. Types keywords "**International Trade Commission Functions"** into Google search box. Scrolls down and clicks on the third title of the search result, but did not get any relevant online research evidence. Clicks on the second pages of the search results but finds no relevant online evidence. GH12 clicks on the "back" button to the first page of the search results and decides to scroll down further. Opens the last but one search result with the title **"A review of the International Trade Administration Commission's Tariff Investigation Role and Capacity"** on a different tab. Clicks the new tab and realises it is a report. Reads the title again and cursorily looks at the table of content to ascertain whether there are/or is section(s) of the document that is relevant for the preparation of the GITC bill. Finds some relevant sections. Saves it in a folder for future thorough reading. Ascertains the validity of an article by determining whether the article has been published in a peer reviewed journal. For instance, NGR indicates that **"I go to the University to look for peer reviewed journals....."**. Also considers articles that have been recommended by experts to Parliament of Ghana. Example	**Information behaviour:** **Starting:** • Starts by typing the full sentence of specific article in Google search engine. • scrolls down. • reads the titles on the first page of the search results. • opens specific articles in different tabs. • saves relevant specific article in a folder and reads at a convenient time. **Chaining:** GH12 copies key references in one relevant document into Google search engine to find more relevant documents. **Browsing:** GH12 scrolls through titles on the first page of the search results. **Differentiating:** GH12 separates relevant search results from the irrelevant ones by opening the relevant ones on a new tab. Saves only relevant article. **Monitoring:** None. **Extracting:** Download and saves relevant articles. **Processing:** See repurpose on the LHS. **Assessing validity/use:**

	GH5 explains that "sometimes some of the members of the parliamentary select committees are experts.....and they recommend papers for consideration.....". Relies on some of these articles that have been recommended to parliament for the preparation of the bill. Shares online research evidence with colleagues through email or WhatsApp. Thirdly, relies on some internationally recognised country specific portals like United States of America International Trade Commission website (https://www.usitc.gov/). **Example use of information:** GH12 was working on a Parliamentary bill. **Output:** Has completed a first draft. **Uptake:** Research evidence on the internet has made research easier, but access to peer reviewed journals remains a challenge. For instance, GH12 indicates that **"for this office I don't get much journal articles because we don't subscribe to them ....I am able to get access to them when I go to the University [University of Ghana]".** Access to internet can also be a major challenge. For instance, **"I don't get internet in my office regularly, and even when it is available, it is not always fast"**.	• Relies more on journal articles and articles. • Articles recommended by experts to the Parliament of Ghana. • Also relies on country specific portals that focus on International Trade Commission Bill. For instance, United States of America International Trade Commission website (https://www.usitc.gov/).

Participant–GH12	Category 7 part 2	
Scidev.net **Steps to find specific article:** 1. Types key words **(i.e. Big data for development)** of specific article into the search box but gets zero result. 2. Modifies the search by typing the full title of the specific article "**Big data for development: Fact and Figures".** 3. Scrolls up and down twice before identifying the article on the first page of the search result. 4. Reads the brief summary below the title of the article. Clicks on the title of the article. Reads the article summary. **Steps to find information:** 1. First moves the cursor on the search titles on the portal. 2. Clicks on the title **"Governance"** and then **"policy"**. Types the sentence **Fiscal regimes of the oil sector in Ghana** into the search box of the article but finds zero results. Gets disappointed. 3. Modifies the sentence by typing **"Fiscal regimes + oil sector + Ghana"** but still gets zero results. 4. Realises Scidev.net is a science bias portal as such decides to type **"health insurance + farmers"** but gets 2 results which are not related to key words that was typed into the search box.	**Eldis.org** **Steps to find specific article:** Portal did not load **Steps to find information:** Types **www.eldis.org** into Google search engine and realised the website was down. GH12 believes it is not an internet issue since the other two portals (i.e. www.scidev.net and R4D.dfid.gov.uk) worked with the same internet source. **Barriers to use:** No data were obtained **Validity/use:** No data were obtained **Downloading:** No data were obtained **Interactive functions:** No data were obtained	**R4D** **Steps to find specific article:** 1. Types the full title of the specific article **(i.e. Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality)** into the search box of R4D. 2. Scrolls down and then up. 3. Begins to read the titles and the brief summaries below the titles on the first page of the search result. 4. Finds the specific article on the first page. Opens it on a different tab. 5. Reads the summary of the article. **Steps to find information:** 1. Glances through the home page of the portal to see what it entails. 2. Types **International trade commission** into the search box of R4D but does not get any relevant information though the total search result is **1590.** 3. However does not go beyond the first page of the search result. 4. Modifies the search by typing **International Trade Commission Dumping** and carefully scrolls down to see if there is something relevant to the GITC bill. Does not find any relevant article. **Barriers to use:** • Finds it frustrating when looking for an article for the GITC bill. As such gets discouraged in going further with the search.

5. Widens the search by typing only **"health insurance"**. Gets 4 search results that are search results. **Barriers to use:** Finds it confusing to navigate through the portal. **Validity/use:** No impact. **Downloading:** No downloading activity. **Interactive functions:** No impact.		• Finds the portal to be slow. **Validity/use:** Focuses on peer reviewed articles. **Downloading:** downloads specific article. **Interactive functions:** No impact.

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

No evidence to show.

Use of information:

Shared online: Shares online articles with friends and colleagues at work through email.

Shared offline: Does not share article offline.

Saved: Saves only relevant articles.

Repurposed: GH13 rewrites in Microsoft Word all online articles in own words.

Read: Reads online articles a lot.

Participant–GH13 Category 3

GH13 is a lawyer who advises on legal related issues and contracts. Also works on policy related issues in the oil and gas industry, government and Non-governmental organisations.

Uses Google for general search. Demonstrates this by searching for petroleum laws that are relevant for contract negotiations. Tries to compare petroleum laws from various sources, both local and international, to ascertain which of them best suits the context of GNPC and Ghana.

Also uses Google search engine to looks for general issues and not necessarily law. This is to get a detailed and general knowledge about a subject. For instance uses Google to look for policy related issues on best practices for policy formation in the petroleum industry. This does not necessarily focus on Ghana.

However, GH13 uses specific portals to search for specific information. For instance, looks for information on petroleum laws on Nigerian Department for petroleum resources, since Nigeria is one of the largest oil producing countries in the world. European Commission is also a very good source of information for European laws.
Also subscribes to databases for relevant data in the petroleum field. For instance, subscribes to the Thompson Reuters portal—gives up to date information on laws. The Thompsons portal is very current and they send emails on various rulings across the globe. The portal contains legal databases where contract samples and published articles can be found.

Likes reading articles that are collected online and then make own judgement on whether the article is credible or not. Also considers articles from recognised portals like that of the European commission.

Also likes PDF documents since they provide the name of the author,

Information behaviour:

Starting: Starts searching for general information so as to get detailed knowledge about a subject. Uses Google search engine to start general search.

Chaining: No evidence.

Browsing: Uses Google for general browsing. However, looks for specific articles or information on specific portals. For instance will prefer to look for law related stories on the Thompson Reuters portal. Will also like to get best practices in the oil and gas industry in Northern economies from the European Commission portal. Uses the Nigerian Department for petroleum resources to ascertain best practices in developing countries.

Differentiating: Saves most relevant online articles.

Monitoring: Subscribes to portals such as Thompsons Reuters so as to get the most relevant and current law related information.

Extracting: Seldom copies online articles. Prefer to read and make own notes.

Processing: Reads and then writes in own words the sections of the article that are relevant.

Assessing validity/use:

- Articles from reliable sources such as

	title, etc. of the article. Also triangulates by enquiring from others to check whether the document is credible. Shares with friends and colleagues through email. Also shares information pertaining to the organisation through the GNPC intranet. Though relies on of portals at the Ministries, Departments and Agencies in Ghana, GH13 does not rely on the GNPC local portal since it contains less information. **Example use of evidence:** Did not show an example of research evidence. **Uptake:** GH13 does not buy books anymore—mainly because almost all research evidence can be accessed online. Makes working easier and also save time. Some websites are updated on daily basis so it helps GH13 to get most recent information. Able to get very current information in every part of the world. Helps to learn new things easier and faster. This builds GH13's confidence.	Thompson Reuters and the European Commission. • Also have confidence in articles in PDF format. • Likes to use own judgement to determine the credibility of an article. • Also triangulates by ascertaining the credibility of an article from others.

Table 37: Ghana Research Diary Summary Table

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
GH01	**find/retrieve**	Found: used an online search engine	Found: used an online search engine/ received info in an email from a person	Retrieved: stored digitally	x	o	o	x	o	Found: used an online search engine Retrieved: stored digitally
	why source	It was a general online search which I normally use when looking for information	To support me with ideas on facilitation of capacity improvement workshop	Because of its relevance in my previous usage	x	N/A	N/A	x	N/A	This was a general internet search which I normally use for information gathering.
	validity	I did directly modify parts of the document to suit my assessment report. For example, relevance of governance structures and public relations of an association. I did use the findings to reflect my introductory write up on governance and public relations for an association's	The document was used for similar purposes for similar audience in North America	Had used portions of it previously and it passed the reliability test already	x	N/A	N/A	x	N/A	Incorporated part of the materials into the PowerPoint Presentation and others into templates for capacity improvement sessions

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		report I was developing								
	use	read/ repurposed	saved/ repurposed/ other: received a shared copy from a colleague	repurposed	x	o	o	x	o	saved
	type/purpose	Took notes and directly used it to update an organisational capacity assessment report	Made notes on a presentation/facilitation of a capacity improvement planning workshop for an association	Made notes on organisational capacity assessment and improvement planning and tailored language to suit my audience.	x	o	o	x	o	Was preparing to facilitate a workshop later in the day on organisational capacity improvement planning and had to rely on internet materials to as part of preparations towards the workshop
	influence				x	o	o	x	o	
	research evidence	NO	NO	NO	x	o	o	x	o	NO
	DFID portals	not used today	not used today	not used today	x	o	o	x	o	not used today
GH5	**find/retrieve**	x [no response recorded for this question]	o	Found: on paper Retrieved: stored on paper	Found: other [no response in free text box]	Found: other: it was an examination question Retrieved: remembered	x [no response recorded for this question]	x	x	Found: on paper Retrieved: contacted a person
	why source	x [no response recorded for this question]	o	N/A	N/A	[???] I read about the it on a printed material.	N/A	x	x	Because the soft copy was not available so I had to

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
										photocopy it
	validity	x [no response recorded for this question]	o	N/A	N/A	By observation and experience in the university.	x [no response recorded for this question]	x	x	It is a national policy which gaining social acceptance /
	use	x [no response recorded for this question]	o	o	o	other: I wrote examination on it	saved	x	x	other: we discussed it in a forum
	type/purpose	x [no response recorded for this question]	o	N/A	N/A	The attitude of Ghanaian University students toward increasing user fees	N/A	x	x	I participated in a conference on discussion of the guideline for the Council for Technical and Vocational Education and Training (COTVET) for Ghana.
	influence							x	x	
	research evidence	NO	NO	YES [???] I read about Ghana's 2016 budget statement and economic policy for the Daily Graphic	NO	YES: The evidence is conspicuous in the slow reaction of university students toward increasing user fees.	NO	x	x	YES [???] I searched for the soft copy of the guide from COTVET website
	DFID portals	not used today	o	not used today	not used today	not used today	not used today	x	x	not used today
GH8	**find/retrieve**	o	Retrieved: repeated	Found:	x	x	o	o	x	Retrieved:

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
			online search	received email from a person						stored digitally
	why source	o	Was discussing the subject with a visitor who had come to my office	I was looking for information on the subject	x	x	o	o	x	I was brainstorming on the subject for some research write up that I am working on, so I needed to reconfirm some thoughts I had gathered from the source
	validity	o	It was a published document that I was previously aware of and knew the author	The author and publisher	x	x	o	o	x	This is a source I had stored from a reliable source. The source is from a respected journal
	use	o	other: downloaded and saved	saved/ read	x	x	o	o	x	read
	type/purpose	Not Applicable: Mondays are my program management meetings so no real research	Report on Abuse of incumbency for electoral/partisan gains	Looked through two journal articles on MPs Accountability and saved them on my desktop	x	x	x [no response recorded for this question]	x [no response recorded for this question]	x	Read through a study on politics of inclusive development
	influence				x	x			x	
	research	o	YES [???] I did an	YES: A	NO	NO	x [no	x [no response	NO	YES: This

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
	evidence		online google search; I typed in the phrase "monitoring abuse of incumbency +Ghana" and found what I was looking for among search returns	friend sent me two research articles from a reputable researcher/ author. I looked through and stored it t return to it at a later date			response recorded for this question]	recorded for this question]		research evidence is based on empirical source to explain a phenomena, leading to some conclusions about it.
	DFID portals	o	not used today	not used today	x	x	not used today	not used today	x	not used today
GH9	**find/retrieve**	o	x	x	x	o	x	Retrieved: stored digitally	x	x
	why source	o	x	x	x	o	x	It was relevant to the discussions and it was the most current information source with respect to the subject matter under consideration.	x	x
	validity	o	x	x	x	o	x	It is a very reliable document	x	x
	use	o	x	x	x	o	x	Other: We had a group discussion around using research for advocacy	x	x
	type/purpose	o	x	x	x	o	x	I used the Ghana Demographic and Health	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
								Survey 2014 report. I used the findings to discuss with my other colleagues to develop strategies for advocacy to improve Comprehensive Sexuality in schools		
	influence		x	x	x		x		x	x
	research evidence	NO	x	x	x	NO	x	NO	x	x
	DFID portals	not used today	x	x	x	not used today	x	not used today	x	x
GH10	**find/retrieve**	Found: other website: Facebook timeline	o	x	Found: received in an automated message	o	o	Found: talked to person/ use online search engine/ used online RPP: statsghana.gov.gh Retrieved: had stored digitally/ remembered	Found: talked to person/ use online search engine	Found: talked to person/ use online search engine Retrieved: remembered/ contacted a person with access to info
	why source	I needed to make references to aspects of it during the chat with said friend since I could not recall all the details of the text off head.	o	x	SciDev.Net provides timely information on global science and development issues. They provide consistent info on antibiotics	o	o	The Ghana Demographic and Health Survey is the standard tool for accessing key information pertaining to demographic trends in Ghana / / The adolescent	They are routine sources used in my line of work	Routine sources used in my line of work

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
					and their efficacy against bacteria.			reproductive health policy and SDGs contain critical information important to my line of work (youth reproductive health) / / All the above sources are deemed as reliable due to the institutions behind the information.		
	validity	I checked the source of the information - alliance for reproductive health and rights (mamaye.org.gh) / I checked the date of publication - March 2015 to determine relevance/timeliness / I compared some details given with my field experience in the district	o	x	x [no response recorded for this question]	o	o	Time for making presentations were limited so there wasn't much scrutiny. However I had used some of this information in the past.	Checked two or more sources to verify information	Cross-checked with information obtained by colleagues with whom I was working with on the assignment
	use	other: discussed it online	o	x	shared online	o	o	shared offline	shared offline	saved/ repurposed
	type/purpose	I discussed an incidence of	o	x	I read an article on	o	o	I used portions of Ghana's	Used information	Used information

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		teenage pregnancy in South Dayi District with a friend who works in another organisation, based on information she earlier shared on Facebook. We tried to understand the thought processes of a 13 year old girl who chose to get pregnant because she wanted more attention from her mother (her sister was pampered when she got pregnant).			SciDev.Net on the negative effects (bacterial resistance) of increased use of antibiotics in developing countries. I shared the link to the story on Twitter.			reproductive health policy, the Demographic and Health Survey, as well as UN SDGs to identify gaps in SRH programming in Ghana during a workshop	from various sources on youth SRH to begin work on an advocacy plan	from various sources on youth SRH in Ghana to begin work on an advocacy/policy engagement plan
	influence			x		o	o			
	research evidence	NO	o	NO	YES: The story on SciDev.Net website comes with links to recent research synthesised by the likes of CDC. This might be worth looking at in detail in the future	NO	NO	YES: The contraceptive prevalence rate and youth SRH trends, including teenage pregnancy rates, were gleaned from the Ghana Demographic and Health Survey. The data were used	YES: We found critical evidence in a study that 50% of parents want children to taught sexual education in schools	NO

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
								for a presentation identifying gaps/opportunities for advocacy in a workshop /		
	DFID portals	not used today	o	x	SciDev.Net	o	o	not used today	not used today	not used today
GH12	**find/retrieve**	Found: talked to person	Retrieved: had stored digitally	x	Found: used online search engine Retrieved: had stored digitally	Found: used online search engine Retrieved: had stored digitally	Found: on paper/ used online search engine Retrieved: had stored digitally	Found: used online search engine Retrieved: had stored digitally	x	Found: on paper/ used other website: ministry of finance Retrieved: had stored digitally
	why source	To get first-hand information on the subject matter	I remembered I had an information of that sort	x	I remembered that it was going to be useful	Remembered it was relevant for my work	Related to the topic	To meet a deadline	x	Found it to be relevant
	validity	Interviewed two more people	x [no response recorded for this question]	x	Went to an International trade Commission website in an advanced country. THIS WAS WHEN I HAD READ THE ARTICLE I DOWNLOADED	Searched other database sites	Browse from IMFwebsite	Searched other organisational website	x	Checked from world bank and bank of Ghana website
	use	read	repurposed	x	saved	repurposed	repurposed	repurposed	x	repurposed
	type/purpose	Made notes about the challenges of	Challenges faced by pubic accounts committee of	x	Functions and structure of International	Made notes about how unequal	Role of parliament in managing	Made notes on role of parliament in	x	Budget information, summarized

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		implementation of public accounts committee of Ghana	parliaments		trade Commission	distribution of wealth is a threat to democracy. I am to write a conference paper for a member of parliament	external debt. A request by a member of parliament for publication	curbing illicit flows of financial and national natural resources from Africa to be delivered at a conference		and analysed it for a publication
	influence			x					x	
	research evidence	NO	NO	x	YES: With this information I communicated to the members of parliament who were considering the bill	YES: Efforts by the government of Ghana to reduce wealth inequality. Found it from various articles to write the paper	YES: Ghana debt level and sustainability. From Google search fir the write up	YES: Illicit flows in the extractive sector to be presented at a conference	x	YES: Ghana shared growth and development strategy, have a hard copy, extracted data from it
	DFID portals	not used today	not used today	x	not used today	not used today	not used today	not used today	x	not used today

B.3.2 Tanzania User Profiles and Diary entries

Table 38 below summarises our Tanzanian participants and the data collected from them. This section contains the user profiles generated from 11 of those participants and a summary of the daily journal entries.

Table 38: Summary of Tanzanian participants

Participant	Gender	Category	Sector	Data collected
TZ1	M	Development worker in civil society	Agriculture	CI, DP, D7
TZ2	M	Academic/Researcher	Economic Growth	P, CI, DP, D3
TZ3	M	Knowledge broker/intermediary for policy makers	Generalist	P, CI, DP, D3
TZ4	F	Academic/Researcher	Gender and Equality	CI, DP, D3
TZ5	M	Civil/Public Servant	Agriculture	CI, DP, D10
TZ6	F	Knowledge broker/intermediary for policy makers	Science and Technology	CI, DP, D7
TZ7	F	Academic/Researcher	Economic Growth	CI, DP, D5
TZ8	F	Media professional	Generalist	P, CI, DP
TZ9	M	Academic/Researcher	Economic Growth	f2f interview on influencing policymakers
TZ10	F	Development worker in civil society	Health	P, CI, DP
TZ11	F	Academic/Researcher	Science and technology	f2f interview on influencing policymakers
TZ12	M	Academic/Researcher	Economic Growth	CI, DP, D1
TZ13	M	Civil/Public Servant	Education	P
TZ14	M	Media Professional	Generalist	P
14 participants overall		**10 complete f2f sessions**		**5 complete diaries (i.e. reflection complete)**

* P = phone interview, CI = contextual inquiry (first half of f2f session), DP = discussion of portals (second half of f2f session), D= Diary, Dx = x number of journal entries returned

Knowledge of:

Scidev.net: none

Eldis.org: received their weekly reports in the past, may still be receiving them but haven't had the time to read them. Sometimes share within organisation. Went to website maybe 2 years ago.

R4D: none.

Portal challenges:

- Many organisational websites and portals are complex, but Google searches within the site and you get the relevant report more directly.
- For R4D, was unsure about the purpose of the portal and found the advanced search complicated.
- **"we understand we have poor connection"** but there may be different reasons; not many people are using it on a Sunday (Eldis).

Use of information:

Shared online: internal within organisation, by email, point to specific interesting information in it.

Shared offline: discourage printing.

Saved: Yes**,** TZ1 saves interesting reports for future reference, keeps address and closes online source.

Repurposed: uses paper to note source, and references, identifies figures, charts and tables to refer to, and if there is a focus on social groups, e.g. poorest groups. Can't

Participant – TZ1 Category 4

TZ1 works in an advocacy organisation, an umbrella for bringing together private companies and local and national NGOs and farmers' organisations. The organisation focuses on academic research but also what is working and what is not, in agricultural policy. They use the learning to influence government and donors.

TZ1 usually downloads documents found that are relevant, e.g. by looking at the title and table of contents, and some interesting figures, skims through the document to look for specific topic of current interest. May also look for reports to compare across years.

In Tanzania, they have problems with the data. Organisations and government have different figures for the same statistic, e.g. for the budget allocated to agriculture. For example in Uganda they **"include the budget for military in the agriculture budget because you need security in order to farm – if there is war then you can't farm"**.

They use the information offered by the government and other sources to see how much of the budget they say has been allocated to agriculture. They also look at regions to see what allocation is there, and resource allocation generally. How many deployed extension staff are there in urban areas compared to rural.

"Reading culture is not that much in the country… people want to see ready-made things. They don't have

Information behaviour:

- Starts new search in Google
- Downloads PDF documents to read
- Will skim the table of contents
- Don't need to save everything you've opened but can open in a different tab to have access before deciding to save
- Several tabs may be open from several institutions

Starting: uses key words in Google search engine, key words may include the source organisation.

Chaining: from Google to found sources; from found sources to download reports; may open several from one institution/organisation.

Browsing: scans titles in Google, didn't move from front page in Google, scans downloaded reports for table of contents, interesting figures, key words.

Differentiating: opens different documents from the same organisational website in different tabs, downloads reports and saves if relevant.

Monitoring: keeps up-to-date with changes in CAADP (comprehensive Africa agriculture development programme).

Extracting: extracts budget figures and also statistics relating to employment in smallholder farming, motor bike availability, how do they offer services etc.

Processing: see repurposed on LH.

Assessing validity/use:

- Title for relevance
- Author
- Credible source organisation: FAO, IFPRI

extract directly if source is a PDF (which it mostly is). **Read:** read through report, table of contents, searches for keywords.	**"time to torture their brains <laughs>"** Sometimes when something has been shared and then followed-up, people say that they haven't seen it. **Example use of evidence:** Ongoing concern with agricultural budget, updated every 6 months or so. Within the organisation they assess this programme to see how much budget is going to rural development etc. The target is 10% budget allocation to agriculture. TZ1 authored reports about this programme. May also look at gender or youth budget. **Uptake:** **"There is a lot of evidence available online, but the question I have in mind is "so what?"".** The politicians don't need the portals to change, they need the information on their table to make decisions and move forward. The research needs to be more accessible in terms of presentation but also in terms of the steps to take to implement change. Politicians want steps for how to change, and unfortunately researchers are not doing that. We need to create a link between the researchers and the politicians – to translate between research and practice. Translating research into practice should be taught. Having evidence online is not sufficient. Also, need to create platforms between research/academic and the practice. For example, ReSaKSS aims to link research and country context.	**"Author and credibility of the organisation has a lot to do with the quality of the information."** It doesn't mean that new sources are not credible but it takes time for people to trust the source.

Participant – TZ1 Category 4 part 2		
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "Big data for development: facts and figures": 1. types in 'data for deve facts figures' to search box on top right 2. returns zero results 4. tries to correct search to 'data for development facts figures' but editing in box only allows deleting the characters from the right and re-entering 5. types in 'data for development facts figures' to search box on top right 6. receives 58 results, scrolls down first page and returns to search box to enter person 7. adds 'emmanuel 2014' to search terms 8. article is the only one in the results list	**Steps to find specific article "Three tools to unlock finance for land-use mitigation and adaptation":** 1. types 'tools unlock finance land-use falconer 2015' into search box on LHS 2. required document is first in the results list of 12915 documents	**Steps to find specific article "Using climate information to achieve long-term development objectives for African ports":** Doesn't usually look for a specific article any more 1. types "climate change long term development Africa woolhouse 2015" into simple search box 2. first item in search results.
Steps to find information: **3.** 1. reads the top strap line and says **"I've never heard about this"** 2. scrolls down the home page 3. points to the main menu items Agriculture (1), Environment (2) and (pause) Governance (3) and says would only concentrate on those 4. looks under the agriculture menu and reads the sub-menus 5. considers whether the 'Home' button refers to home for Agric or home for the site and is confused about it 6. chooses fisheries sub-menu and looks for data 7. looks at RHS refinement menu but nothing there	**Steps to find information:** They have a lot of information on agric, and so TZ1 became interested. TZ1 doesn't remember how they found out about it, maybe 2002. 1. took a while to load (59:16 1:02:0) so over 2 minutes **"very slow"**, so went to SciDev.net 2. returns to Eldis 30 mins later and the home page loads quickly 3. goes to LHS set of menus to see what's of interest: evidence for policy and practice, gender and governance 4. chooses evidence for policy and practice, and comments that this is 'user friendly' 5. identifies 'country profiles' on LHS and sees that it is easy to get information about the country **"I hope the profile is not static"** 6. chooses 'more countries', then Tanzania, and scrolls down the information. 7. highlights the main Tanzania box and says **"outdated"** because the population figure is incorrect 8. chooses governance from LHS menu under	**Steps to find information:** 1. reads home page information to decide whether to continue with portal 2. reads the LHS menu structure 3. observes that this appears to be research for the government 4. chooses the LHS menu item "Browse by country" 5. looks for Tanzania but it's not there – is under United Republic of Tanzania 6. looks at region, but doesn't choose anything 7. opens the document *Searching R4D*, was expecting to see further information on the country **"maybe this is a waste of time reading this"** and scrolls through it. Says will go back and fumble around the website 8. chooses documents 9. looks at titles for the first page of results 10. says would like to be able to add extra information in the browse by country search page. 11. goes to advanced search and enters keywords "agricultural budget gender". Now it's no longer

to help 8. scrolls down the articles **"that looks interesting"** 9. chooses data menu item, and it goes to fisheries articles again 10. goes back and tries again. The page returns articles about data and not data, which they expected 11. **"for me the data would have been something like statistics. Maybe the word 'data' has been misused"** **Barriers to use:** position of 'home', 'data', 'multimedia' buttons is confusing; expects 'data' to be a subset of, e.g. 'agriculture' and to find data for fisheries etc. rather than a main menu for data; navigation and links between different sections **Validity/use:** nothing found of interest, although there was a lot on agriculture **Downloading:** nothing downloaded **Interactive functions:** not used	country profile 9. scans the results and chooses one article on political dynamics of economic growth, downloads PDF **Barriers to use:** country information is out of date **Validity/use:** in abstract, found keywords of interest: "political determinants of economic growth" **Downloading:** downloads as above, using tabs **Interactive functions:** not used	for the country-specific. Includes Tanzania in the keywords 12. Tries to complete the advanced search fields **"this is very complicated"** 13. submits search 14. one project is returned and it is outdated 15. returns to browse by theme and looks for country 16. chooses agricultural innovation, finds a project on farming innovations in sub-sahara but can't see if it includes Tanzania 17. looks for completed reports, and for Tanzanian information **"first and foremost my country"** **Barriers to use:** advanced search is complicated; guidance for searching wasn't what was expected; accessing Tanzania-specific information in the field of interest was hard **Validity/use:** date, completed reports, Tanzania-specific information about agriculture **Downloading:** none attempted **Interactive functions:** not used

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

- Eldis was very slow and TZ2 spent 10 minutes trying to download it and still was not successful.
- Internet connectivity was also a problem—mainly because of erratic power supply.

Use of information:

Shared online: No

Shared offline: Shares online articles with friends and the director of the institution.

Saved: Saves every data downloaded in order to avoid going back to the online source for the same article. Also enables TZ2 to save time.

Repurposed: Reads online article, rewrites it in own words and uses it to prepare the report or the PhD proposal.

Read: Reads article online. Also reads saved articles.

In some instances TZ2 will prefer to print out the article before reading.

Participant–TZ2 Category 5

TZ2 is an academic and researcher in innovation economics. Works for a science and technology think tank in Dar es Salaam, Tanzania.

Relies extensively on the World Bank portal in the preparation of the report.

Collects online economic data from the World Bank portal. Collects online data from the World Bank portal because data on the portal is collected from various sources like the National Bureau of Statistics, Tanzania and has been referenced by several authors. Copies the online data into excel for further processing—present the data in a table format and them uses them to plot a graph.

TZ2 saves the data in order not to lose it. TZ2 said that **"...it is because I want to keep it. I can even come back tomorrow to verify. [I save] because I am even if I lose this report I still have the data to produce it again....also saves time for searching the data again".**

Also uses Google scholar to search for relevant online articles. Prefers open access articles. For instance, tries to download an article from Google scholar and realises it is not for open access. Tries to see if the same article can be downloaded in a different format for free. TZ2 indicates that **"this paper is under purchase and I find that there is another opportunity for this one [a PDF version of the same article] to be downloaded for free, so I download it".**

Downloads the article on Google scholar and then looks for more relevant references in the article. Indicates that those articles that cannot be saved or downloaded on Google scholar are used online. TZ2 reads important sections of the online article that

Information behaviour:

Starting: Starts online search purposively. Searches for quantitative data online through the World Bank, International Monetary Fund and Industrial Statistics or the National Bureau of Statistics portal.

Chaining: Copies references from relevant articles and then pastes them in the search box of Google scholar for further search.

Browsing: Browses with both the Laptop and the mobile phone. Likes to read titles of online articles when looking for relevant articles. Also scrolls up and down during online search.

Differentiating: Copies relevant online quantitative data into excel data and saves it.

Monitoring: No evidence found

Extracting: Extract quantitative data from both international and local internet portals.

Also tries to write in own words articles on Google that cannot be saved.

Looks for relevant online articles on Google scholar. Downloads open access documents.

Processing:

Rewrites articles in own words. Also prefers to use quantitative data for analysis in excel. For instance, TZ2 uses data downloaded from the World Bank portal to plot a graph in excel.

Assessing validity/use:

- The source of the online information must be recommended by someone else who is familiar with the source.

	cannot be saved after scrolling up and down. Writes down the article in own words. TZ2 closes the article on the page. Looks for online article by typing keywords in the title of the article. TZ2 was unable to find the article and so decides to narrow search by further narrowing the research title by adding "developing countries". Scrolls down the first page of the search results and carefully reads the titles of each search result. Finds the most relevant article and then downloads it. TZ2 prefer to read the abstract of the article to find out whether the article is relevant. If relevant, TZ2 proceeds to look for more relevant references in the article for further online search. Indicates other portals such as "Industrial Statistics" and the National Bureau of Statistics of Tanzania are also very important for the report and the PhD proposal. Shares article with the director of the organisation. Internet goes off. Spends 22.04 minutes trying to get the internet connection on again. TZ2 trusts portals that have been recommended by others in the area of study. This boost's TZ2's confidence about the portal. For instance, TZ2 explains that the World Bank and National Bureau of Statistics portals have been tried and tested both with and outside Tanzania and so have confidence in those portals. **Example use of evidence:** Writing a report in the area of innovation in formal and informal micro-enterprises. **Uptake:** Able to learn new concepts/things online. For instance TZ2 indicates that **"I did my masters in development studies but I am now in the area of innovation.... and so I am able to look for information online to help me understand this new area [i.e. innovation]"**	• Is also interested in how relevant the abstract is to an ongoing work.

Participant–TZ2	Category 5	part 2
Scidev.net **Steps to find specific article:** 1. Goes back to the home page of the portal to look for specific article with the title "Big data for development: facts and figures". 2. Types the full title of the specific article into search box of the portal. Reads the first four titles of the portal. Identifies the fifth title as the specific article. 3. Clicks on it, reads the speed read section of the article and indicates that it is the right article. Indicates that the article cannot be saved. **Steps to find information:** 1. Decides to use mobile phone to access internet since the fixed lines are no more working. 2. Scrolls down the home page of the portal after downloading the portal. 3. TZ2 reads each title while scrolling down the home page of the portal. 4. Clicks on an article with the title "Patents fail to boost crop yield". Indicates that was expecting to find such types of articles on the portal. **Barriers to use:** Poor internet connectivity was the main issue during the online search on the specific portal. **Validity/use:** Indicates that the portal may be a DFID portal but will prefer that it is recommended by others who are familiar with the portal. This will boost confidence in the use of the portal. **Downloading:** Clicks on the specific portal but finds no attachment. Indifferent about the fact that there is no attachment to be downloaded.	**Eldis.org** **Steps to find specific article:** No access **Steps to find information:** No access **Barriers to use:** The network goes down just after the specific article for SciDev.Net was found. TZ2 spends another 10 minutes trying to fix the internet so as to be able to download the Eldis portal. Becomes unsuccessful and so decides to move on to R4D. **Validity/use:** No access **Downloading:** No access **Interactive functions:** No access	**R4D** **Steps to find specific article:** 1. Types the URL of the portal into the search box of the portal. This was after the network started working again. 2. Types the full title of the specific article **"using climate information to achieve long term development objectives for African ports"** into the search box of the portal. 3. Realises the network is off again. Reconnects to another network and then continue the online search. 4. Types the specific article into the search box again. 5. Reads the titles on the first page of the search results, and finds the specific article. Downloads the article in PDF format. **Steps to find information:** 1. Types the URL of the portal into the find box on the mobile phone. Able to download R4D. Indicates that Eldis was unable to download partly because the portal is **"not working properly"**. 2. Scrolls down and then reads the titles on the left hand side of the portal. Clicks on "browse by theme" button on the left hand side of the portal. 3. Scrolls and reads topics from up to down and left to right on the portal on the mobile phone. 4. Clicks on the sub topic "economic growth". And then reads the various topics on the

Interactive functions: Likes the interactive functions but will only share a link on the portal if the portal is known to be reliable.		results page. Clicks on the fifth research title. 5. Reads the summary and looks at the source of the online article. Downloads the online article in Microsoft word format. 6. TZ2 does not understand the objective of the portal. 7. Power goes off. Network on the phone also goes off since TZ2 was using a different network. 8. Connects to another network on the phone. **Barriers to use:** Poor network connection as a result of frequent power cuts made online search very difficult. For instance, the power went off twice during the online search. **Validity/use:** Source of the article must be referred by someone who is familiar with the portal. **Downloading:** Downloads specific article. **Interactive functions:** No significant impact.

Knowledge of:

Scidev.net: none

Eldis.org: none

R4D: has been to the portal when someone suggested/shared an article available through the portal but hasn't before studied the portal itself

Portal challenges:

- Universities don't have online platform in place for all of their research
- Some power problems, e.g. if it is rainy, but it doesn't happen often

Use of information:

Shared online: internal within organisation, by email as a link or as an attached document

Shared offline: through presentations

Saved: Yes, in folders on PC

Repurposed: extracts references from one source to conduct a new search on that reference. Uses an empty word document to type in points from the paper including the reference (for the papers that TZ3 has concluded that they are useful for current work). Always write in your own words

Read: reads details for the publication, then the abstract. TZ3 doesn't read all the paper, only the relevant sections, identified through the table of contents

Participant – TZ3 Category 7/5

TZ3 works in a think tank whose research is focused on influencing policy around science, technology and innovation. TZ3 produces policy briefs and follows the research produced by the think tank to determine whether the government makes any changes as a result of what they did. TZ3 works on communication and policy engagement strategies.

TZ3 also conducts their own research and is studying with the local university.

Information is found through Google search or ministries' websites, or websites of other institutions, and universities. TZ3 wants detail and is hence most interested in journal papers. TZ3 search in Google includes "research on…" keyword, or specifically "journal paper". TZ3 reads from different sources to get different perspectives.

"this is a powerpoint presentation, which by itself is not enough"

The organisation has internal quality control processes for documents produced, e.g. internal presentations to colleagues and to Board members, who are senior researchers. Also, they sometimes use plagiarism software to check. Social sciences research is very subjective.

Universities are their partners, there are researchers there, we may share with each other, there are calls for papers, events to attend

Example use of evidence:

Policy brief on export processing zones. Policy says there are a number of incentives for innovation so we do research to see how the incentives work out. We then influence decision making as to how the system can be improved. We then

Information behaviour:

- Starts search in Google, to get background to the topic, e.g. definitions and countries involved
- Identifies research papers
- Looks for organisations' work

Starting: looks for keywords, and/or for organisations, starts with general topic and then becomes more specific in the search, e.g. special economic zones, then China, then household. Uses acronyms in search

Chaining: from Google to PDFs for downloading, checks references and searches for references of interest

Browsing: scans titles in search engine, says that sometimes might have to go to second or third page but doesn't do so in this demonstration

Differentiating: looking for journal papers that are relevant to TZ3's topic

Monitoring: regularly checks Dar es Salaam University website, and other university websites

Extracting: identifies lines of interest in the paper, copies the source reference and uses it in a new search. May influence the focus of investigation. Writes notes in empty word file

Processing: see repurposed on LHS

Assessing validity/use:

- Title, it must be on the topic

	follow whether the government makes any changes as a result of what we did. The government is currently reviewing science and technology policy which is not out yet. We were part of the task force group about the policy. **Uptake:** There is more research work available online, but there are many theses from universities that are not accessible. **"very few universities have managed to put in place online platforms where you can read all their research papers… for some it's just an obligation to fulfil for your graduation but for me it's very important to have access to that research work"** For organisations doing policy research, communicating research is high priority in order to influence decisions. There's one primary target audience and also some secondary targets who can put pressure on the decision-makers. Communication is primarily to people in ministries, but also to partner organisations. Access to online evidence has absolutely changed the way TZ3 interacts with colleagues. Without online access **"where would I get it from? Go to the university?"** People used to visit ministries and other institutions to get information, but that was expensive in time and money, especially because without an online platform the procedures were long. TZ3 is confident because access to information gives power.	• Research papers, published journal articles, or policy documents • Wants full papers and as much detail as possible • Knows that some organisations work in the right area • Academic universities and specific departments • Date – if a more recent document is found that supersedes an existing one being used, then the older one is dropped • Has been sent links on R4D by colleagues and so knows that it has important information

Participant – TZ3	Category 7/5 part 2	
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "How to communicate in an interdisciplinary team": 1. scrolls down home page, scanning, then returns half way up the page 2. chooses the magnifying glass icon at top right 3. enters "Jessica Thomson" in search box, nothing returned 4. deletes the large print "Jessica Thomson" in the middle of the page and enters over the top of it "How to communicate in an interdisciplinary team" The double 't' isn't noticed because the end of the search string is obscured by the magnifying glass icon 5. receives 0 results 6. looks over the screen and chooses the magnifying glass icon that is to the right of the large print in the centre of the screen 7. goes to google and types "Jessica Thompson, 2013, communicating in interdisciplinary team" 8. article is fourth in Google's first page and TZ3 chooses it from there and is taken back to SciDev.Net	**Steps to find specific article "Strengthening of health systems for equity and development in Africa", Africa Health Strategy 2007-2015:** 1. chooses 'global health' from LHS menu 2. scans the titles of the 'latest documents' 3. enters in the search box on LHS, within health "Africa health strategy 2007 – 2015" 4. scrolls down the first page of results looking for the title **"because what I inserted is not the full title"** 5. goes to end page of results (4111 results returned) 6. goes back to page 2, scans it, and then page 3, scans and then page 4 7. edits term in search box (still in Health) to "strengthening health systems for equity and development in Africa" 8. scans first page (still 4111 results) **"where is it hiding? There's no author's name?"** 9. goes to page 2, scans it, and then page 3, scans and then page 4 10. TZ3 shakes head, **"there are 400 pages… you keep looking until you are tired"** 11. edits search box and adds "Africa Health Strategy 2007 – 2015" 12. (same number of results returned) scans first page. **"this is hard to find"** 13. scans page 2 and page 3. **"I would look**	**Steps to find specific article "Using climate information to achieve long-term development objectives for African ports":** 1. chooses "browse theme", then "climate and environment" 2. scans the results list three times 3. goes to page 2 of results, then 3 then 4, reading the titles 4. chooses 'search again' and this goes back to browse by theme. 5. chooses 'climate and environment' and 'search again' **"I want to use a short cut"** 6. waits for 'refine your search' options to load 7. goes to page 2 of search results 8. chooses 'documents' in refine your search option 9. finds article on first page and downloads PDF
Steps to find information: 1. scrolls down the home page 2. returns to the main menu and chooses Agriculture, then Food Security 3. reads the abstracts for several of the articles on this page 4. chooses one article on smallholders gaining from nitrogen-efficient maize		**Steps to find information:** 1. types "PEC" into the simple search box 2. reads the titles of first three results and modified search term to "policy engagement and communications" 3. scans the first page of results, and returns to the top. 4. chooses third item in list "Policy and Research Programme on the role of media and communications in…" 5. scans the details of the project 6. goes back to search list and scans again. Several

5. reads 'speed read' and hovers over the social media icons on LHS 6. reads the article **"this is very summarised. It is good to summarise but it is good to give somebody the option to read further"** 7. clicks on a paragraph that refers to household food security, that appears to be a hyperlink. This goes to the sub-section on food security again. TZ3 is surprised **"I expected it to maybe explain what food security means, but it is a pictorial presentation"** 8. continues to browse by sub-menu, moving next to livestock 9. asks about submitting articles to the portal 10. chooses news article on biosiences research 11. would like to be able to read more through a link or to download but recognises that this is a news site and appreciates the pictorial presentation. **"I picked this article because of the title and because of the photo"** 12. goes back to home page by using the back key because TZ3 did not see the home link **Barriers to use:** likes the summarised nature of the articles but expects to have more links to further information to read in depth **Validity/use:** None observed **Downloading:** chose link and downloaded into new tab, not saved	**for the name of the author"** 14. goes to google and enters "strengthening of African health systems for equity and development, Africa Health Strategy 2007 – 2015" 15. scans first page then edits the search to "strengthening of African health systems for equity and development" 16. opens one PDF and says **"the name of the author is really important"** (in fact TZ3 did find the document in the searches but didn't recognise it as looking for author name) **Steps to find information:** 1. scans the menu items on LHS then goes to Topics tab 2. goes to resource guides on LHS 'evidence for policy and practice' 3. reads the headings in brown/green banner 4. chooses an article called "Food safety in developing countries: an overview" 5. chooses the 'web' view full report option and PDF downloads 6. returns to home using the 'home' menu item **"there is also resource guides. I hope this guides you how to use the site"** 7. chooses resource guides menu item and scans the information there. **Barriers to use:** search function was difficult to use effectively **Validity/use:** wants detailed information **Downloading:** downloads fine **Interactive functions:** not used	items are of interest in his work – e.g. "policy influencing and media engagement training workshop" 7. looks for the home page button and says **"I am experiencing troubles to go back to the home page"** 8. chooses browse research contacts and looks at evidence-based health care centre. TZ3 finds the information available interesting 9. returns to home page and chooses 'browse by theme' 10. looks through the options and reads out several topics 11. returns to home page and chooses 'browse by country/region' **"this portal is nice…. It is simple and it is rich in terms of resources"** **Barriers to use:** none observed **Validity/use:** relevant topic – site is trusted and so all information is good **Downloading:** none – downloading was fine **Interactive functions:** not used

Interactive functions: not used		

Knowledge of:

Scidev.net: None

Eldis.org: Seen and used it before—through the organisation's library, but does not know it is a DFID portal.

R4D: None

Portal challenges:

Scidev.net did not respond well during the CI session.

Use of information:

Shared online: Seldom share information since they may not be relevant to everybody.

Shared offline: No

Saved: Yes, saves most relevant articles.

Repurposed: Copies, only relevant sections of an article. Writes the article in own words. Highlights sections copied.

Read: Reads the abstract, background or conclusion of the article to ascertain whether the article is relevant.

Participant–TZ4 Category 5

TZ4 has being working with a think tank since 2012. Works mainly on gender and social protection issues. TZ4 has written several articles and reports on gender and social protection.

Uses Google search engine for various online search activities. Mostly prefers to use keywords or short phrases when searching for a specific article. TZ4 demonstrates this by typing the keywords of a general article "disability policy in Tanzania" into Google search box.

Reads titles of the articles on the first page of the search result. Identifies a relevant article in PDF format on the first page of the search result.

Clicks on the title of the article to download it. Scrolls down and in the process of doing so, TZ4 glances through the table of contents of the article to find the most relevant section of the article.

Finds the page where the most relevant section of the document can be found. Prefers to work electronically, as such, would like to highlight the most relevant section of an article for immediate or future use. Saves the article after use.

TZ4 copies the highlighted section and rewrites it to suit own words. For instance, TZ4 shows a highlighted section of an article and indicates that **"I copied this part [the highlighted section] from an online paper....I am going to write it in my own words".**

In looking for the most relevant document, TZ4 sometimes prefers to read the abstract, background information and conclusion of the downloaded article.

Indifferent about the age of the article, but considers the author of the article and source as the important for credibility issues. Prefers documents from sources like DFID, World Bank.

Example use of evidence:

Information behaviour:

Starting: Starts online search with Google general search engine. Prefers the start online search by typing keywords (in some cases phrase) of the title of the article.

Chaining: Copies most relevant titles in the reference section of an article into Google search engine for further search.

Browsing: Scrolls up and down the pages of a search results. Reads hurriedly the titles on the page.

Differentiating: Highlights most relevant sections of an article before saving.

Monitoring: No evidence.

Extracting: Copies most relevant section of an article.

Processing: Re-writes copied section of an article.

Assessing validity/use:

In assessing the validity of an article, TZ4 considers the following:

- Name of author
- Source of article
- Also prefers articles in PDF format.

	Has written an article on social inclusion of people with disabilities. **Uptake:** Access to information is now very easy and has saved a lot of time for TZ4. Does not need to go to the library to look for information. Indicates that **"even we don't use our library"**	

Participant–TZ4 Category 5 part 2		
Scidev.net **Steps to find specific article:** No access **Steps to find information:** Types the website address into the find bar, but SciDev.Net did not upload well. TZ4 makes several attempts to no avail. TZ4 gives up the search. **Barriers to use:** Scidev.net did not upload properly. **Validity/use:** No access **Downloading:** No access **Interactive functions:** No access	**Eldis.org** **Steps to find specific article:** 1. Types the keywords of specific article into the search box of the portal. 2. The portal did not respondent on the first attempt. But decides to give it a second attempt. 3. Types the keywords of the specific article into the search box of the portal once more. This time portal uploads. 4. Finds the title of the specific article on the first page of the portal. Clicks on the portal and then reads the summary of the specific article on portal. 5. TZ4 is of the view that the number of search results may not be an issue. Will narrow the search to get fewer search results in case the number of results is many. **Steps to find information:** 1. Looks at the portal again to see if it is the same in the organisation's library. Confirms it is the same. 2. Scrolls down the portal and finds the heading "gender". TZ4 clicks on it. 3. Does not find any relevant article. Decides to find a different article. Types the phrase "cash transfer" into the search box of the portal. Finds **2579** documents and **129** organisations. Indicates the search results are too many. 4. Decides to narrow the search since the search results are not only focused on Tanzania. Selects "Tanzania" on the menu and then clicks on search. Find a relevant article on the first page. TZ4 clicks on it.	**R4D** **Steps to find specific article:** 1. Types the keywords of the specific article into the search box of the portal. 2. Reads the title of the specific article. Realises that the first article on the first page of the search results is the specific article. 3. Clicks on the specific article. Looks carefully at the title, the source, date and the name of the author of the specific article. Realises it is the same article. 4. Downloads the article. **Steps to find information:** 1. Types the website address in the find bar. Looks very closely at the DFID and UK aid logos on the left and right hand sides of the portal respectively. 2. Decides to click on the heading "browse by country/region". Looks for "United Republic of Tanzania". TZ4 then clicks on "current projects". 3. Finds 1735 search results. Decides to narrow down the search. Decides to go back to the home page. 4. Clicks on "theme" on the home page. Under the heading "Education" TZ4 clicks of the "focus on girls' education". Becomes impressed with how the various themes have been arranged on the portal. 5. TZ4 finds 65 search results. Scrolls up and down to find a relevant article. Initially does not realise that the search results comprise of a mixture of "projects" and "documents". 6. Realises later that there are "project" and "documents" on the same page. TZ4 then clicks on "documents" to differentiate the content from that of "project".

	Barriers to use: No evidence found **Validity/use:** Considers the date, name of author, and source of the article. **Downloading:** Downloads and save specific article. **Interactive functions:** No significant impact.	**Barriers to use:** No evidence found. **Validity/use:** see Eldis on the LHS. **Downloading:** Downloads specific article. **Interactive functions:** No significant impact

Knowledge of:

Scidev.net: subscribed to their newsletter and has used portal before, most recently 'yesterday'

Eldis.org: none

R4D: has been to it before, but not frequently

Portal challenges:

- Navigation – this was evident on a think tank website that TZ5 is familiar with, but TZ5 could not find what they were looking for. Also in comments during the session that documents may be "hidden"
- Even if in the field TZ5 has to have connections so uses hotspots through smartphone which is generally reliable
- **"Yesterday we had very big problems with Tigo network"**

Use of information:

Shared online: none reported

Shared offline: waste of resources to print lots of papers, but policy briefs are hard copy. Sharing verbally through presentations

Saved: reads from the screen before saving

Repurposed: re-word to avoid plagiarism, extracts quotations, conceptualises from what they have read

Read: reads title, abstract and conclusion to decide whether to save. Reads whole article thoroughly and quickly.

Participant – TZ5 Category 3/6

TZ5 works in an analytics unit on monitoring and evaluation. Conducts mostly quantitative fieldwork as well as a little bit of qualitative research. In TZ5's unit, credible data is very important. They will use data from other ministries in order to compare and check data, even one unit difference can change your decision.

TZ5 has very focused information behaviour. Reads a lot of material, in a focused manner and in detail **"more than even eating, I like reading"**. TZ5 reads and then conceptualises points for study. TZ5 is very concerned with finding credible reports or papers and will check the publisher's credentials as well as the article itself.

TZ5 is not thrown by finding documents that are not relevant, either by searching or by browsing –they just keep looking. Driven by interest and relevance and is able to process a lot of evidence/reading quickly. TZ5 has a clear approach to evidence and is particularly interested in quantitative data.

Example use of evidence:

A report on the adoption of agricultural innovations in Tanzania, specifically sunflower farming innovations for smallholder farmers: their adoption rates, the challenges, and influential factors, and how do these innovations help the farmers.

Uptake:

"When you go to advise politicians you have evidence in hand. Especially for us we are working closely with our president you need to have the evidence"

The President is interested in evidence if presented in the right form. Politicians are not interested in detailed

Information behaviour:

- Starts new search in Google (favourite search engine)
- Includes 'PDF' as one of the search terms **"most of these (reputable) publications have been published in PDF"**
- Opens each paper in new tab. This saves time while others download

Starting: uses key words in Google; prefers Google as a starting point rather than a portal

Chaining: from Google to found sources; read only the first page of results (in our session), but works through each one systematically. As only wants PDF no further online linking

Browsing: reads titles one by one in Google search results

Differentiating: reading helps to rule out information. After initial assessment, reads the whole article. Treats each article/source according to its type, e.g. blog, journal article. Compares data from several sources to inform work

Monitoring: always online looking for interesting articles

Extracting: works systematically through found resources

Processing: see repurposed on LHS

Assessing validity/use:

- Publications from reputable journals

	quantitative information such as t-test results; these have to be re-packaged and reduced in volume because they don't have the time. Re-packaging means including specific results and justification for how you obtained those results. **"You give them your whole document but sometimes they will assign someone to read this for them and compare it to the brief"** They are very strategic, for example looking through the references to check their credibility. More evidence is available for free online. This has made a difference, because if you have many sources it widens the basis of knowledge. **"Now you have free publications, it doesn't mean they are not credible".** Ministry data is online too; a recent open data conference included government, public and private organisations.	• Publisher **"I go to history of the publisher – I assess what kind of publication it is because there are many that are fake"** • Content: methodology, objectives and highlights of results • Trusts documents in the DFID portal (R4D)

Participant – TZ5	Category 3/6 part 2	
Scidev.net **Steps to find specific article "Big data for development: facts and figures":** 1. types in "big+data+development" to search box. 2. scans down first page of 206 results reading titles, document is not there 3. adds "+facts +figures" and comments that they're not sure if it is PDF 4. target document is number 5 of 13 results 5. checks the article's author before confirming that it is the right document **Steps to find information:** 1. scrolls down the whole home page, reading titles 2. chooses 'browse type' from top menu and reads through them – to see what is there 3. types into search box "agricultural innovations" 4. the first item in the list is titled "Open data underpins equality". Although not distracted by this, TZ5 agrees it not what they expected to see **"No, I keep on searching" "but there may be a word agriculture in the document"** 5. chooses the Q&A "Put farmers at heart of 'adoption revolution'" and reads **"I'm not looking for citation… from here I can just read to get the highlight, and would then go to search for publications"** **Barriers to use:** none observed **Validity/use:** most of the things are credible because they are DFID	**Eldis.org** 1. types in "eldis.org" and receives "No data received" message. **"Sometimes if you are in a rush and you are looking for many documents it can discourage you" "I would just go to another portal"** 2. tries reloading the page 3. after over 4 and a half minutes (1.13.55 to 1.19.08) minutes gives up **Barriers to use:** unable to load **Validity/use:** No access **Downloading:** No access **Interactive functions:** No access	**R4D** **Steps to find specific article "Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality":** 1. types "development+finance+institutions" into simple search box 2. first item in search results **Steps to find information:** 1. scans left hand menu 2. chooses Browse by Theme 3. goes straight to Agriculture and Agricultural Innovation 4. reads titles in detail and considers each in turn as to whether it might be of interest (TZ5 recognises the first project in the list) 5. chooses the top project and reads objectives, then background. This is of interest to TZ5 **Barriers to use:** none observed **Validity/use:** as above **Downloading:** none attempted **Interactive functions:** not used

Downloading: none attempted **Interactive functions:** not used		

Knowledge of:

Scidev.net: subscribes newsletter because it has different types of information "**I find it so informative**". Looks for call for proposals

Eldis.org: knows Eldis well, mentioned it during first half of session without prompting. Uses Eldis if looking for information that is not on Tanzania. TZ6 knows that Eldis is a bit richer. Visits maybe twice a month. Eldis helped them to set up tzonline

R4D: none – seen it listed on DFID website but not used it

Portal challenges:

portals are targeted but Google has everything

Use of information:

Shared online: there is a subscription list for tzonline (30,000) and material is shared via this list. TZ6 is in charge of distributing information within their organisation

Shared offline: The young generation get things online but the old generation still want someone to help to search, to choose keywords

Saved: Saves and then prints

Repurposed: reads policy documents to compare with fieldwork results and make recommendations

Read: title, short abstract, then whole

Participant – TZ6 Category 7

TZ6 works at a not-for-profit institution for research and policy analysis, supported by various organisations including UNDP and the Tanzanian government.

TZ6 also writes assessment reports that make recommendations to the government. TZ6 uses tzonline to search for government strategic plans, implementation plans (e.g. mkukuta), ministerial project manuals, Vision 2025 document and local and national government policies. TZ6 uses these to compare plans, policies and strategies with fieldwork findings, and to make recommendations to the government. All of the relevant documents are available on tzonline so does not use any other portal for this kind of work. This means that TZ6 does not have to go to each different ministerial site to get the documents.

The latest policy documents are scanned and put online. Research documents may be added to tzonline or sometimes will link to another site. If the document is only available in hard copy, it is scanned and put online (with permission). They have a policy document for what to include on the portal and what not.

Tzonline (launched in 2000): 30k mailing list users, 50% in Tanzania and 50% outside. 26k to 28k people visit tzonline per day. Tzonline provides one place for these documents rather than going to ministerial websites. It used to be that when policy needed to be revised, all the

Information behaviour:

Starting: Uses tzonline.org as main source of information. Searches within tzonline. If looking for a new topic, will use google, or will look at categories on portal, e.g. Edis, and if not there then an open search. Mostly uses google because it takes you straight to the article

Chaining: from first page of results to the chosen source

Browsing: none observed

Differentiating: uses their own policy on downloading documents for tzonline. Other differentiating see below

Monitoring: people send documents to put on tzonline. TZ6 also searches for documents online to add to or link to tzonline

Extracting: no direct evidence of systematic searching, but TZ6 does look for materials for tzonline

Processing: see repurposed on LHS

Assessing validity/use:

- Institution, author
- Focuses on Tanzania
- Relevance
- Transferrable/applicable to Tanzania
- Date (wants things in the last 2 or 3 years)

source/document	necessary information was scattered, but now it's in one place (tzonline). TZ6 searches for topics others require, in which case goes to google. TZ6 also helps people to search, particularly the older generation. Finds that people are motivated to look for it. Young generation know how to do it themselves. Most of the research is informing government policy, and most of the information we'd like to review is the policy so look at old policy and related policies. If writing a research proposal then have to take an international perspective, and decide which practices to include. In this case other portals like Eldis may be used. **Example use of evidence:** A report on a study to assess institutional capacity and mapping of best practices and development opportunities in one District in Tanzania. The study included focus groups and fieldwork to produce recommendations to the Government as to what should be done regarding poverty and environment. The study combined fieldwork with searching for evidence online and looking at policy documents. **Uptake:** **"It has been easy because you can search anywhere, at home if you have access to internet... before we used to go to the library and get the physical book"**. Also easy to get information beyond Tanzania, up-to-date information, some people only do it online and	

	not hard copy. This think tank is invited to assess policies and write recommendations.	

Participant – TZ6	Category 7 part 2	
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "How to communicate in an interdisciplinary team": 1. says that they can search by author or by title 2. types 'jessica thompson december 2013' in search box 4. zero results **"then I have to go back and take the title"**. Thinks would have got it first time in Google 5. types 'how to communicate in an interdisciplinary team' 6. 7 results, scrolls down page and finds the article at number 3 **"I don't know why it didn't pick my first one** (search)" **Steps to find information:** 1. scans the first page **"it's more informative"** 2. reads Noticeboard at bottom right of home page, looking for grants 3. scrolls back to the top **"they also put it by subject which makes it easier"** 4. looks at 'Browse type' and then scrolls down the home page **"I don't see where I can search… put it more easy to see"** 5. scrolls back to the top still looking for search 4. 6. types 'tanzania' in search box because the topic menus will take them to generic information rather than for Tanzania 5. 7. gets 'unable to connect' message and checks if the problem is internet connection or SciDev.Net	**Steps to find specific article "Strengthening of health systems for equity and development in Africa", Africa Health Strategy 2007-2015":** 1. comments that they could search by health strategy or put the title 2. types 'Africa health strategy 2007-2015' into search box on LHS. Comments that putting in the title will result in a lot 3. 26357 results 4. opens first item on first page to check it is the right one **Steps to find information:** 1. knows that Eldis has topics, and if the required topic is there then TZ6 would go to the topic menu 2. in this case, looking for youth policy, enters 'youth policy in africa' into the search box 3. 30804 results **"oh, this is good"** 4. scans the first page **"if there is a lot, I search it again"** 5. hits back button 6. changes search to 'youth policy in east Africa' 7. 31978 results **"still this is a lot"** 8. scans first page and then hits back button 9. modifies search to 'youth policy and Africa' 10. 8930 results **"so what I do is go to page one because what it does is it puts the most relevant up, I can go page one, page two".** It puts the most relevant first by keywords	**Steps to find specific article "Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality":** 1. types "development finance institutions and infrastructure" into simple search box 2. opens first item in search results to check it is the right one **Steps to find information:** 1. reads home page information briefly 2. chooses 'Browse by Country' to see what information there is on Tanzania 3. looks for 'Tanzania' then chooses 'United Republic of Tanzania' 4. reads the options (projects, documents, both, all projects, current projects, completed projects) 5. reads the first project summary on results page, says that she's not sure if it is a call for proposals or a finished project 6. reads the next project summary, is looking for things that are of interest 7. reads the project summary information but says **"what is confusing – it could say a bit of what is this"** **Barriers to use:** doesn't recognise where the home page is, i.e. logo **Validity/use:** relevance **Downloading:** not used **Interactive functions:** not used

8. goes back to home page and chooses Enterprise Innovation tab – just to see if internet is working 9. goes back to home page and enters Tanzania in search box again. Looks to see if there is something to link to tzonline or to pass on to a researcher. Sometimes you get new things 10. scrolls down results list reading titles 11. points out a multimedia on Tanzanian farmer 12. scrolls back to top of screen and types 'tanzania and youth' in search box 13. two results but neither has Tanzania in title **"so it means there is no information on Tanzania and youth"** **Barriers to use:** portal is targeted but google has everything **Validity/use:** Tanzania **Downloading:** nothing downloaded **Interactive functions:** newsletter is a good prompt to look at information. **"I always find one or two things that are interesting"**	11. scanning first page and reads titles. 12. chooses one that has employment and youth in the title but doesn't open it 13. scans second page and mentions one but its date is 2005 **"this is a bit too old"** 14. mentions another on page 2 on South Africa **"so far I haven't got a document that I feel is worth opening it and saving it"** 15. hits back button until home page and changes search to 'youth policy and Tanzania' 16. misreads the number of hits as 166, which is the organisation tab 17. scans titles on first page and considers some, based on title and based on year. Opens none of them **Barriers to use:** **Validity/use:** title for relevance. Seems to assume that everything on Eldis is worth considering. Considers date **Downloading:** downloads as above, using tabs **Interactive functions:** not used	

Knowledge of:

Scidev.net: has heard of it, but doesn't recognise it

Eldis.org: "I have used it in the past but not for 10 years"

R4D: none

Portal challenges:

- No challenges observed when TZ7 uses Government portals and other known portals
- Requirement to pay for an article is a barrier
- "the connection is good but sometimes it is a problem"

Use of information:

Shared online: shares with colleagues in the team, using a mailing list. "Every day someone is sharing"

Shared offline: no comments

Saved: on laptop for future reference

Repurposed: reads from several reports and then re-writes in own words. Where specific data is to be used from another publication then that is copied and source cited. Can also learn writing style from others' articles

Read: in detail

Participant – TZ7 Category 5/6

TZ7 is a researcher based in a think tank. Holds a first degree in statistics and a qualification in finance. In current job, gets involved in a range of research areas and currently focuses on research in micro-finance. TZ7 performs research according to the requirements of external stakeholders. Hence TZ7 does strategic research (supply driven) and commissioned research (demand driven).

This think tank was consulted by the Government when they developed the Tanzanian Development Vision 2025. They also conducted a review of the Vision in 2010.

TZ7 demonstrates knowledge of specific portals in the micro-finance area as well as relevant Government policies and strategies and how to access them. TZ7 uses microfinancegateway (CGAP) to get to reports on research/data from others. TZ7 uses a combination of google searches and known portals/websites to obtain relevant evidence.

Example use of evidence:

Review of the National Micro-finance policy, using both primary and secondary studies. Minister of Finance was the client. They conducted a systematic literature review, and used different reports and finance-related policies from the ministry, for example the Co-operative Development policy and National ICT policy.

Uptake:

Agrees that there is more research evidence available online. "I get to know what others have done and mostly in the country and outside for regional studies". It has improved TZ7's confidence "Of course", and the way

Information behaviour:

- tzonline is hosted by this think tank, but TZ7 did not enter the URL, they went to the think tank's website and linked to tzonline from there.
- relies on tzonline for Government policies and strategies.
- uses Google to find the portal even though theyk now its name, e.g. microfinance gateway.

Starting: goes to known portals, e.g. tzonline, Bank of Tanzania, microfinancegateway, National Bureau of statistics, Ministerial websites

Chaining: uses google to get to known portals, then navigates those portals

Browsing: will read on the internet to get ideas for research proposals, tries to link it to Tanzania; others in the team are also tasked with identifying relevant literature for specific projects

Differentiating: no bookmarking evident, googles to find known websites

Monitoring: Government websites and colleagues

Extracting: specific extractions of data

Processing: see repurposed on LHS

Assessing validity/use:

- Known and trusted sources, but not everything on a known source is necessarily trusted
- reads whole document to decide use
- If uses a google search will look for who has published it
- Date of reports – needs up-to-date

	TZ7 interacts with others **"especially for the people that I meet who haven't been to the portals, because my argument will be different from theirs". "If you're in a workshop then they ask you for information and where to get it".**	

Participant – TZ7	Category 5/6 part 2	
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article "How to communicate in an interdisciplinary team": 1. types 'jessica thompson' into search box top right 2. article is returned first in the search results list 3. chooses the article to check it is the right one **Steps to find information:** 1. chooses Enterprise menu and reads sub-categories 2. enters 'microfinance tanzania' in search box 3. reads titles of the 4 results but nothing of interest 4. changes the search to 'microfinance', gets 19 results 5. chooses an article on German plans to boost Africa's higher education teaching 6. recognises that this site only has certain sectors 7. chooses Enterprise 8. changes mind and types 'tanzania' in search box 9. this results in articles on Ghana and East Africa as the top ones, which surprises TZ7 10. chooses to refine topic by Policy 11. scrolls to the bottom **"I can find information linked to my work but not microfinance"**, and chooses an article on nutrition and poverty **"so I can use this website to look at poverty and nutrition"** 12. points out references at the end of the article as being a good thing	**Steps to find information:** 1. scrolls down the home page, looks at LHS topics and scans other elements, scrolls back to the top 2. chooses Evidence for policy and practice **"I thought maybe I could get something on policy issues"** 3. accidentally closes the window, and tries to re-launch eldis.org but gets "The connection has timed out" 4. after several re-freshes over more than 8 minutes, TZ7 gives up **Barriers to use:** unable to load **Validity/use:** No access **Downloading:** No access **Interactive functions:** No access	**Steps to find specific article "Using climate information to achieve long-term development objectives for African ports":** 1. checks to see if there's anything below the search box 2. reads the article title 3. types 'woolhouse' in the simple search box 4. reads the first title and chooses that document 5. goes back to search results and chooses the third title 6. checks the title on the document record **Steps to find information:** 1. uses google to find website (TZ7 was given URL) 2. reads the introductory paragraph 3. chooses Browse by Theme 4. reads the categories and chooses Research into Use (Agriculture) 5. chooses the top document returned **"that's very useful … relevant to my work"** 6. reads the abstract and is disappointed that the full document has to be paid for **Barriers to use:** the article TZ7 chose to look at had to be paid for; there are several articles and projects with similar names and because the full name is not displayed in the search results it can appear to be misleading **Validity/use:** as above **Downloading:** none attempted

Barriers to use: none observed **Validity/use:** no comment **Downloading:** none attempted **Interactive functions:** not used		**Interactive functions:** not used

Knowledge of:

Scidev.net: none **"I haven't seen this, why?"**

Eldis.org: none **"imagine – I haven't seen this before!"**

R4D: thinks has seen the link but does not appear to know it at all

Portal challenges:

- Doesn't use research portals particularly, is registered to news sites and professional bodies, BBC
- Portals have a lot of information. **"going through everything will distract my attention. I may need it later but I don't need it now"**
- Google goes straight to what is wanted now

Use of information:

Shared online: uses email, Facebook, YouTube. Shares links, stories, media monitoring with all the BRN teams, with friends and public

Shared offline: BRN prints less but shares a lot online

Saved: saves and reads in the morning, also uses email as a filing system for some things

Participant – TZ8 Category 8

TZ8 works for the Presidential Delivery Bureau delivering on the "Big Results Now" (BRN) programme. BRN was set up to ensure that priority projects are well-implemented so that they deliver the planned results. BRN was set up in response to a realisation that Tanzania was lagging behind other countries and despite good plans, priority projects had poor delivery mechanisms. The goal is for Tanzania to be a middle-income country before 2025. TZ8 is tasked with making sure BRN communicates to the public and to the implementers, supporting communication officers in the ministries to be proactive. **"Now we are bringing in a new concept of communication".**

BRN has very few people who have worked for the government before, they come from private sector such as telecoms, banking, NGOs, activist groups. **"We were criticising the government and now we have been given the opportunity to show by example".**

TZ8 looks for up-to-date information and news that is relevant to BRN, its teams and its projects – through media monitoring, project reports, google and popular blogs, popular online platforms. TZ8 visits online sources every day for anything new about the government that needs a quick response. TZ8 looks for information for analysis and comparison or monitoring purposes, e.g. comparing examination results before BRN started (2 years ago) and now. TZ8 keeps track of what the President is doing and saying (through michuzi's blog) and shares this information with BRN staff, ministries and public at large, as appropriate. TZ8 shares a lot of information online, using YouTube, Facebook, and

Information behaviour:

- TZ8 looks for recent articles and new items related to BRN and its projects
- TZ8 uses data from ministerial sites
- TZ8 uses sites they know about, such as ministerial sites and michuzi blog

Starting: uses 'michuzi blog' – photographer of the President – to find out what is happening **"if you don't post for a week, everything will get lost"**. Also Facebook; google to find specific sites such as ministerial sites

Chaining: from internal reports

Browsing: skims headlines

Differentiating: chooses items that are about BRN, does comparisons from multiple sources as part of job

Monitoring: every day, through blogs, through media monitoring, through project reports, activist sites, professional bodies, news channels

Extracting: combining/synthesising

Processing: as communication officer much of what they find is passed on, or used to frame further investigation and reporting

Assessing validity/use:

- Focuses on BRN, its activities, what is being said about the programme, and what the President is doing
- Wherever it comes from, whatever is of substance to BRN
- **"we only use reliable websites"**
- Recent (2015) articles and news

Repurposed: compares, copies and pastes to communicate on, uses various channels to send links. Also analyses data for monitoring and to use in reports or for videos etc **Read:** TZ8 reads online, sometimes using phone to read articles	WhatsApp groups to disseminate. Projects share weekly progress reports, TZ8 reads them, identifies interesting stories to follow up, maybe by phone or on WhatsApp. Tanzanian institutions that conduct research, e.g. REPOA, send BRN research as they are on the mailing list, sometimes TZ8 will visit their website. TZ8 becomes tired in the last 20 mins of the session. **Example use of evidence:** The annual report of the BRN programme, specifically some stories related to the projects being supported. Field visit involving interviews with people and photos, using contact information through Facebook. These stories focused on training farmers to use agricultural innovations. Another example is in education where the exam results were compared across the years and they investigated why schools were doing well or poorly. Produced a YouTube video for this topic. **Uptake:** I wouldn't be able to work without the internet because I have to find information and share it. Receiving and sharing of the information online has contributed to the success of BRN. For example in transport sectors we wouldn't know how the Tanzanian port is regarded, e.g. corruption, speed of handling, that the roads are being blocked by trucks.	

Participant – TZ8	Category 8	part 2

Scidev.net

Steps to find specific article "Big data for development: Facts and figures":

1. types 'big data' in search box, only two words because it will take time to type the whole title
2. scrolls the first page of results reading the titles
3. chooses an article on India's drugs war because they have some issues with that
4. reads the 'you might also like' pointers **"it is so much interesting"**
5. types 'big data' in search box again
6. scrolls down the first page looking for it and points out articles on food, on coffee farmers **"I have a lot to read here"**
7. chooses an article on virtual reality
8. scrolls through the images at top of article
9. comments that there's a lot of information here, and so many sectors, and so many people come here **"and so it seems it's a reliable source"**
10. scrolls down the article and looks at the comments from others
11. decides to sign up and enters contact details **"I'm sure they do research on who subscribes. It's important to know who has subscribed"**

Steps to find information:

1. goes to Google to enter the website address

Eldis.org

Steps to find specific article "Strengthening of health systems for equity and development in Africa", Africa Health Strategy 2007-2015":

1. types 'strengthening of health' into search box on LHS
2. 15175 results
3. scrolls down first page of results reading titles **"how to search all. I can't see it here"**
4. goes to search box above 'filter by topic' and adds 'systems for equity'
5. 21972 results
6. points out that there are 21972 results
7. adds the rest of the document title in search box
8. "**oh my**" 28988 results
9. reads the titles on the first page of results and is confused that it's not on the first page

Steps to find information:

It takes time to open so we move on to R4D and then Scidev.net

1. goes straight to 'share' icon and looks at the list of social media
2. goes to Regions and Countries, chooses Tanzania
3. scrolls the Tanzania information and gets to the articles
4. scrolls down the article list and reads the titles
5. opens article on TANESCO, which is 'under' BRN
6. reads the abstract article

R4D

Steps to find specific article "Development Finance Institutions and Infrastructure: A systematic review of evidence for development additionality":

1. types 'development Finance institutions and infrastructure' into simple search box
2. opens first item in search results into a new tab
3. scrolls down the first page of search results to see if there is anything else
4. goes to document record for file found in step 2
5. downloads PDF **"so that when internet is gone, I can read it"**

Steps to find information:

1. goes to Google to enter the website address
2. chooses website as first item on google results
3. points out the logos DFID and UKaid at the top
4. types in 'open government' to simple search box
5. scans the first few items of the results and chooses the third item on Transparency **"because to me transparency is key"**
6. also identifies the fourth in the list as of interest

Barriers to use: lots of information that may be distracting from what doing now
Validity/use: picks up on key words, particularly transparency in government
Downloading: no problem
Interactive functions: not used

2. chooses website as first item on google results 3. goes straight to 'Governance' menu, then chooses 'Sustainability' because TZ8 is concerned that BRN ends soon. How can it be sustainable? 4. scrolls down the results, reads titles, saying that several are interesting but it is not what they need 5. sees that there are 176 results for sustainability 6. scrolls again down the list of results **"the information provided here will be food for everybody"** 7. chooses one article on sustainability science **"what I am looking for is to learn how we can make BRN sustainable but others may be interested in how richer nations are using technology in sustainability"** 8. chooses 'Communication' menu and then 'Influencing' 9. chooses an article on south-south influence and scans it, picking out the quotations **"it's very user friendly"** 10. goes back to the top menus and scans them **Barriers to use:** became distracted from what we were asking and what TZ8 was looking for (comments on this as quoted above) **Validity/use**: interest and relevance to BRN **Downloading:** downloads into new tab and keeps for later reading and maybe saving **Interactive functions:** points out the social media links on LHS and says that this is good because TZ8 likes to share	7. chooses the Tanzania link on RHS to go back to country information 8. chooses 'governance' from LHS menu list 9. scrolls down the articles, reading the titles **"I am looking fro an interesting topic"** 10. goes to page 2 **"I miss really important articles because of poor titles"** 11. looks for 2015 articles and says that there should be somewhere to navigate to get to 2015. Comments that there is a lot come up in the search and it will take a long time to go through 12. goes back to home page and looks for new information in the carousel at the top 13. goes down page to 'other topics' **Barriers to use:** there is a lot of information and TZ8 finds it difficult to navigate and to find recent articles. Behaviour becomes less focused **Validity/use:** title for relevance. Seems to assume that everything on Eldis is worth considering. **Downloading:** downloads as above, using tabs **Interactive functions:** not used	

Knowledge of:

Scidev.net: mistook it for sciencedirect

Eldis.org: recognised the name

R4D: not known

Portal challenges:

Local websites don't include data from other countries, so you need to look at international websites for comparison data sets.

Use of information:

Shared online: not commented on

Shared offline: discussion at technical meetings and briefings, and hard copy

Saved: not commented on

Repurposed: to inform own work, provide theoretical framework

Read: yes

Participant – TZ9 Category 5

Works in a research institute. TZ9's responsibility is in growth and development – economic growth and how that impacts livelihoods of the people: agriculture, constraints to growth and how that can be transformed to accelerate inclusive growth, national resources we have and how those can be used to accelerate economic growth, employment and how that relates to household income and so on.

TZ9 does a combination of looking for others' research and conducting primary research. **"The web facilitates our work by providing the relevant literature to get a theoretical framework to help inform our work."** TZ9 accesses datasets from the National Bureau of Statistics and other government departments and agencies for analytical purposes, and those from World Bank, UNDP, etc. Having online access is a positive thing because if you want to access international data to compare statistics across countries it is easier.

The negatives include if people publish data that is incorrect or inconsistent but as long as the data is credible, it's a good thing and it causes dialogue between people on techniques as well as data. A negative is that we don't go and speak to other people, discussion is a good thing but with online access TZ9 doesn't have to move from the desk. Without it, TZ9 would have to take evidence/data from the hard copy and compile it, but now it can be ready to use in excel format which is easier. **"Overall the power of the technology and the benefit it provides far outweighs the costs and disadvantages".**

Access to information is empowering and helps in promoting a more informed dialogue. We used to be able to talk about Tanzania, but now we can go to a conference and talk about not

Information behaviour:

Not tested as TZ9 did not have sufficient time available to take part in the whole study. Session focused on the process of how research evidence is used to influence policy, programmes and practice.

Assessing validity/use:

- Source, e.g. World Bank, National Bureau of Statistics, UNDP, IMF etc. It depends on the data you want as to where you will go, so each source has its own definitions and specialisms
- Author and institutional affiliation, publisher who is known
- Open access publications preferred but will subscribe if the work is relevant and credible

Uptake:

Availability of evidence on the internet allows work to be done more efficiently and hence to complete more work. **"Now it's online I don't have to move around libraries to look for literature… it definitely makes life a bit easier and makes us work more efficiently, produce more output in a shorter period of time."**

	only Tanzania but many countries because we can get comparable data, e.g. generic statistics such as GNP, population, per capita health spending, age structure are all easily accessible. If we are able to do many more pieces of work than could have been done 10 years ago, it also means that we can engage in many more dialogues than before. **"Information is power so whoever has privilege to information access would definitely be more confident in what they are saying".** In some cases our research findings go straight into plans and strategies. An example is a recent effort to inform the development of industrial policy, because of dramatic differential in productivity between Tanzania and Vietnam e.g. for cashew nut and coffee, which was almost the same in 1970s and is now very different. In Vietnam the government's policy was to proactively promote good farming practices. The Tanzanian government has adopted some of these in its recent initiatives. That happened through policy and forums but also direct engagement by TZ9 and colleagues with those making the policy, specifically they make available their research papers, policymakers ask questions and they provide answers and changes are taken on board. TZ9 is a member of a technical team, where there is direct application of research findings into policy making. The team has working sessions where they take all the materials they have: literature, data, everything. They spend the week reading, discussing, making tables, trends, then coming out with recommendations for what needs to go into the plan. These meetings (a week long) are data and knowledge-intensive. Having access to data and literature is important. **"If you want to make some cross-country correlations between GNP and maybe rates of unemployment or money supply, every data is there… : food production, productivity, yield rates by crop type, for any country.. prices, every type of**	

	data". Definitions vary between different statistic data sets, This impacts how the data can be used and we are very careful to see what definitions have been used, and the impact that has. Sometimes we use both definitions. In Tanzania, management information systems are not so robust, which is why we complement our work with primary data, depending on what is the research question we want to ask. Reliability of the evidence/data depends on the nature of the data, and we wouldn't go to a website that is not credible. **"We know which ones are credible by their very nature. So I am confident when I quote the source."**	

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

- Poor internet connectivity.
- Eldis and to some extent R4D responded slowly.

Use of information:

Shared online: Share information with colleagues through email.

Shared offline: Shares hard copies of document with colleagues.

Saved: Only relevant articles.

Repurposed: Rewrites online articles in a book and later transfers into Microsoft word.

Read: Prints and reads saved online articles.

Participant–TZ10 Category 6

TZ10 holds an MPH degree from the United Kingdom and is a development consultant in Tanzania. Currently undertaking a consultancy work on a communication strategy. Looks for online information to prepare reports on the project.

Signs in to Gmail account and then types the information required into the search box through the Gmail account.

Types in the phrase "communication strategy" into the search box of Gmail account. Search results open on a different tab. Clicks on the first title on the first page of the search result. Has no reason for searching through Gmail account.

Reads the content of the article thoroughly to get ideas for the consultancy work. TZ10 indicates that **"what I want to get from this document is the "specificity"".** In order to confirm what is already known, TZ10 reads through every document downloaded. After that TZ10 copies article onto Microsoft Word and modifies it to suit what is required. TZ10 also acknowledges the source. Save the article on a removable disk for future use.

Relies on basic knowledge and experience in ascertaining the validity and credibility of an online article. Discards irrelevant information and only keeps the relevant ones.

Example use of evidence:

Communication strategy in the health sector.

Uptake: Rate of learning has increased, mainly because there is a lot of useful information on the internet that one can use for meaningful work.

Information behaviour:

Starting: Starts searching by signing in into Gmail account.

Chaining: No evidence.

Browsing: Endeavours to read all titles of the search results so as to ascertain the most relevant article. For instance in search for the specific articles TZ10 took time to read all the title on the first page of the search results.

Differentiating: Discards irrelevant articles. TZ10 sometimes opens new online research articles on a new page. Also shades the most relevant section of any article copied in order to remind TZ10 that the article was copied online.

Monitoring: No evidence.

Extracting: Copies sections of relevant.

Processing: Modifies the copied article to suit desired output.

Assessing validity/use:

- Does not have any specific criteria. Only relies on experience and knowledge.

Participant–TZ10	Category 6	part 2
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. Goes back to the home page. Types specific article into the search box of the portal. Finds 16 search results. 2. Doesn't recognise where the search results are since all the search results are complemented with picture illustration. 3. Finds the article in the middle of the search result.	**Steps to find specific article:** 1. In finding specific article, TZ10 clicks on the heading "global health" on the resource guide column of the portal. Finds 0 search results. 2. Goes to the home page of the portal. Types the full title of the specific article into the search box of the portal. 3. Carefully reads the title of each article on the first page of the search results. 4. Finds the specific article but the wording of the title is not similar to what TZ10 typed. Becomes confused that they are the same article. Clicks on the title.	**Steps to find specific article:** 1. Type the full title of the specific article into the search box. 2. However, the number of characters in the title was more than what the search box takes. 3. Reads each article title on the first page of the search results. 4. The second title gives the result for the specific article.
Steps to find information: 1. Type the URL of SciDev.Net into the find bar of the portal. 2. TZ10 tries to find out what the portal is about by carefully reading the background information on the portal. 3. Asks the question **"what are the issues here?"** 4. Tries to look for a general article. 5. Moves the cursor on each title and reads the various title on the home page carefully and at the same time, scrolls down slowly. 6. Find no relevant article.	**Steps to find information:** 1. Types the URL of the portal into the find bar. In search for online evidence related to "communication strategy", 2. TZ10 thoroughly reads all the titles of articles and headings on the home page of the portal. 3. Indicates that reading the online article in detail is a habit. Finds no information. 4. Finds no result on the home page. Decides to look for information on medicine. 5. Clicks on the heading "health" and decides to look for articles related to communication. Finds a relevant article titled "designing a social and behaviour change communication strategy". Copies the summary page into Microsoft word and then saves it. 6. Unable to realise that the full article is attached to the document. Later on realises that the full article is	**Steps to find information:** 1. Reads the introductory part of the portal thoroughly to ascertain what the portal is about. Types the phrase "maternal newborn child" into the search box of R4D. 2. Reads all the titles of research evidences that appear on the home page of the portal. Finds no relevant article. 3. Goes back to the home page to start searching for the specific article.
Barriers to use: Portal was slow to load. **Validity/use:** relies on experience **Downloading:** downloads specific article. **Interactive functions:** No evidence realised.		**Barriers to use:** Portal was very slow to upload. **Validity/use:** No specific criteria, relies on experience to determine the validity and credibility of the specific article. **Downloading:** Downloads specific article. **Interactive functions:** No impact.

	attached. Clicks on the full article and opens it on a different tab. **Barriers to use:** Portal was slow in responding. **Validity/use:** No specific criteria **Downloading:** Downloads specific article. **Interactive functions:** No evidence.	

Knowledge of:

Scidev.net: not tested

Eldis.org: not tested

R4D: not tested

Portal challenges:

None observed or mentioned.

Use of information:

Shared online: their own research appears on their own website, internally and externally

Shared offline: hard copies of their own policy briefs given to policymakers and other stakeholders

Saved: yes

Repurposed: to inform their own studies

Read: yes

Participant – TZ11 Category 5/7

TZ11 works in a think tank whose research is focused on influencing policy around science, technology and innovation, hence their research is use-oriented. TZ11 and colleagues use online portals to access research, especially those related to science and technology such as Globelics conference website and SPRU at Sussex University. This work is used to inform their own studies.

From their own experience, how policymakers access online resources (which they call policy linkage) is a major area of work and a very problematic one. They have a research department but also a communication department. **"We have a lot of research that is not being used".** So extra work is needed to make sure the research is being used. They have a communication strategy for different stakeholders, both public and private sectors. The research is summarised in order to communicate it to policymakers. To bring the research to their attention, you need to talk to them, entice them, invite them to take part in the research itself, work with them and once finished call them up to tell them about the research, prepare a policy brief, and aggressively sell the ideas to them. **"There is no demand for research outputs to use in their policymaking."**

TZ11 visits the government offices a lot and talks to them about how do you get the objectives to form a policy? TZ11 has been told that it may be based on a directive from the minister or from the president **"It's rare to hear that they have been reading to understand the underlying problem."** The information has to be pushed out there, to be sold to policy makers.

Influencing policy making is a process, not just an end result. Innovation requires a process, and in Tanzania the policy makers know this now. Whenever the government is doing something on science and technology they call this think tank, but it has taken time and effort to get to that situation. Innovation needs to be supported by a

Information behaviour:

Not tested as TZ11 did not have sufficient time available to take part in the whole study. Session focused on the process of how research evidence can be used to influence policy, programmes and practice.

Assessing validity/use:

Source, e.g. relevant conference, university etc.

Uptake:

Availability of evidence on the internet allows work to be done more quickly. Although TZ11 has a good collection of books, **"Now when I'm writing a paper I rarely look at those books because what is current is online"**

	system, and this has been reviewed for Tanzania. A new policy on innovation is being put in place because of this review. An example where their research influenced policy is a project on the role of vocational education, where they wanted to know whether this training increased productivity in the small state enterprises. The research found that it was not, and it identified problems concerning the design of the curricula. Those involved in the education and the policy were invited to see the results and that then influenced what was done. **"It is a process that is so painful, you've got to be persistent … when it happens it's very satisfying".** It was successful because we took the evidence to them, and were persistent. Science, technology and innovation is not a popular research area in Africa.	

Table 39: Tanzania Research Diary Summary Table

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
TZ 01	**find/retrieve**	Found: used online search engine Retrieved: repeated previous online search/ had stored digitally/ remembered	Found: received automated message	x	x [no response recorded for this question]	Found: used online search engine Retrieved: had stored digitally	x	x	x [no response recorded for this question]	x [no response recorded for this question]
	why source	I needed to be sure if the same information is reflected in other sources. But I could not. Also the source is assumed to be credible, at least for now in the country	x [no response recorded for this question]	x	x [no response recorded for this question]	I wanted to substantiate my article and advice/ recommendations with what other sources are saying about similar issue	x	x	x [no response recorded for this question]	x [no response recorded for this question]
	validity	Information from government website (Parliament and Ministries) could be compared with other sources, such	x [no response recorded for this question]	x	x [no response recorded for this question]	It was a video and I remember to have heard the speech but did not have the full Speech by the President	x	x	x [no response recorded for this question]	x [no response recorded for this question]

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		as think tanks and universities whether they bear same data. Also the level of details differ								
	use	shared online/ saved/ read	o [did not do anything with info today]	x	o [did not do anything with info today]	shared online/ saved/ read/ repurposed/ other: writing an article for local newspaper	x	x	o [did not do anything with info today]	o [did not do anything with info today]
	type/purpose	I accessed information on National Budget for agricultural sector (various sources). I digested it and some of these documents were shared with staff	I was attending a meeting, and had limited time to visit the website	x	x [no response recorded for this question]	This was the President's inauguration speech for the 11th Parliament in Tanzania which was made on 20th November in the parliament. The President Hon JP Magufuli underscored the importance of agricultural industrialization. I used his speech to write an article using Jim FitzPatrick/ANSAF (2013) research report, where it was found out that Tanzania losses US$ 110 million annually by exporting Cashew in raw form to India.	x	x	x [no response recorded for this question]	x [no response recorded for this question]
	influence			x			x	x		
	research	YES: It is	NO	x	NO	[YES ????]: I used	x	x	x [no response recorded	x [no

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
	evidence	about ANSAF research on budget allocation, in relation to potential and poverty levels across the regions, and also viewed				the president's speech signifying the commitment to develop industries and linked with the previous research findings and recommendations			for this question]	response recorded for this question]
	DFID portals	not used today ???	not used today	x	not used today	not used today	x	x	x [no response recorded for this question]	x [no response recorded for this question]
TZ5	**find/retrieve**	Found: used online search engine	o	Found: used online search engine	Found: used online search engine/ other website: www.escjournals.net/UAE	Found: SciDev.Net	Found: R4D	Found: R4D	Found: used other website:india.foodsecurityportal.org	Retrieved: had stored digitally
	why source	I was very much interested in the subject as an additional information I had to search for it.	o	I used the source because I found it valid, reliable and accessible.	I used the source because I trust the content within.	This source is rich in information of my interest.	This source is most trusted to me because most of the contents posted are of high research quality, reliable and valid.	This source is full of credible information for researchers like me.	Because I was looking for the information which is credible.	This source is full of information of my interest.
	validity	The information was credible	o	It contained the data	Scientific methodology used in the paper, it is	I trust the http://www.scidev.net/global because I	The information are	The information posted in	The portal is reputable in producing information which are acceptable in	Publications within the source

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		to me because was published by the reputable journal with World known Author.		which were researched under scientific methods.	found in the peer reviewed journal.	have been receiving the credible ideas from this web such as farming systems of various countries and adoption level of smallholder farmers.	well researched with credible data coupled with cases, plus the scientific arguments on technology transfer.	this portal is of high quality, the information is reviewed before being posted, and it is a world wide accepted portal in publishing research papers/reports.	the world of research.	are well researched and peer reviewed.
	use	read	o	saved/ repurposed/	shared offline	read	repurposed	shared offline/ saved/ repurposed	read	shared offline/ saved/ repurposed
	type/purpose	I made notes on the adoption of agricultural innovations for smallholder farmers.	o	I made notes on the need of technology transfer for socio-economic development.	I made notes on use of modern agricultural technologies.	I conceptualize the ideas in the content.	I made some notes from the information I serached.	I shared the data to justify the importance of using research in fast-tracking economic development.	I made notes about rural transformation.	I made some notes on the need to transform rural livelihood to my colleagues.
	influence		o							
	research evidence	YES: Improved maize varieties or hybrids are finding their way to smallholder farmers in sub-Saharan	o	Yes: Technology transfer by FDIs through vertical linkage with buyers and	YES: Even distribution of resources and services to ensure balanced agricultural development...	YES: The foundation (Bill & Melinda Gates Foundation) has huge global influence in different sectors...	YES: Technology transfer and sustainable rural development	Yes: Many inventions have had positive impacts on the poor...	YES: There is an emerging consensus that the well-being of rural households improve with the blending of farm activities with non-farm economic activities...	Yes: The fact that agriculture continues to employ a large proportion of the total labour force

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		Africa...		suppliers...						(about 74%)...
	DFID portals	not used today	o	not used today	not used today	SciDev.Net	R4D	R4D	not used today	not used today
TZ6	**find/retrieve**	Found: used online search engine (google) Retrieved: other: searched using key word	o	o	Found: used different type of website: google and UN websites Retrieved: repeated previous search online	x	x	Found: used online search engine/ used online RPP	o	x
	why source	Since it was a broader topic, google search was perfect, it was able to give me more relevant link	o	o	`i used google because, It direct you other links with relevant information	x	x	I used the source because, I was sure of getting the information I needed, I used Tanzania Online database. www.tzonline.org	o	x
	validity	I just wanted general information/knowledge all information was treated the same	o	o	I used Information from Unated Nations, which is a trusted webside	x	x	Tanzania Online is a national database with information which have passed through a check list procedure for validity	o	x
	use	saved/ read	o	o	repurpose	x	x	shared offline/	o	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
								repurposed		
	type/purpose	I seached information on ICT4D and innovation and saved it, I will use it on my work this week	NOT APPLICABLE	o	I research information on how to localize SDGs,, country and global examples. The information will be part of national communication strategy for SDGs Iam worling on	x	x	I wrote a project report on capacity for Developme nt Results Based	I did not use any information , I was attending meeting	x
	influence		o	o		x	x		o	x
	research info	NO	o	o	YES The information obtained today will inform/ or give input the national SDGs communication strategy	x	x	Yes The information was on impact on different Innovations on Agriculture and Fisheries sector this was a policy document published by the Government	o	x
	DFID portals	not used today	o	o	not used today	x	x	not used today	o	x
TZ7	**find/retrieve**	Found: received in email from person	Retrieved: had stored digitally	o	x	Found: online RPP: R4D	x	x	x	x
	why source	This was part of my activities today. To read through	Was called by funders to defend research	NA	x	This was part of literature review exercise for poverty study in Tanzania	x	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		the report and fill in any technical gaps then submit to the client	proposal							
	validity	This was a report that we wrote and shared with technical team for comments. I received it for a final eye check before we submit to the client	Reliable sources eg from governme nt statistics dept	NA	x	Very reliable	x	x	x	x
	use	read	other: in defending proposal	o	x	saved	x	x	x	x
	type/purpose	Read through a report on the Mining policies/strategies and programmes in Tanzania	Was looking for income statistics for Tanzania population	NA	x	Information that focused on poverty dynamics from R4D site. Was a part of literature review on a study on sustained pathways to poverty in Tanzania	x	x	x	x
	influence			o	x		x	x	x	x
	research evidence	NO	NO	o	x	YES: Source: R4D website. The research was done by ODI, ESRF and REPOA back in 2012	x	x	x	x
	DFID portals	not used today	not used today	o	x	R4D	x	x	x	x
TZ4	**find/retriev**	used online	x	o	x	x	x	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
	e	search engine/ email from a person/ repeated previous search online								
	why source	I already know the website link.	x	o	x	x	x	x	x	x
	validity	I know the website	x	o	x	x	x	x	x	x
	use	read it	x	o	x	x	x	x	x	x
	type/purpose	I read the information about the Commitment to Equity (CEQ), especially the new grant by Bill and Melinda Gate's Foundation and the goals of the grant of reducing inequality in developing countries. I read it because I am involved in a similar study with CEQ from two years ago, of which we are ready to disseminate results. I read	x	o	x	x	x	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		an email about the new CEW grant and decided to go and read it from the CEQ website.								
	influence									
	research evidence	NO	x	o	x	x	x	x	x	x
	DFID portals	not used today	x	o	x	x	x	x	x	x

B.3.3 Nigeria User Profiles and Diary entries

Table 40 below summarises our Nigerian participants and the data collected from them. This section contains the user profiles generated from 8 of those participants and a summary of the daily journal entries.

Table 40: Summary of Nigerian Participants and data collected

Participant	Gender	Category	Sector	Data collected
NIG1	M	Development consultant	Education	P, CI, DP, D4
NIG2	M	Development consultant	Water and Sanitation	CI, DP
NIG3	M	Media professional	Media	P, CI, DP, D4
NIG4	M	Academic/Researcher	Education	P, CI, DP, D10
NIG5	F	Media professional	Media	CI, DP
NIG6	M	Media professional	Media	CI, DP
NIG7	F	Knowledge broker/intermediary for policy makers	Education	CI, DP, D2
NIG8	M	Civil/Public Servant	Education	P, CI, DP, D10
NIG9	M	Development consultant		CI, DP, D2
NIG10	F	Development consultant	Gender and Equality	CI, DP
NIG11	M	Media Professional	Media	CI, DP
NIG12	M	Development worker in civil society	Health	P, CI, DP, D8
NIG13	M	Media Professional	Media	P
NIG14	M	Media Professional	Media	P
NIG15	M	Media Professional	Media	P
NIG16	M	Development worker in civil society	Health	P
16 participants overall		**12 complete face-to-face sessions**		**4 complete diaries**

* P = phone interview, CI = contextual inquiry (first half of f2f session), DP = discussion of portals (second half of f2f session) D= Diary, Dx = x number of journal entries returned

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

- Navigating through the portals can be very confusing.
- For instance it was difficult to find the topic "education" on SciDev.Net.
- Search results on Eldis were mixed up.

Use of information:

Shared online: Shares online research with colleague.

Shared offline: No

Saved: Yes, saves only relevant articles.

Repurposed: Rewrites article in own words and ensures that it is original. Also shades the most relevant article.

Read: Reads the summary/ background section first and then later reads it more thoroughly.

Participant–NIG1 Category 6

NIG1 is currently a development consultant and currently working on a DFID funded project which seeks to improve the quality of education in private schools, especially those serving children from low-income households. As part of the project, NIG1 has written an article on effective learning.

Uses the internet very often for research work. NIG1 searched for online information to write the above article. In demonstrating how the online evidence was searched for, NIG1 types key search terms.

NIG1 looks for how often the online article has been cited, on the search results. Clicks on the first title of the search result. Realises that the article is well cited. Indicates that **"this article is good....a lot of people have cited it".**

Unable to find the article date of publishing but realises that the article is a recent one since most of the references are dated between 2015 and 2011. Reads the first few sentences of the article hurriedly. Indicates that there are very relevant keyword in the summary which means that the article is a relevant one.

Downloads the article and then saves it. NIG1 prefers to read the article thoroughly later. After reading, NIG1 copies the relevant sections and indicates that, will re-write it later. Records the source. For example the website and author's name. Likes to subscribe to portals and other online sources (e.g. the Social Science Research Journal (SSRJ)) so as to be receiving very current information. Also prefers open access portals than those that are paid for.

Is also interested in well updated local internet repositories. Example those of University of Lagos website.

Information behaviour:

- Starts searching by typing keywords of an article into Google search engine.
- Focuses on the date and how well cited the article is.
- Prefers reading the brief summary below the titles of the portal.
- Downloads relevant articles, read the summary or introductory part first and then saves them if relevant.

Starting: Starts online search with Google search engine.

Chaining: Subscribes for newsletters from Journal (e.g. SSRJ and other internet portals.

Browsing: Scrolls up and down and reads the various titles cursorily before deciding on the most relevant article.

Differentiating: Makes sure that no unwanted article is copied or saved. Only saves the most relevant article. In some instances, prints the most relevant article.

Monitoring: Subscribes to portals and journals in order to be informed about the most recent and relevant information on a subject area.

Extracting: Shades and highlights the most relevant section of an online research.

Processing: Copies, rewrites, saves the most

	Example use of evidence: NIG1 has written an unpublished article regarding effective teaching. **Uptake:** NIG1 indicates that internet enables working across different research areas. Can discuss issues related to governance though focuses more on education related issues. NIG1 explains that because NIG1 is able to find information very easily online. Has made demand for more robust research evidence higher.	relevant article. **Assessing validity/use:** • Looks out for the Date, author; and • How well cited the online article is.

Participant–NIG1	Category 6 part 2	
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. Types the full title of the specific article into the search box of the article. Indicates that the search results are also colourful. 2. Asks whether the keyword below each research title is of significant importance for looking for the specific article. Clicks on "data"—one of the keywords. 3. Scrolls up and down the first page. Unable to find the specific article on the first page. 4. Then moves to the second page. Finds the specific article. Clicks on it. Likes the speed read section of the article, but indicates that information on the portal cannot be used if there is no internet since no article on it can be downloaded.	**Steps to find specific article:** 1. Moves the cursor on the menu. Clicks on the search box. 2. Types the specific article into the search box. 3. Finds the specific article on the first page of the portal. Clicks and download the portal. 4. Indicates that the portal is really slow but contains a lot of information.	**Steps to find specific article:** 1. Navigates to R4D home page and then types the full title of the specific article. 2. Finds the specific article on the first page. 3. Clicks and downloads it. Reads the title again to ensure that it is actually the article required.
Steps to find information: 1. Types the URL of SciDev.Net into the find bar. Finds the portal very lively and colourful. 2. Moves the cursor on the menus. Clicks on each topic on the menu. Realises that "education" is under the heading "communication". Gets confused and asks whether there is a reason why "education" is under "communication". 3. Scrolls up and down the front page to see if there is any relevant article. 4. Indicates that the portal is more science and innovation biased and does not focus on area of research. 5. Indicates that there is no attachment and the	**Steps to find information:** 1. Type the URL of Eldis into Google search engine. Finds the portal and wonders what the portal is about. 2. Scrolls up and down twice to browse the content of the portal. 3. Looks for the topic "education" on the menu but unable to find it. Indicates that this portal is among the few I have seen with education not on the main bar [main menu]. 4. Scrolls to the bottom and finds "education" at the bottom section of the portal. Indicates that "I did not expect this here". Clicks on it. 5. Like the content since it is consistent with research area. Download the fourth and seventh articles on the home page of the portal. Scrolls down the two articles. 6. Saves the two articles after cursorily scrolling down the table of content of the two articles.	**Steps to find information:** 1. Type the URL of the portal into Google. 2. Scrolls down and then quickly types two keywords of the prepared article--"Classroom observation"—into the search box of the portal. 3. Gets **318** results. Becomes impressed that the search results are many. 4. Identifies two relevant articles on the first page. Downloads them and quickly saves them. Signs up to the portal.
		Barriers to use: Finds it difficult to navigate to the home page. **Validity/use:** Because it is a DFID funded portal. **Downloading:** Download specific article **Interactive functions:** Signs up for the portal.

referencing. As such cannot be used for any serious job. 6. Finds it difficult to navigate to the home page. **Barriers to use:** Unable to download an article. Unable to navigate to the home page. **Validity/use:** Believes it is a DFID portal but the portal cannot be used for academic purposes. **Downloading:** Unable to download articles on the portal **Interactive functions:** No evidence.	7. Writes the URL in a book. Indicates that the portal contains a lot of "reliable and relevant detailed" information. **Barriers to use:** Slow to respond. **Validity/use:** Indicates that portal contains a lot of articles from "top class" authors. **Downloading:** Download general article. **Interactive functions:** Subscribes to the portal.	

Knowledge of:

Scidev.net: None

Eldis.org: Yes, have search for online information on it for an agriculture project before.

R4D: None

Portal challenges:

Finds it difficult to go back to the home page and also locate the site map.

Use of information:

Shared online: shares the website link of the article with colleagues via email. Also uses WhatsApp or Twitter on mobile phone /or tablet for sharing.

Shared offline: Shares paper flyers and Magazines with colleagues.

Saved: Saves relevant online resources in named folders. For instance, NIG 2 has created a folder for the programme.

Repurposed: After reading relevant articles online, NIG 2 makes note on computer for ongoing work.

Read: Reads online using tablet or phone.

Participant–NIG 2 Category 6

NIG 2 is a development consultant in Nigeria, working on a DFID funded programme that provides support to help Nigeria use its resources more efficiently and effectively.

NIG 2 uses Google for online research evidence search and does not have any favourite website. This is because Google is able to link the relevant document or website for NIG 2.

For instance, NIG 2 types the phrase **results based management** into Google search engine. Reads aloud the various titles that appear on the first page of search results. Clicks on the second title of the search results. Gets the document from the United Nations Development Group website.

Reads the abstract of the document and finds it relevant. Saves the article in a folder.

Clicks on subsequent pages to find more relevant research evidence.

Reads through relevant documents and make paper notes.

NIG 2 reads the abstract or executive summary before deciding to read the entire document. Downloads and occasionally prints the document.

Shares relevant website links of research evidence with colleagues through emails, Twitter and WhatsApp.

Also uses Twitter and WhatsApp on phone to share website links to research evidence. Also follow bloggers for useful research evidence.

NIG 2 prefers not to focus on scholarly articles that usually appear as the first title on the first page of the search results; this is because they are often for sale and are also too technical.

NIG 2 determines the validity of the source of online research evidence by ascertaining how recognised the sources are internationally. Example NIG 2 explains that **"I most rely on international think thanks like UNDP, ODI, UNDG"**.

Also consider the date and the credibility of the author to determine

Information behaviour:

Starting: Uses Google for general search.

Chaining: clicks shared websites of a relevant article on Twitter and WhatsApp on phone.

Browsing: Clicks each page of the search results and also scrolls carefully to find research evidence.

Differentiating: Downloads relevant information on tablet or phone to read; especially when travelling. Discards irrelevant articles by closing or deleting them.

Monitoring: No evidence found

Extracting: Reads online research evidence and the makes paper notes.

Processing: Saves relevant documents.

Assessing validity/use:

- Date of publication
- Reliability of author
- Source of information

	the validity of the article. **Example of research evidence:** NIG 2 is developing a document on Performance Management Systems. **Uptake:** Information access has become relatively easier as a result of improvement in the ICT infrastructure in Nigeria.	

Participant–NIG 2 Category 6 part 2

Scidev.net

Steps to find specific article:

1. Finds it difficult and confusing navigating to the home page. Tries to look for site map for the portal to see if it will help but to no avail.
2. Scrolls up and down to see if the topic "**articles**" can be located on the home page of the portal but was not successful. Click the topic **"communication"** on the home page to find specific article but unable to find any information related to specific article.
3. Type the full title of the specific article (NIG 2 misspells some of the words in the process of typing) into the search box of the portal and gets **0** results.
4. Types the full title of the article into Google search engine. NIG 2 finds the specific article as the first title of the first page of the search result.
5. Clicks on the title and gets to www.scidev.net.
6. Clicks on the article and downloads it.
7. Find the presentation of the specific article very good. The presentation of the specific article contains a brief summary of the article, date, name of author, and the main article.

Steps to find information:

1. Scrolls up and down and concludes that the SciDev.Net is focusing on climate change—mainly because NIG 2 finds more climate change related stories on the portal. For example on the home page alone, NIG 2 finds topics such as **disappearing islands, African climate fever, negotiating climate future.**
2. Scrolls up and down again to see if there is information on results based management. Clicks on **"Governance"** on the home page to see if there is an article on performance

Eldis.org

Steps to find specific article:

1. Types the title of the specific article **(i.e. three tools to unlock finance for land-use mitigation and adaptation)** into the search box of Eldis.
2. Likes the way the content of the specific article is presented. Indicates that **"I like the summary provided....it helps you to decide whether to download the article or not".**
3. Downloads that specific article. Bookmarks the portal.

Steps to find information:

1. Scrolls up and down and the types the phrase **"agric information"** in the search box of Eldis.
2. Scrolls down the first page and realises there are **1075** search results. Finds the number of search results **"packed"** and **"clumsy".**
3. Tries to find an introduction of what the portal is about. Finds the title **"what is Eldis"** at the bottom of the portal. Indicates that **"this topic [what is Eldis] should have been at the top here".**

Barriers to use: number of search results discourages NIG 2.
Validity/use: Documents on DFID related portal will be credible.
Downloading: Downloads specific portal

R4D

Steps to find specific article:

1. NIG 2 finds the specific article by typing it **(i.e. Using climate information to achieve long term development objectives for African ports)** in the search box of r4D.
2. Scrolls up and down three times and then finds the specific article as the fifth title.
3. Becomes surprised that the specific article is not the first title. Also realises that the specific article automatically opens on a different tab.
4. Likes the presentation. Indicates that this provides you open free access, the author, year of publication and the link to the main source of the document. Also presented in different forms in word, PDF, and there is the option for printing.

Steps to find information:

1. Scrolls up and down the r4D portal. Reads the title on the home page aloud.
2. Wonders what the portal is about. Indicates that it is not clear as to what the portal is mainly about. Types in the phrase **"public sector budget in Nigeria"** in the search box for R4D. Scrolls up and down the portal twice.
3. Clicks on the third title of the search results on the first page. It opens on a different tab of which NIG 2 is happy with. Realises that the search results only highlight the keywords in the above phrase in the brief summaries under each title of the search result. Tries to go to the home page but finds it very difficult to navigate.

management. But believes they are all focused on climate change issues. Bookmarks www.scidev.net 3. Likes the pictures and colourful nature of the portal. **Barriers to use:** Finds it difficult to navigate to the home page. Unable to locate the site map of the portal. **Validity/use:** DFID related portals are valid and useful. **Downloading:** downloads specific article on the portal through Google. **Interactive functions:** No impact	**Interactive functions:** No impact	4. NIG 2 gives up. **Barriers to use:** Not clear as to what the portal is about. **Validity/use:** The credibility of the author, the reliability of the source, the date of publication. **Downloading:** None observed **Interactive functions:** No impact

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

- No significant challenge was encountered for Scidev.
- NIG3 could not identify any relevant online article on Eldis because the search results did not show it on the first two pages.
- Navigating to the home page on R4D was difficult for NIG3.

Use of information:

Shared online: Shares information via Facebook wall and Twitter handle.

Shared offline: Does not share information offline.

Saved: Saves online information for six months and after that discards it.

Repurposed: Gathers online evidence, looks for evidence to show that the online research is accurate. Rewrites the story in own way and then broadcasts it.

Read: Mostly reads online articles through Facebook and Twitter.

Participant–NIG 3 Category 8

NIG 3 is a media professional for a radio station in Lagos. Focuses mostly on reporting developmental and political issues in Lagos. Was part of the 'Occupy Nigeria' group, a non partisan group that was formed to fight corruption in Nigeria.

Very active on social media— Facebook and Twitter. Shares information on Twitter and Facebook. Before a story goes on air, NIG 3 checks the Twitter handle and Facebook wall of the FM station to ascertain the most trending stories in Nigeria and beyond. Then looks for the source and those who are commenting and posting further details and pictures on the story.

Makes follow ups with public figures who have commented on the story to get their views for the station. Also finds public figures who are in charge to seek their views and the actions they've taken. Then writes the story, presents it on air and subsequently posts it on Facebook and Twitter for feedbacks.

Quotes the source of information. For example the name of the resource person or public figure. The source can also be an internet portal like the Cable News Network, British Broadcasting Cooperation, Daily Mail, The Punch of Nigeria etc. NIG 3 indicates that international news are ascertained from the above named international news portals. Checks the same story from all the above mentioned portals to ascertain the actual facts before making a story. Indicates that **"the Oscar Pistorius case for example was very tricky to report...I heard from some of the Western media that the man has been freed...so I quickly told my people to hold on for a while.....I checked several western media including the BBC and realised that it wasn't the final judgement...so we then had to put the pieces together before reporting".**

Likes Twitter and Facebook because they suggest the sources of the

Information behaviour:

Starting: Uses Facebook and Twitter as well as local and international portals to start internet search.

Chaining: Moves from one portal to another in search of information. As such picks the source of information and then looks for that source online.

Browsing: Browses for online information on various portals including international and local ones. For instance, looks for international news on the BBC and CNN. Searches for local news from The Punch and The Vanguard website.

Differentiating: Opens every piece of online evidence on a different tab. Copies the most relevant articles and then posts them through the Twitter handle of the FM station.

Monitoring: Checks on several portals to see if there are new stories. Also looks for trending stories on Twitter and Facebook.

Extracting: Copies information from different sources and then redrafts the stories to suit what the station wants.

Processing: Writes own stories, broadcasts them and subsequently posts the story on Twitter and Facebook.

Assessing validity/use:

- Triangulates by checking the same stories on different internet portals.

	information and so it makes follow-ups easier. Helps NIG 3 to access the number of people who absorb information from the FM station in Nigeria and beyond. For instance indicates that **"we posted this news item on our Twitter handle yesterday and you can see that it has reached a total of 7,214 people....1,378 people have also liked the story. This shows how the FM station are [is] performing".** Makes a lot of validity checks by any (or a combination of them) of the following approaches: 1. Checks from different sources to confirm a story before reporting. 2. Looks /or waits for highly placed sources for reports. 3. Checks other news portals both international (see above) and local (e.g. The Punch, The Vanguard, etc). 4. Relates with neutral sources to confirm information. This includes local Non-governmental Organisations (NGOs) like Enough is Enough (a local NGO). Also works closely with international NGOs like the Red Cross and DFID for first-hand information. NIG 3 gave an example **"the Boko Haram case for example.....the abduction of the school girls,.... the government was denying it initially...so we had to confirm the story from the International Red Cross who were on the ground before broadcasting the story".** Also said that **"An accident happens and 10 people die...and the Nigeria police is saying only 3 people died and you have an eye witness who says he counted 10 dead bodies....you will place more premium on the eye witness' story....that eye witness has nothing to lose".** Also looks for online information from various internet portals through Google. For instance, types the full name of the organisation responsible for population data in Nigeria—the National Population	• Relies on independent observers. • Looks for evidence from people in highly placed positions.

	Commission of Nigeria—into the Google search engine. Downloads data from the portal and indicates that the data collected will be used to write a story for broadcasting. Uses Facebook analytics to ascertain how relevant the FM station is on Facebook, and which programme is getting more audience on Facebook. For instance, identifies an afternoon show that has recorded 15,000 likes on Facebook. Saves information for six months and then discards it. Saves only online research evidence. Does not save paper document because it is expensive and occupies space. **Example use of evidence:** No research evidence was produced but exhibited several news items on the radio Twitter handle and Facebook wall. **Uptake:** NIG 3 indicates that uptake of research evidence has made information access cheaper, faster and easier. Able to find, validate and prepare a news item without paying for it. Flow of information from government sources has become easier since the information can be sent with a click of a button. Able to check government activities to ensure good governance.	

Participant–NIG3	Category 8	part 2
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. Types the keywords of the specific article into the search box of the portal. 2. Unable to find the article on the first page. 3. Click the next page and still could not find it. 4. Types two keywords in the title of the specific article into the search box but was unsuccessful. **Steps to find information:** 1. Types the URL of SciDev.Net into the find bar on the computer. 2. Moves the cursor on the main menu bar and then clicks on "Governance". 3. Subsequently clicks on "human rights" under the topic "Governance". 4. Scrolls down and reads the titles aloud to find out which of the general articles is relevant to the work at the FM station. 5. Finds no relevant information. Decides to look for energy related articles. 6. Gets no relevant result. Then types "fossil fuels". 7. NIG3 then scrolls down and reads the titles on the search results page. 8. Clicks on the second title, scrolls down gradually and then reads the "speed read" section.	**Steps to find specific article:** 1. Tries to look for specific article on the Eldis community page but was not successful. 2. Types the full title of the article into the search box of the Eldis Community page. 3. Gets no relevant results. Gives up. **Steps to find information:** 1. Types the URL of the Eldis portal into the find bar of the computer. 2. Downloads the portal. 3. Clicks on Eldis Community on the right hand side of the portal—the interactive page of the portal. 4. Types "Nigeria" into the search box on the "Eldis Community" page. 5. Unable to find relevant stories on Nigeria as expected. 6. Clicks on the profiles of Nigerians who have discussed issues on the page. 7. Clicks on one of the comments to view the content after scrolling down. 8. Will prefer the dates on the comments to be more conspicuous. **Barriers to use:** No evidence found. **Validity/use:** See SciDev.Net on the LHS **Downloading:** Unable to find any relevant article on Eldis as such could not download.	**Steps to find specific article:** 1. Types keywords in the title of the specific article into the search box of a portal. 2. Finds the specific article on the first page of the search results. Indicates that it is good to have the name of the author, the source, and date of the article. Indifferent about the fact that the article is presented in a PDF format. Reads the full title and summary. **Steps to find information:** 1. Types the URL of R4D into Google search box. Downloads the portal. Looks at the website and then types **"Nigeria"** into the search box. Not really interested in the number of search results since NIG3 is only interested in the first two pages of the search result. 2. Unable to find any relevant article. NIG3 then types **"Electricity and Nigeria".** Scrolls down and then clicks on the second title of the article on the first page of the search results. 3. Reads the title and conclusion of the article. 4. Finds it difficult to navigate to the home page. Indicates that this will be very difficult to use on mobile phones since one may not have a full view of the portal at a time on a phone. **Barriers to use:** Navigating to the home page of the portal was a major issue. **Validity/use:** See SciDev.Net on the far LHS

9. Copies the link to the article. 10. Pastes the link on the Twitter handles of the FM station and then posts it. Indicates that **"I will be expecting comments from those who will read it".** 11. Indicates that the portal is friendly and does not look congested. It also has the source and the name of the author of an article. 12. Will prefer not to use the various interactive features including Facebook, Twitter and LinkdedIn on the portal, since there are a lot of spam and virus issues in Nigeria. 13. Clicks on the SciDev.Net logo to go back to the home page. **Barriers to use:** No evidence of barriers **Validity/use:** Validate online information by cross checking on other internet portals. Also interviews resources persons and people in high positions who are in charge. **Downloading:** Download online data from the National Population Commission. **Interactive functions:** Does not use the online interactive features.	**Interactive functions:** Clicked on the Eldis community page to find out the type of people who interact on the Eldis community. Also decides to look at the profiles of those who interact on Eldis.	**Downloading:** Downloads relevant article. **Interactive functions**: Does not use the interactive functions on the portal because of virus and spam issues in Nigeria.

Knowledge of:
Scidev.net: None
Eldis.org: None
R4D: None

Portal challenges:
The objectives of the portals are often not clear.

Use of information:
Shared online: shares unsolicited articles via email.
Shared offline: No
Saved: Saves article on computer desktop.
Repurposed: Makes paper notes after reading the article.
Read: Reads abstract first; reads the whole article in detail when NIG 4 finds article relevant.

Participant–NIG 4 Category 5

NIG 4 is a professor at the University of Lagos. NIG 4 is a member of the academic committee and the senate of the University, and is also doing work for the Department of Education.

Uses www.yahoo.com to search for research evidence online. For instance, NIG 4 types in **Non formal education programmes** into the yahoo search box. NIG 4 scrolls up and down and reads the title of each search result on the first page. Looks for keywords in the title and the brief summaries below the title. NIG 4 clicks on the second page and indicates that **"I go through each page to make sure I get very relevant information".**

Goes back to the first page of the search result. Clicks on the third title on the first page of the search results.

NIG 4 then reads through the abstract to ascertain whether the article is needed for the above stated work, and saves it on the computer desktop after reading the abstract. NIG 4 reads specific article in detail later on and makes paper notes. Occasionally NIG 4 makes notes in Word. Prints the most relevant article. Shares unsolicited online articles via email after reading.

Prefers to visit specific portals (preferably United Nations portals) if already knows that the specific relevant articles can be found on that portal. NIG 4 indicates that **"if I want to search for information on population of education, I will go to UNFPA [United Nations Population Fund]...if I want to look at issues relating to labour, I will want to look at ILO [International Labour Organisation] .....and If I want to look at anything Education, I may want to look at UNESCO [United Nations Educational, Scientific and Cultural Organisation".**

Also considers any article that is institutional based as credible. Finds instructional based articles credible because there are a lot of quality assurance checks before it is published.

NIG 4 also considers the author and date of the article. Ideally, will prefer

Information behaviour:
Starting: starts online search through yahoo search engine.
Chaining: No evidence
Browsing: Searches each page of the search result to ensure that the most relevant articles are identified.
Differentiating: Deletes irrelevant articles and downloads and saves only the relevant ones on the desktop of the computer.
Monitoring: Colleagues share recently published articles with NIG 4.
Extracting: Uses www.yahoo.com to look for online information.
Processing: reads and makes own paper notes. Also shares with colleagues.

Assessing validity/use:
- Date and credibility of the author.
- Through institutions.

	most recent articles. **Example of Research Evidence:** Currently working on a project for the Department of Education at University of Lagos. **Uptake:** Research evidence on the internet has made research easier. NIG 4 explains that **"when I was doing my PhD, ...for me to get information, I will have to travel from here to Lagos Island....I had to be moving from one library to another and from one office to another to look for information....but with my modem, I can even be interacting with my students...generally internet has made learning easier and comfortable"**. It has also made working easier and quicker.	

Participant–NIG 4	Category 5	part 2

Scidev.net	Eldis.org	R4D
Steps to find specific article: 1. NIG 4 finds it very difficult to find the search box. 2. Types keywords of the specific article. 3. Scrolls up and down three times but could not find it. 4. Though the article was the fourth title of the search results of the first page, NIG 4 was unable to identify it. 5. NIG 4 explains that such specific articles are usually the first title of the first page of the search results.	**Steps to find specific article:** 1. Types specific article and finds **15504.** 2. Indicated that the number of search result is good because it gives more options. 3. NIG 4 prefers spending more time looking for documents in order to get the most relevant ones. 4. Find the specific article on the first page of the search results of portal.	**Steps to find specific article:** 1. Types in the title of specific article into the search box of R4D. 2. Realises that the first article on the first page of the search results. 3. Reads the brief summary below the title to ascertain whether what the article entails. 4. Clicks on the article.
Steps to find information: 1. Finds www.scidev.net through the www.yahoo.com search box. 2. Moves cursor from one topic to the other on the home page. 3. Finds the sub-topic **education** under the topic **communication** inconsistent. But believes there is an unexplained rationale for that. 4. Finds the portal beautiful but sceptical whether the relevant online information can be found on it. 5. Clicks on the sub-topic evaluation under the main topic **communication** to see if there is any information related to **"research policy"** there. 6. Finds nothing directly related to research policy. 7. Moves cursor to the bottom right corner of the page to click on the sub-title **how to engage with participants in field research**. 8. Reads through the article cursorily and finds it interesting to the above mentioned project. 9. NIG 4 prefers articles that have a very good analytical rigour and rich in data, but very easy to understand. Finds the portal not rigorous enough for academic work. The portal focuses on short pieces of non-specialist information. Indicates that **"if I am doing a serious academic work like writing a paper, I will not focus on this one [www.scidev.net]".** However, will use it to gain a general	**Steps to find information:** 1. Tries to find the Eldis portal but the website was down. Tries for the second time and was successful. 2. Clicks on the main topic **conflict and security** on the portal. Finds a Nigerian related article titled **Unemployment and security challenges in Nigeria**. 3. Finds the presentation of the article on the portal very good considering the fact that a summary of the article has been provided for a quick read. Also likes the fact that the article can be downloaded in a PDF format. 4. NIG 4 clicks on the article. Then clicks on the document in a PDF format mainly because NIG 4 finds the information that **"view full report".** As an academic, wishes to find a topic called **Publication**. **Barriers to use:** The portal often goes down and also slows. **Validity/use:** See SciDev.Net on LHS	**Steps to find information:** 1. Looks for interesting online evidence on the portal by typing **early childhood development** into the search box of R4D. 2. Happy with the fact that there are **959** search results. Clicks on the fourth title on the second page of the search results. 3. Likes the fact that the document is opened on a different tab. Finds the portal credible since it is DFID funded. NIG 4 believes quality assurance works will be undertaken on any document that is published on the portal. 5. NIG 4 tries to go back to the home page by clicking on the DFID logo on the portal. It opens the main DFID portal. Clicks back to the home page of the portal. 6. Does not understand the key objectives of the portal. Suspects it is a

knowledge about issues that NIG 4 has no specialty in. 10. Tries to go to the home page of www.scidev.net but found too confusing. **Barriers to use: No impact** **Validity/use:** Valid since it is a DFID funded portal. Information from this portal will always be checked for quality before it is published. **Downloading:** No evidence **Interactive functions:** No evidence	**Downloading:** Downloads specific article and an article NIG 4 found. **Interactive functions:** No impact.	DFID portal for projects but doesn't know specifics. Indicates that DFID should network such portals with institutional libraries in countries like Nigeria. **Barriers to use:** The objective of the portal is not clear enough. Confuses NIG 4 as to what exactly to look for as a first timer on the portal. **Validity/use:** See SciDev.Net on LHS **Downloading:** No impact **Interactive functions:** No impact

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

Erratic power supply affects proper functioning of internet, which subsequently slows down the speed of the portals.

Use of information:

Shared online: Shares online evidence with colleagues.

Shared offline: No

Saved: Saves relevant article for future use

Repurposed: Copies articles online and re-writes them to suit the objectives of the project. Saves the article for future use.

Read: Reads online article including the references.

Participant–NIG 8 Category 3

Works in Education at Lagos State. Undertaking a project which aims at creating a platform for bringing together all Lagos schools.

Looks for information on best practices on international portals like UNESCO and the World Bank.

Also looks for information through Google search engine. In doing so NIG 8 focuses mainly PDF. NIG 8 explains that **"most often when I get PDF documents, I get what I am looking for the document".**

Types the full title of an article into Google search engine.

Open several online articles on different tabs.

Scrolls up and down to looks for the most relevant document. Clicks on the most relevant article on the first page of the search results.

Copies the online article into word and rewrites to suit the ongoing work. Make references where necessary.

Also copies the website on which the relevant article was found into Google search engine to see if more relevant online evidences can be found.

Also uses Google drive on phone to look for information.

Uptake:

Availability of internet has made information very accessible; even on mobile phones. One does not need to be located at a particular position like the internet cafe to be able to access information.

Information behaviour:

- types full sentence into a Google search engine.
- scrolls up and down to find the most relevant article.
- opens specific articles in different tabs.
- copies article into word and re-writes it in own words.

Starting: Starts online search using Google search engine.

Chaining: Copies key references into Google search engine to finds more relevant information. NIG 8 types the portal addresses to find more articles.

Browsing: Scans through the front page of a portal and opens all online evidence in PDF format.

Differentiating: Opens online articles on new tabs and then selects the most relevant ones out of them. Closes the irrelevant ones.

Monitoring: Uses the UNESCO internet portal as one of the default portals.

Looks for state of the art practices in Education Management Information Systems.

Extracting: Copies online article from internet portals into Microsoft Word.

Processing: NIG 8 always rewrites online article to suit the objective of the project. After that NIG 8 saves the online article.

Assessing validity/use:

		• Considers online articles from international sources valid. • The credibility of the author is also crucial.

Participant–NIG 8 Category 3 part 2

Scidev.net

Steps to find specific article:

1. Types specific article into the search box of SciDev.Net.
2. Gets **236** search results. Scrolls up and down the first page of the search results.
3. Find the specific article on the first page.
4. Downloads it onto a mobile phone device.

Steps to find information:

1. Types the portal address into Google to find the most relevant article in education.
2. Reads the heading on the home page of the portal.
3. Indicates that it contains a lot of pictures which makes it colourful, but renders the website "heavy"— may consume more internet data on the phone (i.e. has cost implications).
4. Clicks on the heading **"Communication"** on the menu of the portal and then goes on to click on the sub-heading **"education"**.
5. Finds no research evidence.
6. Decides to use the search box of SciDev.Net to look for information on education.
7. Clicks on the second title on the first page after scrolling up and down twice. Downloads the article onto a mobile phone.

Barriers to use: Contains lots of pictures—may end up consuming a lot of internet data. This has cost implications.
Validity/use: Contains lots of current information.
Downloading: Downloads article.

Eldis.org

Steps to find specific article:

1. Types keywords of specific article into the search box of Eldis. Finds a total of **8660** search results.
2. Reads the heading of the portal. Believes the search results are many because only keywords were used.
3. Finds the article on the first page of the search results.
4. Impressed to find a summary of the specific article. Also happy that there is an option for downloading the article in PDF format.
5. Downloads it.
6. Happy to know the portal is for free access. Bookmarks the portal for future use. Indifferent with the way the article is presented on the portal.

Steps to find information:

1. Types the URL of the Eldis portal into Google but power goes off and so had to use mobile phone.
2. Clicks on the heading **"education"** on the home page of the portal. Scrolls up and down to see if there is any relevant article. Realises that the first title on the first page of the search result is relevant to the ongoing work.
3. NIG 8 indicates that articles on DFID related websites like Eldis are credible and can be used for any academic purpose.
4. Downloads the article onto the phone and then saves it.

Barriers to use: No barrier realised.
Validity/use: No clear evidence.
Downloading: Downloads article in PDF format.
Interactive functions: No impact.

R4D

Steps to find specific article:

1. Looks for specific article by first clicking on the home page.
2. Clicks on advance search and realises it requires a lot of information on the article.
3. Comfortable with the advanced search since it will give the most accurate information.
4. Tries to find the article but unable to find the specific article. Gives up.

Steps to find information:

1. Types the address of the website into Google search engine.
2. Looks for education related topics. Reads the introductory part of the portal in detail.
3. Finds articles on the portal by clicking on **"Country or Region"**.
4. Scrolls up and down three times to see if there is any relevant article on **"education"** in Nigeria.
5. Realises that the portal contains a lot of current information. For instance, NIG 8 explains that **"I can see this portal contains a lot of information...if I say I am looking for information, ... I can get a lot of current information from this portal...for my position papers".**
6. NIG 8 notes the address of the portal in a diary.
7. Subscribes to the portal. Finds the subscription simple.

Barriers to use: Finds advance search page too detailed. Requires too many information and time consuming.

Interactive functions: No impact.		**Validity/use:** Detailed portal and contains articles in PDF format (of the view that articles in PDF format are reliable) **Downloading:** Downloads online articles in PDF format. **Interactive functions:** Subscribes to R4D.

Knowledge of:
Scidev.net: None
Eldis.org: Yes, stumbled on it on few occasions.
R4D: None

Portal challenges:

- General search: Indirect effect of erratic supply of power.
- Eldis: Was down
- Scidev.net: Unable to access articles offline.
- R4D: not extensive enough.

Use of information:
Shared online: Rarely share online
Shared offline: No
Saved: Saves online articles for future use.
Repurposed: Copies online article to Microsoft Word and rewrites articles in own words
Read: Yes, reads the summary first, reads the whole article if it is relevant.

Participant–NIG 10 Category 4

NIG 10 is a development worker in civil society who mostly works on gender related issues across the West African sub-region. NIG 10 is currently a member of a Pan-African advocacy group . This advocacy group focuses on expanding the role and influence of women in the financial sector. NIG 10 has published an article on gender and trade**.**

NIG 10 uses Google search engine for online evidence searches. Below describes how NIG 10 searched for online information to write the above article:

Types keywords of the article into Google search engine.

Scrolls up and down, looks at the heading, gender and trade biased sources (i.e. International Trade Centre, Cuts International, World Bank).

Prefers PDF document since it can be downloaded and then read offline.

NIG 10 indicates that PDF documents make reading easier. NIG 10 explains that **"If I see something like this [an article in a PDF format]..... I can download and then when I am the plane 35,000 ft high, I can read it offline on my tablet...I don't have to stay online all day".**

Downloads the online articles on a tablet, so as to move easily with the article. Also, websites can easily disappear—particularly when the operators can no more finance its maintenance.

Clicks on the first title on the first page of the search result. Downloads the article and reads cursorily to ascertain whether it is relevant. In doing so NIG 10 looks at the issues addressed in the article.

Saves articles on computer. Shares article with colleagues.

Looks for facts and figures online for advocacy work. Acknowledges the authors of the article afterwards.

Does not go beyond the first page when looking for a research article.

Information behaviour
Starting: Starts online search with Google search engine.
Chaining: write the addresses of the portals in a diary for future use.
Browsing: Considers the first page of the search results as the most likely page for the search result. Reads cursorily the various titles of the articles on the first page of the search result.
Differentiating: Downloads and saves relevant articles.
Monitoring: Keen on keeping track of articles on the three internet portals. For instance NIG 10 indicates that **"I will come back to this website [R4D].... for more information on DFID projects".**
Extracting: Looks for credible facts and figures online for advocacy work.
Processing: Copies and rewrite online articles to suit a document in preparation.

Assessing validity/use:

- Finds online research from international sources like World Bank more valid.

	Seldom shares hard copies. **Example use of evidence:** Writing an article on gender and trade. **Uptake:** Internet has made it easier for information to be ones finger tips.	

Participant–NIG 10 Category 4 part 2

Scidev.net

Steps to find specific article:

1. Looks for specific article by typing keywords of the article title into the search box of the portal.
2. Finds 0 results.
3. Tries to refine the search by clicking on **browse type** on the menu.
4. Reads the sub-heading but finds **0** results.
5. Clicks on the heading **data**, scrolls through the titles on the search result but unable to find it.
6. NIG 10 gives up.

Steps to find information:

1. Have never seen www.scidev.net before but indicates that the portal is a science bias portal.
2. Types gender into the search box of www.scidev.net.
3. Finds **345** search results. Reads the titles of each search result on the first page aloud.
4. And at the bottom of each result, NIG 10 finds some key words which include gender.
5. Clicks on the keyword gender and realises that a new page has been opened with another set of search results. Becomes confused. NIG 10 indicates that **"I was not expecting this".**
6. But clicks on the first title of the search results on the first page.
7. Tries to look for an attachment to download so as to read it offline. Finds no attachment.
8. NIG 10 indicates that **"... you have to come online to read it... and you know there may be no light [power]".**

Eldis.org

Steps to find specific article:

No access

Steps to find information:

NIG 10 type the web address of Eldis but unable to access it.

Barriers to use:

Validity/use: No access
Downloading: No access
Interactive functions: No access

R4D

Steps to find specific article:

1. Looks for specific article by typing a phrase into the search engine of the portal **"development finance institutions and infrastructure".**
2. Finds the article on the first page of the search result.
3. Clicks on the article and becomes impressed with the fact that the portal provides the web link for the main source of the article.
4. The article is also downloadable in PDF. The name of author and date of publication has also been provided.

Steps to find information:

1. Finds the portal simple. Types the website address into Google search engine.
2. Finds the R4D portal. NIG 10 clicks on **"themes".** Finds out that there are sub-themes.
3. NIG 10 scrolls up and down in search for the topic "**Gender"**, but was not successful. Indicates that the thematic areas are not extensive enough mainly because NIG 10 could not find **gender**.
4. Goes back to the home page using the **"back"** tab. But unable to find one. Decides to look for issues related to peace and security. Clicks on the heading **"Governance and Conflict"**. NIG 10 then clicks on a sub-heading **"building, peace and stability"**.
5. Finds **795** records. Scrolls up and down once. Reads each title on the first page of the search results aloud.
6. Clicks on the fifth title of the search result. Finds

9. Realises that the article can be emailed through the portal. NIG 10 emails the article to own email address. **Barriers to use:** May not be able to read articles on the portal offline since there is no option for downloading the article. **Validity/use:** Finds it valid because the portal is funded by DFID. Explains that DFID does a lot of quality checks before publishing an article on its portals. **Downloading:** No evidence of download **Interactive functions:** Sends the article to own email address.		the presentation of the article on the portal very interesting. This is because the portal shows the author, date, cost of project and article is also in a PDF format. 7. Also very impressed with the way the document was prepared. NIG 10 indicates that **"I am impressed with this [online article] it has a title, name of author, an executive summary...I can decide whether I can go ahead or not".** 8. Also happy to note that it is an open access portal. 9. NIB 10 writes the website address of the portal for future use. **Barriers to use:** Finds the portal not detailed enough. For instance does not cover gender issues much. **Validity/use:** See www.scidev.net on the LHS. **Downloading:** Downloads reports in PDF. **Interactive functions:** No impact.

Knowledge of:
Scidev.net: None
Eldis.org: None
R4D: None

Portal challenges:
Poor internet access and erratic power supply.

Use of information:
Shared online: Shares information through the GEM website and emails.
Shared offline: No
Saved: Saves online evidence in Word in a folder.
Repurposed: copies articles from the portal into Microsoft Word and then re-writes it.
Read: Reads relevant articles in detail.

Participant–NIG 11 Category 8

NIG 11 is a journalist in Nigeria. Looks for information from private and public schools in Nigeria. Looks for information on local portals particularly those of the local print media in Nigeria.
Looks for online information using Google search engine. Types the URL of google to get the search engine of Google. Types the sentence "our emphasis is on basic education Ambode" in order to narrow search.
Scrolls up and down the first page of the portal to find information related to the above title. Clicks on the first title on the search result. Takes NIG 11 to the "Vanguard" newspaper online portal. Indicates that the newspaper is of the most reliable source of information for the education sector.
Copies the article and pastes it in word. NIG 11 glances through and indicates that the editor will have to review the article to ensure that the information is credible. NIG 11 then reviews the article to suit the publication's standards. Saves the article in a folder after use.
Ascertains the validity of the online evidence by accessing information from reliable local internet portals. Do not rely much on information outside the country—mainly because they can be controversial and can make their audience stay away from their work.

Example use of evidence:
General search in education.

Uptake:
Journalism has become easier, as a result of internet access. One is able to access information easily for publication. It has also made verification of information easier.

Information behaviour:
Starting: Start online search using Google search engine.
Chaining: Follows links of other local portals on Twitter, Facebook and WhatsApp.
Browsing: Scans through various titles on the first page of the search result.
Differentiating: Downloads and saves only relevant articles. Occasionally prints.
Monitoring: Visits other local internet portals to get new information for their work.
Extracting: Copies information from online sources into Microsoft word for editing and re-writing.
Processing:
Publishes the most credible information in the news Journals.

Assessing validity/use:
- Reliability of local internet portal.
- Do not rely on international sources since they may controversial.

Participant–NIG11	Category 8	part 2

Scidev.net	Eldis.org	R4D
Steps to find specific article: 1. NIG 11 uses the "back" tab to go to the home page—mainly because NIG 11 could not find the home tab. 2. In orders to narrow the search results, NIG 11 types the full title of the specific article (i.e. Big data for development: facts and figures) into the portal. 3. Finds the article on the first page after scrolling up and down twice. 4. Clicks and copies the article.	**Steps to find specific article:** 1. NIG 11 starts looking for the specific article by typing the full title of the article in the search box of the portal and finds **19,814** search results. 2. NIG 11 scrolls up and down the first page but did not find the article. 3. NIG 11 then clicks on the next page and finds the article on the bottom half of the portal. 4. Indicates that the article should have been on the first page of the portal.	**Steps to find specific article:** 1. Types the full sentence of the specific article. Finds the specific article with the title **"Development finance institutions and infrastructure: A systematic renew of evidence for development additionality".** 2. Finds only the article appearing on the first page of the search result. 3. Becomes happy to find the article but was surprised to realise that search result showed only the specific article. 4. Realises that the main title as it presented above has been briefed in the search result. Reads the summary below the article. 5. However, the full title appeared when NIG 11 clicked on the title. Downloads the specific article.
Steps to find information: 1. As a journalist, NIG 11 tries to find online research evidence that is worth publishing. 2. Scrolls up and down. 3. Slowly scrolls up and down. Moves the cursor on the topics to see if there is/are any relevant article related to the topic **education.** 4. Expected to find education as a topic on the home page of the portal. NIG 11 did not find education so decides to find a closely related relevant article that can be published. 5. Finds an article with the title **"Focus on disability: defend girls' sexual health rights"** at the bottom part of the home page of the portal. 6. Clicks on the title. Likes the brief summary	**Steps to find information:** 1. NIG 11 types the website www.eldis.org into Google search box. 2. NIG 11 first impression is that the portal is very slow. 3. Moves the cursor on the portal looking for **education** related evidence on the portal. 4. Types the topic **education** in the search box and finds **8476** search results. Finds the number of search results impressive. 5. Looks for articles which are related to **"education in Lagos"** on the portal. However, indicates that it can be time consuming looking for information on the portal. For instance NIG 11 explained that **"this can be time consuming and imagine you going to a cyber cafe to look for information from this search results. You may have to**	**Steps to find information:** 1. Types the sentence **"How Lagos State can change the face of education in Nigeria"**. 2. Scrolls down the first page to find the relevant article. 3. Finds article as the 6th title on the first page of the search results. 4. Saves article because the article is teacher education related. 5. Clicks on R4D to go back to the homepage. 6. Clicks on "browse by country" and subsequently selects Nigeria and Africa. Finds **2281** search results.

of the article provided for reader. 7. Without typing any word into the search box of the portal, NIG 11 clicks on search and gets a total search result of **10,965.** Finds it laborious to search through all. 8.NIG 11 then type the topic **education** into the search box and get **2,177** results. **Barriers to use:** Finds it difficult to navigate to the home page, and laborious to search through the **10,965** search results. **Validity/use:** Does not rely on international portals because they can be controversial. **Downloading:** No download was undertaken. **Interactive functions:** No impact	**spend a lot of cash to find such result [online evidence]".** **Barriers to use:** Finds the portal slow. This can be expensive if one is working in an internet cafe. **Validity/use:** See www.scidev.net on LHS **Downloading:** No evidence of download **Interactive functions:** No impact realised	Finds the number of search result irrelevant. 7. NIG 11 is more interested in breaking news for the media work. Does not find any relevant article. **Barriers to use:** No barrier identified **Validity/use:** See www.scidev.net on the LHS **Downloading:** Downloads specific article **Interactive functions:** No impact realised

Knowledge of:

Scidev.net: None

Eldis.org: None

R4D: None

Portal challenges:

Poor internet access

Use of information:

Shared online: through WhatsApp, Twitter, and Dropbox.

Shared offline:

Saved: Saves only relevant documents.

Repurposed: NIG 12 copies sections of relevant articles into word and rewrites it to suit desired output.

Read: Reads saved articles. Also reads online.

Participant–NIG 12 Category 4

NIG 12 is a pharmacist with over 20 years' experience. NIG 12 is an entrepreneur who trades in hospital apparatus like dialysis equipment. NIG 12 is working on a document about childhood killer diseases**.**

NIG 12 uses Google search engine for online research evidence search. In order to narrow search results, NIG 12 types in the full title of any document that is important for the work. For instance, NIG 12 types in the title **"current guidelines for the treatment of childhood pneumonia"**. Hurriedly reads through the search results on the front page of the search results. Pays particular attention to the titles. Also reads the brief summaries below the titles.

Finds the first title on the first page more close to the information needed. Clicks on it to download.

Reads the title again and then the summary of the article. Saves the article in a folder.

Copies sections of relevant article and rewrites it to focus on the objectives of the above stated project.

NIG 12 also searches for free reliable biomedical and life sciences journal articles on health portals like PubMed Central (i.e. http://www.ncbi.nlm.nih.gov/pmc/). NIG 12 considers specific portals like PubMed Central when the article is known to be there. Also considers known journal like The Lancet when looking for evidence online. Example NIG 12 was able get a detailed understanding of the Ebola epidemic in West Africa through The Lancet portal.

NIG 12 also considers highly recognised authors and originality of the document (i.e. avoiding plagiarised documents) when searching for a valid online research evidence.

However, NIG 12 is very careful with research evidence from international portals like World Health Organisation (WHO). NIG 12 explains that **"the WHO's handling of the Ebola crisis was poor and so most people in the medical sector in Nigeria don't rely on it much as a source of research evidence".**

Will only pay for articles online if that is the only alternative.

Also uses Twitter and emails to share and also access information from other

Information behaviour:

Starting: NIG 12 mainly starts online information search with Google search engine.

Chaining: Follows links on Twitter to access more information in the form.

Browsing: Scrolls up and down the first page of the search results. Does not go beyond the first page.

Differentiating: NIG 12 downloads and save relevant online research evidence.

Monitoring: Revisits specialised portals in medicine like **PubMed Central** to see whether there is new information relevant to the above mentioned project.

Extracting: Reads full article and then copies relevant sections into word and then rewrites it to suit desired output. After that the document is saved in a named folder.

Processing: Uses information to prepare power point presentations and also acknowledging the author.

Assessing validity/use:

- NIG 12 focuses on well-known authors and originality (i.e. avoids

	colleagues. **Example use of evidence:** NIG 12 is working on a document titled **"Advocacy for the prevention of Childhood killer diseases".** **Uptake:** It has made it easier to look for information from a wider perspective. It is now easier to cross check information. NIG 12 indicates that **"nowadays, no one can deceive me; I am able to cross check information that has been provided to me to see if it is authentic".** NIG 12 has also gained a huge level of respect from his competitors—medical doctors—since NIG12 is able to make meaningful contributions to medical issues as a result of availability of online information. At the organisational level, access to information has become more inclusive.	plagiarised documents) of the research evidence.

Participant–NIG 12 Category 4 part 2		
Scidev.net	**Eldis.org**	**R4D**
Steps to find specific article: 1. NIG 12 types in the specific article with the title **"big data for development: facts and figures"** into the search box of the portal. 2. NIG 12 expected the first line of the search results to be the title of the article but did not find it—gets disappointed because the article was on the fourth line. 3. NIG 12 believes the algorithm of the portal is not good enough. 4. Reads subsequent title to ascertain the nature of search results on the portal. 5. Clicks on the specific article and gets surprised to hear a voice playing in the downloaded article.	**Steps to find specific article:** 1. Types in the specific article **(i.e. Africa Health Strategy 2007 – 2015)** into the search box of Eldis. 2. Finds only **1** search result with a different title—**"Strengthening of health systems for equity and development in Africa".** 3. NIG 12 clicks on the title as indicated above and finds the title "**Africa Health Strategy 2007 – 2015"** as a sub-title of the general title **"Strengthening of health systems for equity and development in Africa".** 4. NIG 12 finds it very inconsistent. 5. NIG 12 does not read the content but downloads the article in its PDF format.	**Steps to find specific article:** 1. Types the title of specific article **(i.e. Monitory health policy successes in the SADC region)** into the search box of R4D. 2. Unable to find the specific article on the first page. 3. Reads the titles and tries to find keyword in the title. Rather finds the keywords in the brief summaries below the titles of the article. 4. Gets discouraged to continue searching since NIG 12 expects the specific article to appear on the first page of the search result. 5. Clicks on the second and third pages of the search results but to no avail. 6. Did not see how the titles were presented on the second and third pages. 7. Asks **"why are the titles only written as report and others document".**
Steps to find information: 1. Moves the cursor through the various topics on the portal. 2. Types **childhood pneumonia** into the search box and then gets 11 results. 3. Decides to narrow the search by typing **childhood pneumonia diagnosis algorithms** but finds zero results. 4. Looks for pneumonia under the topic **Health** on www.scidev.net portal finds nothing on pneumonia. 5. NIG 12 indicates that the sub topics under health are too limited and does not create the space for more options. 6. NIG 12 decides to look for the same article in Google. Finds the article on www.scidev.net on the first page of the search results of Google.	**Steps to find information:** 1. Scrolls up and down the Eldis portal and then reads the various headings. 2. Decides to click on **Global Health.** 3. Does not find anything relevant to work. 4. Types **Childhood pneumonia** into the search box. 5. Again, does not find anything relevant to the project. But rather indicates that the search results are more related to education than Childhood pneumonia or health.	**Steps to find information:** 1. NIG 12 scrolls up and down the portal and then clicks on country. 2. Selects Nigeria and projects. However the search result was a combination of documents and projects. 3. Scrolls up and down to see if there is research evidence of relevance.
	Barriers to use: Finds it difficult to navigate to the home page after searching for evidence online. **Validity/use:** See SciDev.Net on the LHS	

7. NIG 12 later realises that the portal is focused mainly on simple news items on science and development. 8. NIG 12 decides to type **Sustainable Development Goals** into the search box but the search results are not consistent with what was expected. **Barriers to use:** Finds it difficult navigating through the portal. **Validity/use:** Convinced information on the portal is valid because it is a DFID funded portal. **Downloading:** Downloaded document on pneumonia. **Interactive functions:** NIG 12 finds the interactive links such as Facebook, Twitter very useful.	**Downloading:** downloads specific article in PDF format. **Interactive functions:** Finds the Twitter, Facebook links very useful but seldom uses it.	4. Identifies research evidence on family planning and reproductive health to be of interest. **Barriers to use:** the search results on the specific portals are difficult to identify. **Validity/use:** See Scidev.net on extreme LHS **Downloading:** No evidence **Interactive functions:** No evidence

Table 41: Nigeria Research Diary Summary Table

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
NIG1	**find/retrieve**	x	x	retrieved info stored digitally/ contacted a person	email from a person/ contacted a person	x	email from a person/ contacted a person	x	x	x
	why source	x	x	Had a deadline to review the paper by close of work today, and shares a brief on lessons learnt from the findings to work colleagues.	Reviewing the information and giving feedback is a work deliverable and part of sharing lessons learnt on the programme	x	For a few new research areas, I used google search and read up. Mainly worked with internal programme documents today.	x	x	x
	validity	x	x	The authors are known writers, and I was part of the sample design and data collection. The findings are replicable and valid based on the tables and frequencies.	Found the information reliable and valid since I was part of the data collection, entry, editing, and analysis. The writers are known to me as colleagues.	x	Information was clear, precise, and well articulated when compared to the table values and method of analysis..	x	x	x
	use	x	x	read it	read it	x	read it	x	x	x
	type/purpose	x	x	I spent time reviewing a DFID report whih evaluates the programme using mixed methods. Added	Reviewed an internal DFID report shared by a research organisation, made comments,	x	Received comments on a research report from colleagues and compiled them for authors review	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
				comments to the document and sent to the authors.	collated colleagues comments, and sent back to the authors of the report.					
	influence									
	research evidence	x	x	YES - The report was circulated as a draft by colleagues, and it evaluates the programme using a mic of quantitative and qualitative methods. It has policy recommendations and how best to implement the programme theory of change.	YES -Internal DFID document which uses mixed methods research approach to explain research findings from a survey of 359 schools. The methodology uses random sampling of parents, teachers, and pupils. The qualitative aspect uses purposive selection in the sampling approach. The report uses both an inductive and deductive approach to reach conclusions. // The document was first reviewed for my understanding	x	YES - Mixed methods baseline report. Sent by a sister programme for review by Results & Learning Department.	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
					and learning. Then comments where shared with the authors, since its still a draft document, and allows the authors to finish the write-up.					
	DFID portals	x	x	not used today	???	x	not used today	x	x	x
NIG3	**find/retrieve**	Found: online RPP: R4D Retrieved: other: reviewed a search result from R4D	Found: online RPP: R4D	Retrieved: stored digitally/ remembered	Found: online RPP: R4D	x	x	x	x	x
	why source	Curiosity. Today is also the International Anti Corruption day.	Its a 64 page document, read only the objectives of the report yesterday. Planned to study more and see how better and more transparent election results monitoring may be done in Nigeria	It was relative and showed how the huge numbers of current users of social media in Nigeria may be co-opted to help with the objective (data integrity)	access to credible and verified information	x	x	x	x	x
	validity	I have not done a check to see but a simple glace at the data on the poverty research shows	I was an eye witness to the subject of the research being the 2015 political campaigns and	The figures do not look disproportionate considering the widespread use of smart mobile devices in	Locally available information and scenarios was examined	x	x	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		a correlation with local assumptions on the spread of poverty. The wide of use of social media during the last elections in Nigeria threw up many scenarios that can be exploited to have better and more transparent local elections	elections in Nigeria	Nigeria. few reservations with data collated from blog sites and included in the research. many of such figures are often exaggerated for commercial purposes.						
	use	saved/ read	saved/ read	shared offline	saved/ read	x	x	x	x	x
	type/purpose	1. Multidimensional poverty in Nigeria: First order in dominance approach 2. Monitoring social media content relating to the Nigerian elections . / Checked to see whether the report was in tandem with widely held belief about the spread of poverty in Nigeria. The research ended in 2012 which, New Govt in Nigeria was sworn in 2015	Social Media for Election for Communication and election monitoring in Nigeria. Saved the PDF doc. Further reading later, planning to explore the official use of social media for election results monitoring	Brought it up in a meeting to discuss how to use technology to report events that nay be distorted in official reports. Partisan interests often conflict data integrity in Nigeria	Was looking for a factual report on the corruption issue in Nigeria. The report will guide as to what the causes are and where they are endemic	x	x	x	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		and New policies are being unveiled. 2. looking at the argument for the use of social media to report election results from polling stations to improve transparency.								
	influence					x	x	x	x	x
	research evidence	NO	YES: The prevalence of smart mobile devices meant more Nigerians can be reached through the devices and more Nigerians may be recruited into the electoral process in many ways. one way may be "exit polls" etc	NO	NO	x	x	x	x	x
	DFID portals	R4D	R4D	not used today	R4D	x	x	x	x	x
NIG4	**find/retrieve**	x	Found: used online RPP: www.worlfoundation.org / other: yahoo	x [no response recorded for this question]	Retrieved: other: Materials have been downloaded and saved in the system	Retrieved: other: Already downloaded and stored in the desktop	Found: other: R4D	Found: received in email from a person	Found: received in email from a person Retrieved: had stored on paper	o
	why source	x	It is my normal way of searching for	x [no response recorded for this question]	The information was	The information was crucial	This source is most trusted to me because	Because the information is specifically on	It was gotten from the website	I did not retrieve nor use

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
			materials on the internet when I do not have specific link for what am searching for.		specifically searched for because they relate to what I was doing	to my justification of the need for the establishment of the Research Centre we are proposing	most of the contents posted are of high research quality, reliable and valid.	the area of interest. It is the information provided by the funding agency	of the required organisation	any information today
	validity	x	The first one was from the website of a foundation which can be regarded as an institution or organization. The other is a journal that is institutional based.	x [no response recorded for this question]	The source was UNESCO and an institutionally based journal which made them to be reliable	They are from established institutions	The information are well researched with credible data coupled with cases, plus the scientific arguments on technology transfer.	it was downloaded from the institution's website and it was document from the institution	The institution is a recognized one	x [no response recorded for this question]
	use	x	saved/ read	o [did not do anything with info today]	other: I read it, save and print it. I also use some data in it to add to the work i was doing	repurposed	repurposed	saved	other: We had a meeting in which the document was used.	o
	type/purpose	x	was reviewing a proposal on male inclusion. I wanted to find out what had taken place before. Got materials from World Foundation website. I needed more time to digest the contents. Made brief note	x [no response recorded for this question]	It is a paper presented at UNESCO Institute of Statistics Workshop on Education Statistics in Anglophone Countries, Windhoek 2010. I was looking for issues in education in	It was a continuation of yesterday's work. I added statistics from the material to the proposal for the establishment of a Faculty Research	I made some notes from the information I searched.	I was informed of call for research funding. wanted to have detailed information of the call and the requirements. Downloaded and printed it for further usage	We are to develop a study on local decision making and governance, We used it as a platform for the / development of our research	o

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
			and save others. I also got information on from Australian Education Review.		Nigeria, I used application and admission ratio into Nigerian higher institutions. It is to beef up the rationale for the establishment of a research centre in the Faculty. I also used an Australian Education Review Journal article: Early Childhood Education Pathways to quality and equity for all children. This is to locate research needs areas in childhood education. The work is still ongoing.	Centre I was fine tuning. I also reprocess another document for inclusion of part of it into the same material.			proposal	
	influence	x								o
	research evidence	x	NO	NO	YES: They were data based	NO	YES: Technology transfer and sustainable rural development	NO	YES: In the website of the organization concerned. It was for the purpose of finding out	o

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
									what have been done an what gap our study is supposed to fil	
	DFID portals	x	not used today	not used today	not used today	not used today	R4D	not used today	not used today	o
NIG8	**find/retrieve**	Retrieved: had stored digitally	o	Found: used online search engine/ used online RPP: www.unesco.org Retrieved: repeated previous online search/ had stored digitally/ remembered	o	Found: used online search engine Retrieved: repeated previous online search/ had stored digitally	x	o	o	Found: used online search engine Retrieved: repeated previous online search/ had stored on paper/ had stored digitally/ remembered
	why source	To complete the Performance rating of schools on EKO Project	NOT APPLICABLE	Was able to juxtapose with the concept we have in Lagos State EMIS	N/A	Was able to get enough details of occupations and A - Z of Occupation with pictorial view /	x	N/A	N/A	It goes with my work plan
	validity	Reading through the sources of the information	NOT APPLICABLE	Made references to all reference by reading through the original	N/A	No	x	N/A	N/A	The authors are renowned so

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
				documents and standard documents that were validated by UNESCO, DFID, WORLD BANK						there is no doubt regarding validity of data. Their findings have been testified and acknowledged. Secondly, these findings have been used widely for theoretical and conceptual understanding. So, my work today was simply to identify the share of each author for the activities collected and used
	use	o	o	saved/ read/ repurposed	o	saved/ read/ repurposed	x	o	o	saved/ read

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
	type/purpose	Not Applicable	o	The use of EMIS platform to generate student performance report. / Made note from some reports from UNESCO site on EMIS factors	N/A	Made notes on the topic Occupation / Search for A - Z of Occupation Using Pictorial view / Made notes on A - Z of Occupation /	x	o	N/A	Searched data regarding technological capability building. The intention was to find out the share of each author with regard to innovative activities that form he levels of technological capabilities of firms. I searched and saved data. Then I am comparing with the table I constructed where the

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
										activities are filled in and find out which activities were drawn from which author.
	influence	o	o		o		x	o	o	
	research evidence	o	o	NO	o	NO	x	o	o	YES: Please refer to my comments above
	DFID portals	o	o	???	o	not used today	x	o	o	not used today
NIG12	**find/retrieve**	online search engine	online search engine	o	o	o	o	talked to person	x	x
	why source	In preparation for a Television interview on the 2016 Nigerian Budget and challenges of financing immunisation in Nigeria.	Wanted to point colleagues to a credible source of information on health facilities in Nigeria.	o	o	o	o	To open discussions on modifying a program objective/strategy based on new research evidence.	x	x
	validity	Multiple sources with same/similar information including the WHO website.	It is a Government website.	o	o	o	o	Reliability determined based on the credibility of the reporting organisation.	x	x
	use	saved it	shared offline	o	o	o	o	shared offline	x	x
	type/purpose	In preparation for a Television	Highlighted the existence of a	o	o	o	o	x [no response recorded for	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
		interview, printed information about Nigeria's Immunisation Schedule and the replacement of at least one of the three Oral Polio Vaccine with Injectable inactivated Polio Vaccine.	major health resource portal (http://nmis.mdgs.gov.ng/) during a meeting of 8 major Nigerian Civil Society Organisations "Partnership for Advocacy in Child and Family Health".					this question]		
	influence									
	research evidence	YES - The research evidence was a news publication on the World Health Organisation website (www.afro.who.int/en/nigeria/press-materials). / The document helped to debunk a claim, (in an online blog) that Nigeria was still promoting Oral Polio Vaccine instead of switching to the new Injectable inactivated Polio Vaccine.	NO	o	o	o	o	YES - The research evidence was the result of a study done to determine the prevalence of antibiotic treatment in some Nigerian States, and to determine the attitude and behaviour of Healthcare Providers regarding the use of the antibiotic(s) in the treatment of Childhood Pneumonia. The study outcome was reported at a peer group meeting in November 2015. Information on	x	x

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
								the research evidence was provided at a meeting with a project sponsor, to justify the need to modify advocacy objectives in the light of the research evidence.		
	DFID portals	not used today	o	o	o	o	o	not used today	x	x

KEY			
Code	**Meaning**	**Notes**	**Data source**
x	**missing data**	did not return this journal, or did not respond to questions on this topic	Qualtrics shows no journal returned
o	**null data**	returned daily journal with positive responses to all three options showing that participant did not find, retrieve or use information on this day	Daily Q2_7 "I did not do anything with data today" AND Daily Q4_17 "I did not find any information today" AND Daily Q5_7 "I did not retrieve information today"
find/retrieve	**find/retrieve information**	positive response to at least one question about finding info or retrieving info	At least one response to Daily Q4_2 to Q4_10 (find) OR Daily Q5_1 to Q5_6 (retrieve)
why source	**why this source?**		Daily Q6
validity	**How assess validity?**		Daily Q7
use	**use information**	positive response to at least one question about using info	At least one response to Daily Q2_1 to Q2_6
type/purpose	**type of info and purpose**	descriptive	Daily Q3 (type of info and purpose)

Participant	Evidence Used	Daily01	Daily02	Daily03	Daily04	Daily05	Daily06	Daily07	Daily08	Daily09
research evidence	**used research evidence**	positive response to question about research info		No to Daily Q8: not used	OR Yes to Daily Q8: brief description					
DFID portals	**used R4D, SciDev or Eldis**	stated that used at least one of these portals in the reflective journal		Reflective Q2_1, Q2_2, Q2_3						
influence	**influence others using information**	indicates that participant will use information to influence others in making policy etc.		Reflection Q5 (will it be used and how) what extent (0 to 100)			Reflection Q4_1 to			

B.4 Heuristic evaluation outcomes (second exercise)

The heuristic evaluation technique was used to assess the portals a second time, after the in-country studies had been completed. This section contains the results of this evaluation. The first three tables contain the completed heuristic evaluation form for each of the portals. The next table summarised the issues encountered by the participants, for each portal.

Table 42: Data collection and analysis form for heuristic evaluation: SciDev.Net

Website: SciDev.net **Website URL: www.scidev.net** **Web browser used: Internet Explorer 11**			
Expert evaluator's name: David Morse		Session date: 8th January 2016	
Key issues *Article selection*: articles appear in some searches and then don't appear in others. This doesn't engender confidence because you are left wondering what you are missing and what the website 'knows' that it isn't telling you.			
Relevant task or task step	**Heuristic being assessed**	**Usability defect description**	**Expert evaluator's comments regarding the usability defect**
Where am I now? Where can I go next? Search for relevant information *Q: Are users kept informed about what is going on?* *Q: Is appropriate feedback provided within reasonable time about a user's action?*	**Visibility of system status** The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.	The website doesn't have an obvious Home button, although the logo works as one.	Using the logo as a home button is an often used convention, but there would be space to put a 'Home' button in the top menu.
Is the language used on the website familiar to its intended audience? Website trustworthiness – do I trust the information on the website? Is it clear who owns and maintains the website?	**Match between system and the real world** The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system oriented terms. Follow real-world conventions, making information appear in a natural and logical order.	There are some abbreviations in the menus that I didn't know, such as SDGs. Others, such as R&D and ICTs are sufficiently well known that their use is not a problem. Series (in Browse type > Series) I didn't understand.	Does the website need a glossary of terms and acronyms in the help section of the website? Or should SDGs be removed from the menus or expanded? Is a series intended as a special topic,

Website: SciDev.net **Website URL: www.scidev.net** **Web browser used: Internet Explorer 11**			
Website name in search engine search. Is it clear what the website contains and who produces the website? *Q: Is the language used at the interface simple?* *Q: Are the words, phrases, and concepts used familiar to the user?*		This isn't a usability issue but is an issue about how up to date the website is. The most recent article in the Agriculture section of the website that refers to Ghana is three years old.	
Are there clear ways to navigate around the website (e.g. is there a home button?) *Q: Are there ways of allowing users to easily escape from places they unexpectedly find themselves in?*	**User control and freedom** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Supports undo and redo.	When you select a section from the menu e.g. Enterprise > Data, the website says 'You are looking at articles about Data' and highlights Enterprise. I was looking for a breadcrumb trail of the form 'Enterprise > Data' Interestingly, a breadcrumb trail appears in blue once you select an article, below the main grey menu bar. It is not clear that the words in the grey menu bar are clickable. While 'Browse type' addresses the nature of the content whereas the other options in the menu address the subject matter, there is an argument for giving all the words the same colour background.	The use of highlighting is to indicate your position in the website is OK once I had realised that is what is going on.
Does the website follow discipline conventions? *Q: Are the ways of performing similar actions consistent?*	**Consistency and standards** Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.	Search doesn't seem to allow for different spellings: 'sulphur' versus 'sulfur' for example, but does appear to allow for stemming through wildcards, so 'sulph*' works. 'Browse type' in the menu is called 'content' in the site map and in the 'Refine by'menu block it appears as 'Type'.	In an ideal world, search that allowed for different spellings would be good. 'Content type' might be clearer than 'Browse type'. But it would be good if a consistent term was used.
Incorrect or meaningless search terms *Q: Are error messages helpful?* *Q: Do they use plain language to describe*	**Help users recognise, diagnose, and recover from errors** Error messages should be expressed in plain language (no codes), precisely indicate the problem, and	Another misunderstanding. When looking at the Global edition, I had assumed that articles below the banner heading 'Latest on xxx from our other editions' were from those	Perhaps remove the block completely or make the 'Latest on …' heading more clearly part of the scrolling block.

Website: SciDev.net Website URL: www.scidev.net Web browser used: Internet Explorer 11			
the nature of the problem and suggest a way of solving it?	constructively suggest a solution.	other editions. But no, articles below that banner relate to the subject at the top of the page.	
Data entry (e.g. form filling) during search or contact me *Q: Is it easy to make errors?* *Q: If so, where and why?*	**Error prevention** Even better than a good error message is a careful design which prevents a problem from occurring in the first place.	I don't have a clear understanding of what the website does and how it works. I had assumed that different 'Editions' of the website had a geographical focus, but that doesn't appear to be the case. 1) A search for 'COP21' in the South Asia region yielded an article on the Paris climate change agreement (http://www.scidev.net/south-asia/climate-change/opinion/paris-agreement-green-investments-COP21.html) This article didn't appear in the same search on the Global website. 2) If I looked in the South Asia section of the website, then searched for an article, e.g. on 'tea' (the drink), there were still articles on tea with a region of Sub-Saharan Africa.	Perhaps a help system that explains how articles are selected according to various criteria: editions,
Website search and customisation (e.g. is there type-ahead in search) *Q: Are objects, actions and options always visible?*	**Recognition rather than recall** Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.	The three bars by 'Browse type' could indicate a menu, but all the other words in this bar have the same functionality, so why don't they have the same icon beside them?	A more common icon to indicate a drop-down menu is some form of arrow pointing downwards, such as the arrows used in the 'Refine by' menu block.
Does the website have any shortcuts for proficient users? *Q: Have accelerators (i.e., shortcuts) been provided that allow more experienced user so to carry out tasks more quickly?*	**Flexibility and efficiency of use** Accelerators -- unseen by the novice user – may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.	It has taken me a long time to realise that the search term can be edited and the magnifying glass icon is clickable. These appear immediately above the search results. So there are two search boxes on the search results page which are not connected since edits to one are not reflected in the other. If I click in the lower of the two search boxes	I am not sure that a second search box is particularly helpful unless it could be made more obvious that the search box on the search results page is a search box.

Website: SciDev.net Website URL: www.scidev.net Web browser used: Internet Explorer 11			
		then the magnifying glass changes to an icon that I don't recognise when the page is viewed in Internet Explorer. A magnifying glass with a cross over it?	
Do search forms contain irrelevant information? *Q: Is any unnecessary and irrelevant information provided?*	**Aesthetic and minimalist design** Dialogues should not contain information, which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.		
Documentation on search – is it provided for advanced searches? *Q: Is help information provided that can be easily searched and easily followed?*	**Help and documentation** Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.	As far as I can tell there is no documentation on how to use the website. Consequently it took me a long time to understand how the website works. I assumed that Browse type was orthogonal to Agriculture, Environment etc. but it isn't because 'Type' (not 'Browse type') appears in the 'Refine by' menu block on the right hand side.	Some reference documentation on how the website works would be helpful.

Table 43: Issues identified by country case study participants: scidev.net

Issue	Issue description	User
Navigation	Couldn't find home page and couldn't find site map	NIG2, GH6
	Does 'Home' refer to home for the website or home for a section (e.g. agriculture)?	TZ1
	'Data' section should be renamed 'statistics'? Presumably TZ1 was expecting raw data in the data section of the website? Also confusion over Data as a primary category; was expecting it to be a subset of other categories such as agriculture.	TZ1
Search	Search engine not tolerant of misspellings; Google is better in this respect	NIG2; TZ3
	Difficulty in finding search box and result in returned search list	NIG4, NIG12, TZ6
	Difficult to search through large numbers of results (doesn't seem to refine result set).	NIG11
	Expectation that search result would be the first in the result set.	NIG11, NIG12, TZ6
	Difficulty in using search – results returned do not match expectations.	GH5, GH12
	Poor functionality of search box. Doesn't allow editing of search terms beyond deleting them using backspace and then re-entering them	TZ1

Issue	Issue description	User
Content	Content of home page can influence perception of coverage. If all articles relate to climate change then this can colour visitor perceptions	NIG2
	Useful to gain overview of new field but suspects articles lack the academic depth and rigour that he/she is looking for.	NIG4
	Likes summaries of articles but expected to have links to full-text.	TZ3
Trust	It's DFID funded so it must be valid and trustworthy.	NIG2, NIG4, NIG12, TZ5
	Doesn't trust international portals	NIG11

Table 44: Data collection and analysis form for heuristic evaluation: Eldis

Website: Eldis.org **Website URL: www.eldis.org** **Web browser used: Internet Explorer 11 and Firefox 43**			
Expert evaluator's name: David Morse		Session date: 9th January 2015	
Relevant task or task step	**Heuristic being assessed**	**Usability defect description**	**Expert evaluator's comments regarding the usability defect**
Where am I now? Where can I go next? Search for relevant information *Q: Are users kept informed about what is going on?* *Q: Is appropriate feedback provided within reasonable time about a user's action?*	**Visibility of system status** The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.	What is the ordering of the articles in the blog? It doesn't seem to be alphabetical nor date order. Likewise, what is the order of articles in a search? Presumably some sort of relevance order? The breadcrumb trail for search from the home page says 'Home' whereas most other pages work as expected, so when on a country page, it says 'Home > Country and Region Profiles > Ghana'.	For a blog, date order would be a suitable sort order. For the search, it would be useful if you could select the sort order. Typically relevance and date order would be most usual.
Is the language used on the website familiar to its intended audience? Website trustworthiness – do I trust the information on the website? Is it clear who owns and maintains the website? Website name in search engine search. Is it clear what the website contains and who produces the	**Match between system and the real world** The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system oriented terms. Follow real-world conventions, making information appear in a natural and logical order.	Does search work on full text search whereas article selection through country profiles and topics works on keywords? When I search for 'coffee statistics' I obtained 2172 documents. When I selected the Statistics topic and then searched for coffee, I obtained 3 or 4 hits, or sometimes the error "Element APISEARCH_ORGS.RESULTS.METADATA.	How does search work? It would be really helpful if this was explained in a help system somewhere. Why do you get so many more hits when you search for terms rather than looking through the topic index or searching within a topic? This devalues the topic and country profiles since you get so many more hits when you search. But you get so many hits that it is difficult to work through them and

Website: Eldis.org Website URL: www.eldis.org Web browser used: Internet Explorer 11 and Firefox 43			
website? *Q: Is the language used at the interface simple?* *Q: Are the words, phrases, and concepts used familiar to the user?*		TOTAL_RESULTS is undefined in LOCAL." Is a search of coffee statistics actually searching coffee AND statistics or coffee OR statistics? I suspect the latter since the number of hits goes up as you add words to the search, not down.	there are no tools to help you rank the results.
Are there clear ways to navigate around the website (e.g. is there a home button?) *Q: Are there ways of allowing users to easily escape from places they unexpectedly find themselves in?*	**User control and freedom** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Supports undo and redo.	I liked the country profiles display, which I got to by selecting Ghana in the 'Africa South of Sahara' in the Regions and Countries panel on the right hand side of the home page. But, I thought the 'BLDS Print Collection' was in the wrong place and should be moved to the left or right hand columns since it formed a barrier to reading the 'Latest from ELDIS online' panel. The latest what – articles, news, …? On what subject – any topic or restricted to Ghana? Searching for 'coffee' gave me one article on coffee in Ghana. Fair enough. But how do I get back to the country page? Selecting Ghana in the drop-down Country profiles panel didn't do anything.	
Does the website follow discipline conventions? *Q: Are the ways of performing similar actions consistent?*	**Consistency and standards** Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.	Resource guides and topics appear to be the same thing.	Perhaps call them both topics, or even topic guides, to get the word 'topic' in both?
Incorrect or meaningless search terms *Q: Are error messages helpful?* *Q: Do they use plain language to describe the nature of the problem and suggest a way of solving it?*	**Help users recognise, diagnose, and recover from errors** Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.	There is a broken link to the 'Adaptation Learning Mechanism'. This appears when you select 'Country profiles' (top menu), then a country (right hand drop-downs), then 'Climate change' within the right hand topics menu.	The website went down while I was using it. The last error message I saw was a 503 error. I hope it wasn't anything that I did.
Data entry (e.g. form filling) during search or contact me *Q: Is it easy to make errors?* *Q: If so, where and why?*	**Error prevention** Even better than a good error message is a careful design which prevents a problem from occurring in the first place.	There is a bug in the display on my supplied laptop and web browser (Windows 7; Internet Explorer 11). The Internet Explorer menu bar is repeated in grey below the top bar. The repeated menu bar is active; the	While Internet Explorer isn't an up-to-date browser, it would be nice if the website worked with older browsers as well as new.

Website: Eldis.org Website URL: www.eldis.org Web browser used: Internet Explorer 11 and Firefox 43			
		top one is not. The repeat obscures the Eldis logo and the language selection drop-down. Setting 'Compatibility view' in Internet Explorer appeared to fix the problem.	
Website search and customisation (e.g. is there type-ahead in search) *Q: Are objects, actions and options always visible?*	**Recognition rather than recall** Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.	How do I unselect a topic within a country profile? Suppose I select a country profile through the top menu of Kenya. I then select a topic of 'Agriculture and food'. I can select other topics within the topic menu, but clicking on All key topics does nothing. The only way to deselect a topic is to click on the country name in the topics menu.	The website appears to be remembering things without telling the user that is what they are doing. This can be helpful but it would be better to make this visible rather than assuming that the user will remember or work it out for themselves.
Does the website have any shortcuts for proficient users? *Q: Have accelerators (i.e., shortcuts) been provided that allow more experienced user so to carry out tasks more quickly?*	**Flexibility and efficiency of use** Accelerators -- unseen by the novice user – may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.	The 'latest news' page is too long. Is there any news before 12 April 2013 or does latest news mean 'all news'?	Break the news page into chunks, displaying, say, 20 articles per page, with a 'More', or 'Older news link to display the next set of news items.
Do search forms contain irrelevant information? *Q: Is any unnecessary and irrelevant information provided?*	**Aesthetic and minimalist design** Dialogues should not contain information, which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.	It is normal to have headers and footers to websites. The footer on the Eldis website is larger than it needs to be. Could some of the links be moved to other pages, or could the space be used more efficiently in some way? Could the 'Reporter email bulletins' and the 'Edited by' blocks be moved elsewhere? The Search box jumps around between the home page and other pages. It would be easier to locate if it was always in the same section of each web page.	Could the footer be reformatted, so the 'About' text just links through to the About Eldis page (which, after all, is a separate link. Could the Site and Engage menus be flattened or shortened so that they didn't take up so much extra space. Since the 'Subscribe' button takes you to a page which invites you to fill in the 'Reporter email bulletins' box, you don't really need both on each page.
Documentation on search – is it provided for advanced searches? *Q: Is help information provided that can be easily searched and easily followed?*	**Help and documentation** Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too	The website remembers topic settings between country profiles within a region but not between regions. If I select a country profile (e.g. Tanzania) then select a topic (e.g. Gender) if I select a different country profile such as Nigeria	A manual would be useful, rather than trying to figure out search behaviour by trial and error.

Website: Eldis.org Website URL: www.eldis.org Web browser used: Internet Explorer 11 and Firefox 43		
	large.	using the drop down below the map (as opposed to clicking on the Country profiles item in the main menu) then I will be presented with the articles on the Gender topic for Nigeria. However, if I do choose the country using the Country profiles top menu and then the country, the topic will be reset back to All key topics.

Table 45: Issues identified by country case study participants: Eldis

Issue	Issue description	User
Navigation	Difficult to navigate to the home page after searching for information online	NIG12
	'What is Eldis' link should have been at the top of the page, not the bottom	NIG2
	Difficult to navigate the website and find recent articles	TZ8
Search	Difficult to search through large numbers of results; returns far too many results	GH1, GH6, NIG2, TZ3, TZ8
	Confused by the documents and organisations tabs in search results	GH1
	Expectation that search result would be on the first page of the result set	NIG11, TZ8
	Finds the number of search items returned encouraging – indication that there is a lot of [presumably relevant] information in the portal	NIG4, NIG11, TZ6
Content	Concerned about out of date country information	TZ1
	Likes summaries of articles presented	NIG2, NIG4
Trust	Its DFID funded so it must be valid and trustworthy	NIG2, NIG4, NIG12, TZ6
	Portal often not available.	GH5, GH12, NIG4, TZ5
	The portal can be slow (time is money, particularly in Internet cafes)	NIG11, TZ1

Table 46: Data collection and analysis form for heuristic evaluation R4D

Website: R4D **Website URL: http://r4d.dfid.gov.uk/** **Web browser used: Internet Explorer 11**			
Expert evaluator's name: David Morse		Session date: 10th January 2016	
Relevant task or task step	**Heuristic being assessed**	**Usability defect description**	**Expert evaluator's comments regarding the usability defect**
Where am I now? Where can I go next? Search for relevant information *Q: Are users kept informed about what is going on?* *Q: Is appropriate feedback provided within reasonable time about a user's action?*	**Visibility of system status** The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.	The search other sites tab in simple search is really useful. But the only way to find out which websites are being searched is to look at the link. It would be useful if the source website could be displayed as part of the summary information for each search result. If there were options to group results by website, sort by relevance, date of the source, and so on, that would be powerful too.	A list of external sites that are searched would be useful, rather than just stating that they are chosen by the R4D editors.
Is the language used on the website familiar to its intended audience? Website trustworthiness – do I trust the information on the website? Is it clear who owns and maintains the website? Website name in search engine search. Is it clear what the website contains and who produces the website? *Q: Is the language used at the interface simple?* *Q: Are the words, phrases, and concepts used familiar to the user?*	**Match between system and the real world** The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system oriented terms. Follow real-world conventions, making information appear in a natural and logical order.	The usage dashboard may not be of interest to many users. However, I struggled to understand what it was telling me. I also found the Browse by date box difficult to use since it didn't fit within the window embedded within the web page. Whenever I clicked on the scroll bars, the Browse by date box disappeared. I also found the idea that I could set Browse by date to 'next three days' (or weeks, or years) confusing. What was this telling me?	While the usage dashboard is important to R4D for monitoring and reporting purposes, I am not sure that it needs to be visible to external users. I would have thought summary reports and highlights would be sufficient, placed on a sub-page of an About us page that describes the website and R4D.
Are there clear ways to navigate around the website (e.g. is there a home button?) *Q: Are there ways of allowing users to*	**User control and freedom** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having	Simple search doesn't have different ways to order the research results.	When many results are returned (as they often are with simple search) ordering the results in different ways can help to identify

Website: R4D Website URL: http://r4d.dfid.gov.uk/ Web browser used: Internet Explorer 11			
easily escape from places they unexpectedly find themselves in?	to go through an extended dialogue. Supports undo and redo.		results of interest.
Does the website follow discipline conventions? *Q: Are the ways of performing similar actions consistent?*	**Consistency and standards** Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.	It would be nice if the display of search results was tailored by the type of document displayed. If I select 'document' in the Browse by radio button, the display of search results has the same options for sorting results: Title, Project start date, Project cost, and so on.	Perhaps the number of options for sorting search results could be extended, to include, for example, the author, or the place of publication? This is relevant if you select a Document type of Journal article in the Refine your search box.
Incorrect or meaningless search terms *Q: Are error messages helpful?* *Q: Do they use plain language to describe the nature of the problem and suggest a way of solving it?*	**Help users recognise, diagnose, and recover from errors** Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.	In the browse by country option, I selected browse for documents, in country Nigeria, region Australia and New Zealand. The search documentation explained that this would return documents tagged with 'both the country and the region selected'. There were 748 documents. The 'Refine your search' box says that there are 2 documents.	If you select either a country, or a region then Refine your search appears to report the correct number of documents. If you select both, then it appears not to work, reporting many fewer documents than are found.
Data entry (e.g. form filling) during search or contact me *Q: Is it easy to make errors?* *Q: If so, where and why?*	**Error prevention** Even better than a good error message is a careful design which prevents a problem from occurring in the first place.	The website asks you to see gov.uk for terms and conditions. This website says 'GOV.UK is maintained for your personal use.' I think I am using R4D in my professional capacity so does my use of the website fit these terms and conditions?	
Website search and customisation (e.g. is there type-ahead in search) *Q: Are objects, actions and options always visible?*	**Recognition rather than recall** Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.	The menu in the left-hand margin is potentially confusing because it groups browse and search operations – which are active things that I would want to do with the website every time I visit it, with other, less common activities such as contacting R4D or reading about Open data. There is an 'Advanced search' link in the menu but there isn't a simple search in that menu. You have to remember that simple search is on the home page of the website, every time you decide that an advanced search is more complex than you need.	As part of a refresh of the website, I would break up the left hand menu block, potentially placing some of the links in to headers or footers for the website. I think there should be a separate link to the simple search in the search menu, even though the designers might choose to leave the simple search on the home page of the website.

Website: R4D Website URL: http://r4d.dfid.gov.uk/ Web browser used: Internet Explorer 11			
Does the website have any shortcuts for proficient users? *Q: Have accelerators (i.e., shortcuts) been provided that allow more experienced user so to carry out tasks more quickly?*	**Flexibility and efficiency of use** Accelerators -- unseen by the novice user – may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.	There isn't an easy to find, obvious shortcut to the home page of the website. I kept clicking on the DFID logo in the top left of the website, which takes you to the DFID home page. The UKAid logo at top right is not clickable.	A link to the home page in a menu bar at the top of the web page would be helpful. The 'Research for development (R4D) database' link in the menu to the right of the page isn't obvious enough.
Do search forms contain irrelevant information? *Q: Is any unnecessary and irrelevant information provided?*	**Aesthetic and minimalist design** Dialogues should not contain information, which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.	The website feels dated, as though it could do with a refresh. (Since the source HTML has CSS style sheets for Internet Explorer 6 and 7, it is.) On look and feel alone, it compares unfavourably with SciDev and Eldis. Parts of the website are dated. The systematic reviews page http://r4d.dfid.gov.uk/SystematicReviews.aspx states that 'DFID are currently commissioning 10 new systematic reviews' and gives a link for more information. The call for reviews closed in October 2013.	A refresh of the website so that it looks more modern would help overcome any impression of datedness.
Documentation on search – is it provided for advanced searches? *Q: Is help information provided that can be easily searched and easily followed?*	**Help and documentation** Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.	There is a manual on search and there is a link to it from the home page of the website and the Advanced search page.	The manual is very useful, explaining not only how to perform searches, but also where to find different types of search on the website. If only other websites provided such a manual too. Since there is a manual, I think there ought to be a separate help section of the website, explaining what it is for, who set it up and how to use the website. This could include some of the links in the menu on the left hand side of the page.

Table 47: Issues identified by country case study participants: R4D

Issue	Issue description	User
Navigation	Couldn't find home page because there isn't a button labelled 'Home'	GH4, GH6, NIG2, TZ3, TZ6
Search	Advanced search is confusing, too complex and difficult to use	GH1, GH6, TZ1
	Simple search gave an error	GH1
	Difficult to search through large numbers of results	GH5, GH6, GH12
	Expectation that search result would be the first in the result set	NIG2, NIG12
	Search results on specific portals (presumably when searching other portals in simple search) are difficult to identify	GH12
Content	Not clear what the portal is for (does it contain calls for proposals or finished projects?)	NIG2, NIG4, TZ6
	Likes availability of full-text in different formats	NIG2
Trust	It's DFID funded so it must be valid and trustworthy	NIG4, NIG12
	Focus on peer-reviewed articles is valued	GH12
	Likes the portal, it is simple and rich in resources	TZ3

Issues identified in heuristic evaluation that were supported by case study findings:

This section summarises the principal issues and features of the three portals that were examined in the user profiles. Issues have been included in the list if they were raised by several users. I included myself as one of the users (flagged as UK) so if it is an issue that I encountered or thought important, which was corroborated by other users in the user profiles then I have listed it below.

The issues are summarised by portal. The host country of the users that identified the issue has also been identified.

Table 48: SciDev

Issue	Issue description	Country
Navigation	Users couldn't find the home page and they couldn't find a site map to help them find the home page	Ghana, Nigeria
	The website doesn't have an obvious Home button, although the logo works as one. I didn't look for a site map to help me find the home page since I tried clicking on the logo and that worked. In the absence of a Home button I would try clicking on the logo. Site maps can be useful if they are complete and up to date, but they are a roundabout way of returning to the home page.	UK
Search	**Search is not tolerant of misspellings. The search engine was compared unfavourably with Google which is better in this respect**	Nigeria, Tanzania
	I observed that search doesn't seem to allow for different spellings: 'sulphur' versus 'sulfur' for example, but does appear to allow for stemming through wildcards, so 'sulph*' works. Search worked as expected so I didn't see enhancements such as tolerance of misspellings and variant spellings to represent significant usability problems.	UK
	Users encountered difficulty in finding the search box on the portal (there are two, although it is not that obvious that the one above the search result set is actually a search box)	Nigeria, Tanzania
	I didn't have any problems in finding the search box in the top menu bar of the website since this is where I would normally expect to find a search box. However, it did take me a long time to realise that the heading which forms the title of the search results is in fact an editable search box. I assumed it was simply a heading and the magnifying glass icon was a reminder that this was a search results page, not a search button. Looking at the two search buttons, one is white on blue, the other is reversed. Perhaps if they were both white on blue, that would give a better clue that the heading actually forms an editable search box. And/or putting the heading in a more obvious search box?	UK
	It is difficult to search through large numbers of results (the portal doesn't seem to refine the result set).	Nigeria
	Agreed. I added the observation that SciDev doesn't seem to refine the result set since the portal doesn't provide easy mechanisms for searching within the results. Apart from the options to refine by topic (presumably keywords), the only way forward seems to be to rerun the search with more search terms added.	UK
	It was expected that the search target would be the first item in the result set, or at least towards the top of the first page of results.	Nigeria, Tanzania
	This is a reasonable expectation if you are searching for a very specific article and include sufficient search terms that the search result will rise to the top. However, I	UK

Issue	Issue description	Country
	didn't expect this, but then I tended to search by subject rather than for a particular article.	
	The results returned by the searches do not match users' expectations.	Ghana
	Agreed. A search on 'coffee' yielded 91 records (I was really surprised it was so low). At the bottom of the first page of results was an article 'Is it necessary to teach poor kids to code?' (http://www.scidev.net/global/education/scidev-net-at-large/teach-poor-kids-code-writing.html) which I was surprised was included in the search result set. However, the third paragraph of the article contains the phrase 'after a chat over coffee later'. This is the only mention of the word coffee in the article. That explains why the article was included in the result set but it really wasn't what I was expecting.	UK
Trust	**The portal is DFID funded so the information (articles) it hosts must be valid and trustworthy.**	Nigeria, Tanzania
	In my evaluation I wrote 'I don't have a clear understanding of what the website does and how it works.' I agree that it is important that what the portal holds must be valid and trustworthy, however I don't think this necessarily follows from who the funder is.	UK

Table 49: Eldis

Issue	Issue description	Country
Search	**It is difficult to search through large numbers of results. The portal returns far too many results.**	Ghana, Nigeria, Tanzania
	Agreed, the portal does return a lot of results. Too many? I am less sure that this is an issue, provided that the website provides mechanisms for refining the results, preferably through mechanisms other than simply adding more terms to the search.	UK
	It was expected that the search target would be the first item in the result set, or at least towards the top of the first page of results.	Nigeria, Tanzania
	Agreed, I was expecting relevant results to be towards the top of the first page of results. This is important for inspiring confidence and for encouraging people to use the website.	UK
	Conversely, users find the number of search items returned to be encouraging. This indicates that there is a lot of [presumably relevant] information in the portal.	Nigeria, Tanzania
	Agreed. I took a usability perspective on this so I was concerned with features and facilities for reducing the number of results returned.	UK
Content	**Users like the summaries of articles presented.**	Nigeria
	Yes, this is a very useful feature. I should have appreciated the usability aspects of this but I note that the summaries are truncated and there is no 'More …' button to see the rest of the summary.	UK
Trust	**The portal is DFID funded so the information (articles) it hosts must be valid and trustworthy.**	Nigeria, Tanzania
	For me, it is the number and types of articles that is key to instilling a sense of quality, rather than the funder. But I agree that there is a brand or reputational 'transference' from DFID to the projects and portals it funds.	UK
	The portal is sometimes not available.	Ghana, Nigeria, Tanzania
	I agree, availability is important. And for portals that have an international audience, maintaining availability outside UK core working hours is important too. I had problems with lack of availability when doing my evaluation so I had to switch	UK

Issue	Issue description	Country
	to another website.	
	The portal can be slow (time is money, particularly when accessing the website over a paid-for connection such as in an Internet cafe).	Nigeria, Tanzania
	Agreed, another important point. Slow websites are frustrating to use and tend to discourage people from using them. I noticed that sometimes the website could be slow (and also unavailable).	UK

Table 50: R4D

Issue	Issue description	Country
Navigation	**Users couldn't find the home page because there isn't a button labelled 'Home'. (You have to click on the 'Research for Development (R4D) database' label at the top of the menu in the right hand panel.)**	Ghana, Nigeria, Tanzania
	Agreed. This confused me too. I couldn't work out how to get back to the home page and clicking on the DFID logo takes you to completely the wrong place. [Actually it is the *left*-hand panel.]	UK
Search	**The advanced search is confusing, too complex and difficult to use**	Ghana, Tanzania
	Agreed the form is quite complex and potentially daunting for first-time users. But at least there is an information button for most of the fields and help sheets on the website. I assumed that you didn't have to fill most of the fields in to use the form. If you do (and I saw no evidence that you did) then there are too many options for a search form.	UK
	Simple search occasionally resulted in errors.	Ghana
	I saw no evidence of this (apart from wondering why some results had been returned, which is true of any website).	UK
	It is difficult to search through large numbers of results	Ghana
	Agreed, it is, which is why facilities for refining the search results, including faceted search are important.	UK
	It was expected that the search target would be the first item in the result set, or at least towards the top of the first page of results.	Nigeria
	Agreed, I was expecting to find results relevant to my searches to be towards the top of the first page of results.	UK
	Search results on specific portals are difficult to identify (when searching other portals using the simple search).	Ghana
	Very occasionally it was difficult to find an article having followed a link from the R4D website. However, I found the ability to search other website portals to be really impressive and useful feature of the R4D website.	UK
Content	**It is not clear what the portal is for (does it contain calls for proposals or finished projects?)**	Nigeria, Tanzania
	I can understand this complaint. The home page of the website gives some idea of its contents but it is only by looking at some of the search results, and the options on the advanced search page that I began to understand to what the portal gives access.	UK
	Users like the availability of full-text in different formats	Nigeria
	This I hadn't noticed, but having looked, typically I could only find full text in one format. The document or project record could be downloaded in multiple formats though. Full text, if it is available, is a good and worthwhile addition to the website though.	UK

Issue	Issue description	Country
Trust	**The portal is DFID funded so the information (articles) it hosts must be valid and trustworthy.**	Nigeria
	Again I place greater priority to the quantity and quality of the documents on the portal rather than the funding agency, although the latter does have an influence.	UK
	The focus on peer-reviewed articles is valued by users	Ghana
	Yes, agreed. The fact that R4D flags which articles are peer reviewed (and by implication, which are not) is very helpful.	UK
	Users like the portal, it is simple and rich in resources	Tanzania
	I agree. I liked the portal too, for the same reasons. But simple in looks rather than functionality since the advanced search is very flexible.	UK

David Morse. January 2016; updated February 2016.

Appendix C. The Value for Money Assessment

C.1 VFM Assessment Plan

Table 51: VFM Assessment Plan

	VFM Question	Sub-questions		Sources	Methods/ criteria
Economy	1. Are there effective procurement mechanisms for controlling costs and quality of inputs?	Does procurement follow specified DFID procedures? Is there evidence of negotiation/ adjustment of inputs and prices? Are there any comparisons to be drawn with comparator portals?	***1. Descriptive*** *How are purchases made?* *What are the main cost drivers?* *How and how frequently are costs benchmarked?* ***2. Analytical*** *How could costs be reduced?* *Which costs cannot be reduced/ are beyond your control?* *Can you give examples of cost reductions that have been achieved (without loss of quality)?* ***3. Comparative*** *Do you know of similar projects that could be compared with this project?* *How does the comparator portal do all the above?*	Management views Financial data/meeting notes gathered during inception Additional financial data Sector information gleaned from web Info from Comparator portal interviews	1. Semi-structured interview with Financial Managers 2. Rule of thumb input cost comparisons 3. Short Questionnaire + interview sheets for comparators
	2. Are rates/prices of key cost drivers reasonable?	How do salaries and rates compare with the market? Is there evidence that due care is taken in selecting appropriate staff systems and content?	***1. Descriptive*** *How are salaries/rates of pay determined (and adjusted)?* ***2. Analytical*** *What methods exist to reduce staff costs?* ***3. Comparative*** *How do they compare with rates paid in other parts of your organisation? With other similar organisations?*	Samples of benchmark salary rates, web-hosting prices and content prices	Internet and internal comparison of relevant salaries, web-hosting rates and prices
Efficiency	3. How is efficiency defined?	How are targets set? How relevant are they?	**1. Descriptive** *What constitutes efficiency for a research portal*	Director Project managers	Semi-structured interviews (if feasible

	VFM Question	Sub-questions		Sources	Methods/ criteria
		How stretching are they?	*(and other information supply activities, where relevant?)?*	Financial manager Digital Services manager Additional management information collected on-site Portal logframes, reports, evaluations.	and agreed by team) group session for two-way ranking of components of the project: cost vs. Impact/value
	4. How is efficiency monitored?	How frequently?	*How do you assess/ account for the inherent redundancy and duplication?*		
	5. What management practices are employed to ensure efficiency?	How is duplication avoided? How are less productive activities and services identified? What incentives and competition exist to promote efficiency?	*How are acceptable "hit rates" and other efficiency indicators defined – and levels set?* *What are the greatest obstacles to achieving efficiency and effectiveness?* *Are there examples of changes in ways of working or project activities that made the project more efficient or more effective?*		
	6. Is the programme progressing satisfactorily?	How do programme outputs compare against plans?	**2. Analytical** *How does management increase value/ reduce cost of different components?*		
	7. Is delivery of outputs timely?	Find evidence of efficiency in: (i) starting (ii) progressing (iii) completing delivery of typical outputs	*How does the project make sure it is reaching its core target groups?* *How has the project been modified as a result of learning about results; availability of new methods/technologies etc.?* *How has the project shared ideas with other information projects?*		
	8. How are sector innovations identified?	What do recent innovations indicate about the responsiveness of management to new information? How does this compare with comparator?	**3. Comparative** *Does the project achieve different efficiencies in different geographical areas or with different types of material?* *Are there obvious comparisons with other portals in terms of seeking out efficiencies?*		
Effective-ness	9. How is effectiveness defined?	How is portal use and usability defined? Is it defined in terms of specific audiences? Are wider definitions of effectiveness used?	**1. Descriptive** *What is the overall Theory of Change?* *At what levels do you take responsibility for effectiveness?* *At what level are the Assumptions so great that you cannot take responsibility?* *What are the key assumptions?*	Director Project managers Financial manager Digital Services manager M&E manager Webmetrics Other impact data	(if feasible and agreed by team) Facilitated group discussion Testing of portal webmetrics against new evaluation data
	10. How is effectiveness monitored?	What methods and software are used?			

VFM Question	Sub-questions		Sources	Methods/ criteria
	How timely is reporting? What kind of errors are there? How does this compare with comparators	*What are the best indicators of success of the project?* *How is reach in terms of Southern users and women defined?* *What are the key lessons that project staff take from the webmetrics?* *What changes have been made as a result of learning from webmetrics?*		
11. How are results validated?	How comprehensively and frequently are results tested? How do effectiveness measures stand up to evaluation findings?	**2. Analytical** *Are the web indicators used valid and accurate – in the light of evaluation field work findings regarding Southern users and women?* *Is it possible to put monetary value on outputs/ on the counterfactual of outputs?* *What would project staff consider to be proof (convincing evidence) that the portal is successful?* *What might have been done differently that would have led to greater effectiveness?* *How would this be apparent in the indicators?* **3. Comparative** *Has the portal been more successful in some areas and/or with some types of material and/or types of user?* *Has effectiveness improved over time?* *Has the portal achieved more or less than comparable initiatives? How do staff compare their work with that of other portals?*		
12. How is effectiveness made more sustainable or replicable?	*How is learning on effective information work shared?* *Could this type of project become self-financing?*	**1. Descriptive**. *What can be done to extend or prolong the work of the portal? Could the work become more sustainable after a longer period of funding?* **2. Analytic** *What lessons should be taken forward in information work by others?* **3. Comparative**		

VFM Question	Sub-questions		Sources	Methods/ criteria
		Are there some geographical areas or some types of user or some types of material where the work will continue or where the impact will last longer?		

Appendix D. Webmetrics analysis

D.1 Introduction and methodology

The webmetrics study is conducted to:

- Asses the web performance of Eldis and SciDev.Net
- Identify common trends and outliers between all users and users from the South
- Establish possible benchmarks looking at three 'comparators': Zunia, Pambazuka and GSDRC

The study is conducted using Google Analytics, as the one common monitoring tool used by all websites considered here to report and analyse on their website stats. In particular, Google Analytics is used to extract the following data:

- Overview metrics: Sessions and users; Avg. Session Duration; Pages/Sessions; % new and returning visitors
- Traffic channels: Organic, direct, referral, social, email, other
- Countries
- Devices: desktop, mobile and tablet
- Contents: views and sessions to documents, downloads
- Site speed
- Site search: % of sessions with search, search terms, % of exist after search
- (SciDev.Net only - Goal completions: total and individual goal completions)

For these metrics and dimensions, results are compared between 2014 and 2015, to identify possible trends. Additionally, a custom segment is created and applied in the analysis, to single out Traffic from the South – i.e. excluding traffic from Europe, North America, Russia, Central Asia, China, Taiwan, South Korea, Japan, Australia and New Zealand. In some instances, a second custom segment is applied to analyse Africa traffic alone – i.e. including only traffic from African countries.

When using GA to collect and analyse data it is important to recognize some of its limitations and caveats, such as:

Sampling

"Sampling in Google Analytics or in any web analytics software refers to the practice of selecting a subset of data from your website traffic. Sampling is widely used in statistical analysis because analysing a subset of data gives similar results to analysing all of the data." (Google)
As a consequence, Google Analytics (GA) reports requested at different times may present slightly different data, depending on the sample they are based upon.

Cookies and returning visitor

When users delete their browser's cookies prior to their next Visit to a site, these users will be counted as New Users upon their return to a website. Return Visits metric in GA should therefore be considered under-represented.

Time on Page and visit duration

Both these metrics should be carefully interpreted and used. Time on Page is the time between the start time of a specified Pageview and the start time of a subsequent Pageview or Event. For this reason, sessions that include only one Page View and no subsequent action on the site will have a Time on Page of zero seconds. This is the case for example if a Visitor views a page and then closes the browser. Time on Page calculations of zero seconds do contribute to the average. Likewise, visit Duration is the time between the start time of the first Pageview and the start time of the last Pageview of a Visit. Visits with only one Pageview may be attributed a Visit Duration of zero seconds, which contributes to the average. Finally, the time spent on the last Web page will not be fully captured.

D.2 Summary findings

Between the DFID funded portals and the comparators studied, SciDev.Net makes use of the most **advanced configuration of Google Analytics** to create different reporting views, filter data, track goals and conversions. More in general, SciDev.Net appear to have an advanced understanding of its digital analytics process, with multiple data collection points that feed into a dashboard spreadsheet with formulas and calculations.

We haven't had the possibility to discuss in depth how Google Analytics fits into portals' specific digital analytics framework, and review the overall process to identify strength and gaps.

Both DFID-funded portals have **expanded their reach to Southern audiences** (and African in particular) and year on year trend is positive.

Users from these locations tend to be more engaged and loyal, interact more (longer, more in depth) with the portals, when compared to the aggregate data for all traffic locations.

While reach has been expanded, **year on year user's engagement tends to decrease**, with less time spent on page and less deeper sessions. This is true for both portals and independent of the segment analysed.

For both sites, **traffic from the South is unevenly distributed**. India, Kenya, Philippines and South Africa count for around 50% of Southern traffic to both Eldis and SciDev.Net.

There are substantial differences between the two portals in terms of traffic channels. For Eldis **Organic search**, **Direct** and **Referral traffic** remain the three primary sources of traffic, regardless of the segment analysed. For SciDev.Net, **social traffic** is the primary traffic source for traffic for sessions generated in Africa. Facebook counts for more than 90% of this traffic. Social traffic has been increasingly substantially

over the past two years. On the other hand, the percentage of traffic generated by organic search has been declining in spite the fact it increased in absolute numbers.

In terms of **devices**, desktop remains the most commonly used between users of both Eldis and SciDev.Net. However the use of **mobile and tablet expands** largely between 2014 and 2015. This is particularly noticeable for SciDev.Net, registering a 156% growth in mobile sessions. This trend is even stronger when looking at southern traffic alone with mobile sessions increasing over 105% for Eldis and 221% for SciDev.Net.

For both DFID-funded portals, the percentage of sessions with **internal search** remains very low. Returning users are more likely to make use of site search than first time users.

Finally, looking at the use of contents, data available indicate that users increasingly consume and engage with the contents produced and made available by the two portals.

D.3 Eldis Webmetrics

D.3.1 Overview

- Eldis has an **increasing number of sessions and user** between 2014 and 2015
- Expanding reach: Total number of sessions increases 1% year on year; larger increase in sessions from South (21%)
- **Sessions from the South increases** from 41% to over 50% of the total sessions
- Other metrics tend to indicate a **decreasing user's engagement**, with decreasing average session durations and number of visited pages per session
- **635K users** in 2015, well beyond the goal set in Eldis logframe
- Most of the users are **first time users** - over 81%. No change between 2014 and 2015

Table 52 Eldis Sessions 2014/2015

	Date Range	Eldis-All sessions	Eldis-South
Sessions	Jan 1, 2015 - Dec 31, 2015	768,119	385,261
	Jan 1, 2014 - Dec 31, 2014	759,072	317,404
	% change	1.19%	21.38%
% Sessions from South	Jan 1, 2015 - Dec 31, 2015	50.16%	
	Jan 1, 2014 - Dec 31, 2014	41.81%	
	% change	8.35%	
Users	Jan 1, 2015 - Dec 31, 2015	635,012	326,268
	Jan 1, 2014 - Dec 31, 2014	626,073	263,805
	% change	1.43%	23.68%
% Sessions with Search	Jan 1, 2015 - Dec 31, 2015	3.32%	2.46%
	Jan 1, 2014 - Dec 31, 2014	3.69%	3.02%
	% change	-0.0037	-0.0056
Avg. Session Duration	Jan 1, 2015 - Dec 31, 2015	00:02:01	00:02:04

	Date Range	Eldis-All sessions	Eldis-South
	Jan 1, 2014 - Dec 31, 2014	00:02:09	00:02:33
	% change	-0:00:08	-0:00:29
Pages / Session	Jan 1, 2015 - Dec 31, 2015	2.15	2.00
	Jan 1, 2014 - Dec 31, 2014	2.31	2.34
	% change	-0.16	-0.34
Sessions by New Visitor	Jan 1, 2015 - Dec 31, 2015	81.70%	83.40%
	Jan 1, 2014 - Dec 31, 2014	81.50%	81.50%
	% change	0.002	1.90%

D.3.2 Traffic channels

- **Organic** (73%) remains largest traffic source - small increase between 2014 and 2015.
- Direct, referral and social all decrease between the two years.
- Organic search generates over 75% of all sessions for Southern traffic.
- Increase in direct, social and others for Southern traffic.

Table 53 Eldis Traffic Channels 2014/2015

Default Channel Grouping	Date Range	Eldis	Eldis-South
Organic Search	Jan 1, 2015 - Dec 31, 2015	566,735	290,970
	Jan 1, 2014 - Dec 31, 2014	540,166	235,602
		4.92%	23.50%
Direct	Jan 1, 2015 - Dec 31, 2015	127,640	63,809
	Jan 1, 2014 - Dec 31, 2014	139,659	49,864
		-8.64%	27.97%
Referral	Jan 1, 2015 - Dec 31, 2015	60,270	23,522
	Jan 1, 2014 - Dec 31, 2014	64,042	24,781
		-5.89%	-5.08%
(Other)	Jan 1, 2015 - Dec 31, 2015	7,003	4,233
	Jan 1, 2014 - Dec 31, 2014	4,147	1,788
		68.87%	136.74%
Social	Jan 1, 2015 - Dec 31, 2015	6,401	2,703
	Jan 1, 2014 - Dec 31, 2014	7,286	2,113
		-12.15%	27.92%
Email	Jan 1, 2015 - Dec 31, 2015	43	18
	Jan 1, 2014 - Dec 31, 2014	11	74
		-61.26%	-75.68%
Paid Search	Jan 1, 2015 - Dec 31, 2015	6	6
	Jan 1, 2014 - Dec 31, 2014	3,661	3,182
		-99.84%	-99.81%

Figure 43 Eldis - Traffic channels (2015)

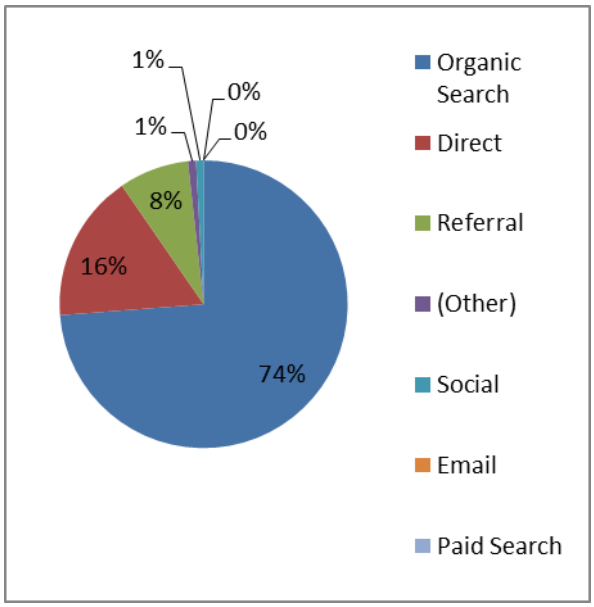
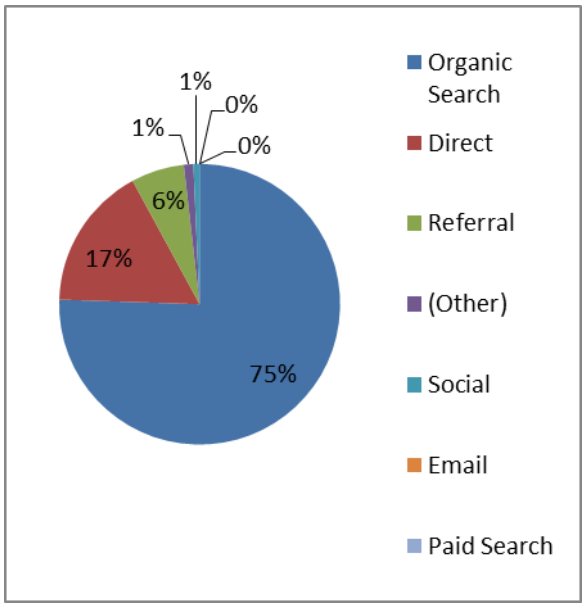


Figure 44 Eldis-Traffic channels (South,2015)

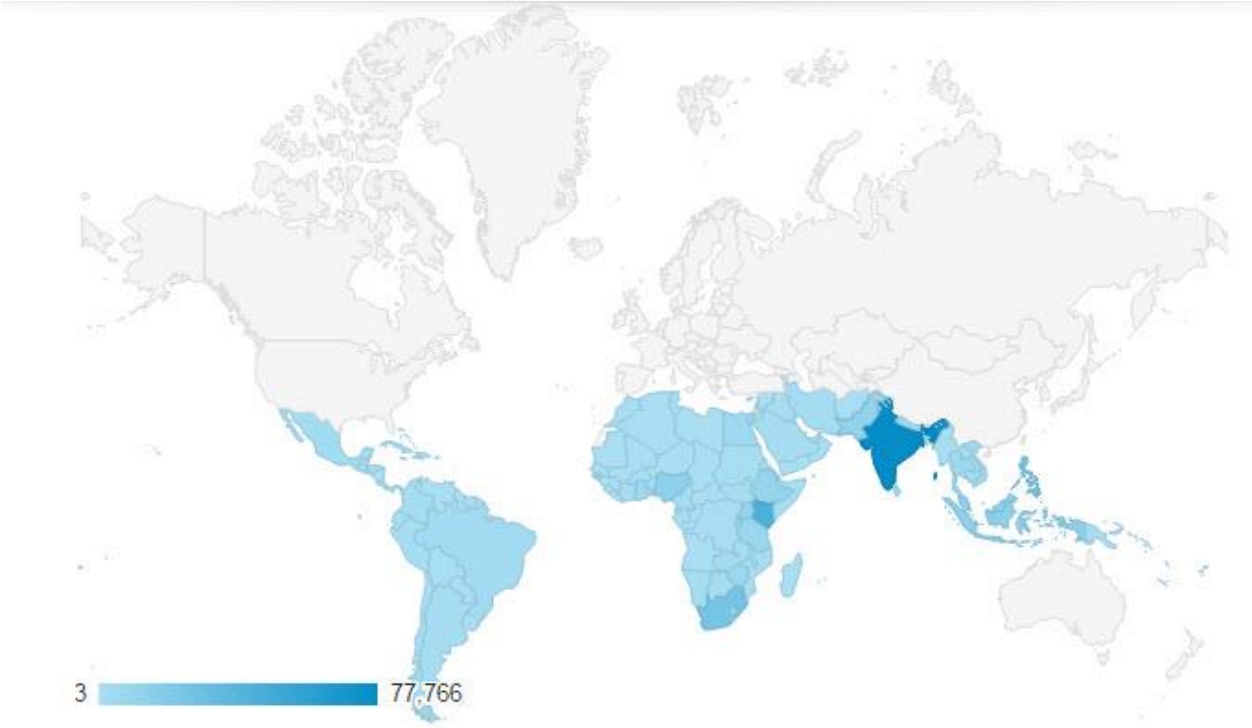


D.3.3 Countries

- Sessions from the South to Eldis in 2015 are generated from 157 countries – see image below.
- **Traffic is unevenly distributed**, with **India (22.20%)**, **Kenya (11.4%)**, **Philippines (6.89%)** and **South Africa (6.41%)** contribute to close to 50% of all Southern traffic.

- Other Southern countries amongst the top 10 include Indonesia, Nigeria, Malaysia, Pakistan, Uganda, and Ethiopia.

Figure 45 Eldis sessions in the South by location



D.3.4 Devices

- Large growth in **mobile (48%)** and **tablet (18%)**
- This is much larger for traffic from the South (mobile +106%; tablet +35%)
- Desktop remains most used device (over 77% of all sessions) but its importance is decreasing, especially for Southern users.
- Mobile sessions are increasing for both new and returning visitors at similar rate from all sessions and Southern traffic.

Table 54 Eldis traffic by device

	Date Range	Eldis	Eldis-South
desktop	Jan 1, 2015 - Dec 31, 2015	596,825	280,920
	Jan 1, 2014 - Dec 31, 2014	638,156	263,197
		-6.48%	6.73%
mobile	Jan 1, 2015 - Dec 31, 2015	141,047	90,495
	Jan 1, 2014 - Dec 31, 2014	95,440	43,942
		47.79%	105.94%

	Date Range	Eldis	Eldis-South
tablet	Jan 1, 2015 - Dec 31, 2015	30,212	13,847
	Jan 1, 2014 - Dec 31, 2014	25,512	10,265
		18.42%	34.90%
New Visitor	Jan 1, 2015 - Dec 31, 2015	119,853	76,574
	Jan 1, 2014 - Dec 31, 2014	80,088	37,172
		49.65%	106.00%
Returning Visitor	Jan 1, 2015 - Dec 31, 2015	21,194	13,920
	Jan 1, 2014 - Dec 31, 2014	15,352	6,770
		38.05%	105.61%

Figure 46 Eldis - % session by devices (2015, all sessions)

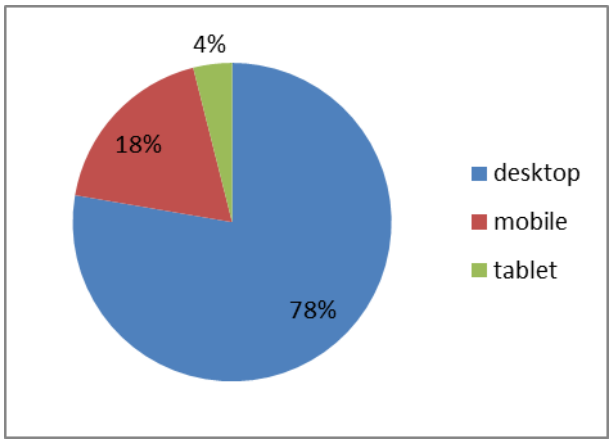
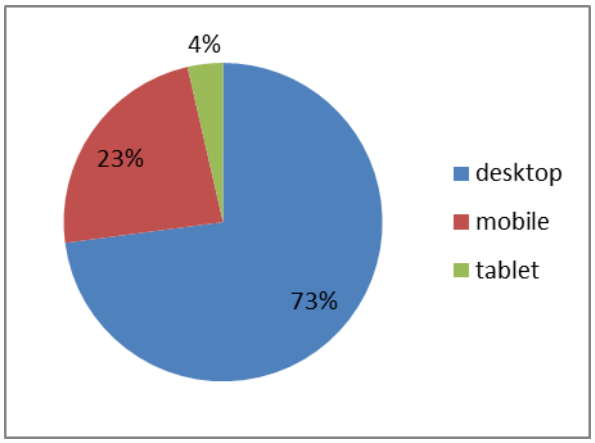


Figure 47 Eldis - % session by device (2015, South sessions)



D.3.5 Site speed

Site speed decreases for all sessions in the year on year comparison.

Figure 48 Eldis site speed by location 2014/2015

	Date Range	Eldis
Avg. Page Load Time (sec)	Jan 1, 2015 - Dec 31, 2015	14.43
	Jan 1, 2014 - Dec 31, 2014	11.19
		28.95%
Tanzania Avg.	Jan 1, 2015 - Dec 31, 2015	23.73
	Jan 1, 2014 - Dec 31, 2014	22.43
		5.79%
Ghana Avg.	Jan 1, 2015 - Dec 31, 2015	18.23
	Jan 1, 2014 - Dec 31, 2014	13.45
		35.53%
Nigeria Avg.	Jan 1, 2015 - Dec 31, 2015	30.49
	Jan 1, 2014 - Dec 31, 2014	19.77
		54.23%

D.3.6 Site search

- Decreasing trend in year-on-year comparison.
- Returning visitors tends to use site search more frequently (8% of sessions with search).

Figure 49 Eldis site searches

	Date Range	Eldis	Eldis-South
%Sessions with site search	Jan 1, 2015 - Dec 31, 2015	3.32%	2.46%
	Jan 1, 2014 - Dec 31, 2014	3.69%	3.02%
		-0.37%	-0.56%
Total Unique Searches	Jan 1, 2015 - Dec 31, 2015	40,126	14,092
	Jan 1, 2014 - Dec 31, 2014	44,078	14,621
		-8.97%	-3.62%
% Search Exits	Jan 1, 2015 - Dec 31, 2015	27.71%	29.17%
	Jan 1, 2014 - Dec 31, 2014	27.34%	27.39%
		0.0037	1.78%
Returning Visitor	Jan 1, 2015 - Dec 31, 2015	8.16%	5.22%
	Jan 1, 2014 - Dec 31, 2014	8.32%	6.67%
		-1.97%	-21.75%
New Visitor	Jan 1, 2015 - Dec 31, 2015	2.50%	1.91%
	Jan 1, 2014 - Dec 31, 2014	2.64%	2.19%
		-5.36%	-12.57%

D.3.7 Contents

- Use of Eldis content increases in year-on-year comparison. In 2015, data show: 46K document records available; 500K unique pageviews to documents (+7.8%); 175K downloads (+15.6%).
- Unique pageviews to resource guides show a slight decrease.

Table 55 Use of Eldis content

	Date Range	Unique Page Views	As % of total	Unique Page Views-South	As % of total traffic
Unique Pageviews to Docs	Jan 1, 2015 - Dec 31, 2015	500,229	38.49%	258,135	19.74%
	Jan 1, 2014 - Dec 31, 2014	463,853	34.01%	226,780	16.67%
		7.84%	4.48%	13.83%	0.0307
Unique Pageviews Resource Guides	Jan 1, 2015 - Dec 31, 2015	187,777	14.45%	99,273	7.59%
	Jan 1, 2014 - Dec 31, 2014	199,760	14.65%	105,902	7.78%
		-6.00%	-0.20%	-6.26%	-0.0019
Event: Download document	Jan 1, 2015 - Dec 31, 2015	175,368	5.11%	89,880	2.60%
	Jan 1, 2014 - Dec 31, 2014	153,198	5.25%	78,933	2.67%
		14.47%	-0.14%	13.87%	-0.0007
Sessions with downloads	Jan 1, 2015 - Dec 31, 2015	109,899	14.31%	52,819	6.88%
	Jan 1, 2014 - Dec 31, 2014	95,001	12.52%	44,581	5.87%
		15.68%	1.79%	18.48%	0.0101
Subject > Content partner su…	Jan 1, 2015 - Dec 31, 2015	34,501	1.00%	17,717	0.51%
	Jan 1, 2014 - Dec 31, 2014	12,737	0.44%	6,966	0.24%
		170.87%	0.56%	154.34%	0.0027
Subject > Service > OKHub	Jan 1, 2015 - Dec 31, 2015	5,032	0.15%	2,592	0.08%
	Jan 1, 2014 - Dec 31, 2014	644	0.02%	295	0.01%
		681.37%	0.13%	778.64%	0.07%

Visits to Eldis pages containing document abstracts are in general longer than the average length of visits to all pages. While the average session duration for the whole of Eldis is just above 2 minutes, sessions to document abstracts last on average three minutes, with time on page going between 1 and 6 minutes for the top 10 documents in 2015.

Figure 50 Trends for Eldis document abstract page views

Plot Rows | Secondary dimension | Sort Type: Default

Page	Pageviews	Unique Pageviews	Avg. Time on Page
	645,758 % of Total: 39.13% (1,650,112)	500,229 % of Total: 38.49% (1,299,549)	00:02:59 Avg for View: 00:01:44 (71.56%)
1. /go/home&id=50387&type=Document	12,546 (1.94%)	10,576 (2.11%)	00:03:56
2. /go/home&id=15006&type=Document	7,457 (1.15%)	5,920 (1.18%)	00:04:08
3. /go/home&id=51725&type=Document	4,549 (0.70%)	3,607 (0.72%)	00:03:46
4. /go/home&id=21267&type=Document	3,633 (0.56%)	2,981 (0.60%)	00:04:09
5. /go/home&id=69651&type=Document	3,515 (0.54%)	3,462 (0.69%)	00:01:09
6. /go/home&id=17521&type=Document	2,882 (0.45%)	2,375 (0.47%)	00:04:28
7. /go/home&id=43978&type=Document	2,847 (0.44%)	2,372 (0.47%)	00:05:04
8. /go/home&id=64908&type=Document	2,696 (0.42%)	2,171 (0.43%)	00:04:47
9. /go/home&id=34132&type=Document	2,636 (0.41%)	2,306 (0.46%)	00:05:59
10. /go/home&id=45068&type=Document	2,627 (0.41%)	2,044 (0.41%)	00:04:40

Trends for document abstract pageviews and downloads are similar between all sessions and Southern session's segment.

Figure 51 Eldis - Unique Pageviews to document abstracts

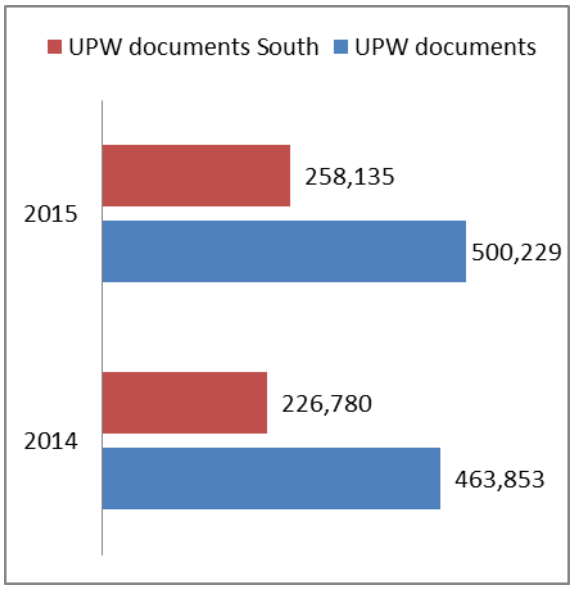
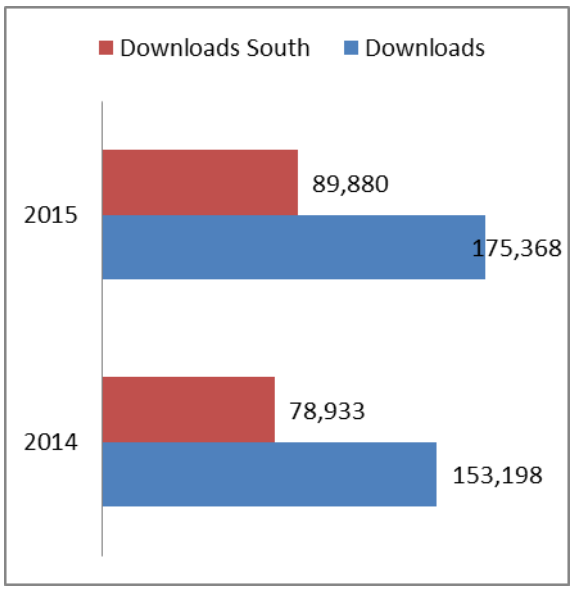
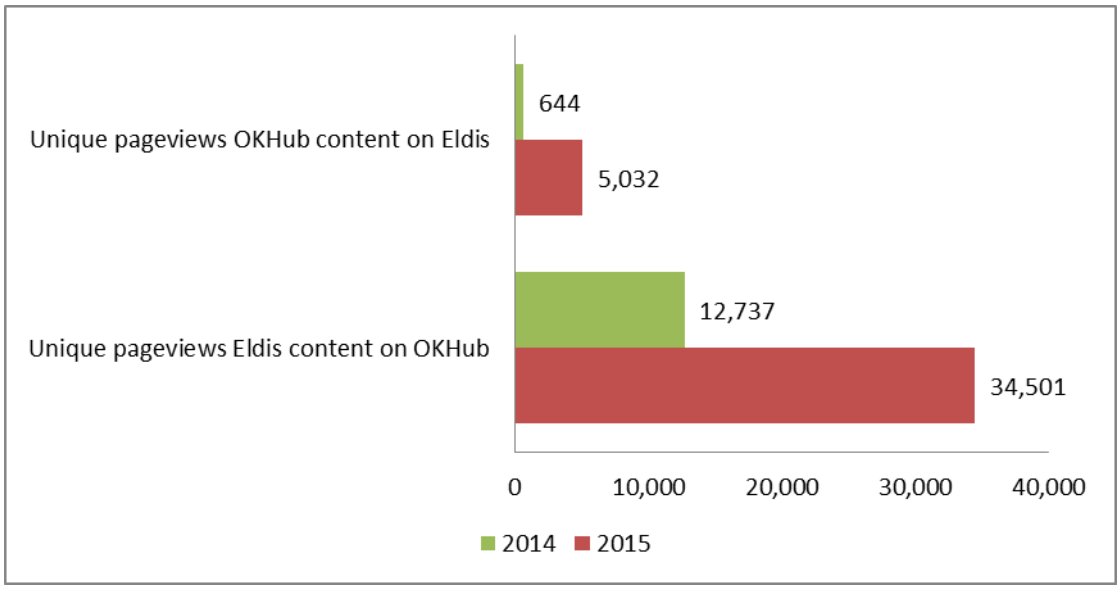


Figure 52 Eldis - Downloads



Increasing access to OKHub content included on Eldis and Eldis content on the OKHub. Total number of unique pageviews remains small percentage of all pageviews.

Figure 53 Unique pageviews OKhub/Eldis content



D.4 SciDev.Net Webmetrics

D.4.1 Overview

- SciDev.Net has a **complex and advanced Google Analytics installation**. This includes:
- use of filters to exclude internal traffic and traffic from robot and spiders;
- creation of multiple 'views' to track traffic more in details, for example to separate data for the different editions in which the portal is available;
- use and tracking of 'goals' and conversions.
- SciDev.Net has the **largest reach** (2.7M sessions and 2.1M users) and **highest year-on-year growth** (+40%). Southern sessions grow more rapidly with 54% increase.
- Sessions from the South in 2015 count for 63% of all sessions.
- Decreasing user's engagement (decreasing average session durations and pages per session).

Table 56 SciDev.Net total sessions

	Date Range	SciDev.Net	SciDev.Net-South
Sessions	Jan 1, 2015 - Dec 31, 2015	2,736,961	1,728,832
	Jan 1, 2014 - Dec 31, 2014	1,954,614	1,122,225
	% change	40.03%	54.05%
% Sessions from South	Jan 1, 2015 - Dec 31, 2015	63.58%	
	Jan 1, 2014 - Dec 31, 2014	57.21%	
	% change	6.37%	
Users	Jan 1, 2015 - Dec 31, 2015	2,107,692	1,324,029
	Jan 1, 2014 - Dec 31, 2014	1,497,421	895,602

	Date Range	SciDev.Net	SciDev.Net-South
	% change	40.75%	47.84%
% Sessions with Search	Jan 1, 2015 - Dec 31, 2015	0.70%	63.58%
	Jan 1, 2014 - Dec 31, 2014	0.94%	57.21%
	% change	-0.0024	6.37%
Avg. Session Duration	Jan 1, 2015 - Dec 31, 2015	00:02:17	0.47%
	Jan 1, 2014 - Dec 31, 2014	00:03:07	0.53%
	% change	-0:00:50	-0.0006
Pages / Session	Jan 1, 2015 - Dec 31, 2015	1.45	00:02:17
	Jan 1, 2014 - Dec 31, 2014	1.55	00:02:56
	% change	-0.1	-0:00:39
Sessions by New Visitor	Jan 1, 2015 - Dec 31, 2015	76%	1.41
	Jan 1, 2014 - Dec 31, 2014	75.60%	1.49
	% change	0.40%	-0.08

D.4.2 Traffic channels

- Year on year increase of sessions for all traffic channels, except paid search and referral.
- Largest increase is for email (110%) and social (96%). The growth of these two traffic channels is higher for sessions from the South (email 180%; social 122%).

Table 57 SciDev.Net traffic channels

Default Channel Grouping	Date Range	SciDev.Net	SciDev.Net-South
Organic Search	Jan 1, 2015 - Dec 31, 2015	1,075,622	628,845
	Jan 1, 2014 - Dec 31, 2014	830,725	449,528
		29.48%	39.89%
Direct	Jan 1, 2015 - Dec 31, 2015	527,289	296,320
	Jan 1, 2014 - Dec 31, 2014	440,040	225,675
		19.83%	31.30%
Referral	Jan 1, 2015 - Dec 31, 2015	138,805	50,190
	Jan 1, 2014 - Dec 31, 2014	147,268	53,757
		-5.75%	-6.64%
(Other)	Jan 1, 2015 - Dec 31, 2015	27,913	14,251
	Jan 1, 2014 - Dec 31, 2014	2,505	1,366
		1014.29%	943.27%
Social	Jan 1, 2015 - Dec 31, 2015	772,148	615,942
	Jan 1, 2014 - Dec 31, 2014	392,526	277,402
		96.71%	122.04%
Email	Jan 1, 2015 - Dec 31, 2015	71,822	38,065
	Jan 1, 2014 - Dec 31, 2014	34,061	13,586
		110.86%	180.18%

Default Channel Grouping	Date Range	SciDev.Net	SciDev.Net-South
Paid Search	Jan 1, 2015 - Dec 31, 2015	105,674	85,219
	Jan 1, 2014 - Dec 31, 2014	114,422	100,911
		-7.65%	-15.55%

There's more diversity in traffic channels, with organic search less dominant than for Eldis (40% of all sessions and 36% of sessions from the South).

For traffic from the South, **social is as important as organic search** (36% of Southern sessions; 48% sessions from Africa).

While **email has the highest year-on-year increase**, it contributes to only 3% of total sessions and 2% of sessions from the South.

Figure 54 SciDev.Net - Traffic channels (2015)

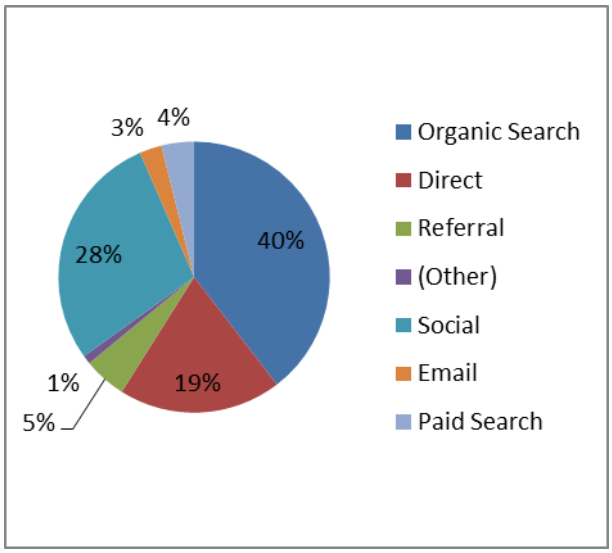
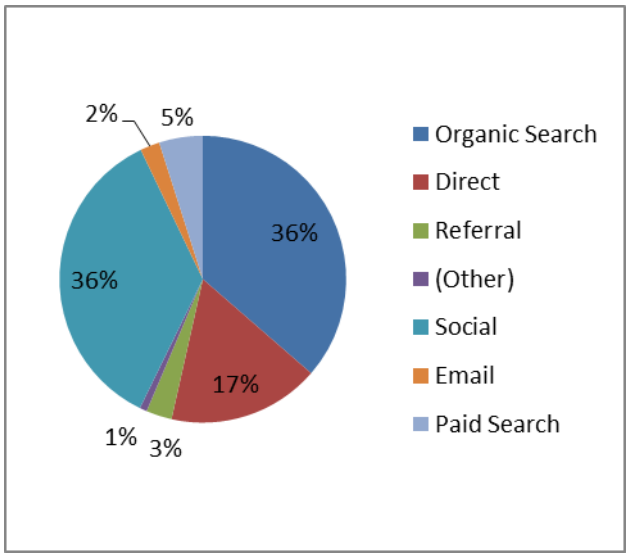


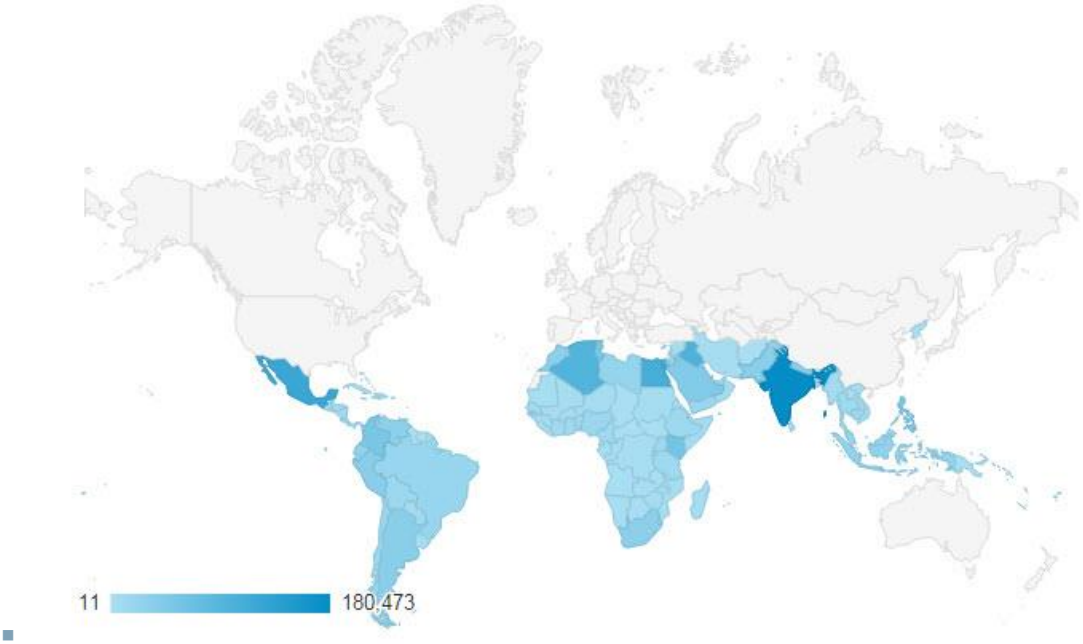
Figure 55 SciDev-Traffic channels (South, 2015)



D.4.3 Countries

- As for Eldis, there's a long tail of Southern countries generating sessions to SciDev.Net.
- There are less 'dominant' countries in terms of number of sessions.
- The top 5 countries generating sessions to SciDev.Net include: India (10.4%), Mexico (7.06%), Egypt (6.21%), Iraq (5.72%), and Algeria (5.51%).

Figure 56 SciDev.Net Southern sessions by location



D.4.4 Devices

- Sessions increase for all devices used, with **mobile showing by far the highest growth in the year on year comparison**, with +156% for all sessions and +221% for sessions from the South.
- Tablet is also increasing more rapidly than desktop.
- Mobile use is increasing for both new and returning visitors.

Figure 57 SciDev.Net sessions by device

	Date Range	SciDev.Net	SciDev.Net-South
desktop	Jan 1, 2015 - Dec 31, 2015	1,642,247	961,333
	Jan 1, 2014 - Dec 31, 2014	1,491,089	853,590
		10.14%	12.62%
mobile	Jan 1, 2015 - Dec 31, 2015	932,092	683,931
	Jan 1, 2014 - Dec 31, 2014	363,28	212,582
		156.57%	221.73%
tablet	Jan 1, 2015 - Dec 31, 2015	144,934	83,568
	Jan 1, 2014 - Dec 31, 2014	107,173	56,053
		35.23%	49.09%
New Visitor (mobile)	Jan 1, 2015 - Dec 31, 2015	700,648	507,232
	Jan 1, 2014 - Dec 31, 2014	285,656	166,282
		145.28%	205.04%

	Date Range	SciDev.Net	SciDev.Net-South
Returning Visitor (mobile)	Jan 1, 2015 - Dec 31, 2015	231,443	176,699
	Jan 1, 2014 - Dec 31, 2014	77,628	46,300
		198.14%	281.64%

Desktop remains most used device (61% of all sessions) but its proportion is decreasing.

Traffic from the South show a larger proportion of mobile users (39% of sessions).

Figure 58 SciDev - % session by devices (2015, all sessions)

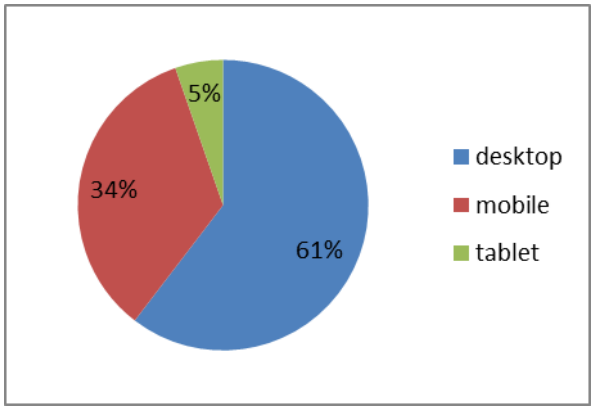
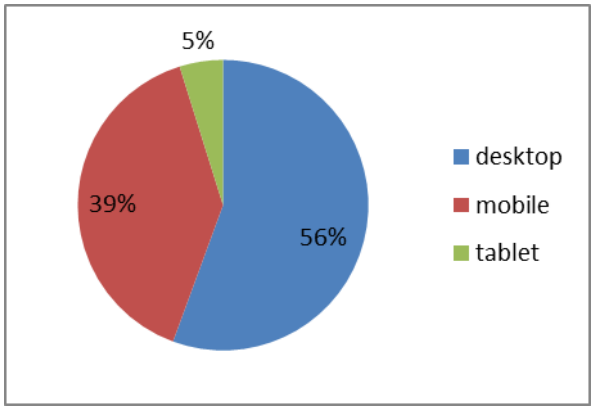


Figure 59 SciDev - % session by device (2015, South sessions)



D.4.5 Site speed

- Similar to Eldis, page speed appears to decrease.

Table 58 SciDev.Net site speed by location

	Date Range	SciDev.Net
Avg. Page Load Time (sec)	Jan 1, 2015 - Dec 31, 2015	17.39
	Jan 1, 2014 - Dec 31, 2014	18.13
		-4.08%
Tanzania Avg.	Jan 1, 2015 - Dec 31, 2015	24.08
	Jan 1, 2014 - Dec 31, 2014	67.96
		-64.57%
Ghana Avg.	Jan 1, 2015 - Dec 31, 2015	34.91
	Jan 1, 2014 - Dec 31, 2014	23.06
		51.38%
Nigeria Avg.	Jan 1, 2015 - Dec 31, 2015	37.44
	Jan 1, 2014 - Dec 31, 2014	43.37
		-13.67%

D.4.6 Site search

Percentage of sessions with search is decreasing for all sessions and sessions from the South.

Table 59 SciDev.Net site searches

	Date Range	SciDev.Net	SciDev.Net-South
%Sessions with site search	Jan 1, 2015 - Dec 31, 2015	0.70%	0.47%
	Jan 1, 2014 - Dec 31, 2014	0.94%	0.53%
		-0.24%	-0.06%
Total Unique Searches	Jan 1, 2015 - Dec 31, 2015	30,403	13,814
	Jan 1, 2014 - Dec 31, 2014	30,098	9,279
		1.01%	48.87%
% Search Exits	Jan 1, 2015 - Dec 31, 2015	19.82%	21.84%
	Jan 1, 2014 - Dec 31, 2014	20.49%	21.88%
		-0.0067	-0.04%
Returning Visitor	Jan 1, 2015 - Dec 31, 2015	1.27%	1.06%
	Jan 1, 2014 - Dec 31, 2014	1.57%	1.20%
		-18.88%	-11.76%
New Visitor	Jan 1, 2015 - Dec 31, 2015	0.40%	0.27%
	Jan 1, 2014 - Dec 31, 2014	0.53%	0.35%
		-25.09%	-9.83%

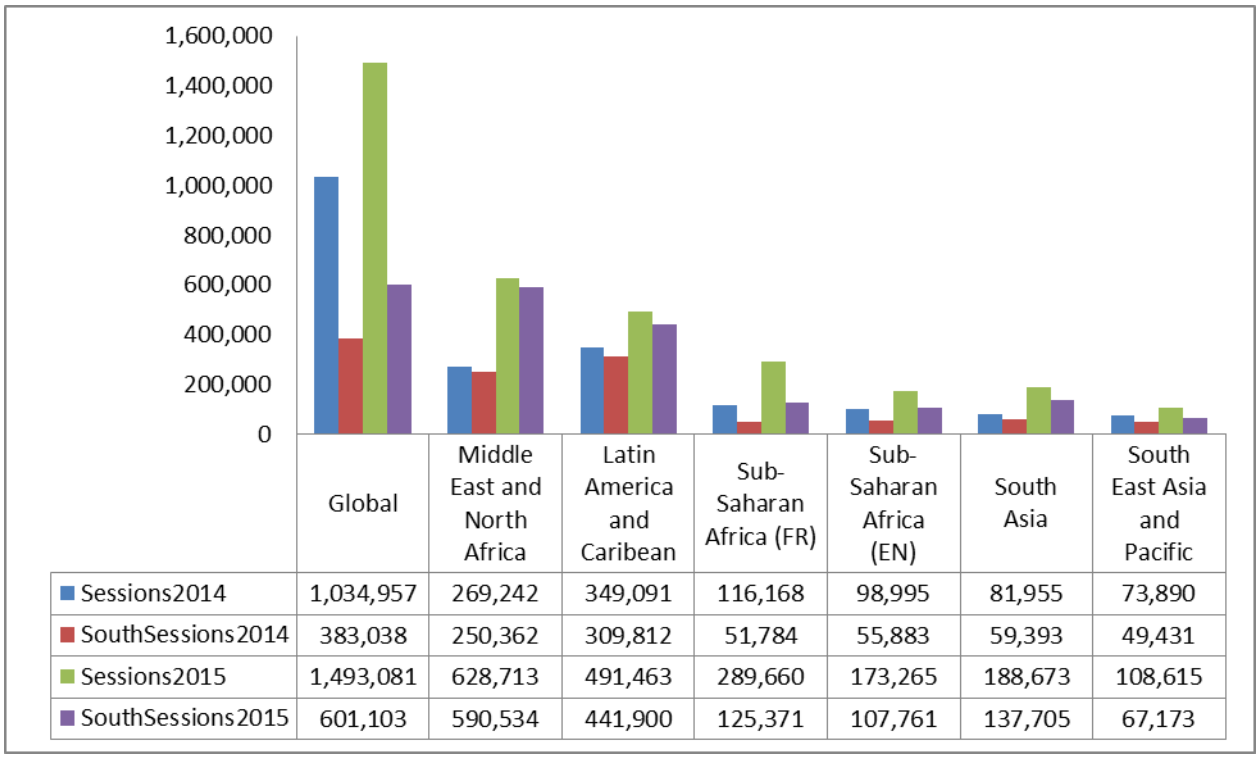
D.4.7 Contents

All SciDev.Net editions see an increase in the number of unique pageviews in the year-on-year comparison

Table 60 Unique page views by SciDev.Net edition

Date Range	Edition	Unique Page Views	% total	Unique Page Views-South	% total-South
Jan 1, 2015 - Dec 31, 2015	Global edition	1,493,081	44%	601,103	17.83%
Jan 1, 2014 - Dec 31, 2014		1,034,957	41%	383,038	15.20%
		44.27%	3%	56.93%	2.62%
Jan 1, 2015 - Dec 31, 2015	Middle East and North Africa	628,713	19%	590,534	17.51%
Jan 1, 2014 - Dec 31, 2014		269,242	11%	250,362	9.94%
		133.51%	8%	135.87%	7.57%
Jan 1, 2015 - Dec 31, 2015	Latin America and Caribbean	491,463	15%	441,900	13.10%
Jan 1, 2014 - Dec 31, 2014		349,091	14%	309,812	12.30%
		40.78%	1%	42.63%	0.81%
Jan 1, 2015 - Dec 31, 2015	Sub-Saharan Africa (FR)	289,660	9%	125,371	3.72%
Jan 1, 2014 - Dec 31, 2014		116,168	5%	51,784	2.06%
		149.35%	4%	142.10%	1.66%
Jan 1, 2015 - Dec 31, 2015	South Asia	188,673	6%	137,705	4.08%
Jan 1, 2014 - Dec 31, 2014		81,955	3%	59,393	2.36%
		130.22%	2%	131.85%	1.73%
Jan 1, 2015 - Dec 31, 2015	Sub-Saharan Africa (EN)	173,265	5%	107,761	3.20%
Jan 1, 2014 - Dec 31, 2014		98,995	4%	55,883	2.22%
		75.02%	1%	92.83%	0.98%
Jan 1, 2015 - Dec 31, 2015		108,615	3%	67,173	1.99%
Jan 1, 2014 - Dec 31, 2014		73,890	3%	49,431	1.96%
	South East Asia and Pacific	47.00%	0%	35.89%	0.03%

Figure 60 SciDev.Net sessions (2014/2015) by edition



	Global	Middle East and North Africa	Latin America and Caribean	Sub-Saharan Africa (FR)	Sub-Saharan Africa (EN)	South Asia	South East Asia and Pacific
Sessions2014	1,034,957	269,242	349,091	116,168	98,995	81,955	73,890
SouthSessions2014	383,038	250,362	309,812	51,784	55,883	59,393	49,431
Sessions2015	1,493,081	628,713	491,463	289,660	173,265	188,673	108,615
SouthSessions2015	601,103	590,534	441,900	125,371	107,761	137,705	67,173

Figure 61 Increase in unique pageviews by contents (2014/2015)

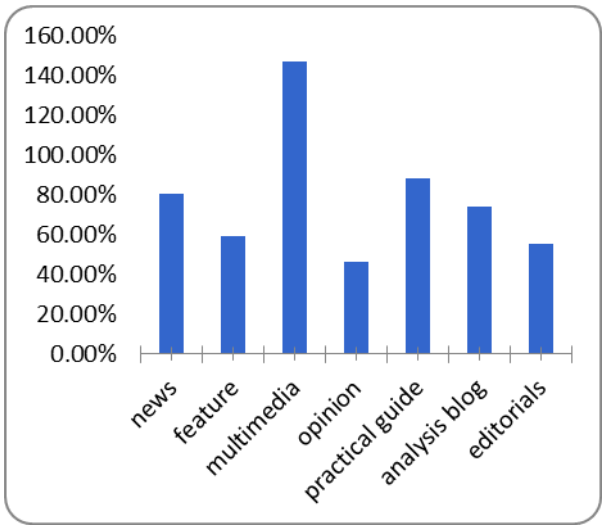
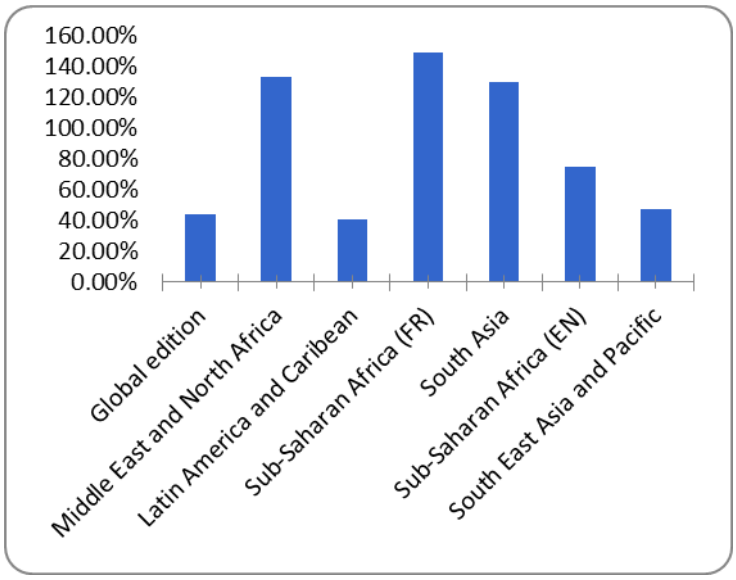


Figure 62 Increase in unique pageviews by edition (2014/2015)



Users engage with the content of the practical guides. The average time on page for the top 10 guides is much higher than the site average, with up to ten minutes spent on the page.

Table 61 Use of Scidev.Net practical guides

Page URL	Unique Pageviews 2015	Avg. Time on Page 2015
/middleeastandnorthafrica/practicalguide/اتصال /submitpaperscientificjournal	14,660 (12.12%)	00:10:23
/global/practicalguide/publishing/howdoiwriteascientificpaper	14,054 (11.62%)	00:08:54
/global/practicalguide/publishing/howdoisubmitapapertoascientificjournal	9,643 (7.97%)	00:09:08
/middleeastandnorthafrica/practicalguide/الى نشر /howwritescientificpaper	5,973 (4.94%)	00:08:47
/global/practicalguide/communication/flashtalksciencevideoguide	5,360 (4.43%)	00:09:38
/global/practicalguide/publishing/targetjournalrightresearchcommunicatepublish	5,253 (4.34%)	00:09:15
/global/practicalguide/journalism/howtoreportfromvisitsandfieldtrips	5,155 (4.26%)	00:10:31
/global/articletype/practicalguide	4,479 (3.70%)	00:01:34
/global/practicalguide/publishing/pitchscienceeditorjournalism	4,371 (3.61%)	00:08:57
/middleeastandnorthafrica/practicalguide/اتصال /flashtalksciencevideoguide1	3,028 (2.50%)	00:09:33

D.4.8 Goals

- SciDev.Net makes extensive use of Goals in its Google Analytics configuration to monitor how users interact with the site. Goals include: view article; read article; scroll; interact with article; share article; etc.

- The overall number of Goal completions is increasing in the year on year comparison for both all sessions (+38%) and Southern sessions (+51%).
- All other goals included in this analysis show a similar year on year increase, indicating an increased effectiveness and efficiency of the site.

Table 62 SciDev.Net goal completion

	Goal completions	All traffic	South
Jan 1, 2015 - Dec 31, 2015	All goals	6,222,221	3,884,496
Jan 1, 2014 - Dec 31, 2014		4,493,695	2,559,614
		38.47%	51.76%
Jan 1, 2015 - Dec 31, 2015	1. View Article (Goal 1 Completions)	2,272,897	1,413,840
Jan 1, 2014 - Dec 31, 2014		1,557,237	867,196
		45.96%	63.04%
Jan 1, 2015 - Dec 31, 2015	2. Read Article (Goal 2 Completions)	1,627,254	1,020,707
Jan 1, 2014 - Dec 31, 2014		1,184,451	681,996
		37.38%	49.66%
Jan 1, 2015 - Dec 31, 2015	4. Scroll (Goal 4 Completions)	909,505	571,995
Jan 1, 2014 - Dec 31, 2014		701,727	410,609
		29.61%	39.30%
Jan 1, 2015 - Dec 31, 2015	6. Interact (Goal 6 Completions)	29,238	13,454
Jan 1, 2014 - Dec 31, 2014		24,474	11,404
		19.47%	17.98%
Jan 1, 2015 - Dec 31, 2015	7. Share Article (Goal 7 Completions	21,744	10,873
Jan 1, 2014 - Dec 31, 2014		17,325	8,387
		25.51%	29.64%

D.5 Comparators: Zunia, Pambazuka, GSDRC

D.5.1 Overview

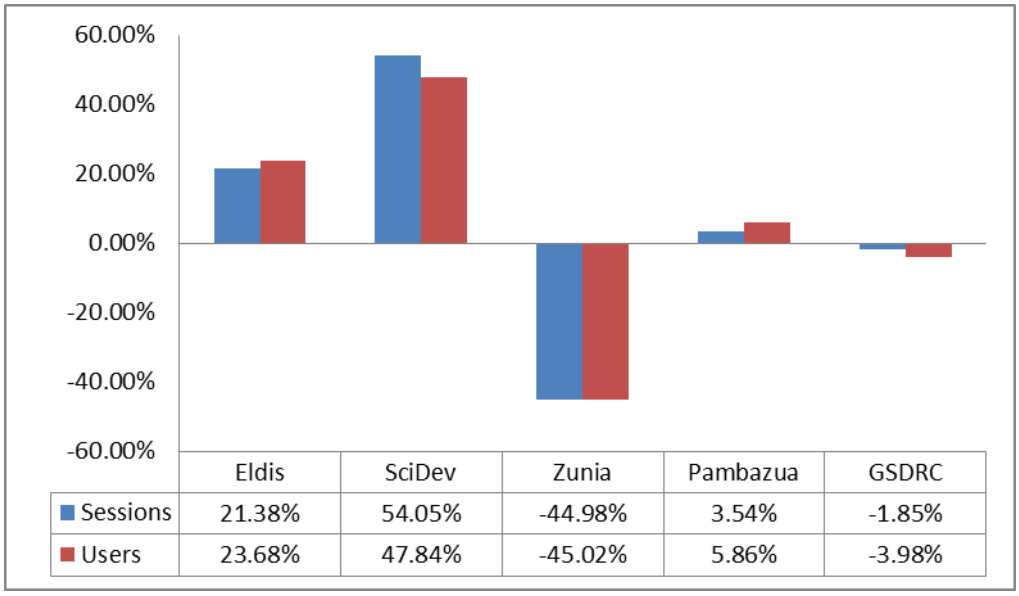
- All comparators analysed don't seem to have the same advanced understanding and knowledge of Google Analytics as applied by Eldis and SciDev.Net in particular.
- The comparators also show a smaller reach than Eldis and SciDev.Net, between the 70K session of Zunia and the 472K sessions of Pambazuka.
- Only Zunia records over 50% of sessions generated in the South.
- All comparators are experiencing a reduction in the number of sessions and users in the comparison between 2014 and 2015. While this is relatively small for Pambazuka and GSDRC, Zunia shows an alarming 50% drop for these metrics.

Table 63 Comparator total sessions

	Date Range	Zunia	Pambazuka	GSDRC
Sessions	Jan 1, 2015 - Dec 31, 2015	71,591	472,444	408,042
	Jan 1, 2014 - Dec 31, 2014	143,735	511,257	442,807
	% change	-50.19%	-7.59%	-7.85%
% Sessions from South	Jan 1, 2015 - Dec 31, 2015	50.75%	46.01%	45.41%
	Jan 1, 2014 - Dec 31, 2014	45.94%	40.95%	42.53%
	% change	4.81%	5.06%	2.88%
Users	Jan 1, 2015 - Dec 31, 2015	59,195	359,334	312,034
	Jan 1, 2014 - Dec 31, 2014	116,584	388,672	346,105
	% change	-49.23%	-7.55%	-9.84%

- These same trends are confirmed when looking at traffic from the South alone, with Eldis and SciDev.Net performing much better than the other websites included in the analysis in terms of reach.

Figure 63 Year-on-year % change in sessions and users (2014/2015, South



D.5.2 Channels

- In terms of traffic channels, Organic, Direct and Referral remain consistently the top three channels for the websites considered. This doesn't change when looking at traffic from the South alone.

Figure 64 Zunia - Traffic channels (2015, all sessions)

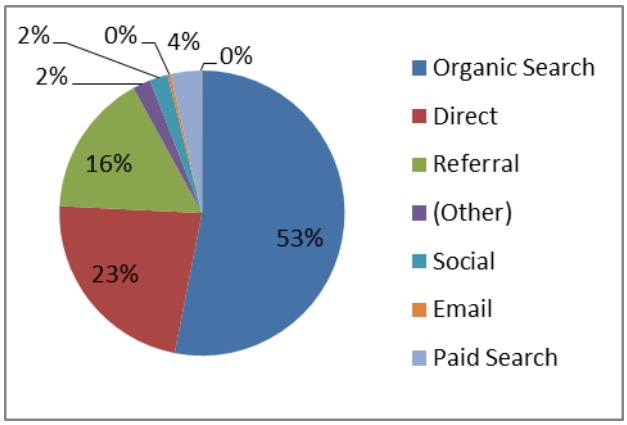


Figure 65 Zunia- Traffic channels (2015, South sessions)

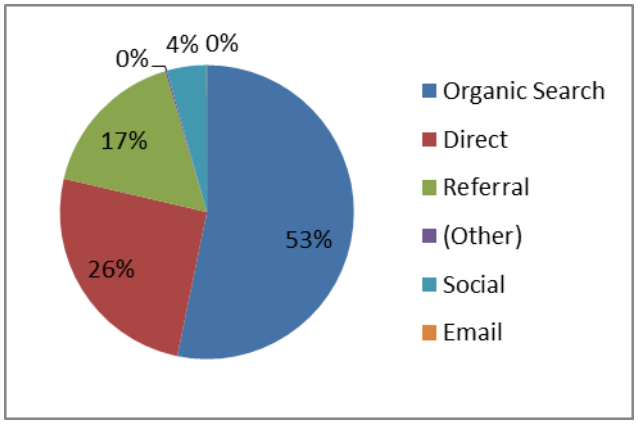


Figure 66 Pambazuka - Traffic channels (2015, all sessions)

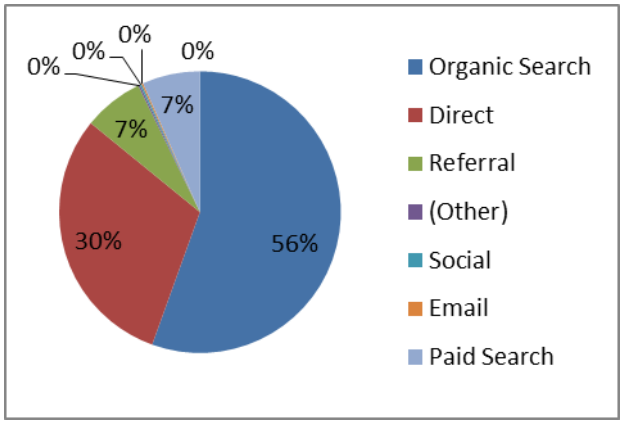


Figure 67 Pambazuka - Traffic channels (2015, South sessions)

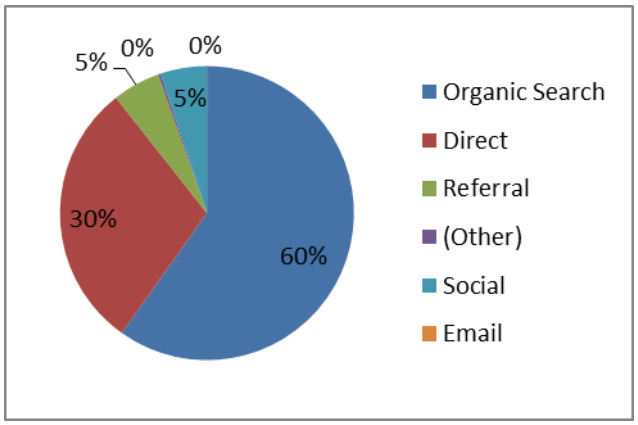


Figure 68 GSDRC - Traffic channels (2015, all sessions)

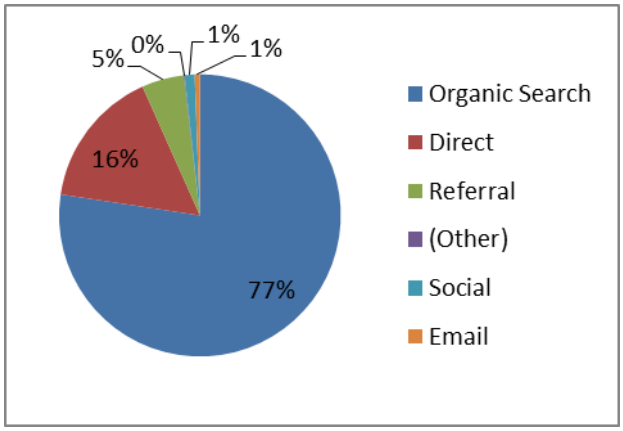
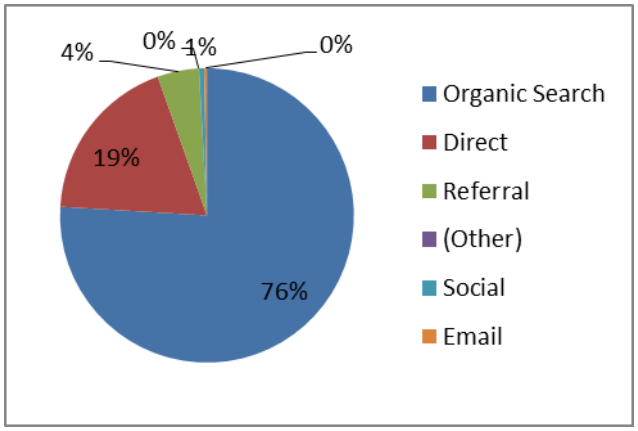


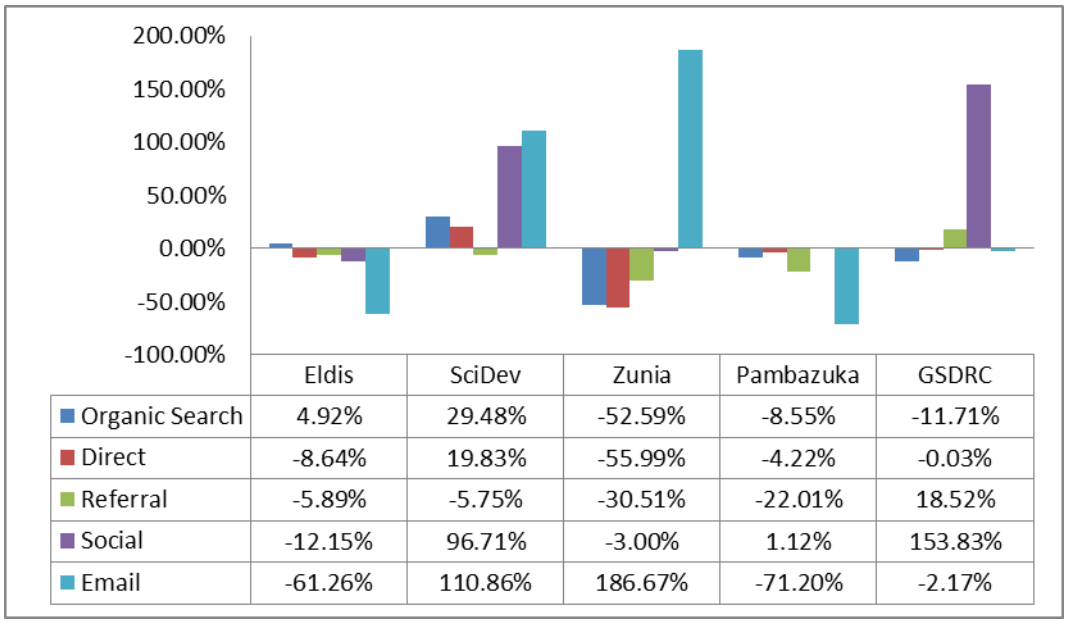
Figure 69 GSDRC-Traffic channels (2015, South sessions)



The year on year trend sees the most of the traffic channels reducing the number of session they contribute to the sites. This is not surprising given the general reduction in traffic observed earlier for the comparators.

The only two channels that grow are Email for Zunia and Social for GSDRC. For Zunia, this is an indication of potential value of push services to maintain a residual user base, while traffic is shrinking.

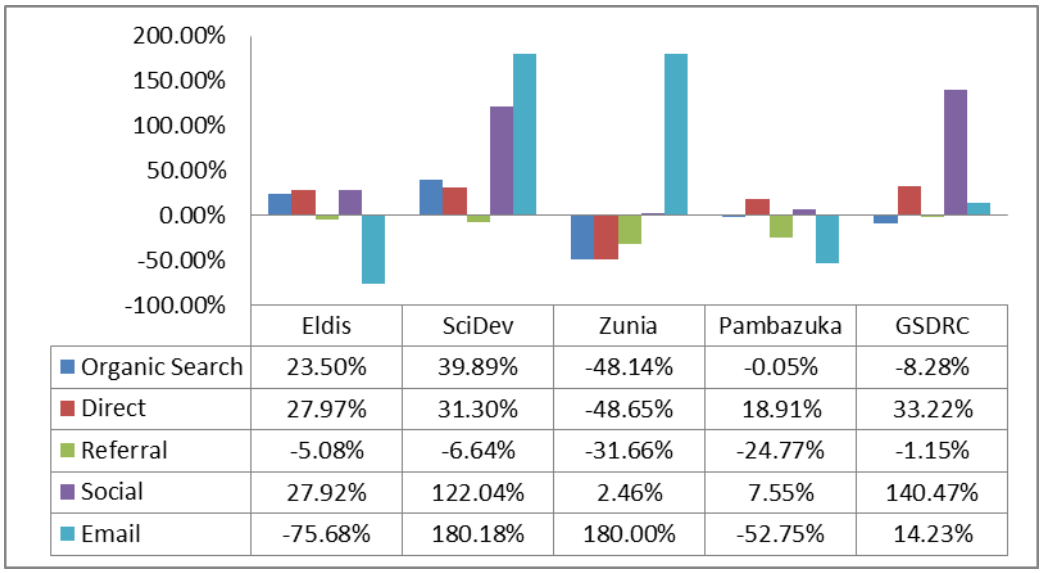
Figure 70 Traffic channels - year-on-year % change (all sessions)



	Eldis	SciDev	Zunia	Pambazuka	GSDRC
Organic Search	4.92%	29.48%	-52.59%	-8.55%	-11.71%
Direct	-8.64%	19.83%	-55.99%	-4.22%	-0.03%
Referral	-5.89%	-5.75%	-30.51%	-22.01%	18.52%
Social	-12.15%	96.71%	-3.00%	1.12%	153.83%
Email	-61.26%	110.86%	186.67%	-71.20%	-2.17%

Looking at traffic from the South alone, the picture is more varied with an increase also in Social for Zunia, Direct and Social for Pambazuka and Direct, Social and Email for GSDRC. This appears more in line with the general trends observed for Eldis and SciDev.Net especially in terms of social traffic.

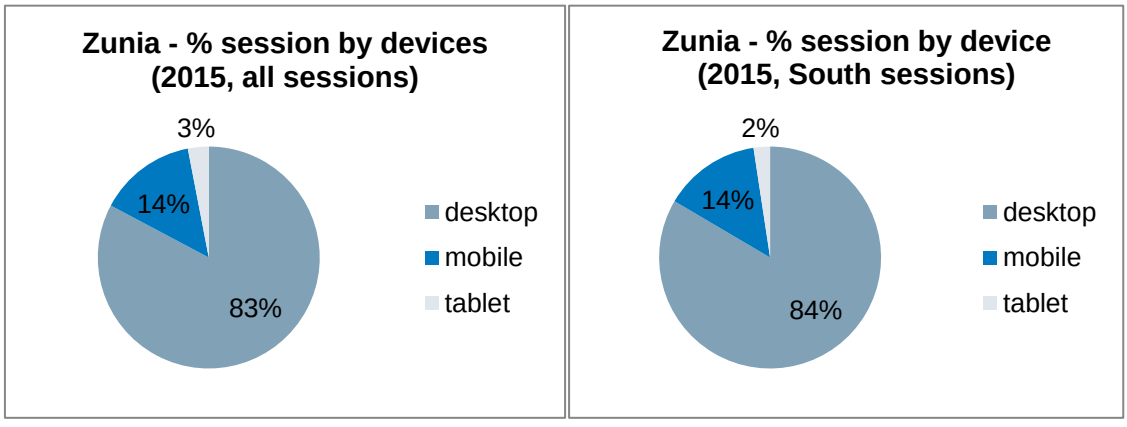
Figure 71 Traffic channels - year-on-year % change (Sessions from South)

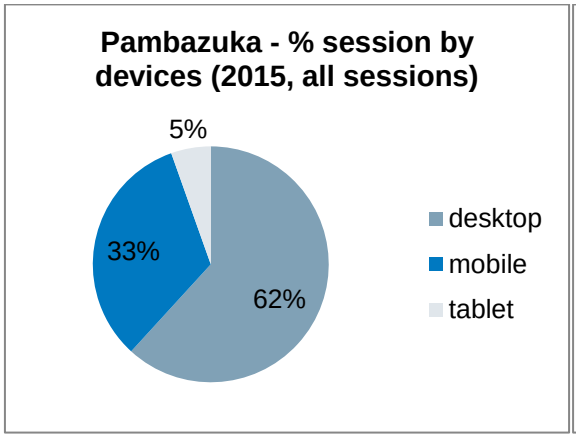


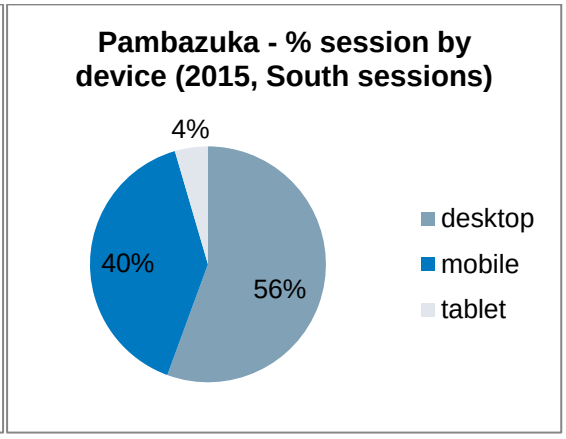
	Eldis	SciDev	Zunia	Pambazuka	GSDRC
Organic Search	23.50%	39.89%	-48.14%	-0.05%	-8.28%
Direct	27.97%	31.30%	-48.65%	18.91%	33.22%
Referral	-5.08%	-6.64%	-31.66%	-24.77%	-1.15%
Social	27.92%	122.04%	2.46%	7.55%	140.47%
Email	-75.68%	180.18%	180.00%	-52.75%	14.23%

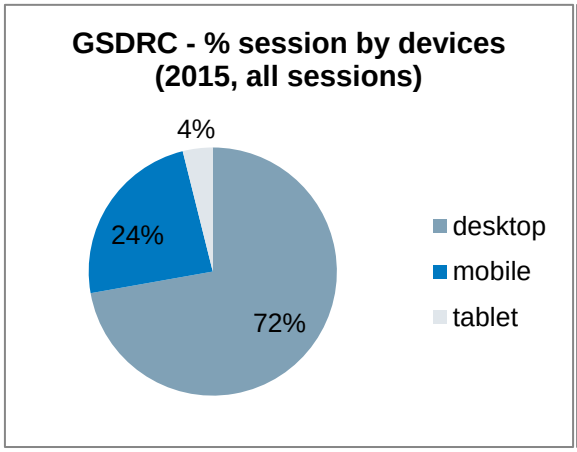
D.5.3 Devices

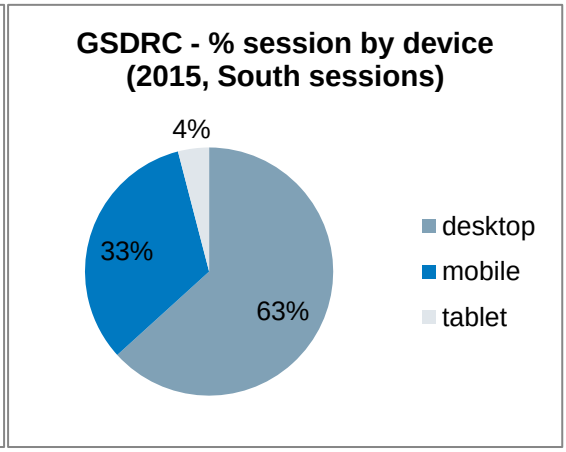
In terms of devices, desktop remains the most common device for users of the three comparators. The percentage of sessions from desktop however is quite different amongst the different sites, with over 83% of sessions to Zunia performed from desktop, while this percentage drops to 62% for Pambazuka. It is confirmed that users from the South tends to use handled devices more frequently. This is noticeable especially for Pambazuka, where 44% of Southern traffic comes from mobile or tablet.



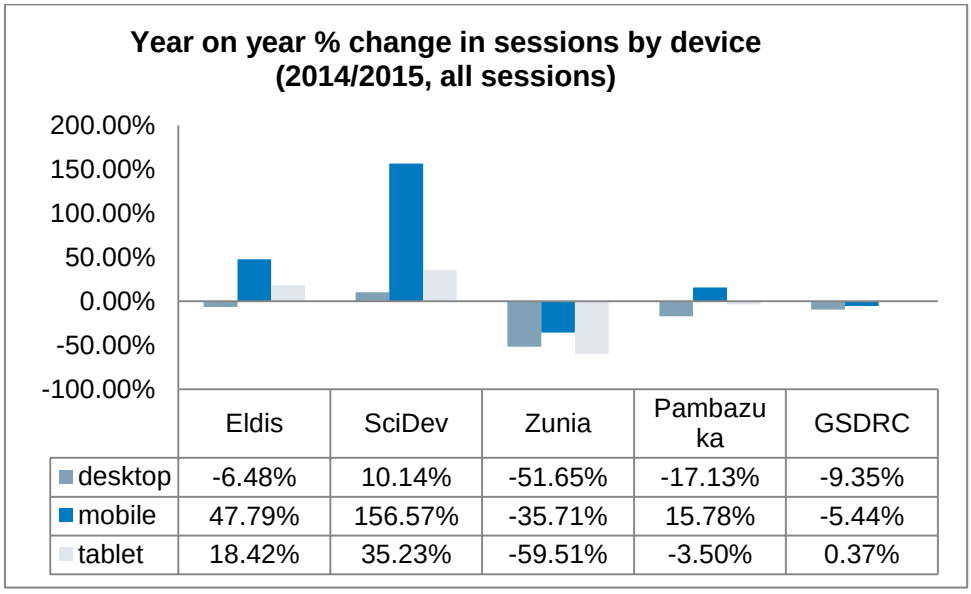






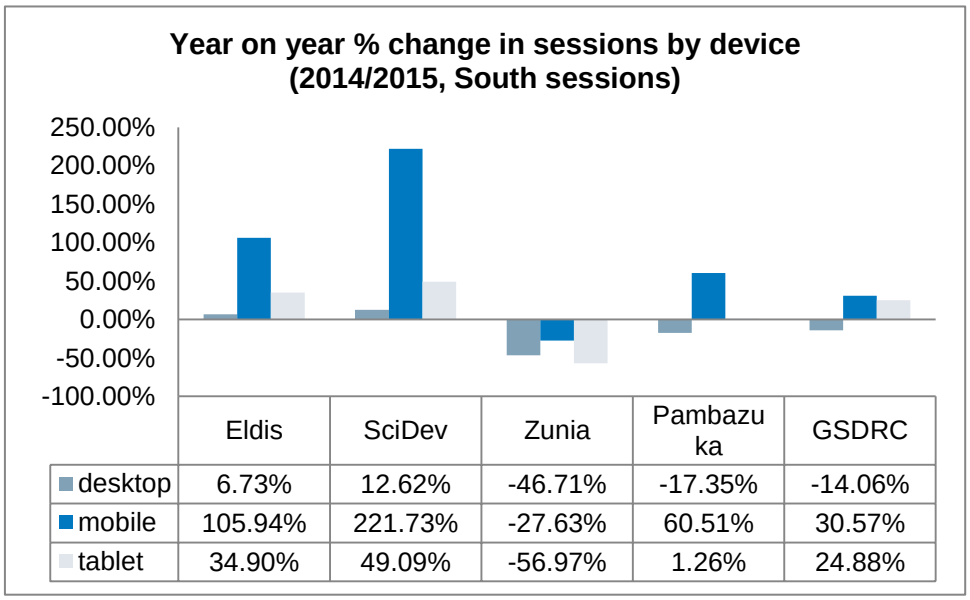


- Year on year trends are in line with what was observed earlier for Eldis and SciDev.Net. Besides Zunia, the other two websites show an increase (even if minimal) in mobile or tablet traffic.



	Eldis	SciDev	Zunia	Pambazuka	GSDRC
desktop	-6.48%	10.14%	-51.65%	-17.13%	-9.35%
mobile	47.79%	156.57%	-35.71%	15.78%	-5.44%
tablet	18.42%	35.23%	-59.51%	-3.50%	0.37%

The trend is more evident when looking at Southern traffic alone. For this segment, mobile and tablet traffic to Pambazuka and GSDRC increases between 60% and 54% between 2014 and 2015.



	Eldis	SciDev	Zunia	Pambazuka	GSDRC
desktop	6.73%	12.62%	-46.71%	-17.35%	-14.06%
mobile	105.94%	221.73%	-27.63%	60.51%	30.57%
tablet	34.90%	49.09%	-56.97%	1.26%	24.88%

D.5.4 Site speed

Table 64 Comparator portal site speed

	Date Range	Zunia	Pambazuka	GSDRC
Avg. Page Load Time (sec)	Jan 1, 2015 - Dec 31, 2015	13.03	11.35	8.58
	Jan 1, 2014 - Dec 31, 2014	11.46	11.01	6.7
		13.70%	3.09%	28.06%
Tanzania Avg.	Jan 1, 2015 - Dec 31, 2015	49.16	29.11	21.81
	Jan 1, 2014 - Dec 31, 2014	20.24	30.45	9.79
		142.90%	4.41%	122.81%
Ghana Avg.	Jan 1, 2015 - Dec 31, 2015	50.56	28.86	7.47
	Jan 1, 2014 - Dec 31, 2014	36.68	58.43	13
		37.86%	50.61%	-42.56%
Nigeria Avg.	Jan 1, 2015 - Dec 31, 2015	40.54	16.23	19.28
	Jan 1, 2014 - Dec 31, 2014	25.44	24.13	22.88
		59.35%	-32.75%	-15.73%

D.5.5 Site search

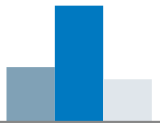
- Amongst the comparators, only GSDRC appears to register sessions with site search. The percentages of sessions with search remain low for the aggregated traffic, as well as for new and returning visitors. Returning visitors use the search function more frequently than first time visitors.

Site search (2015, all sessions)



	Eldis	SciDev	GSDRC
All Sessions	3.32%	0.70%	0.36%
Returning Visitor	8.16%	1.27%	0.80%
New Visitor	2.50%	0.40%	0.17%

Site search (2015, South sessions)



	Eldis	SciDev	GSDRC
All Sessions	2.46%	0.47%	0.22%
Returning Visitor	5.22%	1.06%	0.50%
New Visitor	1.91%	0.27%	0.15%

Volume Two: Appendices

Appendix E. Glossary

Table 65: Glossary

TERM	DEFINITION	SOURCE
Access	The opportunity to use the resources that are available. It depends on personal search and discovery skills, presence of alternative research sources, e.g. research assistants and librarians, as well as the design of the interface with the online resources.	K. Wella & J Harle, ARCADIA Availability, access and use: re-understanding the e-journal problem?
Availability	The existence of the technology, connectivity and online resources. This depends on bandwidth, the quality of the IT, financial resources for online subscriptions, etc.	"K. Wella & J Harle, ARCADIA Availability, access and use: re-understanding the e-journal problem?
Awareness	The knowledge of the resources that are available.	"K. Wella & J Harle, ARCADIA Availability, access and use: re-understanding the e-journal problem?
Discoverability	In internet terms, the ability of services to be found *when needed* (as opposed to more active marketing which aims to distract and disrupt).	www.techopedia.com
Information behaviour	A range of activities or processes that include accidentally encountering, needing, finding, foraging, choosing, organising, sharing, using and avoiding information. Information behaviour encompasses purposive behaviours such as information seeking and passive or unintentional behaviours (including passive searching and passive listening).	Looking for Information: A Survey of Research on Information Seeking, Needs and Behaviour, Donald O. Case, 2012. New Directions in Information Behaviour, ed. Amanda Spink and J Heinstrom, 2011
Information literacy	Knowing when you need information, and are then able to identify, locate, evaluate, organise and effectively use the information to address and help resolve personal, job related, or broader social issues and problems.	UNESCO, US National Commission on Libraries and Information Science et al Goals, objectives and participant responsibilities. Meeting of experts on information literacy 2002.
Information need	A recognition that your knowledge is inadequate to satisfy a goal that you have. Information needs can include: the need for new information to form an opinion, to discover what is happening, or to build knowledge of a subject; or the need for information to confirm information or beliefs already held.	Looking for Information: A Survey of Research on Information Seeking, Needs and Behaviour, Donald O. Case, 2012. Wilson T, Walsh C. Information behaviour: an inter-disciplinary perspective. British Library Research and Innovation Report 10. London: British Library Research and Innovation Centre, 1996;
Information seeking	The conscious effort to acquire information in response to having identified a need or gap in one's knowledge e.g. through active searching or ongoing searching.	Looking for Information: A Survey of Research on Information Seeking, Needs and Behaviour, Donald O. Case, 2012.
Information browsing	Scanning heading and lists on a website and following one interesting link to another, possibly with an objective but without a planned search strategy.	http://www.businessdictionary.com/
Information encountering	'A memorable experience of unexpected discovery of useful or interesting information.	S. Erdelez & K.Rioux, Towards understanding Information Encountering on the Web, 2000

TERM	DEFINITION	SOURCE
North and Northern	Referring to the set of countries categorised as high income by the World Bank.	
Evidence	Credible and verifiable information which you use in making judgements and decisions. It comes from a wide range of sources, including other practitioners, subject experts and policy communities, as well as peer-reviewed journal articles, statistical databases or evaluation reports.	Evaluation team definition used in market research
Online research portals and repositories	Online portals and repositories that make international development research findings available.	DFID ToRs for Evaluation of online research portals and repositories (Final version 5.1.15)
Policy actors	The people who influence the shaping and implementing of policy. They are not responsible for taking substantive decisions, but contribute indirectly by generating and promulgating research and evidence.	L. Shaxson: Developing a strategy for knowledge translation and brokering in public policymaking, 2010
Policy makers	The people who take substantive decisions about how a policy is shaped and implemented. Depending on the type of policy being developed, policy makers are not only senior officials in central line Ministries; they include the network of people and organizations involved in crafting and delivering the policy throughout its lifetime. Policy makers are thus a sub-set of policy actors.	L. Shaxson: Developing a strategy for knowledge translation and brokering in public policymaking, 2010
(Web) Portal	An all-in-one Web site used to find and to gain access to other sites, but also one that provides the services of a guide that can help protect the user from the chaos of the Internet and direct them towards an eventual goal.	Web Portals: The New Gateways to Internet Information and Services, ed. Arthur Tatnall,
Research evidence	A type of evidence which comes from a more limited range of sources, such as peer-reviewed journal articles, statistical databases or evaluation reports. Research evidence is evidence which is systematically obtained and usually quality assured in some way.	Evaluation team definition used in market research
South (and Southern)	Referring to the set of countries categorised as low and middle income by the World Bank.	GDNet Year 3 M&E Report, Gregorowski et al, 2014
Uptake	Findings being applied in international development decision-making, such as by policy-makers or practitioners.	DFID ToRs for Evaluation of online research portals and repositories (Final version 5.1.15)
Use	Reading, downloading, sharing of portal services or material found on the web.	DFID ToRs for Evaluation of online research portals and repositories (Final version 5.1.15)
Intended users	(In general) Target users of the portals. DFID's target population in this instance is all policy actors. (In relation to the Market Research) Participants that fell into one of the 8 priority categories.	
Primary Intended Users	A term specifically used to describe participants in the Market Research that fell into one of the priority category groups that received a minimum of 50 responses.	
Actual users	Actual/current users of online research resources and evidence.	
Potential users	Intended users who are currently not users of online research resources and evidence.	

Appendix F. Terms of Reference

Section 3: Terms of Reference[23]

A. Purpose, Scope and Approach

The primary purpose of this evaluation is to "evaluate the success and value for money of DFID-funded work to communicate research online". For example, the findings in relation to Eldis will provide evidence on the performance of the Global Open Knowledge Hub programme at outcome level. The findings will also inform DFID decision-making about its future portfolio.

The secondary purpose of this evaluation is to "strengthen the evidence base that supports research communication in the interests of international development". The aim, therefore, is to investigate robustly the information needs and information seeking behaviour on the internet of development decision-makers and related actors, as well as the extent to which needs are met by current portals and repositories.

The supplier will carry out a high quality, broad and deep evaluation of online research communication. This will encompass current and recent online programmes funded by DFID and others as well as the existing body of evidence and, selectively, other portals/repositories.

The scope comprises, mainly, four portals and repositories supported by DFID: do they make research available and accessible in a cost-effective way and have impacts on policy and practice decision-making? The anticipated impact, outcomes and outputs, taken from the programmes' logical framework, are listed below, along with relevant indicators. However, the evaluation will not focus solely or indeed primarily on these but encompass meaningful outcomes and impacts more broadly - positive and negative, intended and unintended.

1) **Eldis** portal which is funded through the Global Open Knowledge Hub programme managed by the Institute for Development Studies

ELDIS log frame excerpts		Relevant indicators
Impact	Evidence-information policy making and practice by development actors which will ultimately contribute to improvements in the lives of poor people	

[23] These are the terms of reference as modified for the evaluation contract

Outcome	Content made available via the Hub is accessed and valued by policy makers, practitioners and intermediaries	Indicator 2: Number of visits to Eldis Indicator 4: Eldis valued by users.
Output 1	Development of Open Knowledge Hub, underpinned by open source technology which enables IDS and others to contribute, pull out and repurpose content	
Output 2	IDS GOKH services (BLDS, Eldis, BRIDGE) make research knowledge available by sourcing, synthesising and contributing content to the Hub	Indicator 2: Number of resources made available on Eldis
Output 3	Non-IDS data providers make research knowledge more widely available by contributing content to the Hub, which is then available to all services using the Hub	

2) R4D (Research for Development), DFID's repository for all DFID-funded global public good research.

R4D log frame excerpts		Relevant indicators
Impact	Assist DFID, along with other online services such as .Gov.uk and the Development Tracker, in improving the openness and transparency of DFID funded research	
Outcome	R4D sites, database and services are continually maintained, updated and accessible to all users. R4D makes available open applications, such as the API and R4D open data, to organisations and individuals for use in other websites and/or databases and enable others to develop further applications from them. R4D ensures that DFID's research information and outputs are open to all and continue to enable users in finding, manipulating and understanding the research projects and programmes we fund.	Indicator 1: Access to DFID funded research information by target audiences Indicator 2: Increase in size of R4D database and contains up to date records and information Indicator 3: Website, Database and Platforms (e.g. Linked Development) are maintained and accessible
Output 1	R4D website and database content is accessed/used. Content is accessible to users in a range of forms and formats	Indicator 1: Content is accessed and used (webstats) Indicator 2: Content is accessed and used by users in the North and South Indicator 3: Content is accessible via key search engines/reference services and through feeds
Output 2	Content in the R4D database updated and maintained	Indicator 1: Content updated and added to the R4D database Indicator 2: Content maintained, up to date and accurate Indicator 3: Metatags/metadata cleaned, updated and maintained to improve accuracy of content
Output 3	Website, database, search and open applications maintained and accessible to all users; handover of R4D and involvement in Government Digital Service process	Indicator 1: Website, database and search maintained and accessible Indicator 2: Open data and

	for the new R4D undertaken and completed as specified by DFID	applications (API) maintained and available through R4D

3) **GDNet** portal managed by the Global Development Network

GDNet log frame excerpts		Relevant indicators
Impact	Better policy informed by better research	
Outcome	Diverse research and policy audiences make better use of development research from the global south	Indicator 1: Southern users make use of southern research in their own research Indicator 2: Cases of knowledge–into-use in policy processes in Southern countries
Output 1	Southern research better informed by current ideas and knowledge	Indicator 1: Level of use of, and satisfaction with GDNet research-oriented on-line services Indicator 2: Level of use of, and satisfaction with, themed services
Output 2	Researchers better able to communicate their research to policy	
Output 3	Knowledge networking between researchers and with policy actors Increased	Indicator 1: GDNet user base interaction Indicator 2: Researchers' interactions with the policy domain
Output 4	Lessons about knowledge brokering best practice in the global south learnt and communicated	

4) **SciDev.Net** which communicates journalistic articles about development research

SciDev.Net log frame excerpts		Relevant indicators
Impact	Greater uptake of science and scientific evidence in policies, programmes and projects that reduce poverty, promote well-being and build equitable sustainable economic growth in support of achieving the Millennium Development Goals	
Outcome	Decision makers, science communicators, policy intermediaries and scientific researchers are better able to make use of high quality science-related information	Indicator 1: SciDev.Net readers use science-based information to inform decision-making and development projects Indicator 2: Senior level policy makers and scientists as opinion authors report that there has been an improvement in research networks as well as increased engagement on policy and development as a result of publishing an opinion piece.
Output 1	A well-used, authoritative and responsive service providing timely news and information across stakeholders	Indicator 1: Production of news and features that focus on the role of science in development Indicator 2: Content is well read by global audience Indicator 3: High level of trustworthiness/ authoritiveness of

		SciDev.Net as perceived by its readers Indicator 4: Increase global syndication of SciDev.Net content Indicator 5: Annually introduce an innovation to digital infrastructure to improve reach of content Indicator 6: Increase proportion and numbers of female registrants accessing our content
Output 2	Production of analysis of research findings and tech innovation, exploring socio-economic implications	Indicator 1: Mainstream gender awareness & wellbeing approach in production and delivery Indicator 2: Thematic columns on 5 key topics (gender, private sector, marginalised, disabled and migration) providing news analysis whose readership increases year on year Indicator 3: Number of opinion articles by external contributors - mainly from the developing world
Output 3	Support to capacity building to sustain uptake of research	

Approach

The supplier will conduct the study in two stages. The first stage will both i) undertake a set of discrete tasks, each on a modest scale, in order to produce an early substantive report; and ii) serve as an inception phase for the central and challenging elements of the evaluation to be undertaken in stage two.

Specifically, stage one will include:

- Collation and analysis of available evidence on the use of these portals and others. This evidence comprises data from webmetrics, as well as that from DFID's internal annual reviews, previous external reviews and other literature.
- Review of directly relevant literature.
- Preliminary assessment of the quality, accessibility (to global users) and ease of use of these portals/repositories.
- Interrogation and validation of recent webmetrics data and recommendations on suitable definitions for comparable indicators and the tools to capture these. Note the need for suitable specialist expertise here, due to the rapid current evolution of methods for capturing webmetrics and the high variation in the results produced by different tools.
- Further development of the initial Theory of Change (ToC), drawing in the evidence found in the literature element of this stage as well as through stakeholder engagement.
- Drafting of a substantive report drawing together these elements. This report is intended to be a synthesis/review of currently available evidence around the issues outlined in the

provisional evaluation questions below. In addition to synthesising available evidence, the report should identify where the main gaps in the evidence lie.

- Production of a separate inception report confirming proposals for the second stage of the evaluation, addressing core questions on user populations, uses, impacts and value for money. The inception report should set out detailed methods proposed to fill the gaps identified in the substantive report.

- Secondary evidence synthesis and analysis should be conducted in line with DFID's guidance on "Assessing the Strength of Evidence" (2014).

B. Evaluation Objectives and Questions

The objectives of this evaluation are:

1. To assess the quality and accessibility of online research portals and repositories and to collate and analyse the available evidence on their use.
2. To describe user populations and examine how they interact with online research portals and repositories.
3. To identify and illustrate plausible pathways between portal use and use of evidence in policy and practice. This will involve consideration of what users do with the content once they have read it online or downloaded it. This objective primarily relates to the DFID-funded portals and repositories, but evaluators may choose to also consider those not funded by DFID as this will help to show how DFID-funded work could be improved.
4. To assess whether the DFID-funded portals and repositories present value for money, in their own right and in relation to portals and repositories not funded by DFID. DFID's approach to value for money is outlined in published guidance.[24]
5. To provide recommendations for how the DFID-funded programmes might be improved and better monitored.

Suggested evaluation questions in relation to the objectives included:

Q1.1 How well designed and structured are portals and repositories to promote ease of access and use from locations worldwide, especially those in the global South?

Q1.2 What is known of the current levels of access to and use of portals and repositories; what are the trends and reasonable expectations?

[24] https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/67479/DFID-approach-value-money.PDF

Q1.3 What is the wider evidence in relation to accessibility, potential use and impacts on policy and practice decision-making?

Q2.1. How do potential target populations use the internet to access research information? For example:

- What proportion regularly uses the internet to find research information? What are the characteristics of users? What sites do they use to access information?
- How do they decide which portal or repository to use? To what extent do they consider the quality (research rigour) of the content they find?
- What proportion has heard of each of the DFID-funded portals and repositories under investigation? How did they hear about the research portals / repositories?
- Do they access the portal / repository through an online search facility (e.g. Google or PubMed), and/or do they go directly to them to run the search for evidence, and/or do they subscribe to "push" services such as email updates and feeds?
- What proportion regularly uses each of the portals / repositories under investigation?
- Of those who use the DFID-funded portals / repositories, what was the purpose of their recent use?
- Does the presence of interactive functions (e.g. debates, ability to comment) add value to searchable collections of information?
- If online portals and repositories did not exist, what would be the challenges to their work?
- To what extent do the potential target populations engage with portals and repositories differently? Does use of portals and repositories vary according to the development sector they work in?
- How does use of portals and repositories differ according to gender and disability?

Q2.2. What is the ease or difficulty of use of specific portals and repositories (including those funded by DFID). For example:

- Finding a specific article using each portal / repository?
- Finding adequate information/evidence on a specified topic using each portal / repository?
- Downloading information from each portal / repository?

Q2.3. How do users rate the sufficiency and quality of the research evidence which they are able to access through portals / repositories?

Q2.4. What are the challenges of Southern users accessing portals / repositories and how might these be addressed by the services?

Q3.1. To what extent do online research portals and repositories – collectively and individually - contribute to research uptake?

Q3.2. In what ways do online research portals and repositories enable research uptake that other methods of communicating research cannot? Do portals play any role in enabling users to access a body of evidence to inform their work?

Q3.3. What impacts (positive and negative) do they have, if any, and how and why do these materialise?

Q4.1. At what rates are the portals and repositories accessed relative to their costs? Does profiling of the content (e.g. summary; news story) increase use compared to having the research outputs in a repository with a search facility? Given changing usage of the internet, is it appropriate to expect that the number of visits will increase in the short and/or medium term?

Q4.2. To what extent do the portals / repositories under investigation duplicate content and to what degree is duplication useful and/or wasteful?

Q4.3. What technologies have been adopted over the last two years that have improved the service that portals / repositories provide? What technologies are emerging that may present opportunities over the next 2-5 years? How is use of these technologies affected by gender and disability?

Q4.4. Overall, what value for money does each of the DFID-funded portals and repositories present? How does this compare with other portals and repositories?

Q5.1 What are the recommendations of the evaluation team in respect of:

- Improving design and access
- Improving content and use
- Enhancing impacts on decision-making in policy and practice
- Raising value for money
- A suitable suite of monitoring statistics (webmetrics). This may include but may not be limited to cost per visitor, cost per click, cost per download, percentage of users reporting/demonstrating reading the content, percentage of users reporting/demonstrating impact on policy, programming or practice. In order to make this assessment, portals and repositories not funded by DFID will be considered.

These evaluation questions are indicative of the areas in which DFID had identified information demands. They were used in stage 1 to guide the search for what evidence currently exists. In light of the quality and quantity of evidence found and discussion between the evaluation team and the management group during stage 1, a revised set of evaluation questions has been proposed and is included as Annex X in the inception report.

B. Design and Methodology

A key challenge recognised by DFID is that the portals and repositories are internet-based with a global audience, and so it is not straightforward to identify and contact users in systematic ways to produce a high quality evaluation. None of the programmes does or can know the identity of most of their users. Nevertheless, DFID are committed to a high level of quality and rigour in the study, in line with international good practice in evaluation.

The Supplier will deliver the Services in accordance with the design and methodology presented in their proposal (Annex A) and as subsequently revised by agreement under the Concept Note (Annex B), and their inception report (Annex C), the latter taking precedent.

For the online survey, as indicated DFID recognises the likely challenges in defining the user populations, compiling one or more sample frames, sampling and securing high response rates. DFID and the Supplier have agreed that the evaluation will conduct market research with target groups with 1000 Intended Users worldwide. DFID have recruited an advisory group of research uptake specialists to identify potential participants from a range of Intended User populations, to advise on maximising response rates and offer other advice to the Supplier and DFID.

To help with consideration of user populations and proposals for compiling sample frames, sampling and securing high response rates (which may involve local, in country co-ordination and approaches), DFID have identified a number of known (intended) user groups of particular interest to us:

- Parliamentary committee clerks, or equivalent
- Parliamentary researchers, or equivalent
- Parliamentary librarians, or equivalent
- MPs' assistants, or equivalent
- Civil servants (probably more senior, such as top third of the salary scale, and recognising key ministries for potential research use, such as health, education, finance, agriculture, ministry with responsibility for civil service)
- Ministerial advisors
- Government Ministers

- Members of parliament, or equivalent
- Members of regional (sub-national) legislatures/councils
- Councillors/local council members
- Senior advisors in intergovernmental organisations (probably stratified by grade)
- Staff of think tanks (both southern and northern)
- Staff of NGOs (both southern and northern)
- Community based organisations (CBOs)
- Journalists
- Knowledge intermediaries
- Researchers/academics
- Students

- The design for stages 1 and 2 have addressed well known challenges of evaluating the communication and use of research information. These challenges include:

- Contribution/attribution: uptake of evidence is unlikely to be the only factor impacting on the changes observed. There is a lot of potential for over- or under-estimating the influence of research evidence.
- Context: the evaluation will need to draw lessons from across a wide range of countries and contexts. For this evaluation, there are particular challenges in measuring the use, value and impact of online portals and repositories due to the nature and constituency of the internet. During inception the supplier will provide a credible approach to generalisation.
- Time-lags: The duration between initial access to evidence on a portal / repository, to making some use of it and then to potential impacts on decision-making could well be lengthy.

The evaluation will focus on the communication of research evidence in a broad sense, i.e. published academic research papers; statistical databases; “established” i.e. widely debated and accepted, policy papers and positions; and evaluation findings. It is not expected systematically to encompass experiential evidence (i.e. evidence based on professional insight, skills or experience) or contextual evidence (e.g. evidence based on likely uptake or impact within a given community), though clearly these may be of relevance where portals and repositories contain such evidence, or users access other sites for these types of evidence.

Sources

Sources of data that will be used in the evaluation will, at a minimum, include:

- **Background documentation:** including DFID-funded online portal / repository business cases and the programme’s proposals; annual reports and reviews.

- **Literature:** a document review and analysis of existing evidence and analytical frameworks relating to research communication and online portals / repositories including research/evaluations carried out in relation to international development and where appropriate, evidence generated by the private sector.
- **Primary data gathered by the DFID-funded programmes to be evaluated:** including webstats and other data from the programmes' monitoring frameworks and management systems.
- **Secondary data**, being relevant data from other sources.
- **Primary data gathered by the Evaluation team:** e.g. qualitative interviews with users, intermediaries and researchers, surveys and/or other data collection methods.

In the inception report the supplier sets out the different data sources they expect to collect and use – including types of primary data – and what weighting they plan to attribute to these data when forming their evaluation conclusions.

DFID-funded online research portal / repository programmes are aware of DFID's plans for independent external evaluation. Good levels of participation can be anticipated with regard to reasonable requests to support the evaluation.

C. Timing and Scope

The evaluation will consist of a three-month inception phase (stage one), and a nine-month implementation phase (stage 2).

- DFID reserves the option to scale back funding depending on performance and on-going need. Specifically, the study will proceed to stage two only upon satisfactory completion of stage one (as advised by DFID's external quality assurance service) and with our agreement that proposals for stage two are both technically sound and represent good value for money.

The supplier will be responsible for arranging their own logistical arrangements. However, DFID, the DFID-funded portals and repositories, and the advisory group will provide some support with identifying and contacting key contacts.

DFID envisage that a small number of non-DFID-funded portals / repositories will be included in this evaluation. Suggestions of which portals / repositories to investigate have been discussed during inception.

The evaluation will focus on research communication for the benefit of poverty reduction in low- and middle-income countries.

C. Outputs

The Evaluation team will produce the following outputs:

- **A stage one substantive report (for publication)** incorporating an overarching theory of change, findings and any initial recommendations, together with methodological/statistical appendices.
- **An inception Report focusing on the design on the second stage (for external quality assurance)**. This should include refinements/amendments of evaluation questions and full methodology; identified sources of data and risk management strategy; and communications/dissemination/uptake strategy.
- **Interim report of stage 2,** around November2015, containing as much data as possible to assist DFID decision-making regarding future programming.
- **Draft final report** in January 2016. This report will be commented on by DFID, with areas for revision and further analysis highlighted. The first or second draft (at DFID discretion) will be externally quality-assured.
- **Final evaluation report** for publication agreed in March 2016 comprising the full report (maximum of 150 pages with a maximum six page Executive Summary) that incorporates feedback obtained on the draft report.
- Appendices with details on the methodology, informants, etc.

D. Audience and Management Arrangements

The direct recipients of the services will be DFID's Research and Evidence Division. The published reports are expected to be of value to other donors and practitioners in the research uptake community, including those that support online portals and repositories such as Sida, the World Bank, Irish Aid and IDRC.

The evaluation will be overseen by a Management Group, who will be responsible for approving the evaluation outputs and commenting on draft reports. The management group shall comprise of:

- Rachel McIntosh and Tarah Friend from DFID's Evidence into Action team

- A DFID evaluation adviser and/or research specialist not directly involved in the DFID programmes being evaluated
- An external representative of a joint funder of one of the DFID-funded programmes.

The day-to-day point of contact for the evaluation will be Tarah Friend.

The supplier will meet quarterly with the management group, and will include up to three presentations (including one to present jointly the first substantive and inception reports; and one to present the draft final report). These meetings will take place in London, though it will be possible for more distant, key members of the evaluation team to participate by video-conference.

E. The Evaluation team

As these ToR make clear, a range of expertise will be required to deliver this evaluation. Multi-disciplinarity and innovative perspectives are encouraged, provided that strong core evaluation skills are also incorporated. As securing southern perspectives is vital to this evaluation, DFID would welcome southern participation in the evaluation team.

The supplier will design, co-ordinate and draw together the evaluation findings. They will have sole responsibility for validating the data collected and ensuring an initial level of quality assurance of all outputs.

The DFID-funded online research portal programmes will seek to facilitate access to stakeholders who have direct links with the programmes, but the evaluation team will have to make direct approaches to other stakeholders and beneficiaries who are in scope of their evaluation design.

Skills and qualifications

The essential competencies and experience that the evaluation team will need in order to deliver the work are:

- Extensive knowledge and application of evaluation methods and techniques
- Strong qualitative and quantitative research skills
- Skills in measuring the impacts of research
- Expertise in internet portals including webmetrics and Web 2.0 technologies and a good understanding of relevant social media
- Strong analysis, report writing and communication skills.

Highly desirable competencies and experience are:

- A good understanding of research communication

- Relevant experience in the global South
- Expertise in assessing value for money.

For visits to case study countries, the supplier must complete the duty of care requirements.

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